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MILITARY HANDBOOK

POLYMER MATRIX COMPOSITES

VOLUME III. UTILIZATION OF DATA



AMSC N/A **AREA CMPS**

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DEPARTMENT OF DEFENSE
WASHINGTON, D.C. 20025

1. *This Military Handbook is approved for use by all Departments and Agencies of the Department of* Defense.

2. Every effort has been made to reflect the latest information on polymeric composites. The Handbook is continually reviewed and revised to insure its completeness and currentness. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to : Department of the Army, U.S. Army Research Laboratory, Materials Directorate, Attn: AMSRL-MA-TS, MIL-HDBK-17 Coordinator, Watertown, MA 02172-0001, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

3. Military Handbook 17 (Part 1) provides guidelines and material properties for polymer (organic) matrix composite materials. This handbook encompasses, but is not limited to, polymeric composites intended for aircraft and aerospace vehicles and military combat vehicle applications. Metal matrix composites (MMC) , ceramic matrix composites (CMC), and carbon/carbon composites (C/C) will be covered in separate military handbooks as developments occur.

4. This standardization handbook has been developed and is being maintained as a joint effort of the Department of Defense and the Federal Aviation Administration.

5. The information contained in this handbook was obtained from materials producers, industry, reports on Government sponsored research, the open literature, and by contact with research laboratories and those who participate in the MIL-HDBK-17 coordination activity.

6. All information and data contained in this handbook have been coordinated with industry and the US Army, Navy, Air Force, NASA, and Federal Aviation Administration prior to publication.

7. Copies of this document and revisions thereto may be obtained from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.

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SUMMARY OF CHANGES IN MIL-HDBK-17-3D

Chapter 1

Figure 1.3.1 on mechanical property notation was added.

In Section 1.4, a definition for traveller was added. The definitions for polymerization, addition polymerization, chain-growth polymerization, and step-growth polymerization were revised.

Chapter 2

Section 2.2.1.4, Silicones [matrix materials], was added.

Section 2.3.3, Glass [fibers], was added.

Section 2.3.6, Silicon carbide [fibers], was revised. Sections 2.3.6.1 and 2.3.6.2 were combined as Section 2.3.6, and six paragraphs and two tables were appended.

Chapter 3

In Table 3.2.1, Acceptance and revalidation tests required for suppliers and users, Note 1 was revised to apply more universally.

Chapter 4

A number of changes were made to Chapter 4 to incorporate stacking sequence effects information.

1. The last two paragraphs of Section 4.3.2, Lamination theory, were replaced with a paragraph that enumerates observations on stacking sequence effects.
2. Six paragraphs were appended to Section 4.3.3.2, Bending.
3. Section 4.3.4.1, Symmetric laminates, and Section 4.3.4.2, Unsymmetric laminates, were appended to Section 4.3.4, Thermal and Hygroscopic Analysis.
4. Five paragraphs were appended to Section 4.4.3, Strength.
5. One paragraph was appended to Section 4.4.4, Stress Concentrations.
6. Section 4.4.5, Delaminations, was added.
7. Section 4.4.6, Damage and Failure Modes, was added.
8. Section 4.6.5, Laminate Stacking Sequence Effects, was added.
9. Section 4.7.1.9, Stacking Sequence Effects in Buckling, was added.
10. Six paragraphs and two figures were appended to Section 4.11.1, Damage Tolerance.
11. Section 4.13, Vibration, was added to the handbook outline. Section 4.13.2, Stacking Sequence Effects [Vibration], was added to the handbook.

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Section 4.11.1.3, Evolving Military and FAA Specifications [Damage Tolerance] was revised. The next to last sentence in the first paragraph was added.

Section 4.11.3, Damage resistance, was added.

Section 4.12, Critical fluids, sensitivity and evaluation, was revised to add information on bismaleimides and polyimides, to add Decontamination solution to the Group two fluid exposure classification, and to clarify the exposure procedures.

Section 4.14, Computer Programs, was added.

Chapter 7

Section 7.2.4, Theoretical Property Determination, was added.

Chapter 8

Section 8.1, Introduction, was added.

Section 8.2, Design for Repair, was added.

Section 8.3, Design of Repairs, was added.

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Chapter 1. GENERAL INFORMATION

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1.1 INTRODUCTION. The standardization of a statistically-based mechanical property data base, procedures used, and overall material guidelines for characterization of composite material systems is recognized as being beneficial to both manufacturers and government agencies. A complete characterization of the capabilities of any engineering material system depends on the inherent material physical and chemical composition which are independent of specific applications. Therefore, at the material system characterization level, the data and guidelines contained in this handbook apply to military and commercial products and provide the technical basis for establishing statistically valid design values acceptable to certificating or procuring agencies.

This standardization handbook has been developed and is maintained as a joint effort of the Department of Defense and the Federal Aviation Administration. It is oriented toward the standardization of methods used to develop and analyze mechanical property data on current and emerging composite materials.

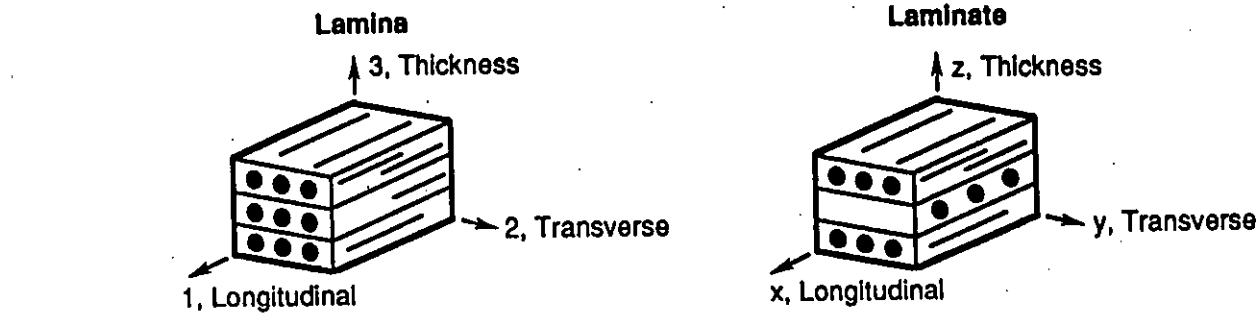
1.2 PURPOSE, SCOPE, AND ORGANIZATION OF VOLUME III. The purpose of this handbook is to provide a standard source of statistically-based mechanical property data for current and emerging composite materials. Volume III provides guidelines for the application of the data which are presented in Volume II of MIL-HDBK-17. Chapter 2 addresses typical composite materials and processes. The matrix materials and reinforcement fibers of interest in MIL-HDBK-17 are reviewed as are the typical product forms. Fabrication methods are also covered for information purposes. Chapter 3 reviews important issues related to quality control for the production of composite materials. Recommended manufacturing inspection procedures are reviewed, along with techniques for material property verification and statistical quality control. Chapter 4 addresses the design and analysis of composite material systems. It provides an overview of the current techniques and describes how the various constituent properties reported in MIL-HDBK-17 are used in the design and analysis of a composite material system.

1.3 SYMBOLS, ABBREVIATIONS, AND SYSTEMS OF UNITS. This section defines the symbols and abbreviations which are used within MIL-HDBK-17 and describes the system of units which is maintained. Common usage is maintained where possible. References 1.3(a), 1.3(b), and 1.3(c) served as primary sources for this information.

1.3.1 Symbols and abbreviations. The symbols and abbreviations used in this document are defined in this section with the exception of statistical symbols. These latter symbols are defined in Chapter 8. The lamina/laminate coordinate axes used for all properties and a summary of the mechanical property notation are shown in Figure 1.3.1.

- The symbols f and m, when used as either subscripts or superscripts, always denote fiber and matrix, respectively.
- The type of stress (e.g., cy - compression yield) is always used in the superscript position.
- Direction indicators (e.g., x, y, z, 1, 2, 3, etc.) are always used in the subscript position.
- Ordinal indicators of laminae sequence (e.g., 1, 2, 3, etc.) are used in the superscript position and must be parenthesized to distinguish them from mathematical exponents.
- Other indicators may be used in either subscript or superscript position, as appropriate for clarity.
- Compound symbols (i.e., basic symbol plus indicators) which deviate from these rules are shown in their specific form in the following list.

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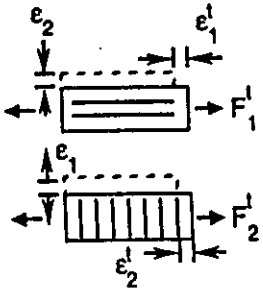
Notation = H^{ik}_{j}

Where,

H = { σ, τ; Applied Normal, Shear Stress
F; Allowable Stress
ε, γ; Extensional, Shear Strain
E, G; Young's, Shear Modulus
ν; Poisson's Ratio

Note: ν^{l}_{12} = Major Poisson's Ratio = $-\frac{\varepsilon_2}{\varepsilon^{l}_1}$

ν^{l}_{21} = Minor Poisson's Ratio = $-\frac{\varepsilon_1}{\varepsilon^{l}_2}$



i = { 1; Longitudinal
2; Transverse
3; Thickness
12, 13, 32; Shear, Poisson's } Lamina
x; Longitudinal
y; Transverse
z; Thickness
xy, xz, zy; Shear, Poisson's } Laminate

j = { c; Compression
t; Tension
s; Shear

k = { y; Yield
u; Ultimate, Not Used for Stiffness

Examples, F^{tu}_{2} = Lamina Ultimate Transverse Tensile Allowable Stress

E^{c}_{z} = Laminate Compressive Young's Modulus, Thickness Direction

FIGURE 1.3.1 *Mechanical property notation.*

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The following general symbols and abbreviations are considered standard for use in MIL-HDBK-17. Where exceptions are made, they are noted in the text and tables.

A - (1) area (m^2,in^2)
- (2) ratio of alternating stress to mean stress
- (3) A-basis for mechanical property values

ASTM - American Society for Testing and Materials

a - (1) length dimension (mm,in)
- (2) acceleration (m/sec^2,ft/sec^2)
- (3) amplitude
- (4) crack or flaw dimension (mm,in)

B - (1) B-basis for mechanical property values
- (2) biaxial ratio

Btu - British thermal unit(s)

b - width dimension (mm,in), e.g. the width of a bearing or compression panel normal to load, or breadth of beam cross-section

C - (1) specific heat (kJ/kg °C,BTU/lb °F)
- (2) Celsius

CF - centrifugal force (N,lbf)

CG - (1) center of mass, "center of gravity"
- (2) area or volume centroid

℄ - centerline

c - column buckling end-fixity coefficient

$\bar{c}$ - honeycomb sandwich core depth (mm,in)

cpm - cycles per minute

D - (1) diameter (mm,in)
- (2) hole or fastener diameter (mm,in)
- (3) plate stiffness (N-m,lbf-in)

d - mathematical operator denoting differential

E - modulus of elasticity in tension, average ratio of stress to strain for stress below proportional limit (GPa,Msi)

E′ - storage modulus (GPa,Msi)

E″ - loss modulus (GPa,Msi)

E_c - modulus of elasticity in compression, average ratio of stress to strain for stress below proportional limit (GPa,Msi)

E_o' - modulus of elasticity of honeycomb core normal to sandwich plane (GPa,Msi)

E^{sec} - secant modulus (GPa,Msi)

E^{tan} - tangent modulus (GPa,Msi)

e - minimum distance from a hole center to the edge of the sheet (mm,in)

e/D - ratio of edge distance to hole diameter (bearing strength)

F - (1) stress (MPa,ksi)
- (2) Fahrenheit

F^b - bending stress (MPa,ksi)

F^{ccr} - crushing or crippling stress (upper limit of column stress for failure) (MPa,ksi)

F^{su} - ultimate stress in pure shear (this value represents the average shear stress over the cross-section) (MPa,ksi)

f - (1) internal (or calculated) stress (MPa,ksi)
- (2) stress applied to the gross flawed section (MPa,ksi)
- (3) creep stress (MPa,ksi)

f^c - internal (or calculated) compressive stress (MPa,ksi)

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f_c - (1) maximum stress at fracture (MPa,ksi)
- (2) gross stress limit (for screening elastic fracture data (MPa,ksi)

ft - foot, feet

G - modulus of rigidity (shear modulus) (GPa,Msi)

GPa - gigapascal(s)

g - (1) gram(s)
- (2) acceleration due to gravity (m/s^2,ft/s^2)

H/C - honeycomb (sandwich)

h - height dimension (mm,in) e.g. the height of a beam cross-section

hr - hour(s)

I - area moment of inertia (mm^4,in^4)

i - slope (due to bending) of neutral plane in a beam, in radians

in. - inch(es)

J - (1) torsion constant (= I_p for round tubes) (m^4,in^4)
- (2) Joule

K - (1) Kelvin
- (2) stress intensity factor (MPa$\sqrt{m}$,ksi$\sqrt{in}$)
- (3) coefficient of thermal conductivity (W/m °C, BTU/ft^2/hr/in/°F)
- (4) correction factor
- (5) dielectric constant

K_{app} - apparent plane strain fracture toughness or residual strength (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_c - critical plane strain fracture toughness, a measure of fracture toughness at point of crack growth instability (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_{Ic} - plane strain fracture toughness (MPa$\sqrt{m}$,ksi$\sqrt{in}$)

K_N - empirically calculated fatigue notch factor

K_s - plate or cylinder shear buckling coefficient

K_t - (1) theoretical elastic stress concentration factor
- (2) t_w/c ratio in H/C sandwich

K_x,K_y - plate or cylinder compression buckling coefficient

k - strain at unit stress (m/m,in/in)

L - cylinder, beam, or column length (mm,in)

L' - effective column length (mm,in)

lb - pound

M - applied moment or couple (N-m,in-lbf)

Mg - megagram(s)

MPa - megapascal(s)

MS - military standard

M.S. - margin of safety

MW - molecular weight

MWD - molecular weight distribution

m - (1) mass (kg,lb)
- (2) number of half wave lengths
- (3) metre
- (4) slope

N - (1) number of fatigue cycles to failure
- (2) number of laminae in a laminate
- (3) distributed in-plane forces on a panel (lbf/in)
- (4) Newton
- (5) normalized

NA - neutral axis

NAS - National Aerospace Standard

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n - (1) number of times in a set
- (2) number of half or total wavelengths
- (3) number of fatigue cycles endured

P - (1) applied load (N,lbf)
- (2) exposure parameter
- (3) probability

P^u - test ultimate load, (N,lb per fastener)

P^y - test yield load, (N,lb per fastener)

p - normal pressure (Pa,psi)

psi - pounds per square inch

Q - area static moment of a cross-section (mm^3,in^3)

q - shear flow (N/m,lbf/in)

R - (1) algebraic ratio of minimum load to maximum load in cyclic loading
- (2) reduced ratio

RA - reduction of area

R.H. - relative humidity

RMS - root-mean-square

RT - room temperature

r - (1) radius (mm,in)
- (2) root radius (mm,in)
- (3) reduced ratio (regression analysis)

S - (1) shear force (N,lbf)
- (2) nominal stress in fatigue (MPa,ksi)
- (3) S-basis for mechanical property values

S_a - stress amplitude in fatigue (MPa,ksi)

S_e - fatigue limit (MPa,ksi)

S_m - mean stress in fatigue (MPa,ksi)

S_{max} - highest algebraic value of stress in the stress cycle (MPa,ksi)

S_{min} - lowest algebraic value of stress in the stress cycle (MPa,ksi)

S_R - algebraic difference between the minimum and maximum stresses in one cycle (MPa,ksi)

S.F. - safety factor

s - (1) arc length (mm,in)
- (2) H/C sandwich cell size (mm,in)

T - (1) temperature (°C,°F)
- (2) applied torsional moment (N-m,in-lbf)

T_d - thermal decomposition temperature (°C,°F)

T_F - exposure temperature (°C,°F)

T_g - glass transition temperature (°C,°F)

T_m - melting temperature (°C,°F)

t - (1) thickness (mm,in)
- (2) exposure time (s)
- (3) elapsed time (s)

V - (1) volume (mm^3,in^3)
- (2) shear force (N,lbf)

W - (1) weight (N,lbf)
- (2) width (mm,in)
- (3) Watt

x - distance along a coordinate axis

Y - nondimensional factor relating component geometry and flaw size

y - (1) deflection (due to bending) of elastic curve of a beam (mm,in)
- (2) distance from neutral axis to given point
- (3) distance along a coordinate axis

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Z - section modulus, I/y (mm^3,in^3)
α - coefficient of thermal expansion (m/m/°C,in/in/°F)
γ - shear strain (m/m,in/in)
Δ - difference (used as prefix to quantitative symbols)
δ - elongation or deflection (mm,in)
ϵ - strain (m/m,in/in)
ϵ - elastic strain (m/m,in/in)
ϵ - plastic strain (m/m,in/in)
η - plasticity reduction factor
$[\eta]$ - intrinsic viscosity
η^* - dynamic complex viscosity
ν - Poisson's ratio
ρ - (1) density (kg/m3,lb/in3)
- (2) radius of gyration (mm,in)
ρ_c' - H/C sandwich core density (kg/m^3,lb/in^3)
Σ - total, summation
σ - standard deviation
σ_{ij}, τ_{ij} - stress in j direction on surface whose outer normal is in i direction (i, j = 1, 2, 3 or x, y, z) (MPa,ksi)
T - applied shear stress (MPa,ksi)
ω - angular velocity (radians/s)
∞ - infinity

1.3.1.1 *Constituent properties.* The following symbols apply specifically to the constituent properties of a typical composite material.

E^f - Young's modulus of filament material (MPa,ksi)
E^m - Young's modulus of matrix material (MPa,ksi)
E_x^g - Young's modulus of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (MPa,ksi)
E_y^g - Young's modulus of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (MPa,ksi)
G^f - shear modulus of filament material (MPa,ksi)
G^m - shear modulus of matrix (MPa,ksi)
G_{xy}^g - shear modulus of impregnated glass scrim cloth (MPa,ksi)
G_{cx}' - shear modulus of sandwich core along X-axis (MPa,ksi)
G_{cy}' - shear modulus of sandwich core along Y axis (MPa,ksi)
ℓ - filament length (mm,in)
α^f - coefficient of thermal expansion for filament material (m/m/°C,in/in/°F)
α^m - coefficient of thermal expansion for matrix material (m/m/°C,in/in/°F)
α_x^g - coefficient of thermal expansion of impregnated glass scrim cloth in the filament direction or in the warp direction of a fabric (m/m/°C,in/in/°F)
α_y^g - coefficient of thermal expansion of impregnated glass scrim cloth transverse to the filament direction or to the warp direction in a fabric (m/m/°C,in/in/°F)
ν^f - Poisson's ratio of filament material
ν^m - Poisson's ratio of matrix material

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ν_{xy}^{g} - glass scrim cloth Poisson's ratio relating to contraction in the transverse (or fill) direction as a result of extension in the longitudinal (or warp) direction

ν_{yx}^{g} - glass scrim cloth Poisson's ratio relating to contraction in the longitudinal (or warp) direction as a result of extension in the transverse (or fill) direction

σ - applied axial stress at a point, as used in micromechanics analysis (MPa,ksi)

τ - applied shear stress at a point, as used in micromechanics analysis (MPa,ksi)

1.3.1.2 *Laminae and laminates.* The following symbols, abbreviations, and notations apply to composite laminae and laminates. At the present time the focus in MIL-HDBK-17 is on laminae properties. However, commonly used nomenclature for both laminae and laminates are included here to avoid potential confusion.

A_{ij} (i,j = 1,2,6) - extensional rigidities (N/m,lbf/in)
B_{ij} (i,j = 1,2,6) - coupling matrix (N,lbf)
C_{ij} (i,j = 1,2,6) - elements of stiffness matrix (Pa,psi)
D_x, D_y - flexural rigidities (N-m,lbf-in)
D_{xy} - twisting rigidity (N-m,lbf-in)
D_{ij} (i,j = 1,2,6) - flexural rigidities (N-m,lbf-in)
E_1 - Young's modulus of lamina parallel to filament or warp direction (GPa,Msi)
E_2 - Young's modulus of lamina transverse to filament or warp direction (GPa,Msi)
E_x - Young's modulus of laminate along x reference axis (GPa,Msi)
E_y - Young's modulus of laminate along y reference axis (GPa,Msi)
G_{12} - shear modulus of lamina in 12 plane (GPa,Msi)
G_{xy} - shear modulus of laminate in xy reference plane (GPa,Msi)
h_i - thickness of ith ply or lamina (mm,in)
M_x, M_y, M_{xy} - bending and twisting moment components (N-m/m, in-lbf/in in plate and shell analysis)
n_f - number of filaments per unit length per lamina
Q_x, Q_y - shear force parallel to z axis of sections of a plate perpendicular to x and y axes, respectively (N/m,lbf/in)
Q_{ij} (i,j = 1,2,6) - reduced stiffness matrix (Pa,psi)
u_x, u_y, u_z - components of the displacement vector (mm,in)
u_x^o, u_y^o, u_z^o - components of the displacement vector at the laminate's midsurface (mm,in)
V_v - void content (% by volume)
V_f - filament content or fiber volume (% by volume)
V_g - glass scrim cloth content (% by volume)
V_m - matrix content (% by volume)
V_x, V_y - edge or support shear force (N/m,lbf/in)
W_f - filament content (% by weight)
W_g - glass scrim cloth content (% by weight)
W_m - matrix content (% by weight)
W_s - weight of laminate per unit surface area (N/m^2,lbf/in^2)
α_1 - lamina coefficient of thermal expansion along 1 axis (m/m/°C,in/in/°F)
α_2 - lamina coefficient of thermal expansion along 2 axis (m/m/°C,in/in/°F)
α_x - laminate coefficient of thermal expansion along general reference x axis (m/m/°C,in/in/°F)
α_y - laminate coefficient of thermal expansion along general reference y axis (m/m/°C,in/in/°F)
α_{xy} - laminate shear distortion coefficient of thermal expansion (m/m/°C,in/in/°F)

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θ - angular orientation of a lamina in a laminate, i.e., angle between 1 and x axes (°)
λ_{xy} - product of ν_{xy} and ν_{yx}
ν_{12} - Poisson's ratio relating contraction in the 2 direction as a result of extension in the 1 direction.
ν_{21} - Poisson's ratio relating contraction in the 1 direction as a result of extension in the 2 direction.
ν_{xy} - Poisson's ratio relating contraction in the y direction as a result of extension in the x direction
ν_{yx} - Poisson's ratio relating contraction in the x direction as a result of extension in the y direction
ρ_c - density of a single lamina (kg/m^3,lb/in^3)
$\overline{\rho}_c$ - density of a laminate (kg/m^3,lb/in^3)
ϕ - (1) general angular coordinate, (°)
- (2) angle between x and load axes in off-axis loading (°)

1.3.1.3 *Subscripts.* The following subscript notations are considered standard in MIL-HDBK-17.

1, 2, 3 - laminae natural orthogonal coordinates (1 is filament or warp direction)
A - axial
a - (1) adhesive
- (2) alternating
app - apparent
c - composite system, specific filament/matrix composition. Composite as a whole, contrasted to individual constituents. Also, sandwich core when used in conjunction with prime (')
- (4) critical
cf - centrifugal force
e - fatigue or endurance
eff - effective
eq - equivalent
f - filament
g - glass scrim cloth
H - hoop
i - i^{th} position in a sequence
L - lateral
m - (1) matrix
- (2) mean
max - maximum
min - minimum
n - (1) n^{th} (last) position in a sequence
- (2) normal
p - polar
s - symmetric
st - stiffener
T - transverse
t - value of parameter at time t
x, y, z - general coordinate system
Σ - total, or summation
o - initial or reference datum

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( ) - format for indicating specific, temperature associated with term in parentheses. RT - room temperature (21°C,70°F); all other temperatures in °F unless specified.

1.3.1.4 *Superscripts.* The following superscript notations are considered standard in MIL-HDBK-17.

b	- bending
br	- bearing
c	- (1) compression
	- (2) creep
cc	- compression crippling
cr	- compression buckling
e	- elastic
f	- filament
g	- glass scrim cloth
is	- interlaminar shear
(i)	- i^{th} ply or lamina
lim	- limit, used to indicate limit loading
m	- matrix
p	- plastic
pl	- proportional limit
rup	- rupture
s	- shear
scr	- shear buckling
sec	- secant (modulus)
T	- temperature or thermal
t	- tension
tan	- tangent (modulus)
u	- ultimate
y	- yield
′	- secondary (modulus), or denotes properties of H/C core when used with subscript c

1.3.2 System of units. To comply with Department of Defense Instructive 5000.2, Part 6, Section M, "Use of the Metric System," dated February 23, 1991, the data in MIL-HDBK-17 are generally presented in both the International System of Units (SI units) and the U. S. Customary (English) system of units. ASTM E-380, Standard for Metric Practice, provides guidance for the application for SI units which are intended as a basis for worldwide standardization of measurement units (Reference 1.3.2(a)). Further guidelines on the use of the SI system of units and conversion factors are contained in the following publications (References 1.3.2(b) - (e)):

(1) DARCOM P 706-470, *Engineering Design Handbook: Metric Conversion Guide*, July 1976.

(2) NBS Special Publication 330, "The International System of Units (SI)," National Bureau of Standards, 1986 edition.

(3) NBS Letter Circular LC 1035, "Units and Systems of Weights and Measures, Their Origin, Development, and Present Status," National Bureau of Standards, November 1985.

(4) NASA Special Publication 7012, "The International System of Units Physical Constants and Conversion Factors", 1964.

English to SI conversion factors pertinent to MIL-HDBK-17 data are contained in Table 1.3.2.

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TABLE 1.3.2 *English to SI conversion factors.*

To convert from	to	Multiply by
Btu (thermochemical)/in^2-s	watt/meter^2 (W/m^2)	1.634 246 E+06
Btu-in/(s-ft^2-°F)	W/(m K)	5.192 204 E+02
degree Fahrenheit	degree Celsius (°C)	T = (T - 32)/1.8
degree Fahrenheit	kelvin (K)	T = (T + 459.67)/1.8
foot	meter (m)	3.048 000 E−01
ft^2	m^2	9.290 304 E−02
foot/second	meter/second (m/s)	3.048 000 E−01
ft/s^2	m/s^2	3.048 000 E−01
inch	meter (m)	2.540 000 E−02
in.^{2}	meter^2 (m^2)	6.451 600 E−04
in.^{3}	m^3	1.638 706 E−05
kilogram-force (kgf)	newton (N)	9.806 650 E+00
kgf/m^2	pascal (Pa)	9.806 650 E+00
kip (1000 lbf)	newton (N)	4.448 222 E+03
ksi (kip/in^2)	MPa	6.894 757 E+00
lbf-in	N-m	1.129 848 E−01
lbf-ft	N-m	1.355 818 E+00
lbf/in^2 (psi)	pascal (Pa)	6.894 757 E+03
lb/in^2	gm/m^2	7.030 696 E+05
lb/in^3	kg/m^3	2.767 990 E+04
Msi (10^6 psi)	GPa	6.894 757 E+00
pound-force (lbf)	newton (N)	4.488 222 E+00
pound-mass (lb avoirdupois)	kilogram (kg)	4.535 924 E−01
torr	pascal (Pa)	1.333 22 E+02

*The letter "E" following the conversion factor stands for exponent and the two digits after the letter "E" indicate the power of 10 by which the number is to be multiplied.

1.4 DEFINITIONS. The following definitions are used within MIL-HDBK-17. This glossary of terms is not totally comprehensive but it does represent nearly all commonly used terms. Where exceptions are made, they are noted in the text and tables. For ease of identification the definitions have been organized alphabetically.

A-Basis (or A-Value) -- The mechanical property value above which 99 percent of the population of values is expected to fall, with a confidence of 95 percent. Note - A-values are not currently presented in MIL-HDBK-17. This is a 95% lower confidence limit on the first percentile.

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A-Stage -- An early stage in the reaction of thermosetting resins in which the material is still soluble in certain liquids and may be liquid or capable of becoming liquid upon heating. (Sometimes referred to as resol.)

Absorption -- A process in which one material (the absorbent) takes in or absorbs another (the absorbate).

Accelerator -- A material which, when mixed with a catalyzed resin, will speed up the chemical reaction between the catalyst and the resin.

Accuracy -- The degree of conformity of a measured or calculated value to some recognized standard or specified value. Accuracy involves the systematic error of an operation.

Addition Polymerization -- Polymerization by a repeated addition process in which monomers are linked together to form a polymer without splitting off of water or other simple molecules.

Adhesion -- The state in which two surfaces are held together at an interface by forces or interlocking action or both.

Adhesive -- A substance capable of holding two materials together by surface attachment. In the handbook, the term is used specifically to designate structural adhesives, those which produce attachments capable of transmitting significant structural loads.

ADK -- Notation used for the k-sample Anderson-Darling statistic, which is used to test the hypothesis that k batches have the same distribution.

Aliquot -- A small, representative portion of a larger sample.

Aging -- The effect, on materials, of exposure to an environment for a period of time; the process of exposing materials to an environment for an interval of time.

Ambient -- The surrounding environmental conditions such as pressure or temperature.

Anelasticity -- A characteristic exhibited by certain materials in which strain is a function of both stress and time, such that, while no permanent deformations are involved, a finite time is required to establish equilibrium between stress and strain in both the loading and unloading directions.

Angleply -- Same as **Crossply**.

Anisotropic -- Not isotropic; having mechanical and/or physical properties which vary with direction relative to natural reference axes inherent in the material.

Aramid -- A manufactured fiber in which the fiber-forming substance consisting of a long-chain synthetic aromatic polyamide in which at least 85% of the amide (-CONH-) linkages are attached directly to two aromatic rings.

Areal Weight of Fiber -- The weight of fiber per unit area of prepreg. This is often expressed as grams per square meter. See Table 1.6.2 for conversion factors.

Artificial Weathering -- Exposure to laboratory conditions which may be cyclic, involving changes in temperature, relative humidity, radiant energy and any other elements found in the atmosphere in various geographical areas.

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Aspect Ratio -- In an essentially two-dimensional rectangular structure (e.g., a panel), the ratio of the long dimension to the short dimension. However, in compression loading, it is sometimes considered to be the ratio of the load direction dimension to the transverse dimension. Also, in fiber micro-mechanics, it is referred to as the ratio of length to diameter.

Autoclave -- A closed vessel for producing an environment of fluid pressure, with or without heat, to an enclosed object which is undergoing a chemical reaction or other operation.

Autoclave Molding -- A process similar to the pressure bag technique. The lay-up is covered by a pressure bag, and the entire assembly is placed in an autoclave capable of providing heat and pressure for curing the part. The pressure bag is normally vented to the outside.

B-Basis (or B-Value) -- The mechanical property value above which at least 90 percent of the population of values is expected to fall, with a confidence of 95 percent, specifically the same as A-Basis except on the tenth percentile. (See Volume 1, Section 8.5.1.1)

B-Stage -- An intermediate stage in the reaction of a thermosetting resin in which the material softens when heated and swells when in contact with certain liquids but does not entirely fuse or dissolve. Materials are usually precured to this stage to facilitate handling and processing prior to final cure. (Sometimes referred to as resitol.)

Bag Molding -- A method of molding or laminating which involves the application of fluid pressure to a flexible material which transmits the pressure to the material being molded or bonded. Fluid pressure usually is applied by means of air, steam, water or vacuum.

Balanced Laminate -- A composite laminate in which all laminae at angles other than 0 degrees and 90 degrees occur only in ± pairs (not necessarily adjacent).

Batch (or Lot) -- For fibers and resins, a quantity of material formed during the same process and having identical characteristics throughout. For prepregs, laminae, and laminates, material made from one batch of fiber and one batch of resin.

Bearing Area -- The product of the pin diameter and the specimen thickness.

Bearing Load -- A compressive load on an interface.

Bearing Yield Strength -- The bearing stress at which a material exhibits a specified limiting deviation from the proportionality of bearing stress to bearing strain.

Bend Test -- A test of ductility by bending or folding, usually with steadily applied forces. In some instances the test may involve blows to a specimen having a cross section that is essentially uniform over a length several times as great as the largest dimension of the cross section.

Binder -- A bonding resin used to hold strands together in a mat or preform during manufacture of a molded object.

Binomial Random Variable -- The number of successes in independent trials where the probability of success is the same for each trial.

Birefringence -- The difference between the two principal refractive indices (of a fiber) or the ratio between the retardation and thickness of a material at a given point.

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Bleeder Cloth -- A nonstructural layer of material used in the manufacture of composite parts to allow the escape of excess gas and resin during cure. The bleeder cloth is removed after the curing process and is not part of the final composite.

Bond -- The adhesion of one surface to another, with or without the use of an adhesive as a bonding agent.

Braiding -- A textile process where two or more strands, yarns or tapes are intertwined in the bias direction to form an integrated structure.

Broadgoods -- A term loosely applied to prepreg material greater than about 12 inches in width, usually furnished by suppliers in continuous rolls. The term is currently used to designate both collimated uniaxial tape and woven fabric prepregs.

Buckling (Composite) -- A mode of structural response characterized by an out-of-plane material deflection due to compressive action on the structural element involved. In advanced composites, buckling may take the form not only of conventional general instability and local instability but also a micro-instability of individual fibers.

Bundle -- A general term for a collection of essentially parallel filaments or fibers.

C-Stage -- The final stage of the curing reaction of a thermosetting resin in which the material has become practically infusable and insoluble. (Normally considered fully cured and sometimes referred to as resite.)

Carbon Fibers -- Fibers produced by the pyrolysis of organic precursor fibers such as rayon, polyacrylonitrile (PAN), and pitch in an inert atmosphere. The term is often used interchangeably with "graphite"; however, carbon fibers and graphite fibers differ in the temperature at which the fibers are made and heat-treated, and the amount of carbon produced. Carbon fibers typically are carbonized at about 2400°F (1300°C) and assay at 93 to 95% carbon, while graphite fibers are graphitized at 3450 to 5450°F (1900 to 3000°C) and assay at more than 99% elemental carbon.

Caul Plates -- Smooth metal plates, free of surface defects, the same size and shape as a composite lay-up, used immediately in contact with the lay-up during the curing process to transmit normal pressure and to provide a smooth surface on the finished laminate.

Censoring -- Data is right (left) censored at M, if, whenever an observation is less than or equal to M (greater than or equal to M), the actual value of the observation is recorded. If the observation exceeds (is less than) M, the observation is recorded as M.

Chain-Growth Polymerization -- One of the two principal polymerization mechanisms. In chain-growth polymerization, the reactive groups are continuously regenerated during the growth process. Once started, the polymer molecule grows rapidly by a chain of reactions emanating from a particular reactive initiator which may be a free radical, cation or anion.

Chromatogram -- A plot of detector response against peak volume of solution (eluate) emerging from the system for each of the constituents which have been separated.

Circuit -- One complete traverse of the fiber feed mechanism of a winding machine; one complete traverse of a winding band from one arbitrary point along the winding path to another point on a plane through the starting point and perpendicular to the axis.

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Cocuring -- The act of curing a composite laminate and simultaneously bonding it to some other prepared surface during the same cure cycle (see **Secondary Bonding**).

Coefficient of Linear Thermal Expansion -- The change in length per unit length resulting from a one-degree rise in temperature.

Coefficient of Variation -- The ratio of the population (or sample) standard deviation to the population (or sample) mean.

Collimated -- Rendered parallel.

Compatible -- The ability of different resin systems to be processed in contact with each other without degradation of end product properties. (See **Compatible**, Volume 1, Section 8.5.1.1)

Composite Class -- As used in the handbook, a major subdivision of composite construction in which the class is defined by the fiber system and the matrix class, e.g., organic-matrix filamentary laminate.

Composite Material -- Composites are considered to be combinations of materials differing in composition or form on a macroscale. The constituents retain their identities in the composite; that is, they do not dissolve or otherwise merge completely into each other although they act in concert. Normally, the components can be physically identified and exhibit an interface between one another.

Compound -- An intimate mixture of polymer or polymers with all the materials necessary for the finished product.

Condensation Polymerization -- This is a special type of step-growth polymerization characterized by the formation of water or other simple molecules during the stepwise addition of reactive groups.

Confidence Coefficient -- See **Confidence Interval**.

Confidence Interval -- A confidence interval is defined by a statement of one of the following forms:

$$(1)\ P\{a<\theta\} \leq 1-\alpha$$
$$(2)\ P\{\theta<b\} \leq 1-\alpha$$
$$(3)\ P\{a<\theta<b\} \leq 1-\alpha$$

where $1-\alpha$ is called the confidence coefficient. A statement of type (1) or (2) is called a one-sided confidence interval and a statement of type (3) is called a two-sided confidence interval. In (1) a is a lower confidence limit and in (2) b is an upper confidence limit. With probability at least $1-\alpha$, the confidence interval will contain the parameter θ.

Constituent -- In general, an element of a larger grouping. In advanced composites, the principal constituents are the fibers and the matrix.

Continuous Filament -- A yarn or strand in which the individual filaments are substantially the same length as the strand.

Coupling Agent -- Any chemical substance designed to react with both the reinforcement and matrix phases of a composite material to form or promote a stronger bond at the interface. Coupling agents are applied to the reinforcement phase from an aqueous or organic solution or from a gas phase, or added to the matrix as an integral blend.

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Crazing -- Apparent fine cracks at or under the surface of an organic matrix.

Creep -- The time dependent part of strain resulting from an applied stress.

Creep, Rate Of -- The slope of the creep-time curve at a given time.

Critical Value(s) -- When testing a one-sided statistical hypothesis, a critical value is the value such that, if the test statistic is greater than (less than) the critical value, the hypothesis is rejected. When testing a two-sided statistical hypothesis, two critical values are determined. If the test statistic is either less than the smaller critical value or greater than the larger critical value, then the hypothesis is rejected. In both cases, the critical value chosen depends on the desired risk (often 0.05) of rejecting the hypothesis when it is true.

Crossply -- Any filamentary laminate which is not uniaxial. Same as Angleply. In some references, the term crossply is used to designate only those laminates in which the laminae are at right angles to one another, while the term angleply is used for all others. In the handbook, the two terms are used synonymously. The reservation of a separate terminology for only one of several basic orientations is unwarranted because a laminate orientation code is used.

Cumulative Distribution Function -- See Volume 1, Section 8.5.1.1.

Cure -- To change the properties of a thermosetting resin irreversibly by chemical reaction, i.e., condensation, ring closure, or addition. Cure may be accomplished by addition of curing (cross-linking) agents, with or without catalyst, and with or without heat. Cure may occur also by addition, such as occurs with anhydride cures for epoxy resin systems.

Cure Cycle -- The schedule of time periods at specified conditions to which a reacting thermosetting material is subjected in order to reach a specified property level.

Cure Stress -- A residual internal stress produced during the curing cycle of composite structures. Normally, these stresses originate when different components of a lay-up have different thermal coefficients of expansion.

Debond -- A deliberate separation of a bonded joint or interface, usually for repair or rework purposes. (See Disbond, Unbond).

Deformation -- The change in shape of a specimen caused by the application of a load or force.
Degradation -- A deleterious change in chemical structure, physical properties or appearance.

Delamination -- The separation of the layers of material in a laminate. This may be local or may cover a large area of the laminate. It may occur at any time in the cure or subsequent life of the laminate and may arise from a wide variety of causes.

Denier -- A direct numbering system for expressing linear density, equal to the mass in grams per 9000 meters of yarn, filament, fiber, or other textile strand.

Density -- The mass per unit volume.

Desorption -- A process in which an absorbed or adsorbed material is released from another material. Desorption is the reverse of absorption, adsorption, or both.

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Deviation -- Variation from a specified dimension or requirement, usually defining the upper and lower limits.

Dielectric Constant -- The ratio of the capacity of a condenser having a dielectric constant between the plates to that of the same condenser when the dielectric is replaced by a vacuum; a measure of the electrical charge stored per unit volume at unit potential.

Dielectric Strength -- The average potential per unit thickness at which failure of the dielectric material occurs.

Disbond -- An area within a bonded interface between two adherends in which an adhesion failure or separation has occurred. It may occur at any time during the life of the structure and may arise from a wide variety of causes. Also, colloquially, an area of separation between two laminae in the finished laminate (in this case the term "delamination" is normally preferred.) (See **Debond, Unbond, Delamination.**)

Distribution -- A formula which gives the probability that a value will fall within prescribed limits. (See **Normal, Weibull,** and **Lognormal Distributions,** also Volume 1, Section 8.5.1.1).

Dry Fiber Area -- Area of fiber not totally encapsulated by resin.

Ductility -- The ability of a material to deform plastically before fracturing.

Elasticity -- The property of a material which allows it to recover its original size and shape immediately after removal of the force causing deformation.

Elongation -- The increase in gage length or extension of a specimen during a tension test, usually expressed as a percentage of the original gage length.

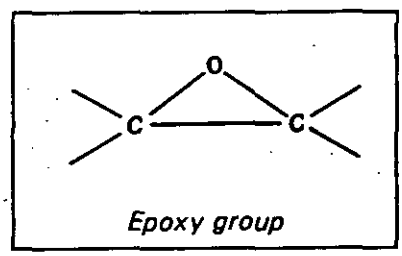
Eluate -- The liquid emerging from a column (in liquid chromatography).

Eluent -- The mobile phase used to sweep or elute the sample (solute) components into, through, and out of the column.

End -- A single fiber, strand, roving or yarn being or already incorporated into a product. An end may be an individual warp yarn or cord in a woven fabric. In referring to aramid and glass fibers, an end is usually an untwisted bundle of continuous filaments.

Epoxy Equivalent Weight -- The number of grams of resin which contain one chemical equivalent of the epoxy group.

Epoxy Resin -- Resins which may be of widely different structures but are characterized by the presence of the epoxy group. (The epoxy or epoxide group is usually present as a glycidyl ether, glycidyl amine, or as part of an aliphatic ring system. The aromatic type epoxy resins are normally used in composites.)



Epoxy group

Extensometer -- A device for measuring linear strain.

F-Distribution -- See Volume 1, Section 8.5.1.1.

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Fabric, Nonwoven -- A textile structure produced by bonding or interlocking of fibers, or both, accomplished by mechanical, chemical, thermal, or solvent means, and combinations thereof.

Fabric, Woven -- A generic material construction consisting of interlaced yarns or fibers, usually a planar structure. Specifically, as used in this handbook, a cloth woven in an established weave pattern from advanced fiber yarns and used as the fibrous constituent in an advanced composite lamina. In a fabric lamina, the warp direction is considered the longitudinal direction, analogous to the filament direction in a filamentary lamina.

Fiber -- A general term used to refer to filamentary materials. Often, fiber is used synonymously with filament. It is a general term for a filament of finite length.

Fiber Content -- The amount of fiber present in a composite. This is usually expressed as a percentage volume fraction or weight fraction of the composite.

Fiber Count -- The number of fibers per unit width of ply present in a specified section of a composite.

Fiber Direction -- The orientation or alignment of the longitudinal axis of the fiber with respect to a stated reference axis.

Fiber System -- The type and arrangement of fibrous material which comprises the fiber constituent of an advanced composite. Examples of fiber systems are collimated filaments or filament yarns, woven fabric, randomly oriented short-fiber ribbons, random fiber mats, whiskers, etc.

Filament -- The smallest unit of a fibrous material. The basic units formed during spinning and which are gathered into strands of fiber, (for use in composites). Filaments usually are of extreme length and of very small diameter. Filaments normally are not used individually. Some textile filaments can function as a yarn when they are of sufficient strength and flexibility.

Filamentary Composites -- A major form of advanced composites in which the fiber constituent consists of continuous filaments. Specifically, a filamentary composite is a laminate comprised of a number of laminae, each of which consists of a nonwoven, parallel, uniaxial, planar array of filaments (or filament yarns) embedded in the selected matrix material. Individual laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Filament Winding -- A reinforced-plastics process that employs a series of continuous, resin-impregnated fibers applied to a mandrel in a predetermined geometrical relationship under controlled tension.

Filament Wound -- Pertaining to an object created by the filament winding method of fabrication.

Fill -- Yarn oriented at right angles to the warp in a woven fabric.

Filler -- A relatively inert substance added to a material to alter its physical, mechanical, thermal, electrical, and other properties or to lower cost. Sometimes the term is used specifically to mean particulate additives.

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Finish (or Size System) -- A material, with which filaments are treated, which contains a coupling agent to improve the bond between the filament surface and the resin matrix in a composite material. In addition, finishes often contain ingredients which provide lubricity to the filament surface, preventing abrasive damage during handling, and a binder which promotes strand integrity and facilitates packing of the filaments.

Fixed Effect -- A common systematic (nonrandom) shift in a group of measurements which is due to a controlled level change of a treatment or condition. (See Volume 1, Section 8.5.1.1.)

Flash -- Excess material which forms at the parting line of a mold or die, or which is extruded from a closed mold.

Fracture Ductility -- The true plastic strain at fracture.

Gage Length -- the original length of that portion of the specimen over which strain or change of length is determined.

Gel -- The initial jelly-like solid phase that develops during formation of a resin from a liquid. Also, a semi-solid system consisting of a network of solid aggregates in which liquid is held.

Gel Coat -- A quick-setting resin used in molding processes to provide an improved surface for the composite; it is the first resin applied to the mold after the mold-release agent.

Gel Point -- The stage at which a liquid begins to exhibit pseudo-elastic properties. (This can be seen from the inflection point on a viscosity-time plot.)

Gel Time -- The period of time from a pre-determined starting point to the onset of gelation (gel point) as defined by a specific test method.

Glass -- An inorganic product of fusion which has cooled to a rigid condition without crystallizing. In the handbook, all reference to glass will be to the fibrous form as used in filaments, woven fabric, yarns, mats, chopped fibers, etc.

Glass Cloth -- Conventionally-woven glass fiber material (see **Scrim**).

Glass Fibers -- A fiber spun from an inorganic product of fusion which has cooled to a rigid condition without crystallizing.

Glass Transition -- The reversible change in an amorphous polymer or in amorphous regions of a partially crystalline polymer from (or to) a viscous or rubbery condition to (or from) a hard and relatively brittle one.

Glass Transition Temperature -- The approximate midpoint of the temperature range over which the glass transition takes place.

Graphite Fibers -- See **Carbon Fibers**.

Greige -- Fabric that has received no finish.

Hand Lay-up -- A process in which components are applied either to a mold or a working surface, and the successive plies are built up and worked by hand.

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Hardness -- Resistance to deformation; usually measured by indention. Types of standard tests include Brinell, Rockwell, Knoop, and Vickers.

Heat Cleaned -- Glass or other fibers which have been exposed to elevated temperatures to remove preliminary sizings or binders which are not compatible with the resin system to be applied.

Heterogeneous -- Descriptive term for a material consisting of dissimilar constituents separately identifiable; a medium consisting of regions of unlike properties separated by internal boundaries. (Note that all nonhomogeneous materials are not necessarily heterogeneous).

Homogeneous -- Descriptive term for a material of uniform composition throughout; a medium which has no internal physical boundaries; a material whose properties are constant at every point, i.e., constant with respect to spatial coordinates (but not necessarily with respect to directional coordinates).

Horizontal Shear -- Sometimes used to indicate interlaminar shear. This is not an approved term for use in this handbook.

Humidity, Relative -- The ratio of the pressure of water vapor present to the pressure of saturated water vapor at the same temperature.

Hybrid -- A composite laminate comprised of laminae of two or more composite material systems. Or, a combination of two or more different fibers such as carbon and glass or carbon and aramid into a structure (tapes, fabrics and other forms may be combined).

Hygroscopic -- Capable of absorbing and retaining atmospheric moisture.

Hysteresis -- The energy absorbed in a complete cycle of loading and unloading.

Inclusion -- A physical and mechanical discontinuity occurring within a material or part, usually consisting of solid, encapsulated foreign material. Inclusions are often capable of transmitting some structural stresses and energy fields, but in a noticeably different manner from the parent material.

Integral Composite Structure -- Composite structure in which several structural elements, which would conventionally be assembled by bonding or with mechanical fasteners after separate fabrication, are instead laid up and cured as a single, complex, continuous structure; e.g., spars, ribs, and one stiffened cover of a wing box fabricated as a single integral part. The term is sometimes applied more loosely to any composite structure not assembled by mechanical fasteners.

Interface -- The boundary between the individual, physically distinguishable constituents of a composite.

Interlaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., shear stress) referenced as existing or occurring between two or more adjacent laminae.

Interlaminar Shear -- Shearing force tending to produce a relative displacement between two laminae in a laminate along the plane of their interface.

Intermediate Bearing Stress -- The bearing stress at the point on the bearing load-deformation curve where the tangent is equal to the bearing stress divided by a designated percentage (usually 4%) of the original hole diameter.

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Intralaminar -- Descriptive term pertaining to some object (e.g., voids), event (e.g., fracture), or potential field (e.g., temperature gradient) existing entirely within a single lamina without reference to any adjacent laminae.

Isotropic -- Having uniform properties in all directions. The measured properties of an isotropic material are independent of the axis of testing.

Knuckle Area -- The area of transition between sections of different geometry in a filament wound part.

k-Sample Data -- A collection of data consisting of values observed when sampling from k batches.

Lamina -- A single ply or layer in a laminate made up of a series of layers.

Laminae -- Plural of lamina.

Laminate -- A product made by bonding together two or more layers or laminae of material or materials.

Laminate Orientation -- The configuration of a crossplied composite laminate with regard to the angles of crossplying, the number of laminae at each angle, and the exact sequence of the lamina lay-up.

Lattice Pattern -- A pattern of filament winding with a fixed arrangement of open voids.

Lay-up -- A process of fabrication involving the assembly of successive layers of resin-impregnated material.

Lognormal Distribution -- A probability distribution for which the probability that an observation selected at random from this population falls between a and b ($0 < a < b < B$) is given by the area under the normal distribution between ℓog a and ℓog b. The common (base 10) or the natural (base e) logarithm may be used. (See Volume 1, Section 8.5.1.1.)

Lower Confidence Bound -- See **Confidence Interval**.

Macro -- In relation to composites, denotes the gross properties of a composite as a structural element but does not consider the individual properties or identity of the constituents.

Macrostrain -- The mean strain over any finite gage length of measurement which is large in comparison to the material's interatomic distance.

Mandrel -- A form fixture or male mold used for the base in the production of a part by lay-up or filament winding.

Mat -- A fibrous material consisting of randomly oriented chopped or swirled filaments loosely held together with a binder.

Material Acceptance -- The testing of incoming material to ensure that it meets requirements.

Material Qualification -- The procedures used to accept a material by a company or organization for production use.

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Material System -- A specific composite material made from specifically identified constituents in specific geometric proportions and arrangements and possessed of numerically defined properties.

Material System Class -- As used in this handbook, a group consisting of material systems categorized by the same generic constituent materials, but without defining the constituents uniquely; e.g., the carbon/epoxy class.

Material Variability -- A source of variability due to variations in the spatial, consistency, and processing aspects of the material. (See Volume 1, Section 8.5.1.1.)

Matrix -- The essentially homogeneous material in which the fiber system of a composite is embedded.

Mean -- See **Sample Mean** and **Population Mean**.

Mechanical Properties -- The properties of a material that are associated with elastic and inelastic reaction when force is applied, or the properties involving the relationship between stress and strain.

Median -- See **Sample Median** and **Population Median**.

Micro -- In relation to composites, denotes the properties of the constituents, i.e., matrix and reinforcement and interface only, as well as their effects on the composite properties.

Microstrain -- The strain over a gage length comparable to the material's interatomic distance.

Modulus, Chord -- The slope of the chord drawn between any two specified points on the stress-strain curve.

Modulus, Initial -- The slope of the initial straight portion of a stress-strain or load-elongation curve.

Modulus, Secant -- The slope of the secant drawn from the origin to any specified point on the stress-strain curve.

Modulus, Tangent -- The ratio of change in stress to change in strain derived from the tangent to any point on a stress-strain curve.

Modulus, Young's -- The ratio of change in stress to change in strain below the elastic limit of a material. (Applicable to tension and compression).

Modulus of Rigidity (also Shear Modulus or Torsional Modulus) -- The ratio of stress to strain below the proportional limit for shear or torsional stress.

Modulus of Rupture, in Bending -- The maximum tensile or compressive stress (whichever causes failure) value in the extreme fiber of a beam loaded to failure in bending. The value is computed from the flexure equation:

$$F^b = \frac{Mc}{I} \qquad 1.4(a)$$

where M = maximum bending moment computed from the maximum load and the original moment arm,

c = initial distance from the neutral axis to the extreme fiber where failure occurs,

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I = the initial moment of inertia of the cross section about its neutral axis.

Modulus of Rupture, in Torsion -- The maximum shear stress in the extreme fiber of a member of circular cross section loaded to failure in torsion calculated from the equation:

$$F^{s} = \frac{Tr}{J} \qquad 1.4(b)$$

where T = maximum twisting moment,
r = original outer radius,
J = polar moment of inertia of the original cross section.

Moisture Content -- The amount of moisture in a material determined under prescribed condition and expressed as a percentage of the mass of the moist specimen, i.e., the mass of the dry substance plus the moisture present.

Moisture Equilibrium -- The condition reached by a sample when it no longer takes up moisture from, or gives up moisture to, the surrounding environment.

Mold Release Agent -- A lubricant applied to mold surfaces to facilitate release of the molded article.

Molded Edge -- An edge which is not physically altered after molding for use in final form and particularly one which does not have fiber ends along its length.

Molding -- The forming of a polymer or composite into a solid mass of prescribed shape and size by the application of pressure and heat.

Monolayer -- The basic laminate unit from which crossplied or other laminates are constructed.

Monomer -- A compound consisting of molecules each of which can provide one or more constitutional units.

NDE -- Nondestructive evaluation. Broadly considered synonymous with NDI.

NDI -- Nondestructive Inspection. A process or procedure for determining the quality or characteristics of a material, part, or assembly without permanently altering the subject or its properties.

NDT -- Nondestructive Testing. Broadly considered synonymous with NDI.

Necking -- A localized reduction in cross-sectional area which may occur in a material under tensile stress.

Negatively Skewed -- A distribution is said to be negatively skewed if the distribution is not symmetric and the longest tail is on the left.

Nominal Specimen Thickness -- The nominal ply thickness multiplied by the number of plies.

Nominal Value -- A value assigned for the purpose of a convenient designation. A nominal value exists in name only.

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Normal Distribution -- A two parameter (μ,σ) family of probability distributions for which the probability that an observation will fall between a and b is given by the area under the curve

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right] \qquad 1.4(c)$$

between a and b. (See Volume 1, Section 8.5.1.1.)

Normalized Stress -- Stress calculated by multiplying the raw stress value by the ratio of measured fiber volume to the nominal fiber volume. This ratio is often approximated by the ratio of the measured specimen thickness to the nominal specimen thickness. Stresses for fiber-dominated failure modes are often normalized.

Observed Significance Level (OSL) -- The probability of observing a more extreme value of the test statistic when the null hypotheses is true.

Oligomer -- A polymer consisting of only a few monomer units such as a dimer, trimer, etc. or their mixtures.

One-Sided Tolerance Limit Factor -- See Tolerance Limit Factor.

Orthotropic -- Having three mutually perpendicular planes of elastic symmetry.

Oven Dry -- The condition of a material that has been heated under prescribed conditions of temperature and humidity until there is no further significant change in its mass.

PAN Fibers -- Reinforcement fiber derived from the controlled pyrolysis of poly(acrylonitrile) fiber.

Parallel Laminate -- A laminate of woven fabric in which the plies are aligned in the same position as originally aligned in the fabric roll.

Peel Ply -- A layer of resin free material used to protect a laminate for later secondary bonding.

pH -- A measure of acidity or alkalinity of a solution, with neutrality represented by a value of 7, with increasing acidity corresponding to progressively smaller values, and increasing alkalinity corresponding to progressively higher values.

Pick Count -- The number of filling yarns per inch of woven fabric.

Pitch Fibers -- Reinforcement fiber derived from petroleum or coal tar pitch.

Plastic -- A material that contains one or more organic polymers of large molecular weight, is solid in its finished state, and, at some state in its manufacture or processing into finished articles, can be shaped by flow.

Plasticizer -- A material of lower molecular weight added to a polymer to separate the molecular chains. This results in a depression of the glass transition temperature, reduced stiffness and brittleness, and improved processability. (Note, many polymeric materials do not need a plasticizer.)

Poisson's Ratio -- The absolute value of the ratio of transverse strain to the corresponding axial strain resulting from uniformly distributed axial stress below the proportional limit of the material.

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Polymer -- An organic material composed of molecules characterized by the repetition of one or more types of monomeric units.

Polymerization -- A chemical reaction in which the molecules of monomers are linked together to form polymers via two principal reaction mechanisms. Addition polymerizations proceed by chain growth and most condensation polymerizations through step growth.

Population -- The set of measurements about which inferences are to be made or the totality of possible measurements which might be obtained in a given testing situation. For example, "all possible ultimate tensile strength measurements for carbon/epoxy system A, conditioned at 95% relative humidity and room temperature". In order to make inferences about a population, it is often necessary to make assumptions about its distributional form. The assumed distributional form may also be referred to as the population. (See Volume 1, Section 8.5.1.1.)

Population Mean -- The average of all potential measurements in a given population weighted by their relative frequencies in the population. (See Volume 1, Section 8.5.1.1.)

Population Median -- That value in the population such that the probability of exceeding it is 0.5 and the probability of being less than it is 0.5. (See Volume 1, Section 8.5.1.1.)

Population Variance -- A measure of dispersion in the population.

Porosity -- A condition of trapped pockets of air, gas, or vacuum within a solid material, usually expressed as a percentage of the total nonsolid volume to the total volume (solid plus nonsolid) of a unit quantity of material.

Positively Skewed -- A distribution is said to be positively skewed if the distribution is not symmetric and the longest tail is on the right.

Postcure -- Additional elevated temperature cure, usually without pressure, to improve final properties or complete the cure or both.

Pot Life -- The period of time during which a reacting thermosetting composition remains suitable for its intended processing after mixing with a reaction initiating agent.

Precision -- The degree of agreement within a set of observations or test results obtained. Precision involves repeatability and reproducibility.

Precursor (for Carbon or Graphite Fiber) -- Either the PAN or pitch fibers from which carbon and graphite fibers are derived.

Preply -- A composite material lamina in the raw material stage ready to be fabricated into a finished laminate. The lamina is usually combined with other raw laminae prior to fabrication. A preply includes all of the fiber system placed in position relative to all or part of the required matrix material that together will comprise the finished lamina. An organic matrix preply is called a prepreg.

Prepreg -- Ready to mold or cure material in sheet form which may be fiber, cloth, or mat impregnated with resin and stored for use. The resin is partially cured to a B-stage and supplied to the fabricator for lay-up and cure.

Pressure -- The force or load per unit area.

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Probability Density Function -- See Volume 1, Section 8.5.1.1.

Proportional Limit -- The maximum stress that a material is capable of sustaining without any deviation from the proportionality of stress to strain (also known as Hooke's law).

Quasi-Isotropic Laminate -- A laminate approximating isotropy by orientation of plies in several or more directions.

Random Effect -- A common shift in a group of measurements due to a random level change of a usually uncontrollable factor. (See Volume 1, Section 8.5.1.1.)

Random Error -- That part of data variation due to level changes in uncontrolled factors which affect each observation separately and independently. (See Volume 1, Section 8.5.1.1.)

Reduction of Area -- The difference between the original cross sectional area of a tension test specimen and the area of its smallest cross section, usually expressed as a percentage of the original area.

Refractive Index - The ratio of the velocity of light (of specified wavelength) in air to its velocity in the substance under examination. Also defined as the sine of the angle of incidence divided by the sine of the angle of refraction as light passes from air into the substance.

Reinforced Plastic -- A plastic with relatively high stiffness or very high strength fibers embedded in the composition. This improves some mechanical properties over that of the base resin.

Release Agent -- See **Mold Release Agent**.

Resilience -- A property of a material which is able to do work against restraining forces during return from a deformed condition.

Resin -- A solid or pseudo-solid organic material usually of high molecular weight which exhibits a tendency to flow when subjected to stress, usually has a softening or melting range, and fractures conchoidally.

Resin Content -- The amount of matrix present in a composite either by percent weight or percent volume.

Resin Starved Area -- Area of composite part where the resin has a non-continuous smooth coverage of the fiber.

Resin System -- A mixture of resin, with ingredients such as catalyst, initiator, diluents, etc. required for the intended processing and final product.

Roving -- A number of strands, tows, or ends collected into a parallel bundle with little or no twist.

S-Basis (or S-Value) -- The mechanical property value which is usually the specified minimum value of the appropriate government specification or SAE Aerospace Material Specification for this material.

Sample -- A small portion of a material or product intended to be representative of the whole. Statistically, a sample is the collection of measurements taken from a specified population. (See Volume 1, Section 8.5.1.1.)

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Sample Mean -- The arithmetic average of the measurements in a sample. The sample mean is an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Median -- Order the observation from smallest to largest. Then the sample median is the value of the middle observation if the sample size is odd; the average of the two central observations if n is even. If the population is symmetric about its mean, the sample median is also an estimator of the population mean. (See Volume 1, Section 8.5.1.1.)

Sample Standard Deviation -- The square root of the sample variance. (See Volume 1, Section 8.5.1.1.)

Sample Variance -- The sum of the squared deviations from the sample mean, divided by n-1. (See Volume 1, Section 8.5.1.1.)

Sandwich Construction -- A structural panel concept consisting in its simplest form of two relatively thin, parallel sheets of structural material bonded to, and separated by, a relatively thick, light-weight core.

Saturation -- An equilibrium condition in which the net rate of absorption under prescribed conditions falls essentially to zero.

Scrim (also called Glass Cloth, Carrier) -- A low cost fabric woven into an open mesh construction, used in the processing of tape or other B-stage material to facilitate handling.

Secondary Bonding -- The joining together, by the process of adhesive bonding, of two or more already-cured composite parts, during which the only chemical or thermal reaction occurring is the curing of the adhesive itself.

Selvedge -- The woven edge portion of a fabric parallel to the warp.

Set -- *The strain remaining after* complete release of the force producing the deformation.

Shear Fracture (for crystalline type materials) -- A mode of fracture resulting from translation along slip planes which are preferentially oriented in the direction of the shearing stress.

Shelf Life -- The length of time a material, substance, product, or reagent can be stored under specified environmental conditions and continue to meet all applicable specification requirements and/or remain suitable for its intended function.

Significant -- Statistically, the value of a test statistic is significant if the probability of a value at least as extreme is less than or equal to a predetermined number called the significance level of the test.

Significant Digit -- Any digit that is necessary to define a value or quantity.

Size System -- See **Finish**.

Skewness -- *See* **Positively Skewed, Negatively Skewed**.

Slenderness Ratio -- The unsupported effective length of a uniform column divided by the least radius of gyration of the cross-sectional area.

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Sliver -- A continuous strand of loosely assembled fiber that is approximately uniform in cross-sectional area and has no twist.

Solute -- The dissolved material.

Specific Gravity -- The ratio of the weight of any volume of a substance to the weight of an equal volume of another substance taken as standard at a constant or stated temperature. Solids and liquids are usually compared with water at 4°C (39°F).

Specific Heat -- The quantity of heat required to raise the temperature of a unit mass of a substance one degree under specified conditions.

Specimen -- A piece or portion of a sample or other material taken to be tested. Specimens normally are prepared to conform with the applicable test method.

Standard Deviation -- See **Sample Standard Deviation.**

Staple -- Either naturally occurring fibers or lengths cut from filaments.

Step-Growth Polymerization -- One of the two principal polymerization mechanisms. In sep-growth polymerization, the reaction grows by combination of monomer, oligomer, or polymer molecules through the consumption of reactive groups. Since average molecular weight increases with monomer consumption, high molecular weight polymers are formed only at high degrees of conversion.

Strain -- the per unit change, due to force, in the size or shape of a body referred to its original size or shape. Strain is a nondimensional quantity, but it is frequently expressed in inches per inch, meters per meter, or percent.

Strand -- Normally an untwisted bundle or assembly of continuous filaments used as a unit, including slivers, tow, ends, yarn, etc. Sometimes a single fiber or filament is called a strand.

Strength -- the maximum stress which a material is capable of sustaining.

Stress -- The intensity at a point in a body of the forces or components of forces that act on a given plane through the point. Stress is expressed in-force per unit area (pounds-force per square inch, megapascals, etc.).

Stress Relaxation -- The time dependent decrease in stress in a solid under given constraint conditions.

Stress-Strain Curve (Diagram) -- A graphical representation showing the relationship between the change in dimension of the specimen in the direction of the externally applied stress and the magnitude of the applied stress. Values of stress usually are plotted as ordinates (vertically) and strain values as abscissa (horizontally).

Surfacing Mat -- A thin mat of fine fibers used primarily to produce a smooth surface on an organic matrix composite.

Symmetrical Laminate -- A composite laminate in which the sequence of plies below the laminate midplane is a mirror image of the stacking sequence above the midplane.

Tack -- Stickiness of the prepreg.

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Tape -- Prepreg fabricated in widths up to 12 inches wide for carbon and 3 inches for boron. Cross stitched carbon tapes up to 60 inches wide are available commercially in some cases.

Tenacity -- The tensile stress expressed as force per unit linear density of the unstrained specimen i.e., grams-force per denier or grams-force per tex.

Tex -- A unit for expressing linear density equal to the mass or weight in grams of 1000 meters of filament, fiber, yarn or other textile strand.

Thermal Conductivity -- Ability of a material to conduct heat. The physical constant for quantity of heat that passes through unit cube of a substance in unit time when the difference in temperature of two faces is one degree.

Thermoplastic -- A plastic that repeatedly can be softened by heating and hardened by cooling through a temperature range characteristic of the plastic, and when in the softened stage, can be shaped by flow into articles by molding or extrusion.

Thermoset -- A plastic that is substantially infusible and insoluble after having been cured by heat or other means.

Tolerance -- The total amount by which a quantity is allowed to vary.

Tolerance Limit -- A lower (upper) confidence limit on a specified percentile of a distribution. For example, the B-basis value is a 95% lower confidence limit on the tenth percentile of a distribution.

Tolerance Limit Factor -- The factor which is multiplied by the estimate of variability in computing the tolerance limit.

Toughness -- A measure of a material's ability to absorb work, or the actual work per unit volume or unit mass of material that is required to rupture it. Toughness is proportional to the area under the load-elongation curve from the origin to the breaking point.

Tow -- An untwisted bundle of continuous filaments. Commonly used in referring to man-made fibers, particularly carbon and graphite fibers, in the composites industry.

Transformation -- A transformation of data values is a change in the units of measurement accomplished by applying a mathematical function to all data values. For example, if the data is given by x, then $y = x + 1$, x, $1/x$, $\log x$, and $\cos x$ are transformations.

Transition, First Order -- A change of state associated with crystallization or melting in a polymer.

Transversely Isotropic -- Descriptive term for a material exhibiting a special case of orthotropy in which properties are identical in two orthotropic dimensions, but not the third; having identical properties in both transverse directions but not the longitudinal direction.

Traveller -- A small piece of the same product (panel, tube, etc.) as the test specimen, used for example to measure moisture content as a result of conditioning.

Twist -- The number of turns about its axis per unit of length in a yarn or other textile strand. It may be expressed as turns per inch (tpi).

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Twist, Direction of -- The direction of twist in yarns and other textile strands is indicated by the capital letters S and Z. Yarn has S twist if, when held in a vertical position, the visible spirals or helices around its central axis are in the direction of slope of the central portion of the letter S, and Z twist is in the other direction.

Typical Basis -- A typical property value is a sample mean. Note that the typical value is defined as the simple arithmetic mean which has a statistical connotation of 50% reliability with a 50% confidence.

Unbond -- An area within a bonded interface between two adherends in which the intended bonding action failed to take place. Also used to denote specific areas deliberately prevented from bonding in order to simulate a defective bond, such as in the generation of quality standards specimens. (See **Disbond, Debond**).

Unidirectional Laminate -- A laminate with nonwoven reinforcements and all layers laid up in the same direction.

Upper Confidence Limit -- See **Confidence Interval**.

Vacuum Bag Molding -- A process in which the lay-up is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible sheet placed over it and sealed at the edges.

Variance -- See **Sample Variance**.

Viscosity -- The property of resistance to flow exhibited within the body of a material.

Void -- A physical and mechanical discontinuity occurring within a material or part which may be two-dimensional (e.g., disbonds, delaminations) or three-dimensional (e.g., vacuum-, air-, or gas-filled pockets). Porosity is an aggregation of micro-voids. Voids are essentially incapable of transmitting structural stresses or nonradiative energy fields. (See **Inclusion**.)

Warp -- The longitudinally oriented yarn in a woven fabric (see **Fill**); a group of yarns in long lengths and approximately parallel.

Weibull Distribution (Two - Parameter) -- A probability distribution for which the probability that a randomly selected observation from this population lies between a and b ($0 < a < b < \infty$) is given by Equation 1.4(d) where α is called the scale parameter and β is called the shape parameter. (See Volume 1, Section 8.5.1.1.)

$$\exp\left[-\left(\frac{a}{\alpha}\right)^{\beta}\right] - \exp\left[-\left(\frac{b}{\alpha}\right)^{\beta}\right] \qquad 1.4(d)$$

Wet Lay-up -- A method of making a reinforced product by applying a liquid resin system while the reinforcement is put in place.

Wet Strength -- The strength of an organic matrix composite when the matrix resin is saturated with absorbed moisture. (See **Saturation**).

Wet Winding -- A method of filament winding in which the fiber reinforcement is coated with the resin system as a liquid just prior to wrapping on a mandrel.

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Whisker -- A short single crystal fiber or filament. Whisker diameters range from 1 to 25 microns, with aspect ratios between 100 and 15,000.

Work Life -- The period during which a compound, after mixing with a catalyst, solvent, or other compounding ingredient, remains suitable for its intended use.

Woven Fabric Composite -- A major form of advanced composites in which the fiber constituent consists of woven fabric. A woven fabric composite normally is a laminate comprised of a number of laminae, each of which consists of one layer of fabric embedded in the selected matrix material. Individual fabric laminae are directionally oriented and combined into specific multiaxial laminates for application to specific envelopes of strength and stiffness requirements.

Yarn -- A generic term for strands or bundles of continuous filaments or fibers, usually twisted and suitable for making textile fabric.

Yarn, Plied -- Yarns made by collecting two or more single yarns together. Normally, the yarns are twisted together though sometimes they are collected without twist.

Yield Point -- The first stress in a material at which an increase in strain occurs without an increase in stress. (The stress is less than the maximum attainable.) It should be noted that only materials that exhibit the unique phenomenon of yielding have a "yield point".

Yield Strength -- The stress at which a material exhibits a specified limiting deviation from the proportionality of stress to strain. (The deviation is expressed in terms of strain such as 0.2 percent for the Offset Method or 0.5 percent for the Total Extension Under Load Method.)

X-Axis -- In composite laminates, an axis in the plane of the laminate which is used as the 0 degree reference for designating the angle of a lamina.

X-Y Plane -- In composite laminates, the reference plane parallel to the plane of the laminate.

Y-Axis -- In composite laminates, the axis in the plane of the laminate which is perpendicular to the x-axis.

Z-Axis -- In composite laminates, the reference axis normal to the plane of the laminate.

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Chapter 2. MATERIALS AND PROCESSES

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2.1 INTRODUCTION. Composites, in general, are cure and process dependent resulting in variation of glass transition (service temperature), corrosion stability, susceptibility to microcracking, general strength, and fatigue and service life. The information in this chapter is presented as a general guide to material type and the necessity for processing controls.

2.2 RESIN MATERIALS.

2.2.1 Thermoset materials.

2.2.1.1 *Epoxy.* The term epoxy is a general description of a family of polymers which are based on molecules that contain epoxide groups. An epoxide group is an oxirane structure, a three-member ring with one oxygen and two carbon atoms. Epoxies are polymerizable thermosetting resins containing one or more epoxide groups curable by reaction with amines, acids, amides, alcohols, phenols, acid anhydrides, or mercaptans. The polymers are available in a variety of viscosities from liquid to solid.

Epoxies are used widely in resins for prepregs and structural adhesives. The advantages of epoxies are high strength and modulus, low levels of volatiles, excellent adhesion, low shrinkage, good chemical resistance, and ease of processing. Their major disadvantages are brittleness and the reduction of properties in the presence of moisture. The processing or curing of epoxies is slower than polyester resins. The cost of the resin is also higher than the polyesters. Processing techniques include autoclave molding, filament winding, press molding, vacuum bag molding, resin transfer molding, and pultrusion. Curing temperatures vary from room temperature to approximately 350°F (180°C). The most common cure temperatures range between 250° and 350°F (120° and 180°C). The use temperatures of the cured structure will also vary with the cure temperature. Higher temperature cures generally yield greater temperature resistance. Cure pressures are generally considered as low pressure molding from vacuum to approximately 100 psi (700 kPA).

2.2.1.2 *Polyester (thermosetting).* The term thermosetting polyester resin is a general term used for orthophthalic polyester resin or isophthalic polyester resin. Polyester resins are relatively inexpensive and fast processing resins used generally for low-cost applications. In combination with certain fillers, they can exhibit resistance to breakdown under electrical arc and tracking conditions. Isophthalic polyester resins exhibit higher thermal stability, dimensional stability, and creep resistance. In general, for a fiber-reinforced resin system, the advantage of a polyester is its low cost and its ability to be processed quickly.

Fiber-reinforced polyesters can be processed by many methods. Common processing methods include matched metal molding, wet lay-up, press (vacuum bag) molding, injection molding, filament winding, pultrusion, and autoclaving. Polyesters can be formulated to cure more rapidly than do phenolics during the thermoset molding process. While phenolic processing, for example, is dependent on a time/temperature relationship, polyester processing is primarily dependent on temperature. Depending on the formulation, polyesters can be processed from room temperature to 350°F (180°C). If the proper temperature is applied, a quick cure will occur. Without sufficient heat, the resin/catalyst system will remain plasticized. Compared to epoxies, polyesters process more easily and are much tougher, whereas phenolics are more difficult to process and brittle, but have higher service temperatures.

2.2.1.3 *Phenolic.* Phenol-formaldehyde resins and their direct precursors were first produced commercially in the early 1900's for use in the commercial market. Ureaformaldehyde and melamine-formaldehyde appeared in the 1920 - 1930's as a less expensive alternative for lower temperature use. Phenolics, in general, cure by a condensation route with the off-gassing of water. The resulting matrix

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is characterized by both chemical and thermal resistance as well as hardness, and low smoke and toxic degradation products.

The phenolic polymers, often called either phenolic resole or novolacs resins, are condensation polymers based upon either a reaction of excess formaldehyde with a base catalyst and phenol (resole), or a reaction of excess phenol with an acidic catalyst and formaldehyde (novolac). The basic difference between resoles and novolacs consist of no methylol groups in the novolacs and the resulting need for an extension agent of paraformaldehyde, hexamethylenetetraamine, or additional formaldehyde as a curative. These resins have higher molecular weights and viscosities than either parent material. Consequently, they are optimal for processing parts of unusual conformations and complex curvature. The resins allow either press or autoclave cure and allow relatively high temperature free-standing postcures.

2.2.1.3.1 *Resoles.* The reaction of phenol and excess formaldehyde in the presence of base is characterized by low-molecular-weight prepolymers that are soluble in base and contain a large degree of methylol groups ($-CH_2OH$). These prepolymers are processed to a workable viscosity (resites) and then cured to an intractable solid of high crosslink density. Water is lost as a volatile (as much as 10-12% of the resin by weight).

2.2.1.3.2 *Novolacs.* The second type of phenolic consists of excess phenol reacted in the presence of an acid catalyst with formaldehyde. These prepolymer resins are complex mixtures of low molecular weight materials slightly soluble in acids and exhibiting random methylene ($-CH_2$) at the ortho-, para-, and ortho-para-positions on the aromatic ring. Unless a large excess of phenol is present, the material will form an infusible resin. The excess phenol used to moderate the processing viscosity can be varied as the application requires. Both water and formaldehyde are volatile products.

2.2.1.4 *Silicones.* The silicones are a synthetic resin, composed primarily of organosilicon. The term silicone resin is a general term used for poly methyl siloxane high T. Silicone resins are available from a low viscosity liquid to a solid friable resin.

The silicone resin is used where high temperature stability, weatherability, good electrical properties and moisture resistance are required. These excellent properties have allowed the silicone resin to be used in laminates, varnishes, mineral filled molding compounds, and long glass fiber molding compounds. The silicone resin has been used as an impregnant for mica paper, flexible glass tape, glass cloth, and mica products. The molding compounds may be processed by conventional methods: injection, compression, and transfer molding. The cure temperature varies from 250°F to 450°F (120°C to 230°C). The cure time varies from 30 minutes to 24 hours, depending upon cure temperature, wall thickness of molded part, and the desired cured properties. In some applications, additional post cure will be required.

2.2.1.5 *Bismaleimides.* Bismaleimides are a class of thermosetting resins only recently available commercially in prepreg tapes, fabrics, rovings, and sheet molding compound (SMC). Bismaleimide resins, as the term implies, are the maleimide formed from the reaction of a diamine and maleic anhydride. Typically the diamine is aromatic, with methylenedianiline (MDA), the most common by far.

Bismaleimides form useful polymers by homopolymerization or by polymerization with diamines, epoxies, or unsaturated compounds, singular or in mixtures. A wide range of materials like allyl-, vinyl-, acrylate-, epoxy-, and polyester-, and phenolic-type reactive diluents and resins can be used to tailor the properties of the bismaleimide system. However, attention to the specific components is required for useful polymers.

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The physical form of the bismaleimide resin depends on the requirement of the final application. The form can vary from a solid to a pourable liquid at room temperature. For aerospace prepregs, sticky resins are required resulting in proprietary specific formulations.

The advantages of BMI resins are best discussed in the relation to epoxy resins. Emerging data suggests that BMI's are versatile resins with many applications in the electronic and aerospace industries. Their primary advantage over epoxy resins is their high glass transition temperature, in the 500-600°F range (260-320°C). Glass transition temperatures for high temperature epoxies are generally less than 500°F (260°C). The second advantage of BMI resins is high elongation with the corresponding high service temperature capabilities. While the high temperature epoxies have approximately one percent elongation when cured with diaminodiphenylsulfone (DDS), BMI's can have two-three percent elongation. Thus, bismaleimide resins deliver higher temperature capability and higher toughness providing excellent performance at ambient and elevated temperatures.

The processing of bismaleimide resins are essentially like that of epoxy resins. BMI's are suitable for standard autoclave processing, injection molding, resin transfer molding, and SMC, among others. The processing time of BMI's are similar to epoxies, except that for the additional higher service temperature, a free-standing post-cure is required. The only limitation is that room temperature curing BMI's have not yet been developed.

The cost of current BMI's is generally higher than the high temperature epoxies. The main disadvantage of bismaleimide resins is their recent commercial introduction. This results in few literature sources or authoritative reviews. Additionally, the suppliers are as limited as the types of BMI's. This latter disadvantage is partially offset by the wide variety of suitable co-monomers.

2.2.1.6 *Polyimides.*

2.2.1.7 *Polybenzimidazoles.*

2.2.2 Thermoplastic materials.

2.2.2.1 *Semi-crystalline.* Semi-crystalline thermoplastics are so named because a percentage of their volume consists of a crystalline morphology. The remaining volume has a random molecular orientation termed amorphous, the name given to thermoplastics containing no crystalline structure. The total percentage of volume which can become crystalline depends on the polymer. Low density polyethylene, for example, can be as high as 70% crystalline (Reference 2.2.2.1(a)). Semi-crystalline thermoplastics are characterized by the ability of their molecules to form three-dimensionally ordered arrays (Reference 2.2.2.1(b)). This is in contrast to amorphous polymers that contain molecules which are unable to pack in an ordered crystalline structure. A partial list of semi-crystalline thermoplastics includes polyethylene, polypropylene, polyamides, polyphenylene sulfide, polyetheretherketone, (polyetherketoneketone) and polyarylketone.

Semi-crystalline thermoplastics can be converted into several physical forms, including films, powders and filaments. Combined with reinforcing fibers, they are available in injection molding compounds, compression-moldable random sheets, unidirectional tapes, towpregs, and woven prepregs. Fibers impregnated include carbon, nickel-coated carbon, aramid, glass, quartz, and others.

Semi-crystalline thermoplastics reinforced with short fibers have been used for over two decades in the injection molding industry. The inherent speed of processing, ability to produce complicated, detailed parts, excellent thermal stability, and corrosion resistance have enabled them to become established in the automotive, electronic, and chemical processing industries (Reference 2.2.2.1(c)).

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The combination of long and continuous fibers with higher performance semi-crystalline thermoplastics is a recent development, but these composites have already shown several advantages over existing materials. The chemical stability of the materials provides for unlimited shelf life. Pot life problems and the need for cold storage are eliminated. The semi-crystalline materials usually possess better corrosion and solvent resistance than amorphous polymers, exceeding that of thermosets in some cases (Reference 2.2.2.1(c)). This corrosion resistance is exploited in chemical processing industry equipment. Another benefit of the crystal structure is retention of properties above the glass transition temperature (T_g) of the material. These materials may be used in applications above their T_g depending on loading requirements. One example is down-hole oil field sucker rod guides (Reference 2.2.2.1(d)).

Some semi-crystalline thermoplastics possess properties of inherent flame resistance, superior toughness, good mechanical properties at elevated temperatures and after impact, and low moisture absorption which have led to their use in the aerospace industry in secondary and primary structures (References 2.2.2.1(e)-(f)). Inherent flame resistance has made these materials good candidates for aircraft interiors and for ship and submarine applications. The superior toughness makes them viable candidates for aircraft leading edges and doors where impact damage resistance is required (Reference 2.2.2.1(g)). Low moisture absorption and low outgassing has stimulated interest in space structures where moisture swelling is a problem (Reference 2.2.2.1(h)). Also nickel-coated carbon/thermoplastic systems are finding uses in EMI shielding applications.

The primary disadvantages of semi-crystalline thermoplastic composites are lack of a design data base, 0° compression properties that are lower than those of 350°F (180°C) epoxy systems, and creep resistance (Reference 2.2.2.1(c)). The creep resistance of semi-crystalline thermoplastics is superior to that of amorphous thermoplastics. Creep resistance in the fiber direction of a laminate is not expected to be a problem.

Processing speed is the primary advantage of thermoplastic materials. Chemical curing of the material does not take place during processing. Therefore, reduced cycle times compared to thermoset composites are experienced (References 2.2.2.1(i) and (j)). However, thermoplastic prepregs are typically boardy and do not exhibit the tack and drape of thermosets. Forms are available that consist of thermoplastic and reinforcing fibers interlaced together, known as commingled which are drapeable. The present costs of high performance engineering thermoplastic materials are slightly higher than equivalent performance epoxies, and tooling costs may be higher. However, final part cost may be reduced, due to the decreased processing time. The ability to postform or reprocess molded parts also offers cost saving advantages.

A wide variety of methods and techniques are available for processing semi-crystalline thermoplastics, including stamp molding, thermoforming, autoclave molding, diaphragm forming, roll forming, filament winding, and pultrusion. Semi-crystalline thermoplastics differ from amorphous ones in that the morphology can change based on the time/temperature history of the material during molding. Therefore, the degree of crystallinity can be controlled by controlling the cooling rate. The material must be processed above its melt temperature, which requires temperatures ranging from 500 to 700°F (260 - 370°C) for the higher performance materials. Thermal expansion differences between the tool and the thermoplastic material should be addressed, due to the high processing temperature. The actual pressure required varies with the process, but can be as high as 5000 psi (34 MPa) for stamp molding and as low as 100 psi (0.7 MPa) for thermoforming. Once formed, semi-crystalline thermoplastics can be joined by a variety of methods, including ultrasonic welding, infrared heating, vibration, hot air and gas, resistance heating, and conventional adhesives.

2.2.2.2 *Amorphous.* The majority of thermoplastic polymers are composed of a random molecular orientation and are termed amorphous. The molecules are unable to align themselves in an ordered

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manner, since they are non-uniform or composed of units which have large side groups. In contrast, semi-crystalline thermoplastics have molecules that form ordered three-dimensional arrays (Reference 2.2.2.1(b)). Some amorphous thermoplastics include polysulfone, polyamide-imide, polyphenylsulfone, polyphenylene sulfide sulfone, polyether sulfone, polystyrene, polyetherimide, and polyarylate.

Amorphous thermoplastics are available in several physical forms, including films, filaments, and powders. Combined with reinforcing fibers, they are also available in injection molding compounds, compression moldable random sheets, unidirectional tapes, woven prepregs, etc. The fibers used are primarily carbon, aramid, and glass.

Amorphous thermoplastics are used in many applications; the specific use depends on the polymer of interest. Their applications are well established in the medical, communication, transportation, chemical processing, electronic, and aerospace industries. The majority of applications use the unfilled and short fiber form. Some uses for the unfilled polymers include cookware, power tools, business machines, corrosion resistant piping, medical instruments, and aircraft canopies. Uses for short-fiber-reinforced forms include printed circuit boards, transmission parts, under-the-hood automotive applications, electrical connections, and jet engine components (Reference 2.2.2.2(a)).

The use of amorphous thermoplastics as matrix materials for continuous fiber reinforced composites is a recent development. The properties of composites have led to their consideration for primary and secondary aircraft structures, including interior components, flooring, fairings, wing skins, and fuselage sections (References 2.2.2.2(b) and (c)).

The specific advantages of amorphous thermoplastics depend upon the polymer. Typically, the resins are noted for their processing ease and speed, high temperature capability, good mechanical properties, excellent toughness and impact strength, and chemical stability. The stability results in unlimited shelf life, eliminating the cold storage requirements of thermoset prepregs. Several amorphous thermoplastics also have good electrical properties, low flammability and smoke emission, long term thermal stability, and hydrolytic stability (Reference 2.2.2.2(a)).

Amorphous thermoplastics generally have higher temperature capabilities than semi-crystalline thermoplastics. Polymers with glass transition temperatures as high as 500°F (260°C) are available. Also, processing is simplified, because the formation of a crystalline structure is avoided, resulting in less shrinkage due to their lower melt viscosities. Amorphous polymers generally have lower solvent and creep resistances and less property retention above the glass transition temperature than semi-crystalline thermoplastics (Reference 2.2.2.1(f)).

The primary advantages of amorphous thermoplastics in continuous fiber reinforced composites are potential low cost process at high production rates, high temperature capability, good mechanical properties before and after impact, and chemical stability. High temperature capability and retention of mechanical properties after impact have made amorphous thermoplastics attractive to the aerospace industry. A service temperature of 350°F and toughness two to three times that of conventional thermoset polymers are typical (Reference 2.2.2.1(f)). The most significant advantage of thermoplastics is the speed of processing, resulting in lower costs. Typically, cycle times in production are less than for thermosets since no chemical reaction occurs during the forming process (References 2.2.2.1(i) and (j)).

Amorphous thermoplastics share many of the disadvantages of semi-crystalline thermoplastics, such as a lack of an extensive database and reduced 0° compression properties compared to 350°F (180°C) cure thermosets. Solvent resistance, which is good for semi-crystalline thermoplastics, is a concern for most amorphous ones. They can be attacked to varying degrees, depending on the polymers and solvents of interest. The creep resistance of the polymer is a concern, but should be

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good for composite forms loaded in the fiber direction. The materials do not have tack and drape as thermosets do; however, some amorphous thermoplastics are available in commingled forms, which are drapable.

The costs of amorphous thermoplastics prepreg used for advanced composites are higher than equivalent performance epoxies. Finished part costs may be lower due to the processing advantages discussed above. Reprocessibility of material results in reduced scrap rates, translating into additional cost savings. For example, the same sheet laminate can be thermoformed several times until the desired configuration is achieved. In addition, certain forms can be recycled.

The processes used with continuous reinforced composites include stamp molding, thermoforming, autoclave molding, diaphragm forming, roll forming, filament winding, and pultrusion. The high melting temperatures require process temperatures ranging from 500°F to 700°F (260 to 370°C). Thermal expansion differences between the tool and the thermoplastic material should be addressed due to the high processing temperatures. Forming pressures range from 100 psi (0.7 MPa) for thermoforming to 5000 psi (35 MPa) for stamp molding. Several amorphous thermoplastics that are hygroscopic must be dried before processing. Hot molds are also recommended to increase material flow. The materials can be joined by several methods, including common adhesives, or fusion bonding such as; ultrasonic welding, infrared heating, hot air and gas, and resistance heating. Surface preparation techniques for using adhesives can be different from those for thermosets. Solvent bonding techniques can be used for joining amorphous thermoplastics but not most semi-crystalline thermoplastics.

2.2.2.3 *Condensation-cure.*

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2.3 REINFORCEMENT FIBERS.

2.3.1 Carbon.

2.3.2 Aramid. In the early 1970's, Du Pont Company introduced Kevlar™ aramid, an organic fiber with high specific tensile modulus and strength. This was the first organic fiber to be used as a reinforcement in advanced composites. Today this fiber is used in various structural parts including reinforced plastics, ballistics, tires, ropes, cables, asbestos replacement, coated fabrics, and protective apparel. Aramid fiber is manufactured by extruding a polymer solution through a spinneret. Major forms available from Du Pont are continuous filament yarns, rovings, chopped fiber, pulp, spun-laced sheet, wet-laid papers, thermoplastic-impregnated tows, and thermoformable composite sheets.

Important generic properties of aramid fibers are: low density, high tensile strength, high tensile stiffness, low compressive properties (non-linear), and exceptional toughness characteristics. The density of aramid is 0.052 lb/in^3 (1.44 gm/cc). This is about 40% lower than glass and about 20% lower than commonly used carbon. Aramids do not melt and they decompose at about 900°F (500°C). Tensile strength of yarn, measured in twisted configuration, can be varied from 500 - 600 ksi (3.4 - 4.1 GPa) by choosing different types of aramids. The nominal coefficient of thermal expansion is $3x10^{-6}$ in/in/°F ($-5x10^{-6}$ m/m/°C) in the axial direction. Aramid fibers, being aromatic polyamide polymers, have high thermal stability and dielectric and chemical properties. Excellent ballistic performance and general damage tolerance is derived from fiber toughness. Aramid is used, in fabric or composite form, to achieve ballistic protection for humans, armored tanks, military aircraft, etc.

Composite systems, reinforced with aramid, have excellent vibration-damping characteristics. They resist shattering upon impact. Temperature of use, in composite form with polymer matrix, range from -33 to 390°F (-36 - 200°C), The nominal tensile properties of composites reinforced with aramid are listed in Table 2.3.2(a) - in thermoset (Reference 2.3.2(a)) and thermoplastic (Reference 2.3.2(b)) resin matrix. At 60% fiber volume fraction, composites of epoxy reinforced with aramid fibers have nominal tensile strength (room temperature) of 200 ksi (1.4 GPa) and nominal tensile modulus of 11 Msi (76 GPa). These composites are ductile under compression and flexure. Ultimate strength, under compression or flexure, is lower than glass or carbon composites. Composite systems, reinforced with aramid, are resistant to fatigue and stress rupture. In the system of epoxy reinforced with aramid, under tension/tension fatigue, unidirectional specimens (V_f ~ 60%) survive 3,000,000 cycles at 50%

Table 2.3.2(a) *Nominal composite properties reinforced with aramid fiber (V_f ~ 60%)*

		Thermoset (epoxy)		Thermoplastic (J2)	
Tensile Property	Units	unidirectional	fabric[1]	unidirectional	fabric[1]
Modulus	Msi (GPa)	11 (68.5)	6 (41)	10.5-11.5 (73-79)	5.1-5.8 (35-40)
Strength	ksi (GPa)	200 (1.4)	82 (0.56)	180-200 (1.2-1.4)	77-83 (0.53-0.57)

[1] Normalized from V_f = 40%; fabric style S285

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of their ultimate stress (Reference 2.3.2(a)). Recently, thermoplastic resin composites reinforced with aramid have been developed. These thermoplastic composite systems have exhibited equivalent mechanical properties compared to similar thermoset systems (references 2.3.2(b)). In addition, thermoplastic systems provide potential advantages in economical processing (Reference 2.3.2(c)), bonding and repair, etc. A unique thermoformable sheet product, in thermoplastic matrix reinforced with aramid fibers, is available (Reference 2.3.2(d)). These composite systems are also used to achieve low coefficient of thermal expansion or high wear resistance. They are non-conductive and exhibit no galvanic reaction with metals. Aramid fibers are available in several forms with different fiber modulus (Table 2.3.2(b)). Kevlar™29 has the lowest modulus and highest toughness (strain to failure ~ 4%). These fibers are used mostly in ballistics and other soft composite systems such as cut- and slash- resistance protective apparel, ropes, coated fabric, asbestos replacement, pneumatic tires, etc. These are also used for composites where maximum impact and damage tolerance is critical and stiffness is less important. Kevlar™49 is predominantly used in reinforced plastics - both in thermoplastic and thermoset resin systems. It is also used in soft composites like core of fiber optic cable and mechanical rubber good systems (e.g. high pressure flexible hose, radiator hose, power transmission belts, conveyor belts, etc.). An ultra-high modulus Type 149 has been made available recently. It has 40% higher modulus than Kevlar™49. Kevlar™29 is available in fiber yarn sizes and two rovings sizes. Kevlar™49 is available in six yarn and two rovings sizes. Kevlar™149 is available in three yarn sizes. Yarn sizes range from the very fine 55 denier (30 filaments) to 3000 denier (1300 filaments). Rovings are 4560 denier (3072 filaments) and 7100 denier (5000 filaments). Composite thermoplastic tows, several types of melt-impregnated thermoplastic reinforced with different Kevlar™ yarns and deniers, are also available.

Aramid composites were first adopted in applications where weight savings were critical - e.g. aircraft components, helicopters, space vehicles, and missiles. Armor applications resulted from the superior ballistic and structural performance. In marine recreational industries, light weight, stiffness, vibration damping, and damage tolerance are valued. Composites reinforced with aramids are used in the hulls of canoes, kayaks, and sail and power boats. These same composite attributes have led to use in sports equipment. Composite applications of aramid continue to grow as systems are developed to capitalize on other properties. The stability and frictional properties of aramids at high temperatures have led to brake, clutch, and gasket uses; low coefficient of thermal expansion is being used in printed wiring boards; and exceptional wear resistance is being engineered into injection-molded thermoplastic industrial parts. Melt-impregnated thermoplastic composites, reinforced with aramids, offer unique processing advantages - e.g. in-situ consolidation of filament-wound parts. These can be used for manufacturing thick parts where processing is otherwise very difficult (Reference 2.3.2(e)).

Table 2.3.2(b) *Nominal properties of aramid fiber*

Tensile Property	Units	Type of Kevlar™		
		29	49	149
Modulus	Msi (GPa)	12 (83)	18 (124)	25 (173)
Strength	ksi (GPa)	525 (3.6)	525-600 (3.6-4.1)	500 (3.4)

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Aramid fiber is relatively flexible and tough. Thus it can be combined with resins and processed into composites by most of the methods established for glass. Yarns and rovings are used in filament winding, prepreg tape, and in pultrusion. Woven fabric prepreg is the major form used in thermoset composites. Aramid fiber is available in various weights, weave patterns, and constructions; from very thin lightweight (0.005 mm, 275 gm/m^2) to thick heavy (0.66 mm, 2.8 gm/m^2) woven roving. Thermoplastic-impregnated tows can be woven into various types of fabrics to form prepregs. These composites demonstrate good property retention under hot and humid conditions (reference 2.3.2(f)). Chopped aramid fiber is available in lengths from 6 mm to 100 mm. The shorter lengths are used to reinforce thermoset, thermoplastic, and elastomeric resins in automotive brake and clutch linings, gaskets, and electrical parts. Needle-punched felts and spun yarns for asbestos replacement applications are made from longer fiber staple. A unique very short fiber (2 - 4 mm) with many attached fibrils is available (aramid pulp). It can provide efficient reinforcement in asbestos replacement uses. Aramid short fibers can be processed into spun-laced and wet-laid papers. These are useful for surfacing veil, thin-printed wiring boards, and gasket material. Uniform dispersion of aramid short fiber in resin formulations is achieved through special mixing methods and equipment. Inherent fiber toughness necessitates special types of tools for cutting fabrics and machining aramid composites.

2.3.3 Glass. Glass in the forms used in commerce has been produced by many cultures since the early Etruscan civilization. Glass as a structural material was introduced early in the seventeenth century and became widely used during the twentieth century as the technology for flat pane was perfected. Glass fibrous usage for reinforcement was pioneered in replacement of metals and used for both commercial and military uses with the advent of formulation control and molten material which is die or bushing pulled into continuous filaments. These events lead to a wide range of aerospace and commercial high performance structural applications still in use today.

Military S-glass*, a more costly structural material was replaced by the general forms of S-2[1] in the 1960's. With that change, more stable surface finishes evolved. These glass types designated as CS were originally used for both polyester and epoxy composite systems. Other matrix systems have since been found to be compatible.

2.3.3.1 *Chemical Description*. Glass is derived from one of our most abundant natural resources--sand. Other than for, possibly, transport and the melting process, it is not petro chemical dependent. For purposes of this handbook the typical glass compositions are for electrical/Grade "E" glass, a calcium aluminoborosilica composition with an alkali content of less than 2%, chemical resistant "C" glass composed of soda-lime-borsilicates and high strength S-2 glass which is a low-alkali magnesi-alumina-silicate composition (See Table 2.3.3.1). Surface treatments (binders/sizing) can be applied directly to the filaments during the pulling step. Organic binders, such as starch oil, are applied to provide optimum weaving and strand protection during weaving of fabrics or "greige goods". These type binders are then washed and heat cleaned off the fabrics for finishing or sizing at the weaver with coupling agents to improve compatibility with resins. (See Figure 2.3.3.1) The exception to this process for fabrics is when they are heat treated or "caramelized", which converts the starch to carbon (0.2 - 0.5%). Glass roving products (untwisted) type yarns are most often directly finished with the final coupling agents during the filament manufacturing step. Therefore, the products will be identified with the glass manufacturer's product codes and the desizing step is not necessary as common with fabric "greige goods" forms. Heat cleaned products are also available where the product is essentially pure glass. These products, which are subject to damage, are commonly utilized for silicone laminates. Another finish designation is applicable to the heat cleaned product when it is followed with a demineralized water wash (neutral pH). More common for structural applications are the coupling

[1]Owens/Corning Fiberglas Designations

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TABLE 2.3.3.1 *Typical chemical compositions of glass fiber.*

	%(wt)	E-Glass	S-2 Glass (Nominals)	H_R Glass(B)
Silicon Dioxide (SiO_2)		52-56 (A)	65	63.5 - 65.0
Aluminum Oxide (Al_2O_3)		12-16 (A)	25	24.0 - 25.5
Boron Oxide (B_2O_3)		5-10 (A)		
Calcium Oxide (CaO)		16-25 (A)		<0.5
Magnesium Oxide (MgO)		0-5 (A)	10	9.5 - 10.5
Lithium Oxide (Li_2O)				
Potassium Oxide (K_2O)	O.C.	0.0-0.2		
Sodium Oxide (Na_2O)	O.C.	0-2		
Titanium Oxide (TiO_2)	O.C.	0-1.5		
Cerium Oxide (CeO_2)				
Zirconium Oxide (Zr_2O_3)				
Beryllium Oxide (BeO)				
Iron Oxide (Fe_2O_3)	O.C.	0.0-0.8		
Fluorine (F_2)	O.C.	0.0-0.1		
Sulfate (SO_2)				
Alkaline Oxides	PPG	0.5-1.5		
Calcium Fluoride (CAF)	PPG	0.0-0.8		
Finishes/Binders		0.5/3.0		

agents which are applied for use with standard organic polymers. During the 1940's Volan[2] finishes were introduced. Since then, many variations/improvements identified with various company designations have appeared. Perhaps the most recognized is Volan A. This finish provides good wet and dry strength properties in use with polyester, epoxy, and phenolic resins. Prior to the application of this finish the clean(ed) glass is saturated with methacrylate chromic chloride so that the chrome content of the finish is between 0.03% and 0.06%. This addition enhances wet-out of the resin during cure. Perhaps more typically called out for use, but not limited to, with epoxy are the silane finishes. Most all are formulated to enhance laminate wet-out. Some also produce high laminate clarity or good composite properties in aqueous environments. Others improve high-pressure laminating, or resist adverse environment or chemical exposures. Although other finishes are used in combination with matrix materials other than epoxy, finishes may have proprietary formulations or varied designations relative to the particular glass manufacturer or weaver, it is believed the compositions are readily available to the resin compounders (prepreggers) to determine compatibility

[2]E. I. DuPont de Nemours

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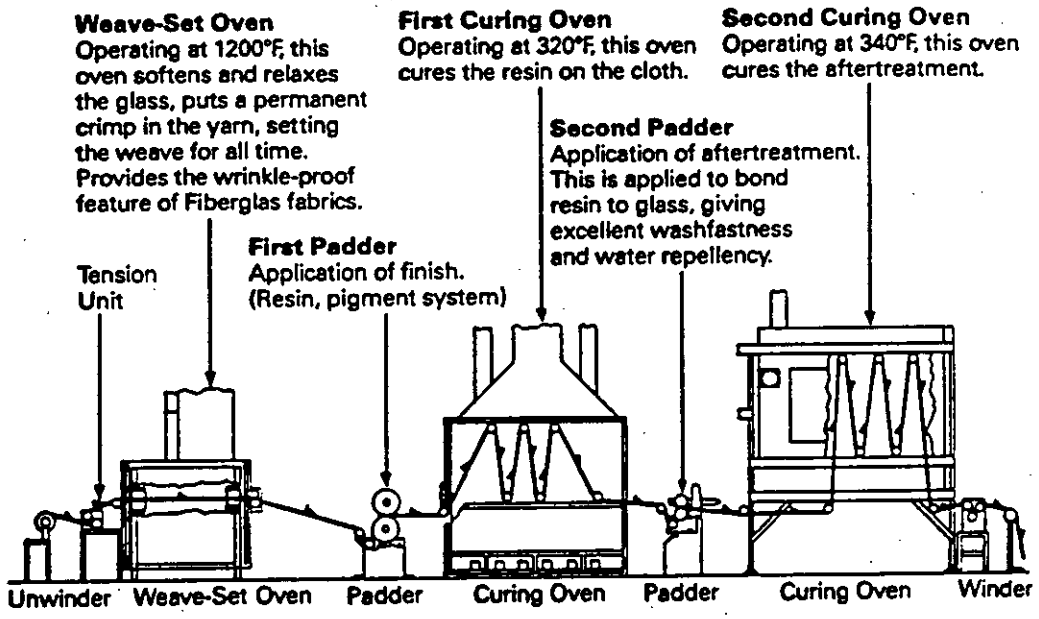


FIGURE 2.3.3.1 *Fabric finishing (Reference 2.3.3.1(c))*

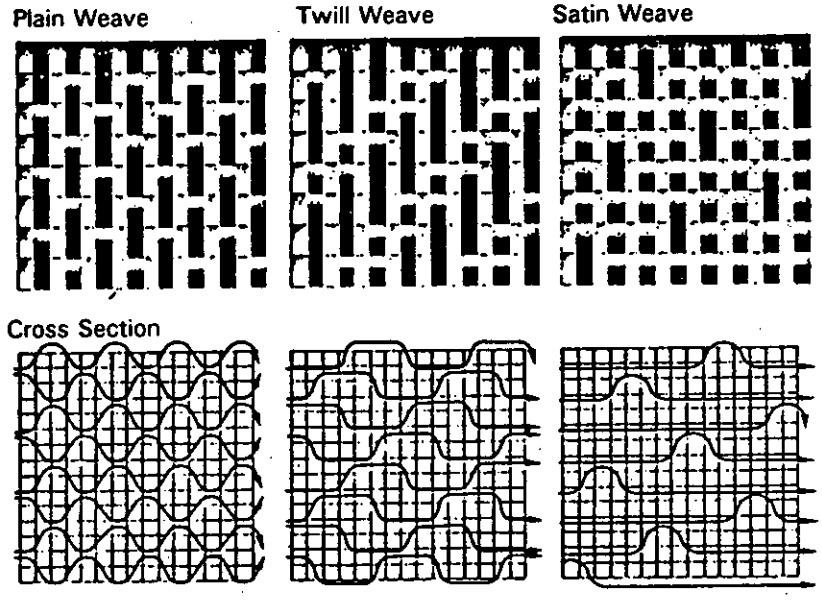


FIGURE 2.3.3.2 *Common types of weaves for glass fabrics (Reference 2.3.3.1(c)).*

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and end use purposes. Note that, non-compatible finishes are purposely applied for ornament applications.

2.3.3.2 *Physical Forms Available.* Due to the high quantity of commercial applications for glass products, there are many product forms available. For purposes of this publication glass forms will be limited to continuous filament product forms. These forms fall into four major categories. They are continuous rovings, yarn for fabrics or braiding, mats, and chopped strand. (See Figure 2.3.3.2 and ASTM Specification D 579, Reference 2.3.3.2(a) for information on glass fabrics.) They are available with a variety of physical surface treatments and finishes. Most structural applications utilize fabric, roving, or rovings converted to unidirectional tapes. Perhaps the most versatile fiber type to produce glass product forms is "E" glass. "E" glass is identified as such for electrical applications. This type or grade of glass has eight or more standardized filament diameters available. These range from 1.4 to 5.1 mils (3.5 to 13 micrometers). (See Table I, ASTM Specification D 578, Reference 2.3.3.2(b).) This facilitates very thin product forms. The "S" glasses are identified as such to signify high strength. The S-2 type glasses are available with but one filament diameter. This does not limit the availability of basic structural fabric styles for S-2 glass however. Although there are more "E" roving products, as to yields, available, this has not noticeably restricted the use of S-2 type roving products or roving for unidirectional tape. S-2 type rovings are available in yields of 250, 750, and 1250 yards per lb (500, 1500, and 250 m/kg). Although woven rovings may be considered a fabric product form it should be noted for its importance for military applications. Also, there are glass product forms which could be considered as complimentary products for advanced structures. These would include milled fibers and chopped strand.

2.3.3.3 *Advantages and Disadvantages.* For many years glass composites have had a distinct strength to weight advantage. Although the rapid evolution of carbon and aramid fibers have gained advantages, glass composite products have still prevailed in certain applications. Cost per weight or volume, certain armament applications, chemical or galvanic corrosion resistance, electrical properties, and availability of many product forms remain as examples of advantage. Coefficient of thermal expansion and modulus properties compared to carbon composites may be considered as typical disadvantages. When compared to aramid composites, glass has a disadvantage as to tensile properties but an advantage as to ultimate compression, shear properties, and moisture pick-up.

Commercial uses for glass products are many-fold. These include filtration devices, thermal and electrical insulation, pressure and fluid vessels, and structural products for automotive and recreation vehicles. Many uses are applicable to military and aerospace products as well. A partial listing would include: asbestos replacement, circuitry, optical devices, radomes, helicopter rotor blades, and ballistic applications. Because of the many product forms, structural applications are limitless to fabricate. If there are limitations, compared to other fibers, they may include low thermal and electrical conductivity or perhaps melting temperatures when compared to carbon fibers.

Typical properties for glass fibers and composite materials reinforced with continuous glass fibers are shown in Tables 2.3.3.3(a)-(d).

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TABLE 2.3.3.3(a) *Typical glass fiber electrical properties.*

	E	S-2	H_R
Density			
Lbs/Cu. In.	0.094	0.089	
g/cm^3	2.59	2.46	2.49
Tensile Strength			
ksi	500	665	665
MPa	34,450	45,818	45,818
Modulus of Elasticity			
Msi	10.5	12.6	12.6
GPa	72.35	86.81	86.81
% Ult. Elongation	4.8	5.4	5.4
Dielectric Constant			
73°F @ 1 MHz	6.3-6.7	4.9-5.3	NA

TABLE 2.3.3.3(b) *Typical glass fiber thermal properties.*

	E	S-2	S_R
Coeff. Thermal Expan. 10^6			
in/in/°F	2.8	1.3	
m/m/°C	5.1	2.6	
Softening Point °F (°C)	1530 (832)	1810 (988)	1778 (970.)
Annealing Point °F (°C)	1210 (654)	1510 (821)	1490 (810.)

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TABLE 2.3.3.3(c) *Typical corrosion resistance of glass fibers (Wt. Loss %).*

Fluid	E	S-2	S_R
10% H_2SO_4	42	6.8	NA
10% HCL	43	4.4	NA
10% HNO_3	43	3.8	NA
H_2O (Distilled)	0.7	0.7	NA
10% Na OH	29	66	NA
10% KOH	23	66	NA

Conditions: 200°F (96°C) - one week immersion

Unburdened costs vary pending product forms and glass types. Typical yield certified "E" glass rovings cost $1.40 per lb., whereas certified S-2 type 750 yield rovings average $6.30 per lb. Lower costing for rovings are experienced with rail car purchases. Typical unburdened fabric costs also vary by weave and fiber type. "E" glass 120 style averages $13.10 per lb., 7781 averages $4.35 per lb., S-2 type 6781 style is $8.40 per lb.

2.3.3.4 *Common manufacture methods and variables.* Most often raw products (and/additives) are mixed and are premelted into marbles. This form facilitates sampling for analysis but, more important, presents a raw product form for automated feeding to the individual melt furnaces. Another method is to feed, via hoppers, dried raw products directly to batch cans. Regardless of the raw form, the material is fed into furnaces to become molten at approximately 2800°F (1500°C). The molten mass flows onto plates which contain many bushings with small orifices from which the individual filaments are drawn. In some cases the individual bushings are heat controlled within <1F° (0.6C°). The diameter of the filaments is controlled by the viscosity of the glass melt and the rate of extrusion. Cooling or solidification occurs rapidly as the glass leaves the bushings in filament form under ambient conditions. Cooling is often added by water spray and/or application of the binders. The individual untwisted filaments are gathered and high speed wound on tubes or "cakes". Sometimes finishes are applied after the strands are wound on the tubes then conditioned (dried). For products common to this document the strands are "C" (continuous) filaments--not "S" (staple) filament. To produce rovings the strands are then creeled, unwound and gathered again to form ends or multiple untwisted strands. (See Table 2.3.3.4(a).) This process of gathering or combining is again repeated to form rovings of desired yields (yards per pound). For weaving of fabrics and braiding, the strands are twisted to form yarns. (See Table 2.3.3.4(b).) Single yarns are composed of single strands twisted by itself. Two (etc.) strand construction is two strands twisted to produce a single yarn. Plied yarns are made from twisting two or more yarns together. Twisting and plying is often referred to as "throwing". A variable in processing "C" filament products is the repeated tensioning required during the numerous product forms fabrication. Tensioning devices are used--such as: disc-type or "whirls", gate-type, tension bars or "S" bars, and compensating rolls in the delivery from the creels. Humidity is another controlled variable in the twisting, plying, braiding, warping, slashing, gulling and weaving areas. These operations are facilitated to maintain a relative humidity of 60 to 70 percent range. During the glass processing operations surface abrasion is a factor which must be monitored. The many devices such as: guide eyes, spacer bars, rollers and such are subject to wear and must be

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maintained. Wear could also affect tensioning. These contact devices are manufactured from materials including: stainless steel, chromium plating, and ceramics.

Additional information can be found in References 2.3.3.4(a) - (c).

2.3.4 Boron. Elemental boron fiber is formed as a deposition reaction on a hot tungsten wire which is continuously drawn through a reactor containing BCl_3 and H_3. The tungsten wire substrate also reacts to form tungsten boride in the core. The crystalline structure of the deposited boron is considered amorphous due to its small size (20Å). Boron is available as a cylindrical fiber in two

TABLE 2.3.3.3(d) *Typical cured epoxy/glass mechanical properties.*

E Glass, Woven 7781 Style	Standard Structural	Dual Purpose Structural/Adhesive
Tensile Strength ksi (MPa)	63 (430)	48 (330)
Tensile Modulus Msi (GPa)	3.8 (36)	2.8 (19)
Compressive Strength ksi (MPa)	60. (410)	50. (340)
Compressive Modulus Msi (GPa)	3.6 (25)	3.2 (22)
Flexural Strength ksi (MPa)	80. (550)	65 (450)
Flexural Modulus Msi (GPa)	3.7 (26)	3.3 (23)
Interlaminar Shear ksi (MPa)	2.6 (18)	3.8 (26)
Sandwich Peel lb/in Width (N/m width)	N.A.	30. (3.4)
Metal-to-Metal Peel lb/lin. in. (N/lin. m)	N.A.	55 (6.3)
Specific Gravity gm/cc (lb/in^3)	1.8 (0.065)	1.6 (0.058)
Cured Resin Content % Wt.	33	48

Reference: Fabric MIL-C-9084, VIII B
Resin MIL-R-9300, Ty I MIL-A-25463, Ty I, C1 2

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TABLE 2.3.3.4(a) *Basic strand fiber designations and strand counts (Reference 2.3.3.1(c)).*

Filament Diameter Designation		Strand Count (Number)		
		TEX	U.S. Customary	
SI (μm)	U.S. Customary (Letter)	g/km	100 Yd. Cuts/Lb.	Yds./Lb.
5	D	11	450	45,000
7	E	22	225	22,500
9	G	33	150	15,000
10	H	45	110	11,000
13	K	68	75	7,500

nominal diameters, 4- and 5.6-mil (0.,10 and 0.14 mm), which have a density of 2.57 and 2.49 g/cm^3 (0.0929 and 0.0900 lb.in^3), respectively. Chemical etching of the fiber surface produces a higher strength, but the process is not used commercially.

Boron fiber is unmatched for its combination of strength, stiffness, and density. The tensile modulus and strength of boron fiber are 60 x 10^6 psi and 0.52 x 10^6 psi (40 GPa and 3600 MPa). Thermal conductivity and thermal expansion are both low, with a coefficient of thermal expansion of 2.5-3.0 x 10^{-6}/°F (4.5-5.4 x 10^{-6}/°C). Typical end-use properties are shown in Table 2.3.4. Currently, the cost of boron fiber is approximately an order of magnitude higher than standard carbon fiber.

Available almost exclusively in filament or epoxy matrix prepreg form, boron fiber has been used for aerospace applications requiring high strength and/or stiffness, and for selective reinforcement in sporting goods. The most notable use of this fiber is the stabilizer sections of the F-14 and F-15 military aircraft, dorsal longeron of the B-1B bomber, and the repair of metallic airframe structures. High modulus (HM) or high strength (HS) carbon/epoxy composites can match either the tensile modulus or strength of boron composites at a more economical price, but boron/epoxy composites offer twice the composite strength. Additional information can be found in References 2.3.4(a) through (g).

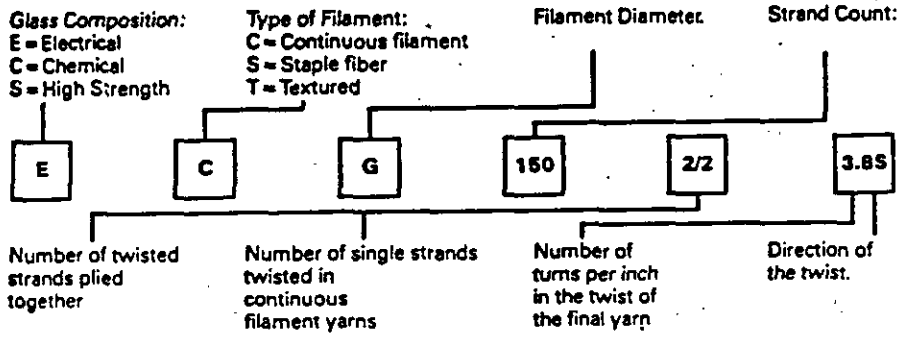
2.3.5 Alumina. Continuous polycrystalline alumina fiber is ideally suited for the reinforcement of a variety of materials including plastics, metals, and ceramics. Alumina is prepared in the form of continuous yarn containing a nominal 200 filaments. It is supplied in bobbins containing continuous filament yarn, and alumina/aluminum and alumina/magnesium plates. Alumina staple is also available for short fiber reinforcement.

Fibers that are more than 99% purity α-alumina have excellent chemical resistance, and have higher modulus and temperature capabilities than ceramic fibers containing silica. The high modulus of 55 Msi (380 GPa) is comparable to that of boron and carbon. The average filament tensile strength is 200 ksi (1.4 GPa) minimum. Since alumina is a good insulator, it can be used in applications where conducting fibers cannot. Nominal properties of alumina are listed in Table 2.3.5(a). Cost projections

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TABLE 2.3.3.4(b) *Typical yarn nomenclature (Reference 2.3.3.1(c)).*

Filament Designation	Nominal Filament Diameter (inches)	Strand Count (x100 = yds/lb)	Approximate Number of Filaments
D	0.00021	1800	51
D	0.00021	900	102
B	0.00015	450	408
D	0.00021	450	204
D	0.00021	225	408
E	0.00029	225	204
B	0.00015	150	1224
C	0.00019	150	750
DE	0.00025	150	408
G	0.00036	150	204
H	0.00043	110	204
C	0.00019	75	1500
DE	0.00025	75	816
G	0.00036	75	408
K	0.00053	75	204
H	0.00043	55	408
DE	0.00025	37	1632
G	0.00036	37	816
K	0.00053	37	408
H	0.00043	25	816
K	0.00053	18	816
G	0.00036	15	2052



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Table 2.3.4 *Typical end-use properties of a unidirectional boron-epoxy laminate (V_f = 0.5)*

	Value, ksi (MPa)
Moduli	
Tensile, longitudinal	30 (207)
Tensile, transverse	2.7 (19)
Strength	
Tensile, longitudinal	192 (1323)
Tensile, transverse	10.4 (72)
Compressive, longitudinal	353 (2432)

for alumina are competitive with carbon.

Alumina, in continuous form, offers many advantages for composite fabrication including ease of handling, the ability to align fibers in desired directions, and filament winding capability. The fact that alumina is an electrical insulator combined with its high modulus and compressive strength make it of interest for polymer matrix composite applications. For example, alumina/epoxy and aramid/epoxy hybrid composites reinforced with alumina and aramid fibers have been fabricated and are of potential interest for radar transparent structures, circuit boards, and antenna supports. Typical properties of unidirectional composites are listed in Table 2.3.5(b).

Table 2.3.5(a) *Nominal properties of alumina*

Composition	> 99% α-Al_2O_3	Filaments/yarn	200, nominal
Melting Point	3713°F (2045°C)	Tensile Modulus	55 Msi (385 GPa)
Filament Diameter	0.8×10^{-3} in. (20μm)	Tensile Strength	200 ksi (1.4 GPa) minimum
Length/Weight	(~4.7 m/gm)	Density	0.14 lb/in^3 (3.9 gm/cc)

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Table 2.3.5(b) *Nominal properties of alumina composite (V_f ~ 50-55%)*

Moduli		
Tensile, axial		30-32 Msi (210-220 GPa)
Tensile, transverse		20-22 Msi (140-150 GPa)
Shear		7 Msi (50 GPa)
Strength		
Tensile, axial		80 ksi (600 MPa)
Tensile, transverse		26-30 ksi (130-210 MPa)
Shear		12-17 ksi (85-120 GPa)
Fatigue - Axial Endurance Limit	10^7 cycles at 75% of static ultimate (tension-tension, R = 0.1, and rotating-bending)	
Average Thermal Expansion	68-750° (20-400°C)	
Axial		4.0 μin/in/°F (7.2 μm/m/°C)
Transverse		11 μin/in/°F (20 μm/m/°C)
Thermal Conductivity	68-750° (20-400°C)	22-29 BTU/hr-ft-°F (38-50 J/m-s-°C)
Specific Heat	68-750° (20-400°C)	0.19-0.12 BTU/lbm-°F (0.8-0.5 J/gm-°C)
Density		0.12 lbm/in^3 (3.3 gm/cc)

2.3.6 ***Silicon carbide.*** *Various super-refractory* fibers were first produced in the early 1950's based upon work by the Arthur D. Little Co. by various production methods. The primary of these based upon:

1. Evaporation for polycrystalline fiber process.

2. HITCO continuous process for polycrystalline fibers.

3. Vapor deposition of aluminum oxide single crystals (Reference 2.3.6(a)).

The most recent advances in the CVD type process in use by AVCO consist of substrate wires drawn through glass reaction tubes at high temperature.

Silicon carbide fibers are produced with a nominal 140 μm filament diameter and are characteristically found to have high strength, modulus and density. Fiber forms are oriented toward the strengthening of aluminum or titanium alloys for both longitudinal and transverse properties.

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Additional forms are also produced as polycrystalline fiber whiskers of varying length and diameters (Reference 2.3.6(b)).

Several systems for describing the material morphology exist, the alpha and beta forms designated by Thibault and Lindquist being the most common (Reference 2.3.6(c)).

Practically all silicon carbide monofilament fibers are currently produced for metal composite reinforcement. Alloys employing aluminum, titanium, and molybdenum have been produced (Reference 2.3.6(b)).

General processing for epoxy, bisimide, and polyimide resin can be either via a solvated or solventless film impregnation process, with cure cycles equivalent to those provided for carbon or glass reinforced products. Organic matrix silicon carbide impregnated products may be press, autoclave, or vacuum bag oven cured. Lay-up on tooling proceeds as with carbon or glass composite products with all bleeding, damming, and venting as required for part fabrication. General temperature and pressure ranges for the cure of the selected matrix resins used in silicone carbide products will not adversely affect the fiber morphology.

Silicon carbide ceramic composites engineered to provide high service temperatures (in excess of 1450°C) are unique in several thermal properties. The overall thermal resistance is determined by the through conductivity, thermal expansion, thermal shock and creep resistance. Thermal conductivities of silicon carbide ceramics have a range in W/m • °K of 60 at room temperature to 48 at 800°C. Expansion values range, in percentage of original dimension, from 0.05 at 200°C to 0.30% at 800°C. The creep resistance of the silicon carbide ceramic will vary as the percentage of intra-granular silicon phase increases. In general, the creep rate is very low when compared to aluminum oxide or zirconium oxide materials.

Mechanical properties of silicon carbide materials are shown in Table 2.3.6(a). Fracture toughness as measured by double torsion analysis has reported literature values in K_{Ic} ($MPa\sqrt{m}$) of 0.6 for monocrystalline SiC/Si to 6.0 for hot pressed SiC ceramics (Reference 2.3.6(g)). Corrosion resistance, of consideration in advanced structural material design, has been evaluated with a variety of mineral acids on the basis of corrosive weight loss as shown in Table 2.3.6(b).

General cost ranges for the CVD processed fibers are currently in the $100.00 /lbs, with the control in crystalline form requiring additional expense (Reference 2.3.6(e)).

2.3.7 Quartz. Quartz fiber is very pure (99.95%) fused silica glass fiber. Typical fiber properties are shown in Table 2.3.7(a). Quartz is produced as continuous strands consisting of 120 or 240 individual filaments of 9 micron nominal diameter. These single strands are twisted and plied into heavier yarns. Quartz fibers are generally coated with an organic binder containing a silane coupling agent which is compatible with many resin systems. Strands for rovings are combined into multiple ends without applied twist. These strands are coated with a "direct size" which is compatible with many resins. Woven fabrics may be used as woven or may be "scoured" (washed) to remove the nonfunctional components of the binder and some, but not all, of the silane coupling agent. Following scouring, the fabric may be finished with a variety of silane coupling agent finishes having specific resin compatibility.

Quartz fiber nomenclature is the same as that for E or S glass fibers except that the glass composition is designated by the letter Q as shown in Table 2.3.7(b). Commonly used quartz fabrics are listed in Table 2.3.7(c). Quartz rovings are continuous reinforcements formed by combining a number of 300 2/0 zero twist strands. End counts of 8, 12, and 20 are available having yields from

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TABLE 2.3.6(a) *Material properties of silicon carbide materials.*

Property	Reported Values	Reference Information
FLEXURAL STRENGTH (ksi)	100-1000	single crystal, 99 + % purity (1)
	10-60	polycrystalline materials, 78-99% purity, with < 12 + % free silicon, sintered (1)
	5-8	sintered SiC - graphite composites - epoxy, imide, polyimide matrix. (2)
COMPRESSIVE STRENGTH (ksi)	500-1000	single crystal, 99 + % purity (1)
	10-25	polycrystalline materials, 78-99% purity, with < 12 + % free silicon, sintered.(2)
	14-60	Sintered SiC - graphite composites - epoxy, imide, polyimide matrix. (2)
TENSILE STRENGTH (ksi)	~20	single crystal, 99 + % purity (1)
	5-20	polycrystalline materials, 78-99% purity, with < 12 + % free silicon, sintered.(2)
	2.5-25	sintered SiC - graphite composites - epoxy, imide, polyimide matrix. (2)
MODULUS OF ELASTICITY (Msi)	~9.5	single crystal, 99 + % purity (1)
	~7.0	Polycrystalline materials, 78-99% purity, with < 12 + % free silicon, sintered.(2)

(1) Reference 2.3.6(b)
(2) Reference 2.3.6(d)

750 to 1875 yards per pound. Quartz fibers are also available in the form of chopped fiber in cut lengths from 1/8 inch to 2 inches.

Quartz fibers with a filament tensile strength of 850,000 psi (5,900 MPa) have the highest strength-to-weight ratio, virtually exceeding all other high temperature materials. The quartz fibers can be used at temperatures much higher than "E" glass or "S" glass fiber with service temperatures up to 1920°F (1050°C) possible. Quartz fibers do not melt or vaporize until the temperature exceeds 3000°F (1650°C), providing potential in ablative applications. Additionally, these fibers retain virtually all of the characteristics and properties of solid quartz.

The quartz fibers are chemically stable. They are not affected by halogens or common acids in the liquid or gaseous state with the exception of hydrofluoric and hot phosphoric acids. Quartz fibers should not be used in environments where strong concentrations of alkalies are present.

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TABLE 2.3.6(b) *Corrosive weight loss in mg/cm² • yr (Reference 2.3.6(e)).*

TEST REAGENT	TEMP °C	Si/SiC COMPOSITES 12% Si	SiC NO FREE Si
98% SULFURIC ACID	100	55	1.8
50% SODIUM HYDROXIDE	100	complete within days	2.5
53% HYDROFLUORIC ACID	100	7.9	< 0.2
70% NITRIC ACID	100	0.5	< 0.2
25% HYDROCHLORIC ACID	100	0.9	< 0.2

Quartz fibers, when combined with certain matrix systems, offer potential advantages in stealth application due to their high electrical resistivity properties. Quartz does not form paramagnetic centers, nor does it capture neutrons in high energy applications. These fibers offer a low dielectric constant and loss tangent providing excellent properties as electrical insulators. Typical properties for quartz fibers combined with three different polymer matrix systems are shown in Tables 2.3.7(d) - (f). Quartz products are relatively expensive compared to "E" or "S_2" glass products, with prices ranging from $45 to $150 per pound. Additional information can be found in Reference 2.3.7.

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Table 2.3.7(a) *Properties of quartz fiber.*

Specific gravity	2.20
Density, lb/in^3	0.0795
Tensile strength	
Monofilament, ksi	870
GPa	6.0
Roving, ASTM D2343 Impregnated	
Strand Test -	
Astroquartz II 9779, ksi	530.5
GPa	3.6
Modulus, Msi	10.0
GPa	72.0
Elongation, percent	
$\frac{\text{Monofilament Tensile Strength x 100}}{\text{Modulus}}$	8.7
Thermal	
Coefficient of expansion	
10^{-6} in/in/°F	0.3
10^{-6} cm/cm/°C	0.54
Thermal conductivity	
Cal/sec/cm/°C	0.0033
BTU/hr/ft/°F	0.80
BTU/hr/sq ft/in/°F	9.5
Specific heat	
Joules/Kg/°C	7500
BTU/lb/°F	1.80
Electrical	
Dielectric constant, 10 GHz, 75°F	3.78
Loss tangent, 10 GHz, 75°F	0.0001

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TABLE 2.3.7(b) *Quartz continuous strands.*

Strand Number	Number of filaments	Nominal yds/lb	Filament Diameter 10^{-5} in.
QCG 300 1/0	119(1)	30,000	45
QCG 300 2/0	240(2)	15,000	35
QCG 300 1/2	240(1)	15,000	35
QCG 300 2/2	480(1)	7,500	35
QCG 300 2/8	1920(1)	1,875	35

(1) Used for fabric yarns.
(2) Used for roving and fabric yarns.

TABLE 2.3.7(c) *Construction of woven fabrics for aerospace applications.*

Style	Count	Warp Fill	Fill Yarn	Weave	Weight Oz/Sq.Yd.
503	50x50	300 1/2	300 1/2	plain	3.5
507	27x25	300 1/2	300 1/2	plain	2.0
525	50x 50	300 1/0	300 1/0	plain	2.0
527	42x32	300 2/2	300 2/2	plain	5.6
531	68x65	300 1/2	300 1/2	8HS	5.1
557	57x31	300 2/2	300 1/0	crowfoot	5.0
570	38x24	300 2/8	300 2/8	5HS	19.3
572	17x16	300 2/8	300 2/8	plain	9.9
581	57x54	300 2/2	300 2/2	8HS	8.4
593	49x46	300 2/2	300 2/2	5HS	7.5

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TABLE 2.3.7(d) *Typical properties for quartz/epoxy.*

PROPERTY	RESULTS
Tensile Strength, psi	
Room Temp.	74,900 - 104,200
1/2 Hr. @ 350°F	65,400 - 92,200
Tensile Modulus, psi	
Room Temp.	3.14×10^6 - 4.09×10^6
1/2 Hr. @ 350°F	2.83×10^6 - 3.67×10^6
Flexural Strength, psi	
Room Temp.	95,500 - 98,900
1/2 Hr. @ 350°F	53,900 - 75,900
Flexural Modulus, psi	
Room Temp.	3.27×10^6 - 3.46×10^6
1/2 Hr. @ 350°F	2.78×10^6 - 3.08×10^6
Compressive Strength, psi	
Room Temp.	66,400 - 72,400
1/2 Hr. @ 350°F	42,600 - 49,900
Compressive Modulus, psi	
Room Temp.	3.43×10^6 - 3.75×10^6
1/2 Hr. @ 350°F	3.10×10^6 - 3.40×10^6
Laminate Resin Content	
wt%	33.5 - 32.0
Specific Gravity	
g/cm^3	1.73 - 1.77

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TABLE 2.3.7(e) *Typical properties for quartz/toughened epoxy.*

PROPERTY	RESULTS
Flexural Strength, ksi	
Room Temp.	129.0
@ 180°F	111.7
Flexural Modulus, Msi	
Room Temp	4.0
@ 180°F	3.9
Compressive Strength, ksi	
Room Temp.	88.2
@ 180°F	77.5
Room Temp. Wet	76.6
@ 180°F Wet	70.8
Compressive Modulus, Msi	
Room Temp.	4.2
@ 180°F	3.8
Room Temp. Wet	3.7
@ 180°F Wet	4.0
Short Beam Shear, ksi	
Room Temp.	13.2
@ 180°F	11.8
Room Temp. Wet	9.2
@ 180°F Wet	9.3
Resin Content, wt%	32.0
Ply Thickness, in	0.009

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TABLE 2.3.7(f) *Typical properties for quartz/polyimide.*

PROPERTY	RESULTS
Tensile Strength, psi	
Room Temp.	79,100 - 105,100
Tensile Modulus, psi	
Room Temp.	- 3.9×10^6
Flexural Strength, psi	
Room Temp.	93,700 - 101,700
1/2 Hr. @ 350°F	62,400 - 68,300
Flexural Modulus, psi	
Room Temp.	3.2×10^6 - 3.2×10^6
1/2 Hr. @ 350°F	2.8×10^6 - 2.6×10^6
Compressive Strength, psi	
Room Temp.	67,400 - 67,000
1/2/Hr. @ 350°F	45,200 - 38,600
Compressive Modulus, psi	
Room Temp.	3.5×10^6 - 3.7×10^6
1/2 Hr. @ 350°F	2.8×10^6 - 2.8×10^6
Laminate Resin Content	
wt%	36.2 - 36.2

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2.4 PRODUCT FORMS.

2.4.1 Unidirectional tapes. All product forms generally begin with spooled unidirectional raw fibers packaged as continuous strands. Normally, untwisted tows or ends are specified for unidirectional product forms to obtain ultimate fiber properties. This particular product form depends on the proper fiber wet-out and the tenacity of the uncured resin to maintain proper fiber placement until the tape reaches the curing procedure.

2.4.1.1 *Conventional unidirectional tapes.* This particular form has been the standard within the user industry for many years and is common with thermosetting resins. The most common method of manufacture is to draw collimated raw (dry) strands into the impregnation machine where hot melted resins are combined with the strands using heat and pressure. The combination of fibers and resin usually travels through the machine between coated carrier papers or films for ease of release. The tapes are usually trimmed to specified widths in line. One side of the carrier is usually removed prior to the roll-up position to facilitate continuous visual inspection. The remaining carrier is usually left in place with the tape to serve as a separator on the roll and as a processing aid for fabrication purposes. The tape manufacturing process is continuous within the linear limits of the raw strands creeled to the machine or specified lot size or availability of resin. Most impregnation machines are designed to permit in-line change over to new rolls (take-ups) without interruption. Raw strand collimation is adjusted to control specified areal weight (dry weight/area). Resin filming for tape machine operations is often done as a separate controlled operation. Some machines accommodate in-line filming that permit resin content adjustments during the impregnation process. Tapes as wide as 60 inches are commercially available.

2.4.1.2 *Two-step unidirectional tapes.* Although not a general practice within the prepreg industry, there are unidirectional tapes manufactured from preimpregnated rovings. The collimation of these rovings to make tapes allow the use of solution impregnated resins, rather than hot melt systems. Although the product form may be similar to conventional tapes, thin uniform flat tapes may be difficult to produce.

2.4.1.3 *Supported unidirectional tapes.* To enhance specific mechanical properties or part manufacturing handling operations, it is sometimes advantageous to add product form during the manufacture of unidirectional tapes. Generally, these added fibrous forms are lightweight to be accommodated during the normal tape manufacture operation. The added form may be combined in the machine dry or preimpregnated prior to the tape production. More common added forms are lightweight mats or scrim fabrics of the same or unlike fiber type. The added product form will affect material properties compared to tapes without the supporting material.

2.4.1.4 *Coated unidirectional tapes.* Some tape suppliers offer the option of added tape surface coating. These resinous coatings of films are usually of different rheology or viscosities from the fiber impregnation resin to remain as distinct boundaries between plies of the cured tapes. As with supported unidirectional tapes, the added layer may be combined during the tape manufacturing operation.

2.4.1.5 *Preplied unidirectional tapes.* These tapes originate as any of the above-described tape forms in single-ply form. Then through a process of stacking, two or more layers of individual tapes are oriented at predetermined angles in relation to the centerline of the new progressively generated tape or broadgoods form. The original individual tapes are located side to side in each angled layer to form a continuous linear form. The original single-ply tapes are usually precut in segments at angles to correspond to the new product form's edges. The progressive stacking sequence usually takes place on a continuous carrier (paper or film) atop a flat surface much like the fabrication process. The carrier, with the preplied form in place, is utilized to take up the preplied tapes onto a shipping/handling

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core. The predetermined length of the individual precut segments will generally regulate the width of the preplied tapes. However, a final trim of both edges to control specified widths can be incorporated during the take-up step. For economic purposes the preplying operation usually is done in widths of approximately 24 inches or greater. Should narrow widths be required, they can be accommodated with a secondary slitting operation. To some extent the retention of this product form's continuity is, like single ply tapes, dependent on the tack or tenacity of the uncured resin.

2.4.2 Woven fabrics. Woven or knitted fabric product forms, unlike tapes and rovings, are in most circumstances produced prior to the resin impregnation step. Therefore, these product forms, in most part, offer product continuity or retention of fiber placement prior to, during, and after the impregnation step. Most fabric constructions offer more flexibility for lay-up of complex shapes than straight unidirectional tapes offer. Fabrics offer the option for resin impregnation either by solution or the hot melt process. Generally, fabrics used for structural applications use like fibers or strands of the same weight or yield in both the warp (longitudinal) and fill (transverse) directions. However, this is not a set rule as the number of combinations of reinforcement fibers and weave styles are essentially unlimited for custom applications. Also some fabrics are produced which incorporate thermoplastic strands that then become the resin matrix when the fabric is processed to its final state.

Woven fabric selections for structural applications have several parameters which may be considered. These variables are strand weight, tow or strand count, weave pattern, and fabric finish. The variables for glass fabrics are considerably greater than carbon fabrics due to the availability of a greater range of yarn weights. The availability of carbon tow weights or filament count tows are few in comparison. Generally, the lighter or thinner the fabric, the greater the fabric cost. Also factored into the cost is the complexity of the weave pattern or machine output for heavy fabrics. For aerospace structures, tightly woven fabrics are usually the choice for areal weight considerations, minimizing resin void size, and maintaining fiber orientation during the fabrication process.

2.4.2.1 *Conventional woven fabrics.* Woven structural fabrics are usually constructed with reinforcement tows, strands, or yarns interlocking upon themselves with over/under placement during the weaving process. The more common fabrics are plain or satin weaves. The plain weave construction results from each fiber alternating over and then under each intersecting strand (tow, bundle, or yarn). With the common satin weaves, such as 5 harness or 8 harness, the fiber bundles traverse both in warp and fill directions changing over/under position less frequently. (See Figures 2.4.2.1(a) and (b).)

These satin weaves have less crimp and are easier to distort than a plain weave. With plain weave fabrics and most 5 or 8 harness woven fabrics the fiber strand count is equal in both warp and fill directions. Example: 3K plain weave often has an additional designation such as 12 x 12, meaning there are twelve tows per inch in each direction. This count designation can be varied to increase or decrease fabric areal weight or to accommodate different fibers of varying weight.

2.4.2.2 *Stitched or knitted fabrics.* These fabrics can offer many of the mechanical advantages of unidirectional tapes. Fiber placement can be straight or unidirectional without the over/under turns of woven fabrics. The fibers are held in place by stitching with fine yarns or threads, after preselected orientations or one or more layers of dry plies. This product form, much like preplied unidirectional tapes, offers a wide range of multi-ply orientations. Although there may be some added-weight penalties or loss of some ultimate reinforcement fiber properties, some gain of interlaminar shear and toughness properties may be realized. Some common stitching yarns are polyester, aramid, or thermoplastics.

2.4.2.3 *Specialty fabrics.* To list all the possible woven or knitted fabric forms would require space beyond the scope of this document. As an example, there are in excess of one hundred glass fabrics

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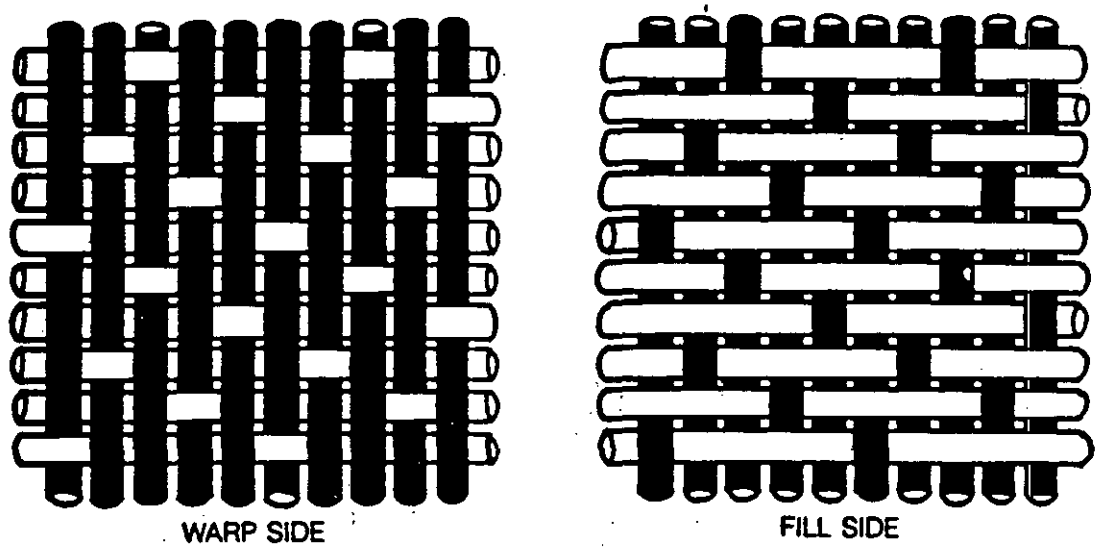


FIGURE 2.4.2.1(a) *5 Harness Satin weave construction. In this weave construction each yarn goes over 4 and under 1 yarn in both directions.*

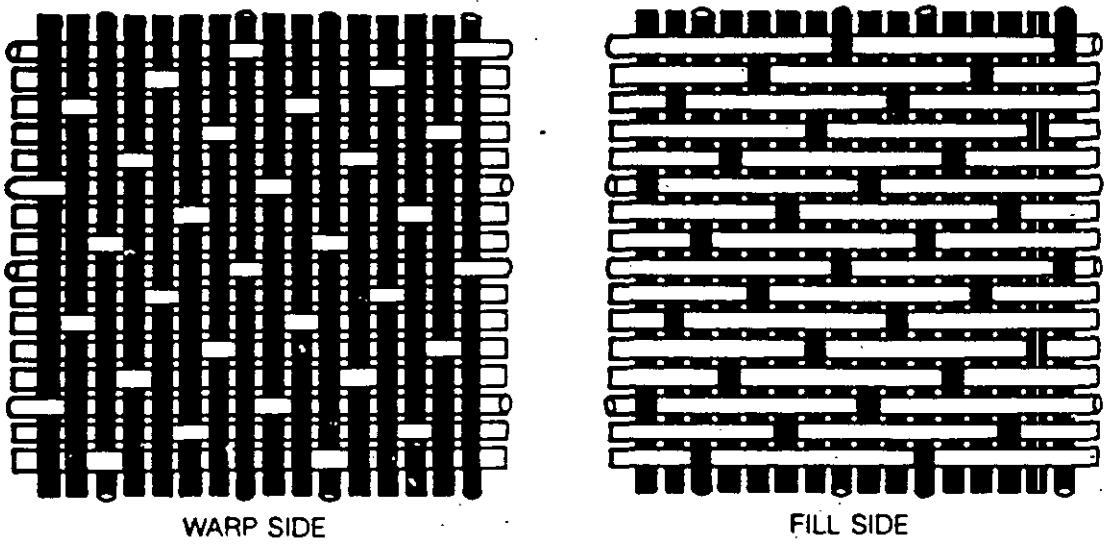


FIGURE 2.4.2.1(b) *8 Harness Satin weave construction. In this weave construction each yarn goes over 7 and under 1 yarn in both directions.*

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listed in a standard weaver's handbook. These fabrics vary in weight from 0.55 oz./square yard (18.65 gm/m^2) to 53 oz./square yard (1796 gm/m^2) and vary in thickness from 0.0012 in (0.0305 mm) to 0.0450 in (1.143 mm). Such an industrial listing is limited to but a few basic patterns such as plain, basket, Leno, harness, and twill weaves. There are many other fabrics such as triaxial, orthogonal, knitted bidirectional, stitched multilayer, and angle interlock, to name a few. From these also arise combinations and three-dimensional weaves.

2.4.3 Rovings. This impregnated product form generally applies to a single grouping of filament or fiber ends, such as 20 end or 60 end glass rovings. Carbon rovings are usually identified as 3K, 6K, or 12K rovings. Other counts are available. It is possible, preferably during the resin impregnation step, to combine two or more counts or filaments or ends to increase the rovings weight, width, etc. per linear length. For mechanical testing purposes individual rovings are usually wound, side by side, to form single ply tapes and processed as such. The roving product form, with its packaging on individual spools, offers the means for automated fiber placement during the manufacture of parts. The rovings can be placed in a unidirectional pattern, like tapes, or to generate a crossover interlocking effect. Most applications for roving products utilize mandrels for filament winding and then resin cure to final configuration. In addition, this product form is used for efficient build-up of oriented filaments to create preforms. The preforms are combined with other lay-ups or processed individually in closed tools rather than the conventional mandrel cure process. Most rovings are supplied untwisted, in nearly flat continuous bands. Band widths can be controlled to a degree during the impregnation step. Compared to tapes or fabrics, roving areal weights for individual plies or wraps are more dependent on the winding process than the impregnation step. However, resin control of the preimpregnated rovings shares a like degree of accuracy.

2.4.4 Braiding. The braiding process fabricates a preform or final shape at the same time that it generates the woven form. This product form is a unique fiber reinforcement which can use preimpregnated yarn as well as dry fibers. The main advantage of the braiding process is its ability to conform to odd shapes and maintain fiber continuity while developing high damage tolerance compared to unidirectional and laminated products. This allows formation of square, oval, and other constant cross-section shapes. The three-dimensional form of braiding has evolved to the point of allowing the non-uniform cross-sections to be fabricated while maintaining weaving in all three planes.

The uses of braiding have varied during its development. The best known example of braided structure is the fiberglass and carbon fishing rods that became popular in the 1980's. Braiding has also found uses in pressurized piping and complex ducting. A demonstration of its versatility is the open-wheel race car body which was fabricated by braiding. The process has also been used in rocket applications for motorcases and launchers.

2.5 FABRICATION. This section on fabrication includes methods generally used in the fabrication of fiber-reinforced composite parts. No attempt has been made to cover all fabrication methods being used in the composite industry.

2.5.1 Autoclave curing. This process uses a pressurized vessel to apply both pressure and heat to parts that have been sealed in a vacuum bag. Autoclaves in general operate at 10 - 300 psi (70-2000 kPa) and up to 800°F (420°C). The following steps are typically used in autoclave curing:

1. Composite material required for the part is placed on the mold.
2. Cure monitoring devices are installed.
3. Bleeder and breather materials are placed over the part.
4. The vacuum bag is placed over the part and sealed. Bag is then checked for any leaks.
5. Parts are placed in autoclave for curing. Pressure, temperature, and time are monitored during the cure cycle to ensure curing parameters are met.

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The autoclave process produces high quality parts with low void content.

2.5.2 Press curing. Press curing uses heated platens to apply both pressure and heat to the part. Presses, in general, operate at 20 - 1000 psi (140 - 7000 kPa) and up to 600°F (320°C). Press curing is very economical for flat parts and high production rates. Tooling requires matched die molds for contoured parts. The following steps are used in press molding:

1. Composite material is placed in the mold cavity.
2. Cure monitoring devices are installed.
3. Parts are placed into press and cured. Pressure, temperature, and time are monitored during the cure cycle to ensure curing parameters are met.

Press curing produces high quality parts with low void content.

2.5.3 Pultrusion process. The pultrusion process consists of passing a continuous fiber bundle through a resin bath (would not be required for a prepreg thermoset or thermoplastic) and then through a heated die for part shape and cure. This process is limited to constant cross-sections such as rods, tubes, I-beams, and channels. The pultrusion process works well with quick-curing resins and is a very low-cost method for high-production parts with constant cross-sections.

2.5.4 Filament winding. Filament winding is an automated process in which a continuous fiber bundle (or tape), either preimpregnated or wet impregnated with resin, is wound on a removable mandrel in a pattern. The filament winding process consists of winding onto a male mandrel that is rotating while the winding head moves along the mandrel. The speed of the winding head as it moves along the mandrel in relation to the rotation of the mandrel controls the angular orientation of the fiber reinforcement. Filament winding can be done using wet resin winding, preimpregnated yarns and tapes. The following general steps are used for filament winding:

1. Program winder for correct winding pattern.
2. Place required number of fiber spools on the creel for required band width.
3. When wet winding, check resin bath by pulling the bundle through the bath and check fiber-to-resin ratio.
4. Pull fiber through winding eye and start winding.
5. When winding is complete, remove mandrel with part from winder.
6. Cure as required at room temperature or elevated temperature. During cure the mandrel is often rotated to maintain resin distribution.
7. After cure, remove mandrel from part.

2.5.5 Resin transfer molding (RTM). The RTM process resin is pumped into a matched die mold which contains the fiber reinforcement. The process requires low viscosity resins and tools designed with bleed holes to allow air to escape during resin transfer. The holes are closed when the cavity is filled to allow pump to apply pressure to the resin. The following general steps are used for resin transfer molding:

1. Place fiber reinforcement into the mold.
2. Close mold and check that all air vent holes are open.
3. Pump resin into mold until full, then close air vents and pressurize resin.
4. Cure at room temperature or elevated temperature.
5. After cure, remove part from mold.

RTM can yield good quality parts similar to compression molded types.

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2.5.6 Vacuum bag molding. Vacuum bag molding is a process in which the lay-up is cured under pressure generated by drawing a vacuum in the space between the lay-up and a flexible sheet placed over it and sealed at the edges. In the vacuum bag molding process, the reinforcement is generally placed in the mold by hand lay-up using prepreg or wet resin. High flow resins are preferred for vacuum bag molding. The following steps are used in vacuum bag molding:

1. Place composite material for part into mold.
2. Install bleeder and breather material.
3. Place vacuum bag over part.
4. Seal bag and check for leaks.
5. Place tool and part in oven and cure as required at elevated temperature.
6. Remove part from mold.

Parts fabricated using vacuum bag oven cure have lower fiber volumes and higher void contents. Vacuum bag molding is a low-cost method of fabrication and uses low-cost tooling for short production runs.

2.5.7 Thermoforming fiber-reinforced thermoplastics. The thermoforming process, as applied to thermoplastic composite materials, capitalizes on the rapid processing characteristics of thermoplastics. The composite thermoforming process can be broken down to four basic steps:

1. The material is heated to its processing temperature external to the forming tool. This can be accomplished with radiant heat.
2. The oven-heated material is rapidly and accurately transferred to the forming tool.
3. The heated material is pressure-formed with matched die set tooling into desired shape.
4. The formed laminate is cooled and its shape is set by sinking the heat into the tooling.

The processing time for thermoforming is governed by the rates at which heat can be added to the material and then removed. This is primarily a function of the material thermal properties, material thickness, forming temperature, and tooling temperature. The pressures required to shape the material are dependent on various factors including part geometry, material thickness, and formability.

2.5.8 Sandwich construction.

2.5.9 Adhesive bonding.

2.5.10 Braiding. The braiding process fabricates a final shape at the same time that it generates the woven form. In biaxial and triaxial braiding, a mandrel is usually used to form the braid. The mandrel also acts as the mold for the final product. The braiding machine controls the rate of feed of the mandrel and the rotational speed of the carriers. The combination of these parameters and the size of the mandrel controls the braid angle. The braid angle, along with the effective yarn, tape, or tow width (width of the specific size yarn, tape, or tow on the mandrel as placed by the braiding process), ultimately controls the coverage of the braid on the surface of the fabricated form. As the braid angle increases, the maximum size of the mandrel which can be covered with a specific yarn, tape, or tow size decreases. For complicated forms, expendable mandrels may be used. These include mandrels made from low melting temperature metal alloys and water-dissolvable casting materials, and collapsible mandrels.

In three-dimensional (3D) braiding, the weaving process itself is used to control the shape of the fabricated product. The typical 3D braiding process involves a bed of cops, or weaving loops, which are moved in a systematic manner. This systematic movement creates an interwoven product in the x-y plane. As the yarns, tapes, or tows are pulled into the weaving process, the z-direction is also

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intertwined. The resulting product is essentially self-supporting due to interweaving in three directions. *For precision exterior dimension, matched metal molds can* be used during the resin matrix curing process. The following are the general steps involved in the braiding process:

1. Set the feed speed, cop speed, and weave pattern (3D braiding).
2. Run the braiding machine until the product is finished.
3. If prepreg material is not being used, use an appropriate resin impregnation process - RTM, wet resin impregnation, etc.
4. Cure according to the appropriate process determined by the impregnation method - autoclave cure, vacuum bag, RTM, etc.
5. Remove the part from the mold or mandrel.

2.5.11 Fiber placement. Fiber placement is the automated application of multiple preimpregnated or coated fibers directly to a tool surface at zero tension. The application roller remains in intimate contact with the tool surface to provide ply compaction and allows lay-up on complex contours. The use of multiple fibers and the ability to drop off and restart individual fiber forms allows for significant changes in part contour and size. In addition, fiber placement may refer to three dimensional preform fabrication which may utilize dry fibers.

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Chapter 3. QUALITY CONTROL OF PRODUCTION MATERIALS

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3.1 INTRODUCTION. Quality conformance tests are needed to assure the continued integrity of a previously characterized material system. The tests performed must be able to characterize each batch/lot of material so a proper assessment of critical properties of a material system can be made. These critical properties provide information on the integrity of a material system with regard to material properties, fabrication capability, and usage. Additionally, the test matrix must be designed to economically and quickly evaluate a material system.

Quality control in a production environment involves inspection and testing of composites in all stages of prepreg manufacture and part fabrication. Tests must be performed by the material supplier on the fiber and resin as separate materials, as well as on the composite prepreg material. The user of the prepreg must perform receiving inspection and revalidation tests, in-process control tests, and nondestructive inspection tests on finished parts. These tests are described in the following sections and normal industry practice is discussed.

3.2 QUALITY ASSURANCE PROCEDURES.

3.2.1 Receiving inspection. The composite material user typically prepares material specifications which define incoming material inspection procedures and supplier controls that ensure the materials used in composite construction will meet the engineering requirements. These specifications are based on material allowables generated by allowables development programs. The acceptance criteria for mechanical tests must be specified to assure that production parts will be fabricated with materials which have properties above the allowables values.

The user material specifications typically require the suppliers to provide evidence that each production lot of material in each shipment meets the material specification requirements. This evidence will include test data, certification, affidavits, etc., depending upon the user quality assurance plan and purchase contract requirements for a particular material. The test reports contain data to verify the conformance of material properties to the user specifications and acceptance standards.

Acceptance test requirements may vary from user to user. However, the tests must be sufficient to assure the material will meet or exceed the engineering requirements. A typical example of acceptance tests required for carbon/epoxy unidirectional tape is shown in Table 3.2.1. Note that Table 3.2.1 is divided into two parts. The first part concerns uncured prepreg properties. The purpose of these tests is to assure that the resin and fibers materials are within acceptable limits. The second part involves tests on cured laminates or lamina. The 0° tension test evaluates the fiber strength and modulus and the 0° compression test evaluates the reinforcement fiber/resin combination. The short beam shear test is used to evaluate the resin in shear.

3.2.2 Process verification. The quality assurance department for the user generally has the responsibility for verifying that the fabrication processes are carried out according to engineering process specification requirements. This encompasses a wide range of activities described below to control the fabrication process.

Material Control: The user process specifications must set the material control for the following items as a minimum.

1. Materials are properly identified by name and specification.
2. Materials are stored and packaged to preclude damage and contamination.
3. Perishable materials, prepregs and adhesives, are within the allowable storage life at the time of release from storage and the allowed work life at time of cure.

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TABLE 3.2.1 *Acceptance and revalidation tests required for suppliers and users.*

PROPERTY	TESTING REQUIRED			SPECIMENS REQUIRED PER SAMPLE
	PRODUCTION ACCEPTANCE (SUPPLIER)(3)	PRODUCTION ACCEPTANCE (USER)(3)	REVALIDATION (USER)(3)	
Prepreg Properties				
Visual & Dimensional	X	X		-
Volatile Content	X	X		3
Moisture Content	X	X	X	3
Gel Time	X	X	X	3
Resin Flow	X	X	X	2
Tack	X	X	X	1
Resin Content	X	X		3
Fiber Areal Weight	X	X		3
Infrared Analysis	X			1
Liquid Chromatograph	X	X	X	2
Differential Scanning *Calorimetry*	X	X	X	2
Lamina Properties				
Density	X			3
Fiber Volume	X			3
Resin Volume	X			3
Void Content	X			3
Per Ply Thickness	X	X	X	1
Glass Transition Temp	X	X	X	3
Short Beam Shear	X(1)	X(1)	X(2)	6
0° Compression Strength & Modulus	X(2)	X(2)	X(2)	6
0° Tension Strength & Modulus	X(2)	X(2)	X(2)	6

(1) Tests shall be conducted at RT/Dry and Maximum Temperature/Dry (See Volume I, Section 2.2.2).

(2) Tests shall be conducted RT/Dry.

(3) Supplier is defined as the prepreg supplier. User is defined as the composite part fabricator. Production acceptance tests are defined as tests to be performed by the supplier or user for initial acceptance. Revalidation tests are tests performed by the user at the end of guaranteed storage life or room temperature out time to provide for additional use of the material after expiration of the normal storage or out time life.

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4. Prepackaged kits are properly identified and inspected.
5. Acceptance and reverification tests are identified.

Materials Storage and Handling: The user material and process specifications set procedures and requirements for storage of prepregs, resin systems and adhesives to maintain acceptable material quality. Storing these materials at low temperatures, usually 0°F or below, retards the reaction of the resin materials and extends their useful life. Negotiations between the supplier and user result in an agreement on how long the supplier will guarantee the use of these perishable materials when stored under these conditions. This agreed to time is incorporated as one of the requirements in the user material specification.

Materials are generally stored in sealed plastic bags or containers to prevent moisture from condensing on the cold material and migrating into the polymer when it is removed from the freezer and allowed to warm up to ambient temperature. The time interval between material removal from the freezer and when the material bag or container may be opened is generally empirically determined. Physical characteristics such as material roll, stacking height thickness, or material type (e.g., tape vs broadgoods) are considered when determining this time interval. Therefore, the user should have procedures that prevent premature removal of materials from storage bags or containers before material temperature stabilization occurs.

Tooling: The tooling (molds) to be used for lay-up are subject to tool proofing/qualification procedures. This demonstrates that the tooling is capable of producing parts that conform to drawing and specification requirements, when used with the specified materials, lay-up and bagging methods, and cure profile. Also, cured material specimens made from the tool should be tested to ensure they meet specified mechanical and physical properties. Tool surfaces must be inspected before each use to ensure the tool surface is clean and free of conditions which could contaminate or damage a part.

Facilities and Equipment: The user will establish requirements to control the composite work area environment. These requirements are a part of the user's process specifications. The requirements should be commensurate with the susceptibility of materials to contamination by the shop environment. Inspection and calibration requirements for autoclaves and ovens must be defined.

Contamination restrictions in environmentally-controlled areas typically prohibit the use of uncontrolled sprays (e.g., silicon contamination), exposure to dust, handling contamination, fumes, oily vapors, and the presence of other particulate or chemical matter which may affect the manufacturing process. Conditions under which operators may handle materials should also be defined. Lay-up and clean room air filtrations and pressurization systems should be capable of providing a slight positive overpressure.

In-Process Control: During lay-up of composite parts, certain critical steps or operations must be closely controlled. Requirements and limits for these critical items are stated in the user process specifications. Some of the steps and operations to be controlled are listed below:

1. Verification that the release agent has been applied and cured on a clean tool surface.

2. Verification that perishable materials incorporated into the part comply with the applicable material specifications.

3. Inspection of prepreg lay-ups to assure engineering drawing requirements for number of plies and orientation are met.

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4. Inspection of honeycomb core installation, if applicable, and verification that positioning meets the engineering drawing requirements.

5. The user paperwork should contain the following information.

   a. Material supplier, date of manufacturer, batch number, roll number, and total accumulated hours of working life.

   b. Autoclave or oven pressure, part temperatures, and times.

   c. Autoclave or oven load number.

   d. Part and serial number.

Part Cure: Requirements must be defined in user process specifications for the operating parameters for autoclaves and ovens used for curing parts. These include heat rise rates, times at temperature, cool-down rates, temperature and pressure tolerance, and temperature uniformity surveys in the autoclave or ovens.

Process Control Specimens: Many manufacturers require special test panels to be laid up and cured along with production parts. After cure, these panels are tested for physical and mechanical properties to verify the parts they represent meet the engineering properties.

The requirements for physical and mechanical testing are frequently defined by drawing notes which designates a type or class for each part. Non-critical or secondary structure may require no test specimens and no testing. Critical or safety-of-flight parts may require complete physical and mechanical testing.

During early composite material production, most users required tests for 0° flexure strength and modulus and short beam shear strength. However, in recent years these tests have been changed by many manufacturers to require glass transition temperature, per ply thickness, fiber volume, void content, and ply count on samples taken from designated areas on the production part.

3.2.3 Final inspection. Having assured in-process control, the detail composite parts must also be inspected for conformance to dimensional and workmanship requirements and nondestructively inspected for processing-induced defects and damage.

Assembly Inspection: Laminates are prone to particular types of defects unless they are machined and drilled properly. Workmanship standards, required by manufacturer's process specifications, are needed to control the quality of trimmed edges and drilled holes. These standards establish visual acceptance/rejection limits for the following typical defects: splintering, delamination, loose surface fibers, overheating, surface finish, off-axis holes, and surface cratering. Typical defects in the drilling operations are delaminations and broken fibers which start at the hole boundary. Since these defects are internal in nature, an evaluation of the seriousness of the flaws is not possible by visual inspection alone. It should be backed up by nondestructive inspection techniques. Internal defect acceptance and rejection limits must be established for nondestructive inspection.

3.2.4 Nondestructive inspection. The extent of nondestructive (NDI) inspection on composite parts is dependent on whether the parts are primary structure, safety-of-flight or secondary structure, non-safety-of-flight. The type or class of part is usually defined on the engineering drawing. The engineering drawing also references a process specification which defines the NDI tests and the

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accept/reject criteria. The NDI tests are used to find flaws and damage such as voids, delaminations, inclusions, and micro-cracks in the matrix.

NDI techniques commonly used in production include visual, ultrasonic and X-ray inspection. Other methods, such as infrared, holographic, and acoustic inspection are being developed and may be used in production applications in the future.

Visual inspection is an NDI technique involving checks to assure the parts meet drawing requirements and to evaluate the surface and appearance of the part. The inspection includes examination for blisters, depressions, foreign material inclusions, ply distortions and folds, surface roughness, surface porosity, and wrinkles. Accept/reject criteria for such defects are given in the manufacturer's process specifications.

The most widely used nondestructive inspection technique for composites production is ultrasonic thru-transmission C-scan inspection, followed by ultrasonic pulse echo A-scan inspection. Since the subject is so broad, the engineering requirements and criteria are usually contained in a document that is referenced in the user's process specification. The principal defects evaluated by ultrasonics are internal voids, delaminations, and porosity. These inspections require fabrication of standards with built-in known defects. The output is in the form of charts which shows the sound attenuation variations over the entire part. The charts are compared to the part to show the locations of the sound attenuation variations. If defects are found outside the limits allowed by the specification, the parts are rejected and dispositioned by Engineering. Parts may be dispositioned 1) acceptable as is, 2) subjected to further rework or repair to make the part acceptable or 3), scrapped.

X-ray inspection is frequently used in NDI testing to evaluate bonding of inserts in laminate panels and honeycomb core to facesheet bonds in sandwich panels. The extent of testing required is designated on the engineering drawing by type or class of inspection. The type or class is usually defined in a separate document that is referenced in the manufacturer's process specification. As with ultrasonic inspection, standards with built-in defects are usually required to evaluate the radiographic film properly.

3.2.5 Specifications and documentation. The specification for materials, fabrication processes, and material testing techniques must ensure compliance with the engineering requirements.

Chapters 3, 4, and 6 in Volume 1 of this handbook describe acceptance test methods for characterizing fiber, matrix, and resin-impregnated fiber materials by their chemical, physical, and mechanical properties. Sections 3.3 and 3.4 of this volume provide information on variable statistical sampling plans that are based on MIL-STD-414 (Reference 3.2.5(a)). These plans control the frequency and extent of material property verification testing to achieve targeted quality levels.

The specifications for destructive and nondestructive test equipment and test methods should contain test and evaluation procedures. These procedures need to describe the means by which the equipment will be calibrated to maintain the required accuracy and repeatability; they should also establish the calibration frequency. Information on the standards to be used in the calibration of chemical analysis equipment will be found in preceding sections of this handbook which deal with the particular test technique.

The standards for quality control documentation requirements are found in military and federal specifications such as MIL-Q-9858A used by the Department of Defense in their hardware procurement contracts and Federal Aviation Regulation Part 21 "Certification Procedures for Products and Parts" used by the Federal Aviation Administration production approval holders (References 3.2.5(b) and (c)).

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3.3 MATERIAL PROPERTY VERIFICATION.

3.3.1 Destructive tests.

3.3.2 Frequency and quantity of destructive tests.

3.4 STATISTICAL PROCESS CONTROL.

3.4.1 Monitoring the production process.

3.4.2 Batch-to-batch variability.

3.5. STATISTICAL PROCEDURES.

3.5.1 $\bar{X}$ **chart including batch effect.** Assume that the data are a sample from a one-way, balanced random effects analysis of variance model (See Volume 1, Section 8.6.7.):

$$X_{ij} = \mu + b_i + e_{ij}, \quad i=1,...,k \quad j=1,...,k \qquad 3.5.1(a)$$

where k is the number of accepted batches, n is the number of specimens in each batch, and X_{ij} represents the jth specimen in the ith batch.

This ANOVA model represents each observation as the sum of three components; μ is the overall average of the population, b_i is the population average for the ith batch, and e is a random error term which represents variation within each batch. The error terms, e_{ij}, are assumed to be independently distributed normal random variables with a mean of zero and a variance of σ_e^2 (the within-batch variance). The batch means, b_i, are assumed to be independent random variables following a normal distribution with a mean of zero and a variance of σ_b^2 (the between-batch variance).

The acceptability of a new batch is to be tested using the data in the k previously accepted batches. This new batch is referred to as the (k+1)th batch. Denote the grand mean on the basis of k batches as

$$\bar{X}^{(k)} = \sum_{i=1}^{k} \sum_{j=1}^{n} X_{ij} /(kn) \qquad 3.5.1(b)$$

The batch means are computed as

$$\bar{X}_i = \sum_{j=1}^{n} X_{ij} /n \quad \text{for } i=1,...,k \qquad 3.5.1(c)$$

In this section, a superscript in parentheses indicates the number of batches of data used to calculate a statistic. For example, $\bar{X}^{(k)}$ is the grand mean based on all batches up to and including the kth batch. From these quantities, the required sums of squares can be computed. The between-batch mean square is computed as

$$MSB^{(k)} = \frac{1}{k-1} \sum_{i=1}^{k} n(\bar{X}_i - \bar{X}^{(k)})^2 \qquad 3.5.1(d)$$

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Assume that the (k + 1)th batch is also described by the model in Equation 3.5.1(a). The mean of the (k + 1)th batch has a normal distribution with mean μ and a variance of

$$\frac{1}{n}\left(n\,\sigma_b^2 + \sigma_e^2\right)$$

It follows that the difference between the grand mean and the (k + 1)th mean,

$$\bar{X}^{(k)} - \bar{X}_{k+1}$$

has a normal distribution with a mean of zero and a variance of

$$\frac{k+1}{k}\,\frac{\left(n\,\sigma_b^2 + \sigma_e^2\right)}{n}$$

Also,

$$\frac{k+1}{kn}\,MSB^{(k)} \sim \frac{k+1}{k(k-1)}\,\frac{\left(n\,\sigma_b^2 + \sigma_e^2\right)}{n}\,\chi^2_{k-1} \qquad 3.5.1(e)$$

where the ~ indicates "is distributed as" and χ^2_{k-1} denotes the χ^2 distribution with k-1 degrees of freedom. Dividing the difference between the grand mean and the (k + 1)th batch mean by the left hand side of Equation 3.5.1(e) gives

$$V^{(k+1)} = \frac{\bar{X}^{(k)} - \bar{X}_{k+1}}{\left[\frac{k+1}{k(k-1)}\sum_{i=1}^{k}\left(\bar{X}_i - \bar{X}^{(k)}\right)^2\right]^{1/2}} \sim t_{k-1} \qquad 3.5.1(f)$$

where t_{k-1} denotes the central t-distribution with k-1 degrees of freedom. This last relationship is the basis of the control chart. $V^{(k+1)}$, calculated from the new mean and all previously accepted batch means, is compared to the t-distribution limits. Specifically, $V^{(k+1)}$ is compared to the a quantile of the central t-distribution with k-1 degrees of freedom, $t_{k-1,\alpha}$, from Table 8.8.18, Volume 1. If the absolute value of $V^{(k+1)}$ exceeds $t_{k-1,\alpha}$, the (k + 1)th batch is not accepted. These limits approach the normal distribution limits as the number of batches increases. Because of the variable control limits, it should be possible to start using this chart after very few batches. It may be reasonable to use it after data from four or five batches have been obtained.

If the (k + 1)th batch is accepted, the grand mean and the between-batch mean square are updated as follows:

$$\bar{X}^{(k+1)} = \frac{k\,\bar{X}^{(k)} + \bar{X}_{k+1}}{k+1} \qquad 3.5.1(g)$$

$$MSB^{(k+1)} = \frac{k-1}{k}\,MSB^{(k)} + \frac{n}{k+1}\left(\bar{X}^{(k)} - \bar{X}_{k+1}\right)^2 \qquad 3.5.1(h)$$

Finally, note that this procedure can fail if there is a trend in the means. Such a trend would inflate the estimate of the variance and result in limits which are too wide. Because of this, the above procedure is used with a "runs" test for trends. Example charts are shown in Figure 3.5.1(a) and (b). These figures show the limits, $t_{k-1}(a)$ and $-t_{k-1}(a)$, and $V^{(k+1)}$ for each successive batch.

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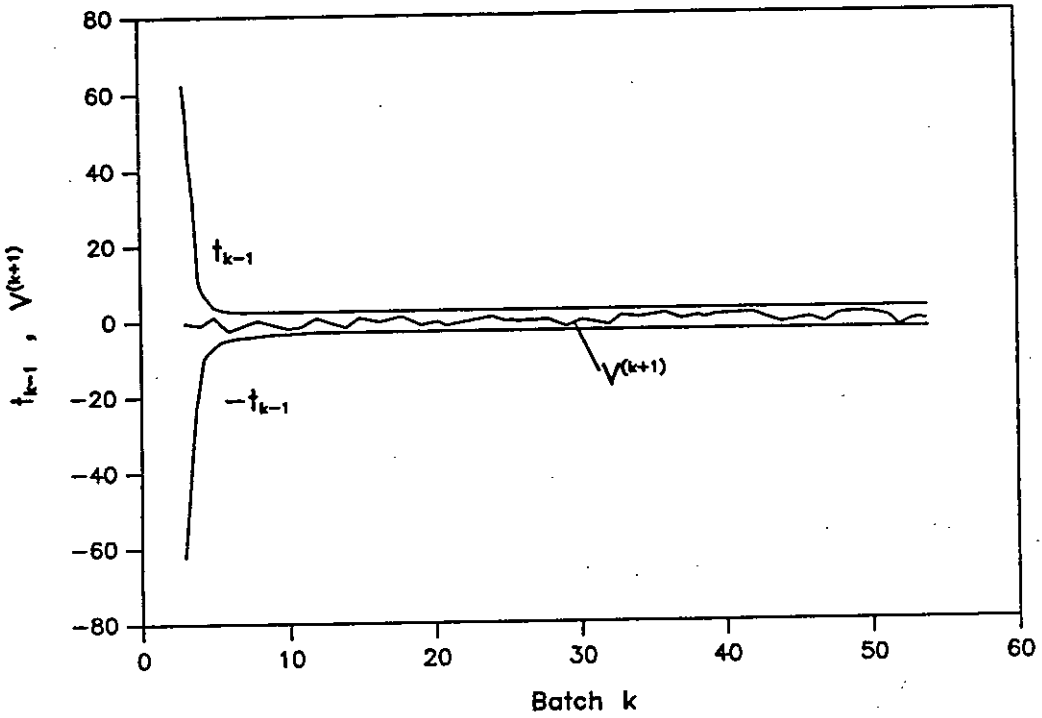


FIGURE 3.5.1(a) *Example control chart for means beginning with the third batch (0.01 level).*

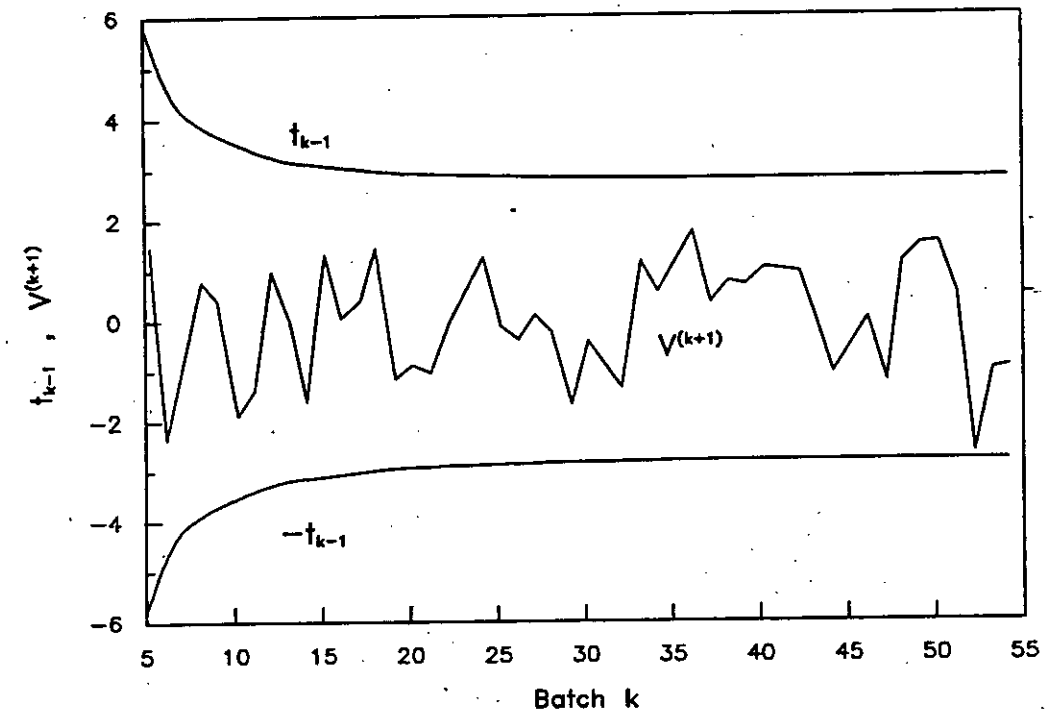


FIGURE 3.5.1(b) *Example control chart for means beginning with the fifth batch (0.01 level).*

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3.5.2 S^2 chart for the within batch component of variance. Let S^2 be the sample variance of the jth accepted batch. Test the (k + 1)th batch variance using the following statistic:

$$U_{k+1} = \frac{S^2_{k+1}}{\sum_{j=1}^{k} S_j^2/k} \sim F \qquad 3.5.2$$

where F denotes the standard F-distribution with n -1 numerator degrees of freedom and k(n-1) denominator degrees of freedom. Compare U_{k+1} to the a quantile of the F-distribution for n-1 numerator degrees of freedom and k(n-1) denominator degrees of freedom, F_α. These values are provided in Table 8.8.5, Volume 1 for a 0.95 level. If the statistic U_{k+1} exceeds F_α, the (k + 1)th batch is not accepted. The control limits will approach constants as the denominator degrees of freedom for the F statistic become large. As with the means chart, this variance chart should be useful after data have been obtained from a few batches. An example S^2 chart is shown in Figure 3.5.2. This figure shows the limit, F_α, and U for each successive batch.

3.5.3 Test for trend in batch means. The $\bar{X}$ chart including batch effect of Section 3.5.1 may not perform properly if there is a systematic trend, either upward or downward, in the batch means for the initial batches received. Such a trend would make $V^{(k+1)}$ (Equation 3.5.1(f)) too small by inflating the denominator. This could result in batches being accepted which would have been rejected had there been no trend. If a trend is detected in the batch means before the control limits have leveled off (eg., before the 25th batch; see Figure 3.5.1(b)) then caution should be used when accepting batches,

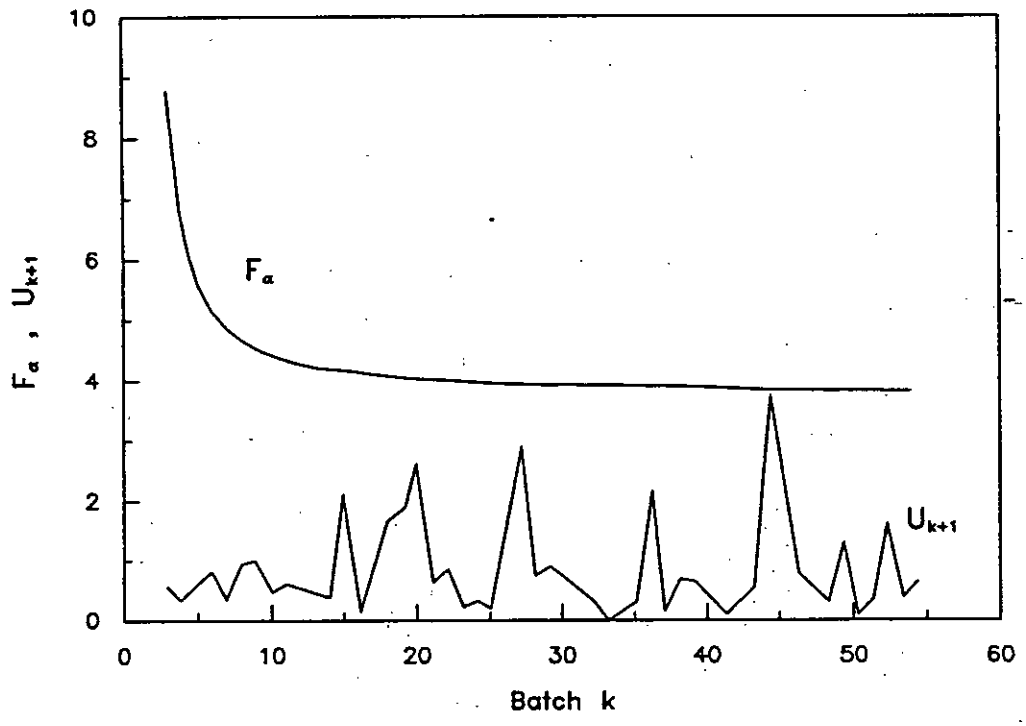


FIGURE 3.5.2 *Example control chart for variances beginning with the third batch (0.01 level).*

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especially if the trend can be seen to be downward. A diagnostic test is given in this subsection which should be performed on the first 25 batches received in order to determine whether a statistically significant trend exists. After 25 batches, this diagnostic should be discontinued, since the control limits will no longer be substantially effected by any trend which might be present, so the validity of the test in Section 3.5.1 need no longer be questioned. The idea behind this test is to fit a straight line through the batch means by least squares, and to determine if this line has a statistically significant slope.

Let the means for the ith accepted batch be denoted $\bar{X}_i$, and let t_i be a time associated with the arrival of this batch. For example, the time of arrival of the first batch may be represented by $t_1=0$, and the remaining t_i may be the number of days which have elapsed since the first batch arrived. Assume that k batches have been accepted thus far, and that the following quantities have been calculated and therefore are available:

$$\bar{X}^{(k)} = \sum_{i=1}^{k} \bar{X}_i / k \qquad 3.5.3.1(a)$$

$$\bar{t}^{(k)} = \sum_{i=1}^{k} t_i / k \qquad 3.5.3.1(b)$$

$$S_{tt}^{(k)} = \sum_{i=1}^{k} \left(t_i - \bar{t}^{(k)}\right)^2 \qquad 3.5.3.1(c)$$

$$S_{xx}^{(k)} = \sum_{i=1}^{k} \left(\bar{X}_i - \bar{X}^{(k)}\right)^2 \qquad 3.5.3.1(d)$$

$$S_{tx}^{(k)} = \sum_{i=1}^{k} \left(t_i - \bar{t}^{(k)}\right)\left(\bar{X}_i - \bar{X}^{(k)}\right) \qquad 3.5.3.1(e)$$

$$b^{(k)} = \frac{S_{tx}^{(k)}}{S_{tt}^{(k)}} \qquad 3.5.3.1(f)$$

$$S_R^{(k)} = S_{xx}^{(k)} - 2b^{(k)} S_{tx}^{(k)} + \left[b^{(k)}\right]^2 S_{tt}^{(k)} \qquad 3.5.3.1(g)$$

The slope of the least squares line based on k batches is $b^{(k)}$, and the standard deviation of this slope is given by

$$SD^{(k)} = \left[S_R^{(k)}/(k-2)\right]^{1/2} \qquad 3.5.3.2$$

When the (k + 1)st batch arrives, at time t_{k+1}, the following steps should be performed.

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Step 1 Update $S_{tt}^{(k)}$, $S_{tt}^{(k)}$, and $S_{tx}^{(k)}$.

$$S_{tt}^{(k+1)} = S_{tt}^{(k)} + \frac{k}{k+1}\left(\bar{t}^{(k)} - t_{k+1}\right)^2 \qquad 3.5.3.3(a)$$

$$S_{xx}^{(k+1)} = S_{xx}^{(k)} + \frac{k}{k+1}\left(\bar{X}^{(k)} - \bar{X}_{k+1}\right)^2 \qquad 3.5.3.3(b)$$

$$S_{tx}^{(k+1)} = S_{tx}^{(k)} + \frac{k}{k+1}\left(\bar{t}^{(k)} - t_{k+1}\right)\bar{X}_{k+1} \qquad 3.5.3.3(c)$$

Step 2 Calculate $b^{(k+1)}$ and $S_R^{(k+1)}$.

$$b^{(k+1)} = \frac{S_{tx}^{(k+1)}}{S_{tt}^{(k+1)}} \qquad 3.5.3.4(a)$$

$$S_R^{(k+1)} = S_{xx}^{(k+1)} - 2b^{(k+1)} S_{tx}^{(k+1)} + b^{(k+1)^2} S_{tt}^{(k+1)} \qquad 3.5.3.4(b)$$

Step 3 Calculate the trend statistic.

$$U^{(k+1)} = b^{(k+1)} \sqrt{\frac{(k-1) S_{tt}^{(k+1)}}{S_R^{(k+1)}}} \qquad 3.5.3.5$$

Step 4 Determine $t_{k-1,\alpha/2}$, the α quantile of the central t distribution with k-1 degrees of freedom (Table 8.8.18, Volume 1). If $|U^{(k+1)}|$ is greater than $t_{k-1,\alpha/2}$, then a statistically significant trend has been detected and should be investigated.

The level α of the test is somewhat arbitrary, but probably should be taken to be small (e.g. 0.001) in order to make small the probability of making an error by declaring a trend when no trend exists.

Step 5 Update the means.

$$\bar{X}^{(k+1)} = \frac{k\bar{X}^{(k)} + \bar{X}_{k+1}}{k+1} \qquad 3.5.3.6(a)$$

$$\bar{t}^{(k+1)} = \frac{k\bar{t}^{(k)} + t_{(k+1)}}{k+1} \qquad 3.5.3.6(b)$$

The following remarks should be made concerning this test:

1) The test should not be performed once the control limits in the $\bar{X}$ chart including batch effect (Section 3.5.1) have leveled off. This should occur at or before 25 batches have been accepted.

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2) This trend test is only designed to detect situations where the validity of the test in Section 3.5.1 is called into question. It is not suitable as a general purpose trend test.

3) The updating Equations 3.5.3.3(a - c) and 3.5.3.6(a - c) make it unnecessary to use Equations 3.5.3.1(a - e) *after each batch*. *Calculating quantities* for a test on the (k + 1)st batch based on the results from the test on the kth batch requires only a hand calculator.

4) A control chart analog to Figure 3.5.1 may be prepared for the trend test as well, *and it could* provide useful information.

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REFERENCES

3.2.5(a) MIL-STD-414, *Sampling Procedures and Tables for Inspection by Variables for Percent Defective*, Change Notice 1, 8 May 1968.

3.2.5(b) MIL-Q-9858A, *Quality Program Requirements*

3.2.5(c) Federal Aviation Regulation Part 21 "Certification Procedures for Products and Parts"

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Chapter 4. DESIGN AND ANALYSIS

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4.1 INTRODUCTION. The concept of designing a material to yield a desired set of properties has received impetus from the growing acceptance of composite materials. Inclusion of material design in the structural design process has had a significant effect on that process, particularly upon the preliminary design phase. In this preliminary design, a number of materials will be considered, including materials for which experimental materials property data are not available. Thus, preliminary material selection may be based on analytically-predicted properties. The analytical methods are the result of studies of micromechanics, the study of the relationship between effective properties of composites and the properties of the composite constituents. The inhomogeneous composite is represented by a homogeneous anisotropic material with the effective properties of the composite.

4.1.1 Purpose. The purpose of this chapter is to provide an overview of techniques for analysis in the design of composite materials. Starting with the micromechanics of fiber and matrix in a lamina, analyses through simple geometric constructions in laminates are considered.

4.1.2 Scope. The analysis in this chapter deals primarily with symmetric laminates. Section 4.2 describes the micromechanics of basic lamina properties. Section 4.3 leads from these properties in the lamina coordinate system to the analysis of laminates in an arbitrary coordinate system. The basic principles of lamination theory are presented in this section. Failure theories are described in Section 4.4, and the reaction of laminates to more complex loads are discussed in Section 4.5. These strength considerations are extended to notch laminates in Section 4.5. Section 4.6 highlights considerations when predicting the behavior of laminates based on lamina results. Buckling of composite plates and other structures is discussed in Section 4.7. The use of carpet plots to represent material properties of families of laminates is described in Section 4.8. The time-dependent properties of creep and relaxation, and fatigue are described in Sections 4.9 and 4.10. Information on other structural properties, in Section 4.11, includes the topics of damage tolerance and durability. Section 4.12 provides information on computer programs used for the prediction of composite material properties. The last section, Section 4.13, describes the certification requirements for polymer matrix composite applications.

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4.2 BASIC LAMINA PROPERTIES AND MICROMECHANICS. The strength of any given laminate under a prescribed set of loads is probably best determined by conducting a test. However, when many candidate laminates and different loading conditions are being considered, as in a preliminary design study, analysis methods for estimation of laminate strength become desirable. Because the stress distribution throughout the fiber and matrix regions of all the plies of a laminate is quite complex, precise analysis methods are not available. However, reasonable methods do exist which can be used to guide the preliminary design process.

Strength analysis methods may be grouped into different classes, depending upon the degree of detail of the stresses utilized. The following classes are of practical interest:

1. Laminate level. Average values of the stress components in a laminate coordinate system are utilized.

2. Ply, or lamina, level. Average values of the stress components within each ply are utilized.

3. Constituent level. Average values of the stress components within each phase (fiber or matrix) of each ply are utilized.

4. Micro-level. Local stresses of each point within each phase are utilized.

Micro-level stresses could be used in appropriate failure criteria for each constituent to determine the external loads at which local failure would initiate. However, the uncertainties, due to departures from the assumed regular local geometry and the statistical variability of local strength make such a process impractical.

At the other extreme, laminate level stresses can be useful for translating measured strengths under single stress component tests into anticipated strength estimates for combined stress cases. However this procedure does not help in the evaluation of alternate laminates for which test data do not exist.

Ply level stresses are the commonly used approach to laminate strength. The average stresses in a given ply are used to calculate first ply failure and then subsequent ply failure leading to laminate failure. The analysis of laminates by the use of a ply-by-ply model is presented in Section 4.3 and 4.4.

Constituent level, or phase average stresses, eliminate some of the complexity of the micro level stresses. They represent a useful approach to the strength of a unidirectional composite or ply. Micromechanics provides a method of analysis, presented in Section 4.2, for constituent level stresses. Micromechanics is the study of the relations between the properties of the constituents of a composite and the effective properties of the composite. Starting with the basic constituent properties, Sections 4.2 through 4.4 develop the micromechanical analysis of a lamina and the associated ply-by-ply analysis of a laminate.

4.2.1 Assumptions. Several assumptions have been made for characterizing lamina properties.

4.2.1.1 *Material homogeneity.* Composites, by definition, are heterogeneous materials. Mechanical analysis proceeds on the assumption that the material is homogeneous. This apparent conflict is resolved by considering homogeneity on microscopic and macroscopic scales. Microscopically, composite materials are certainly heterogeneous. However, on the macroscopic scale, they appear homogeneous and respond homogeneously when tested. The analysis of composite materials uses effective properties which are based on the average stress and average strain.

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4.2.1.2 *Material orthotropy.* Orthotropy is the condition expressed by variation of mechanical properties as a function of orientation. Lamina exhibit orthotropy as the large difference in properties between the 0° and 90° directions. If a material is orthotropic, it contains planes of symmetry and can be characterized by four independent elastic constants.

4.2.1.3 *Material linearity.* Some composite material properties are non-linear. The amount of non-linearity depends on the property, type of specimen, and test environment. The stress-strain curves for composite materials are frequently assumed to be linear to simplify the analysis.

4.2.1.4 *Residual stresses.* One consequence of the microscopic heterogeneity of a composite material is the thermal expansion mismatch between the fiber and the matrix. This mismatch causes residual strains in the lamina after curing. The corresponding residual stresses are often assumed not to affect the material's stiffness or its ability to strain uniformly.

4.2.2 Fiber composites: physical properties. A unidirectional fiber composite (UDC) consists of aligned continuous fibers which are embedded in a matrix. The UDC physical properties are functions of fiber and matrix physical properties, of their volume fractions, and perhaps also of statistical parameters associated with fiber distribution. The fibers have, in general, circular cross-sections with little variability in diameter. A UDC is clearly anisotropic since properties in the fiber direction are very different from properties transverse to the fibers.

Properties of interest for evaluating stresses and strains are:

Elastic properties
Viscoelastic properties - static and dynamic
Thermal expansion coefficients
Moisture swelling coefficients
Thermal conductivity
Moisture diffusivity

A variety of analytical procedures may be used to determine the various properties of a UDC from volume fractions and fiber and matrix properties. The derivations of these procedures may be found in References 4.2.2(a) and (b).

4.2.2.1 *Elastic properties.* The elastic properties of a material are a measure of its stiffness. This information is necessary to determine the deformations which are produced by loads. In a UDC, the stiffness is provided by the fibers; the role of the matrix is to prevent lateral deflections of the fibers. For engineering purposes, it is necessary to determine such properties as Young's modulus in the fiber direction, Young's modulus transverse to the fibers, shear modulus along the fibers and shear modulus in the plane transverse to the fibers, as well as various Poisson's ratios. These properties can be determined in terms of simple analytical expressions.

The effective elastic stress-strain relations of a typical transverse section of a UDC, based on average stress and average strain, have the form:

$$\begin{aligned} \bar{\sigma}_{11} &= n^* \bar{e}_{11} + \ell^* \bar{e}_{22} + \ell^* \bar{e}_{33} \\ \bar{\sigma}_{22} &= \ell^* \bar{e}_{11} + (k^* + G_2^*)\bar{e}_{22} + (k^* - G_2^*)\bar{e}_{33} \\ \bar{\sigma}_{33} &= \ell^* \bar{e}_{11} + (k^* - G_2^*)\bar{e}_{22} + (k^* + G_2^*)\bar{e}_{33} \end{aligned} \qquad 4.2.2.1(a)$$

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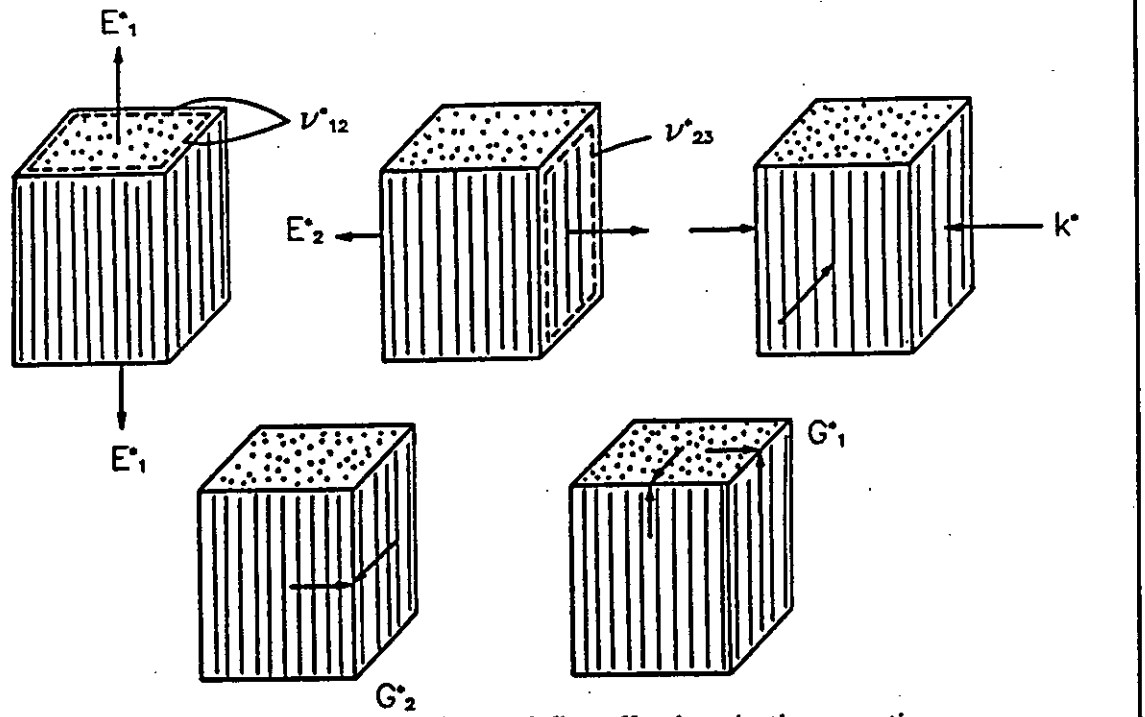


FIGURE 4.2.2.1 *Basic loading to define effective elastic properties.*

$$\bar{\sigma}_{12} = 2G_1^* \bar{e}_{12}$$
$$\bar{\sigma}_{23} = 2G_2^* \bar{e}_{23} \qquad 4.2.2.1(b)$$
$$\bar{\sigma}_{13} = 2G_1^* \bar{e}_{13}$$

with inverse

$$\bar{e}_{11} = \frac{1}{E_1^*}\bar{\sigma}_{11} - \frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{22} - \frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{33}$$

$$\bar{e}_{22} = -\frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{11} + \frac{1}{E_2^*}\bar{\sigma}_{22} - \frac{\nu_{23}^*}{E_2^*}\bar{\sigma}_{33} \qquad 4.2.2.1(c)$$

$$\bar{e}_{33} = -\frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{11} - \frac{\nu_{23}^*}{E_2^*}\bar{\sigma}_{22} + \frac{1}{E_2^*}\bar{\sigma}_{33}$$

where an asterisk (*) denotes effective values. Figure 4.2.2.1 illustrates the loadings which are associated with these properties.

The effective modulus k^* is obtained by subjecting a specimen to the average state of stress $\bar{e}_{22} = \bar{e}_{33}$ with all other strains vanishing in which case it follows from Equations 4.2.2.1(a) that

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$$(\bar{\sigma}_{22} + \bar{\sigma}_{33}) = 2k^*(\bar{\varepsilon}_{22} + \bar{\varepsilon}_{33}) \qquad 4.2.2.1(d)$$

Unlike the other properties listed above, k^* is of little engineering significance but is of considerable analytical importance.

Only five of the properties in Equations 4.2.2.1(a-c) are independent. The most important interrelations of properties are:

$$n^* = E_1^* + 4k^*\nu_{12}^{*2} \qquad 4.2.2.1(e)$$

$$\ell^* = 2k^*\nu_{12}^* \qquad 4.2.2.1(f)$$

$$\frac{4}{E_2^*} = \frac{1}{G_2^*} + \frac{4\nu_{12}^{*2}}{E_1^*} \qquad 4.2.2.1(g)$$

$$\frac{2}{1-\nu_{23}^*} = 1 + \frac{k^*}{\left(1 + 4k^*\dfrac{\nu_{23}^{*2}}{E_1^*}\right) G_2^*} \qquad 4.2.2.1(h)$$

$$G_2^* = \frac{E_2^*}{2(1+\nu_{23}^*)} \qquad 4.2.2.1(i)$$

Computation of effective elastic moduli is a very difficult problem in elasticity theory and only a few simple models permit exact analysis. One type of model consists of periodic arrays of identical circular fibers, e.g., square periodic arrays or hexagonal periodic arrays (References 4.2.2.1(a) - (c)). These models are analyzed by numerical finite difference or finite element procedures. Note that the square array is not a suitable model for the majority of UDCs since it is not transversely isotropic.

The composite cylinder assemblage (CCA) permits exact analytical determination of effective elastic moduli (Reference 4.2.2.1(d)). Consider a collection of composite cylinders, each with a circular fiber core and a concentric matrix shell. The size of the cylinders may vary but the ratio of core radius to shell radius is held constant. Therefore, the matrix and fiber volume fractions are the same in each composite cylinder. One strength of this model is the randomness of the fiber placement, while an undesirable feature is the large variation of fiber sizes. It can be shown that the latter is not a serious concern.

The analysis of the CCA gives closed form results for the effective properties, k^*, E_1^*, ν_{12}^*, n^*, ℓ^*, and G_1^* and closed bounds for the properties G_2^*, E_2^*, and ν_{23}^*. Such results will now be listed for isotropic fibers with the necessary modifications for transversely isotropic fibers (References 4.2.2(a) and 4.2.2.1(e)).

$$k^* = \frac{k_m(k_f + G_m)v_m + k_f(k_m + G_m)v_f}{(k_f + G_m)v_m + (k_m + G_m)v_f}$$

$$= k_m + \frac{v_f}{\dfrac{1}{(k_f - k_m)} + \dfrac{v_m}{(k_m + G_m)}} \qquad 4.2.2.1(j)$$

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$$E_1^* = E_m v_m + E_f v_f + \frac{4(\nu_f - \nu_m)^2 v_m v_f}{\frac{v_m}{k_f} + \frac{v_f}{k_m} + \frac{1}{G}} \qquad 4.2.2.1(k)$$

$$\approx E_m v_m + E_f v_f$$

The last is an excellent approximation for all UDC.

$$\nu_{12}^* = \nu_m v_m + \nu_f v_f + \frac{(\nu_f - \nu_m)\left(\frac{1}{k_m} - \frac{1}{k_f}\right) v_m v_f}{\frac{v_m}{k_f} + \frac{v_f}{k_m} + \frac{1}{G_m}} \qquad 4.2.2.1(l)$$

$$G_1^* = G_m \frac{G_m v_m + G_f(1+\nu_f)}{G_m(1+\nu_f) + G_f v_m}$$

$$= G_m + \frac{v_f}{\frac{1}{(G_f - G_m)} + \frac{v_m}{2G_m}} \qquad 4.2.2.1(m)$$

As indicated earlier, the result for G_2^* is a pair of bounds on the actual value. One or the other of these bounds is recommended, depending on the ratios of the constituent properties. For polymer-matrix composites where $G_f > G_m$ and $k_f > k_m$, the upper bound $G_{2(+)}^*$ is recommended:

$$G_{2(+)}^* = G_m\left[1 + \frac{v_f(1+\beta_m)}{\rho - v_f\left(\frac{1+3\beta_m^2 v_m^2}{\alpha v_f^3 + 1}\right)}\right]$$

$$\alpha = \frac{\beta_m - \gamma\beta_f}{1 + \gamma\beta_f} \qquad \rho = \frac{\gamma + \beta_m}{\gamma - 1} \qquad 4.2.2.1(n)$$

$$\beta_m = \frac{1}{3 - 4\nu_m} \qquad \beta_f = \frac{1}{3 - 4\nu_f}$$

$$\gamma = \frac{G_f}{G_m}$$

To compute the resulting E_2^* and ν_{23}^*, use Equations 4.2.2.1(g-h).

For transversely isotropic fibers, the following modifications are necessary (References 4.2.2(a) and 4.2.2.1(e)):

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For k^* k_f is the fiber transverse bulk modulus

For E_1^*, ν_{12}^* $E_f = E_{1f}$
$\nu_f = \nu_{1f}$
k_f as above

For G_1^* $G_f = G_{1f}$

For G_2^* $G_f = G_{2F}$

$\beta_f = k_f/(k_f + 2G_{2f})$

A rational approximate evaluation of G_2^* of the CCA model is due to Christensen and Lo (Reference 4.2.2(b)). They have assumed that any composite cylinder behaves as if it were embedded in the effective medium with effective shear modulus G_2^*. Numerical analysis of the effective elastic properties of the hexagonal array model reveals that their values are extremely close to those predicted by the CCA model as given by the above equations.

The simple analytical results given here predict effective elastic properties with sufficient engineering accuracy. They are of considerable practical importance for two reasons. First, they permit easy determination of effective properties for a variety of matrix properties, fiber properties, volume fractions, and environmental conditions. Secondly, they provide the only approach known today for experimental determination of carbon fiber properties.

For purposes of laminate analysis, it is important to consider the plane stress version of the effective stress-strain relations. Let x_3 be the normal to the plane of a thin unidirectionally-reinforced lamina. The plane stress condition is defined by

$$\bar{\sigma}_{33} = \bar{\sigma}_{13} = \bar{\sigma}_{23} = 0 \qquad 4.2.2.1(o)$$

Then from Equations 4.2.2.1(b-c)

$$\bar{e}_{11} = \frac{1}{E_1^*}\bar{\sigma}_{11} - \frac{\nu_{12}^*}{E_2^*}\bar{\sigma}_{22}$$

$$\bar{e}_{22} = \frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{11} + \frac{1}{E_2^*}\bar{\sigma}_{22} \qquad 4.2.2.1(p)$$

$$2\bar{e}_{12} = \frac{\bar{\sigma}_{12}}{G_1^*}$$

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The inversion of Equation 4.2.2.1(p) gives

$$\begin{aligned} \bar{\sigma}_{11} &= C_{11}^{*}\,\bar{e}_{11} + C_{12}^{*}\,\bar{e}_{22} \\ \bar{\sigma}_{22} &= C_{12}^{*}\,\bar{e}_{11} + C_{22}^{*}\,\bar{e}_{22} \\ \bar{\sigma}_{12} &= 2G_{2}^{*}\,\bar{e}_{12} \end{aligned} \qquad 4.2.2.1(q)$$

where

$$\begin{aligned} C_{11}^{*} &= \frac{E_{1}^{*}}{1 - \nu_{12}^{*2} E_{2}^{*}/E_{1}} \\ C_{12}^{*} &= \frac{\nu_{12}^{*} E_{2}^{*}}{1 - \nu_{12}^{*2} E_{2}^{*}/E_{1}} \\ C_{22}^{*} &= \frac{E_{2}^{*}}{1 - \nu_{12}^{*2} E_{2}^{*}/E_{1}} \end{aligned} \qquad 4.2.2.1(r)$$

For polymer matrix composites, at the usual 60% fiber volume fraction, the square of ν_{12}^{*} is close enough to zero to be neglected and the ratio of E_{2}^{*}/E_{1}^{*} is approximately 0.1 - 0.2. Consequently, the following approximations are often useful.

$$C_{11}^{*} \approx E_{1}^{*} \qquad C_{12}^{*} \approx \nu_{12}^{*} E_{2}^{*} \qquad C_{22}^{*} \approx E_{2}^{*} \qquad 4.2.2.1(s)$$

4.2.2.2 *Viscoelastic properties.* The simplest description of time-dependence is linear viscoelasticity. Viscoelastic behavior of polymers manifests itself primarily in shear and is negligible for isotropic stress and strain. This implies that the elastic stress-strain relation

$$\sigma_{11} + \sigma_{22} + \sigma_{33} = 3K\,(e_{11} + e_{22} + e_{33}) \qquad 4.2.2.2(a)$$

where K is the three-dimensional bulk modulus, remains valid for polymers. When a polymeric specimen is subjected to shear strain e_{12}^{*} which does not vary with time, the stress needed to maintain this shear strain is given by

$$\sigma_{12}(t) = 2G(t)\,e_{12}^{*} \qquad 4.2.2.2(b)$$

and G(t) is defined as the shear relaxation modulus. When a specimen is subjected to shear stress, σ_{12}^{*}, constant in time, the resulting shear strain is given by

$$e_{12}(t) = \frac{1}{2} g(t)\;\sigma_{12}^{*} \qquad 4.2.2.2(c)$$

and g(t) is defined as the shear creep compliance.

Typical variations of relaxation modulus G(t) and creep compliance g(t) with time are shown in Figure 4.2.2.2. These material properties change significantly with temperature. The relaxation

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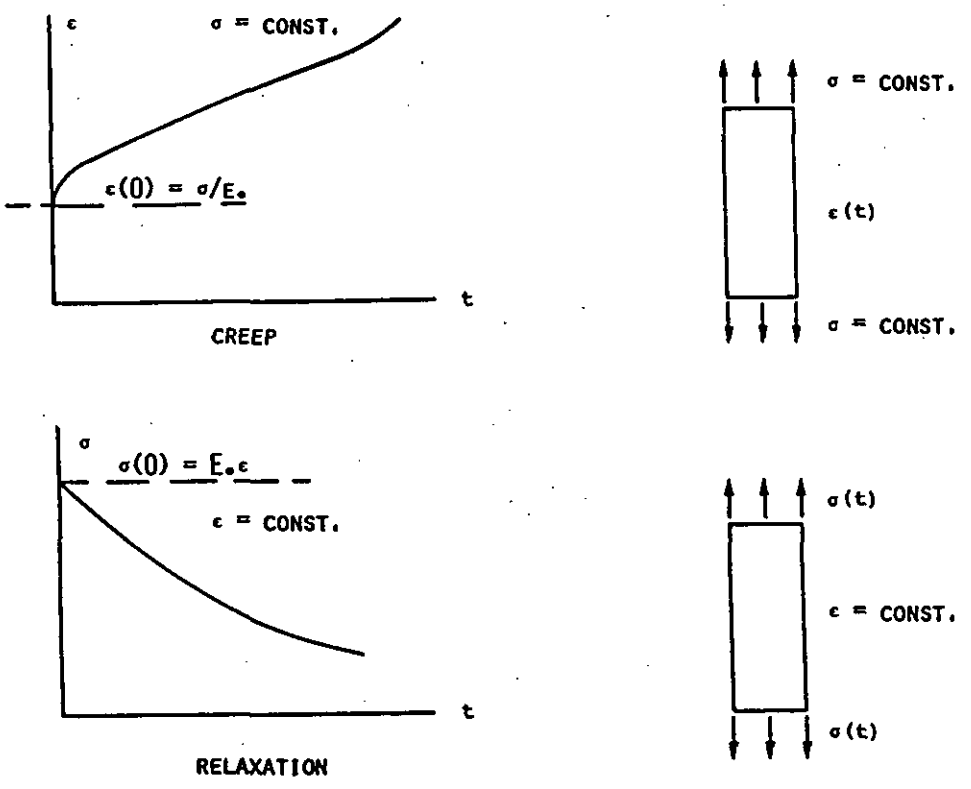


Figure 4.2.2.2 *Typical viscoelastic behavior.*

modulus decreases with increasing temperature and the creep compliance increases with increasing temperature, which implies that the stiffness decreases as the temperature increases. The initial value of these properties at "time-zero" are denoted G_o-and g_o and are the elastic properties of the matrix. If the applied shear strain is an arbitrary function of time, commencing at time-zero, Equation 4.2.2.2(b) is replaced by

$$\sigma_{12}(t) = 2G(t)\,e_{12}(0) + 2\int_o^t G(t-t')\frac{de_{12}}{dt'}\,dt' \qquad 4.2.2.2(d)$$

Similarly, for an applied shear stress which is a function of time, Equation 4.2.2.2(c) is replaced by

$$e_{12}(t) = \frac{1}{2}g(t)\,\sigma_{12}(0) + \frac{1}{2}\int_o^t g(t-t')\frac{d\sigma_{12}}{dt'}\,dt' \qquad 4.2.2.2(e)$$

The viscoelastic counterpart of Young's modulus is obtained by subjecting a cylindrical specimen to axial strain $\dot{\epsilon}_{11}$ constant in space and time. Then

$$\sigma_{11}(t) = E(t)\,\dot{\epsilon}_{11} \qquad 4.2.2.2(f)$$

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and E(t) is the Young's relaxation modulus. If the specimen is subjected to axial stress, $\sigma_{11}^{\cdot}$, constant is space and time, then

$$e_{11}(t) = e(t)\,\sigma_{11}^{\cdot} \qquad 4.2.2.2(g)$$

and e(t) is Young's creep compliance. Obviously E(t) is related to K and G(t), and e(t) is related to k and g(t). (See Reference 4.2.2.2(a).)

The basic problem is the evaluation of the effective viscoelastic properties of a UDC in terms of matrix viscoelastic properties and the elastic properties of the fibers. (It is assumed that the fibers themselves do not exhibit any time-dependent properties.) This problem has been resolved in general fashion in References 4.2.2.2(b) and (c). Detailed analysis shows that the viscoelastic effect in a UDC is significant only for axial shear, transverse shear, and transverse uniaxial stress.

For any of average strains $\bar{e}_{22}$, $\bar{e}_{23}$, and $\bar{e}_{12}$ constant in time, the time-dependent stress response will be

$$\begin{aligned} \bar{\sigma}_{22}(t) &= E_2^{\cdot}(t)\,\bar{e}_{22} \\ \bar{\sigma}_{23}(t) &= 2G_2^{\cdot}(t)\,\bar{e}_{23} \\ \bar{\sigma}_{12}(t) &= 2G_1^{\cdot}(t)\,\bar{e}_{12} \end{aligned} \qquad 4.2.2.2(h)$$

For any of stresses $\bar{\sigma}_{22}$, $\bar{\sigma}_{23}$, and $\bar{\sigma}_{12}$ constant in time, the time-dependent strain response will be

$$\begin{aligned} \bar{e}_{22}(t) &= e_2^{\cdot}(t)\,\bar{\sigma}_{22} \\ \bar{e}_{23}(t) &= \frac{1}{2}\,g_2^{\cdot}(t)\,\bar{\sigma}_{23} \\ \bar{e}_{12}(t) &= \frac{1}{2}\,g_1^{\cdot}(t)\,\bar{\sigma}_{12} \end{aligned} \qquad 4.2.2.2(i)$$

where material properties in Equations 4.2.2.2(h) are effective relaxation moduli and the properties in Equations 4.2.2.2(i) are effective creep functions. All other effective properties may be considered elastic. This implies in particular that if a fiber composite is subjected to stress $\bar{\sigma}_{11}(t)$ in the fiber direction, then

$$\begin{aligned} \bar{\sigma}_{11}(t) &= E_1^{\cdot}\,\bar{e}_{11}(t) \\ \bar{e}_{22}(t) &= \bar{e}_{33}(t) = \nu_{12}^{\cdot}\,\bar{e}_{11}(t) \end{aligned} \qquad 4.2.2.2(j)$$

where $E_1^{\cdot}$ and $\nu_{12}^{\cdot}$ are the elastic results of Equations 4.2.2.2(k) with matrix properties taken as initial (elastic) matrix properties. Similar considerations apply to the relaxation modulus $k^{\cdot}$.

The simplest case of the viscoelastic properties entering into Equations 4.2.2.2(h-i) is the relaxation modulus $G_1^{\cdot}(t)$ and its associated creep compliance $g_1^{\cdot}(t)$. A very simple result has been obtained for fibers which are infinitely more rigid than the matrix (Reference 4.2.2(a)). For a viscoelastic matrix, the results reduce to

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$$G_1^*(t) = G_m(t)\frac{1+v_f}{1-v_f}$$

$$g_1^*(t) = g_m(t)\frac{1-v_f}{1+v_f} \qquad 4.2.2.2(k)$$

This results in an acceptable approximation for glass fibers in a polymeric matrix and an excellent approximation for boron fibers in a polymeric matrix. However, the result is not applicable to the case of carbon or graphite fibers in a polymeric matrix since the axial shear modulus of these fibers is not large enough relative to the matrix shear modulus. In this case, it is necessary to use the correspondence principle mentioned above (References 4.2.2(a) and 4.2.2.2(b)). The situation for transverse shear is more complicated and involves complex Laplace transform inversion. (Reference 4.2.2.2(c)).

All polymeric matrix viscoelastic properties such as creep and relaxation functions are significantly temperature dependent. If the temperature is known, all of the results from this section can be obtained for a constant temperature by using the matrix properties at that temperature. At elevated temperatures, the viscoelastic behavior of the matrix may become nonlinear. In this event, the UDC will also be nonlinearly viscoelastic and all of the results given here are not valid. The problem of analytical determination of nonlinear properties is, of course, much more difficult than the linear problem (See Reference 4.2.2.2(d)).

4.2.2.3 *Thermal expansion and moisture swelling.* The elastic behavior of composite materials discussed in Section 4.2.2.1 is concerned with externally applied loads and deformations. Deformations are also produced by temperature changes and by absorption of moisture in two similar phenomena. A change of temperature in a free body produces thermal strains while moisture absorption produces swelling strains. The relevant physical parameters to quantify these phenomena are thermal expansion coefficients and swelling coefficients.

Fibers have significantly smaller thermal expansion coefficients than do polymeric matrices. The expansion coefficient of glass fibers is 2.8×10^{-6} in/in/F° (5.0×10^{-6} m/m/C°) while a typical epoxy value is 30×10^{-6} in/in/F° (54×10^{-6} m/m/C°). Carbon and graphite fibers are anisotropic in thermal expansion. The expansion coefficients in the fiber direction are extremely small, either positive or negative of the order of 0.5×10^{-6} in/in/F° (0.9×10^{-6} m/m/C°). To compute these stresses, it is necessary to know the thermal expansion coefficients of the layers. Procedures to determine these coefficients in terms of the elastic properties and expansion coefficients of component fibers and matrix are discussed in this section.

When a laminate absorbs moisture, there occurs the same phenomenon as in the case of heating. Again, the swelling coefficient of the fibers is much smaller than that of the matrix. Free swelling of the layers cannot take place and consequently internal stresses develop. These stresses can be calculated if the UDC swelling coefficients are known.

Consider a free cylindrical specimen of UDC under uniform temperature change ΔT. Neglecting transient thermal effects, the stress-strain relations (Equation 4.2.2.1(c)) assume the form

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$$\bar{e}_{11} = \frac{1}{E_1^*}\sigma_{11} - \frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{22} - \frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{33} + \alpha_1^*\Delta T$$

$$\bar{e}_{22} = -\frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{11} + \frac{1}{E_2^*}\bar{\sigma}_{22} - \frac{\nu_{23}^*}{E_2^*}\sigma_{33} + \alpha_2^*\Delta T \qquad 4.2.2.3(a)$$

$$\bar{e}_{33} = -\frac{\nu_{12}^*}{E_1^*}\bar{\sigma}_{11} - \frac{\nu_{23}^*}{E_2^*}\bar{\sigma}_{22} + \frac{1}{E_2^*}\bar{\sigma}_{33} + \alpha_2^*\Delta T$$

where

α_1^* - effective axial expansion coefficient

α_2^* - effective transverse expansion coefficient

It has been shown by Levin (Reference 4.2.2.3(a)) that there is a unique mathematical relationship between the effective thermal expansion coefficients and the effective elastic properties of a two-phase composite. When the matrix and fibers are isotropic

$$\alpha_1^* = \alpha_m + \frac{\alpha_f - \alpha_m}{\frac{1}{K_f} - \frac{1}{K_m}}\left[\frac{3(1-2\nu_{12}^*)}{E_1^*} - \frac{1}{K_m}\right]$$

$$\alpha_2^* = \alpha_m + \frac{\alpha_f - \alpha_m}{\frac{1}{K_f} - \frac{1}{K_m}}\left[\frac{3}{2k^*} - \frac{3(1-2\nu_{12}^*)}{E_1^*} - \frac{1}{K_m}\right] \qquad 4.2.2.3(b)$$

where

α_m, α_f - matrix, fiber isotropic expansion coefficients

K_m, K_f - matrix, fiber three-dimensional bulk modulus

E_1^*, ν_{12}^*, k^* - effective axial Young's modulus, axial Poisson's ratio, and transverse bulk modulus

These equations are suitable for glass/epoxy and boron/epoxy. They have also been derived in References 4.2.2.3(b) and (c). For carbon and graphite fibers, it is necessary to consider the case of transversely isotropic fibers. This complicates the results considerably as shown in Reference 4.2.2.1(c) and (e).

Frequently thermal expansion coefficients of the fibers and matrix are functions of temperature. It is not difficult to show that Equations 4.2.2.3(b) remain valid for temperature-dependent properties if the elastic properties are taken at the final temperature and the expansion coefficients are taken as secant at that temperature.

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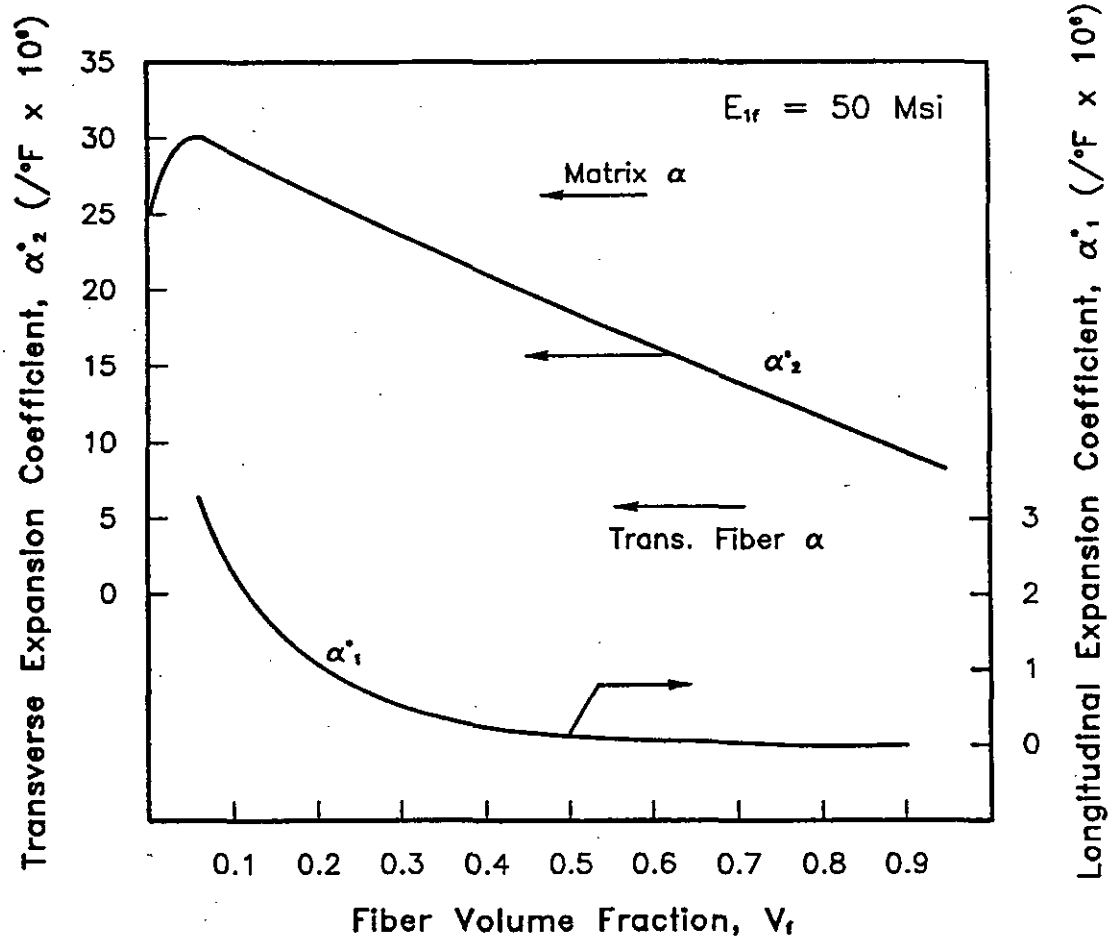


Figure 4.2.2.3 *Effect of fiber volume on thermal expansion for representative carbon/epoxy composite.*

To evaluate the thermal expansion coefficients from Equation 4.2.2.3(b) or (c), the effective elastic properties, k^*, E_1^*, and ν_{12}^* must be known. These may be taken as the values predicted by Equations 4.2.2.1(j-l) with the appropriate modification when the fibers are transversely isotropic. Figure 4.2.2.3 shows typical plots of the effective thermal expansion coefficients of graphite/epoxy.

When a composite with polymeric matrix is placed in a wet environment, the matrix will begin to absorb moisture. The moisture absorption of most fibers used in practice is negligible; however, aramid fibers alone absorb significant amounts of moisture when exposed to high humidity. The total moisture absorbed by an aramid/epoxy composite, however, may not be substantially greater than other epoxy composites.

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When a composite has been exposed to moisture and sufficient time has elapsed, the moisture concentration throughout the matrix will be uniform and the same as the boundary concentration. It is customary to define the specific moisture concentration c by

$$c = C/\rho \qquad 4.2.2.3(c)$$

where ρ is the density. The swelling strains due to moisture are functions of Δc and the swelling coefficients, β_{ij}

$$e_{ij} = \beta_{ij}\,\Delta c \qquad 4.2.2.3(d)$$

If there are also mechanical stresses and strains, then the swelling strains are superposed on the latter. This is exactly analogous to the thermoelastic stress-strain relations of an isotropic material. The effective swelling coefficients β_{ij}^* are defined by the average strains produced in a free sample subjected to a uniform unit change of specific moisture concentration in the matrix. For discussions of other aspects of moisture absorption, both transient and steady state, see References 4.2.2.3(d) and (e).

Finally, simultaneous moisture swelling and thermal expansion, or hygrothermal behavior can be considered. The simplest approach is to assume that the thermal expansion strains and the moisture swelling strains can be superposed. For a free specimen,

$$\bar{e}_{11} = \alpha_1^*\,\Delta T + \beta_1^*\,\Delta c$$
$$\bar{e}_{22} = \bar{e}_{33} = \alpha_2^*\,\Delta T + \beta_2^*\,\Delta c \qquad 4.2.2.3(e)$$

In this event, the matrix elastic properties in Equations 4.2.2.3(a) and (b) may be functions of the final temperature and moisture concentration. This dependence must be known to evaluate α_1^*, α_2^*, β_1^*, and β_2^* in Equation 4.2.2.3(e).

4.2.2.4 *Thermal conduction and moisture diffusion.* The thermal conduction analysis has many similarities with the analyses for moisture diffusion, as well as electrical conduction, and dielectric and magnetic properties. Since these conductivity problems are governed by similar equations, the results can be applied to each of these areas.

Let T(x) be a steady state temperature field in a homogeneous body. The temperature gradient is given by

$$H_i = \frac{\partial T}{\partial x_i} \qquad 4.2.2.4(a)$$

and the heat flux vector by

$$D_i = \mu_{ij} H_j \qquad 4.2.2.4(b)$$

where μ_{ij} is the conductivity tensor. It may be shown (Reference 4.2.2(a)) that for isotropic matrix and fibers, the axial conductivity μ_1^* is given by

$$\mu_1^* = \mu_m V_m + \mu_f V_f \qquad 4.2.2.4(c)$$

and for transversely isotropic fibers

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$$\mu_1^* = \mu_m V_m + \mu_{1f} V_f \qquad 4.2.2.4(d)$$

where μ_{1f} is the longitudinal conductivity of the fibers. The results of Equations 4.2.2.4(c) and (d) are valid for any fiber distribution and any fiber cross-section.

The problem of transverse conductivity is mathematically analogous to the problem of longitudinal shearing (Reference 4.2.2(a)). All results for the effective longitudinal shear modulus G_1^* can be interpreted as results for transverse effective conductivity μ_2^*. In particular, for the composite cylinder assemblage model

$$\mu_2^* = \mu_m \left[\frac{\mu_m V_m + \mu_f (1 + V_f)}{\mu_m (1 + V_f) + \mu_f V_m} \right]$$
$$= \mu_m + \frac{V_f}{\frac{1}{(\mu_f - \mu_m)} + \frac{V_m}{2\mu_m}} \qquad 4.2.2.4(e)$$

These results are for isotropic fibers. For carbon and graphite fibers μ_f should be replaced by the transverse conductivity μ_{2f} of the fibers (Reference 4.2.2.1(e)). As in the elastic case, there is reason to believe that Equation 4.2.2.4(e) accurately represents all cases of circular fibers which are randomly distributed and not in contact. Again the hexagonal array numerical analysis results coincide with the number predicted by Equation 4.2.2.4(e).

To interpret the results for the case of moisture diffusivity, the quantity μ_m is interpreted as the diffusivity of the matrix. Since moisture absorption of fibers is negligible, μ_f is set equal to zero. The results are then

$$\mu_1^* = \mu_m V_m$$
$$\mu_2^* = \mu_m \frac{V_m}{1 + V_f} \qquad 4.2.2.4(f)$$

These equations describe the moisture diffusivity of a composite material.

4.2.3 Fiber composites: strength and failure. The mathematical treatment of the relationships between the strength of a composite and the properties of its constituents is considerably less developed than the analysis for the other physical property relationships discussed in Section 4.2.2. Failure is likely to initiate in a local region due to the influence of the local values of constituent properties and the geometry in that region. This dependence upon local characteristics of high variability makes the analysis of the composite failure mechanisms much more complex than the analyses of the physical properties previously discussed.

Because of the complexity of the failure process, it may be desirable to regard the strength of a unidirectional fiber composite subjected to a single principal stress component as a quantity to be measured experimentally, rather than deduced from constituent properties. Such an approach may well be the practical one for fatigue failure of these composites. Indeed, the issue of determining the degree to which heterogeneity should be considered in the analysis of composite strength and failure is a matter for which there exists a considerable degree of difference of opinion. At the level of unidirectional composites, it is well to examine the effects upon failure of the individual constituents

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to develop an understanding of the nature of the possible failure mechanisms. This subject is discussed in the following sections. The general issue of the approach to failure analysis is treated further in laminate strength and failure.

The strength of a fiber composite clearly depends upon the orientation of the applied load with respect to the direction in which the fibers are oriented as well as upon whether the applied load is tensile or compressive. The following sections present a discussion of failure mechanisms and composite-constituent property relations for each of the principal loading conditions.

4.2.3.1 *Axial tensile strength.* One of the most attractive properties of advanced fiber composites is high tensile strength. The simplest model for the tensile failure of a unidirectional fiber composite subjected to a tensile load in the fiber direction is based upon the elasticity solution of uniform axial strain throughout the composite. Generally, the fibers have a lower strain to failure than the matrix, and composite fracture occurs at the failure strain of the fibers alone. This results in a composite tensile strength, F_1^{tu}, given by:

$$F_1^{tu} = v_f F_f^{tu} + v_m \sigma_m^t \qquad 4.2.3.1$$

where F_f^{tu} - the fiber tensile strength

σ_m^t - the stress in the matrix at a strain equal to the fiber failure strain

The problem with this approach is the variability of the fiber strength. Non-uniform strength is characteristic of most current high-strength fibers. There are two important consequences of a wide distribution of individual fiber strengths. First, all fibers will not be stressed to their maximum value simultaneously. Secondly, those fibers which break earliest during the loading process will cause perturbations of the stress field near the break, resulting in localized high fiber-matrix interface shear stresses. These shear stresses transfer the load across the interface and also introduce stress concentrations into adjacent unbroken fibers.

The stress distribution at each local fiber break may cause several possible failure events to occur. The shear stresses may cause a crack to progress along the interface. If the interface is weak, such propagation can be extensive. In this case, the strength of the composite material may differ only slightly from that of a bundle of unbonded fibers. This undesirable mode of failure can be prevented by a strong fiber-matrix interface or by a soft ductile matrix which permits the redistribution of the high shear stresses. When the bond strength is high enough to prevent interface failure, the local stress concentrations may cause the fiber break to propagate through the matrix, to and through adjacent fibers. Alternatively, the stress concentration in adjacent fibers may cause one or more of such fibers to break before failure of the intermediate matrix. If such a crack or such fiber breaks continue to propagate, the strength of the composite may be no greater than that of the weakest fiber. This failure mode is defined as a weakest link failure. If the matrix and interface properties are of sufficient strength and toughness to prevent or arrest these failure mechanisms, then continued load increases will produce new fiber failures at other locations in the material. An accumulation of dispersed internal damage results.

It can be expected that all of these effects will occur before material failure. That is, local fractures will propagate for some distance along the fibers and normal to the fibers. These fractures will initiate and grow at various points within the composite. Increasing the load will produce a statistical accumulation of dispersed damage regions until a sufficient number of such regions interact to provide a weak surface, resulting in composite tensile failure.

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4.2.3.1.1 *Weakest link failure.* The weakest link failure model assumes that a catastrophic mode of failure is produced with the occurrence of one, or a small number of, isolated fiber breaks. The lowest stress at which this type of failure can occur is the stress at which the first fiber will break. The expressions for the expected value of the weakest element in a statistical population (e.g., Reference 4.2.3.1.1(a)) have been applied by Zweben (Reference 4.2.3.1.1(b)) to determine the expected stress at which the first fiber will break. For practical materials in realistic structures, the calculated weakest link failure stress is quite small and, in general, failure cannot be expected in this mode.

4.2.3.1.2 *Cumulative weakening failure.* If the weakest link failure mode does not occur, it is possible to continue loading the composite. With increasing stress, fibers will continue to break randomly throughout the material. When a fiber breaks, there is a redistribution of stress near the fracture site. The treatment of a fiber as a chain of links is appropriate to the hypothesis that fracture is due to local imperfections. The links may be considered to have a statistical strength distribution which is equivalent to the statistical flaw distribution along the fibers. Additional details for this model are given in References 4.2.3.1.1(a) and 4.2.3.1.2. The cumulative weakening model does not consider the overstress on adjacent fibers or the effect of adjacent laminae.

4.2.3.1.3 *Fiber break propagation failure.* The effects of stress perturbations on fibers adjacent to broken fibers are significant. The load concentration in the fibers adjacent to a broken fiber increases the probability that a second fiber will break. Such an event will increase the probability of additional fiber breaks, and so on. The fiber break propagation mode of failure was studied by Zweben (Reference 4.2.3.1.1(b)). The occurrence of the first fracture of an overstressed fiber was proposed as a measure of the tendency for fiber breaks to propagate, and, hence, as a failure criterion for this mode. Although the first multiple break criterion may provide good correlations with experimental data for small volumes of material, it gives very low failure stress predictions for large volumes of material. Additional work in this area can be found in References 4.2.3.1.3(a) and (b).

4.2.3.1.4 *Cumulative group mode failure.* As multiple broken fiber groups grow, the magnitude of the local axial shear stress increases and axial cracking can occur. The cumulative group mode failure model (Reference 4.2.3.1.4) includes the effects of the variability of fiber strength, load concentrations in fibers adjacent to broken fibers, and matrix shear failure or interfacial debonding which will serve to arrest the propagating cracks. As the stress level increases from that at which fiber breaks are initiated to that at which the composite fails, the material will have distributed groups of broken fibers. This situation may be considered as a generalization of the cumulative weakening model. In practical terms, the complexity of this model limits its use.

Each of these models has severe limitations for the quantitative prediction of tensile strength. However, the models show the importance of variability of fiber strength and matrix stress-strain characteristics upon composite tensile strength.

4.2.3.2 *Axial compressive strength.* Both strength and stability failures must be considered for compressive loads applied parallel to the fibers of a unidirectional composite. Microbuckling is one proposed failure mechanism for axial compression (Reference 4.2.3.2(a)). Small wave-length micro-instability of the fibers occurs in a manner analogous to the buckling of a beam on an elastic foundation. It can be demonstrated that this instability can occur even for a brittle material such as glass. Analyses of this instability were performed independently in References 4.2.3.2(b) and (c). The energy method for evaluation of the buckling stress has been used for these modes. This procedure considers the composite as stressed to the buckling load. The strain energy in this compressed but straight pattern (extension mode) is then compared to an assumed buckling deformation pattern (shear mode) under the same load. The change in strain energy in the fiber and the matrix can be compared to the change in potential energy associated with the shortening of the distance between the applied

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loads at the ends of the fiber. The condition for instability is given by equating the strain energy change to the work done by the external loads during buckling.

The results for the compressive strength, F_1^{cu}, for the extension mode is given by

$$F_1^{cu} = 2v_f\sqrt{\frac{v_f E_m E_f}{3(1-v_f)}} \qquad 4.2.3.2(a)$$

The result for the shear mode is

$$F_1^{cu} = \frac{G_m}{1-v_f} \qquad 4.2.3.2(b)$$

The compressive strength of the composite is plotted as a function of the fiber volume fraction, v_f, in Figure 4.2.3.2 for E-glass fibers embedded in an epoxy matrix. The compressive strength of glass-reinforced plastic, with a fiber volume fraction of 0.6 to 0.7, is on the order of 460 to 600 ksi (3100 to 4100 MPa). Values of this magnitude do not appear to have been measured for any realistic specimens. However, the achievement of a strength of half a million psi in a composite of this type would require an average shortening greater than 5%. For the epoxy materials used in this calculation, such a shortening would result in a decrease in the effective shear stiffness of the matrix material since the proportional limit of the matrix would be exceeded. Hence, it is necessary to modify the analysis to consider the inelastic deformation of the matrix. As a first approximation, the matrix modulus in Equations 4.2.3.2(a) and (b) can be replaced by a reduced modulus. A more general result can be obtained by modeling the matrix as an elastic, perfectly plastic material. For this matrix, the secant value at each axial strain value may be assumed to govern the instability. These assumptions (Reference 4.2.3.2(d)) yield the following result for the shear mode:

$$F_1^{cu} = \sqrt{\frac{v_f E_f F^{cpl}}{3(1-v_f)}} \qquad 4.2.3.2(c)$$

where F^{cpl} is the matrix yield stress level.

For the generally dominant shear mode, the elastic results of Equation 4.2.3.2(b) are independent of the fiber modulus, yet the compressive strength of boron/epoxy is much greater than that of glass/epoxy composites. One hypothesis to explain this discrepancy, is that use of the stiffer boron fibers yields lower matrix strains and less of a strength reduction due to inelastic effects. Thus, the results of Equation 4.2.3.2(c) show a ratio of $\sqrt{6}$ or 2.4 for the relative strengths of boron compared to glass fibers in the same matrix.

All of the analytical results above indicate that compressive strength is independent of fiber diameter. Yet different diameter fibers may yield different compressive strengths for composites because large diameter fibers such as boron (0.005 inch, 0.13 mm D) are better collimated than small diameter fibers, such as glass (0.0004 in, 0.010 mm D). For small diameter fibers, such as aramid and carbon, local out-of-straightness can introduce matrix shear stresses, cause fiber debonding, and produce lower instability stress levels (References 4.2.3.2(e) and 4.2.2.1(d)). Carbon and aramid fibers are anisotropic and have extremely low axial shear moduli. As a result, the elastic buckling stress in the shear mode is reduced to:

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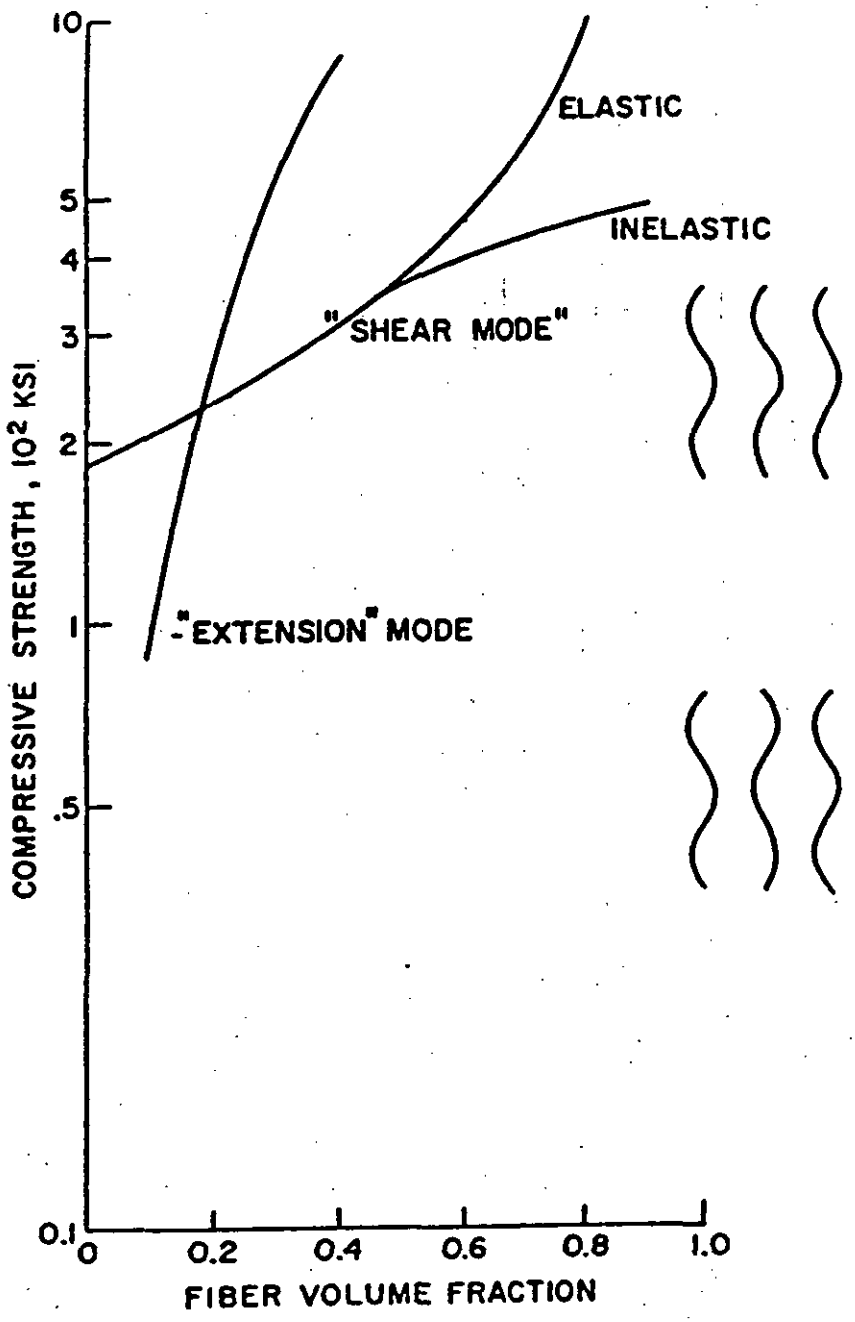


Figure 4.2.3.2 *Compressive strength of glass-reinforced epoxy composite.*

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$$F^{cr} = \frac{G_m}{1-v_f(1-G_m/G_{1f})} \qquad 4.2.3.2(d)$$

where G_{1f} is the fiber longitudinal shear modulus (Reference 4.2.3.2(e)). For high fiber shear moduli, this equation reduces to Equation 4.2.3.2(b).

Another failure mechanism for oriented polymeric fibers such as aramid fibers (Reference 4.2.3.2(e)) is a kink-band formation at a specific angle to the direction of compressive stress. The formation of kink-bands is attributed to the fibrillar structure of the highly anisotropic fiber and poor fiber shear strength. Breakup of the fiber into very small diameter fibrils results in degradation of shear stiffness and hence the compressive strength.

The results of the compressive strength analyses indicate that for the elastic case, the matrix Young's modulus is the dominant parameter. For the inelastic case, however, there are strength limitations which depend both upon the fiber modulus and upon the matrix strength. For some materials, performance is limited by a matrix yield strength at a given fiber modulus. For other materials, a gain in compressive strength can be achieved by improving the matrix modulus.

4.2.3.3 *Matrix mode strength.* The remaining failure modes of interest are transverse tension and compression and axial shear. For each of these loading conditions, material failure can occur without fracture of the fibers, hence the terminology "matrix-dominated" or "matrix modes of failure". Micromechanical analyses of these failure modes are complex because the critical stress states are in the matrix, are highly non-uniform, and are very dependent upon the local geometry. As a result, it appears that the most fruitful approaches will be those that consider average states of stress.

There are two types of shearing stresses which are of interest for these matrix-dominated failures: (1) in a plane which contains the filaments, and (2) in a plane normal to the filaments. In the first case, the filaments provide very little reinforcement to the composite and the shear strength depends on the shear strength of the matrix material. In the second case, some reinforcement may occur; at high volume fractions of filaments, the reinforcement may be substantial. It is important to recognize that filaments provide little resistance to shear in any surfaces parallel to them.

The approach to shear failure analysis is to consider that a uniaxial fibrous composite is comprised of elastic-brittle fibers embedded in an elastic-perfectly plastic matrix. For the composite, the theorems of limit analysis of plasticity (e.g. References 4.2.3.3(a) and (b)) may be used to obtain upper and lower bounds for a composite limit load (Reference 4.2.3.3(c)). The limit load is defined as the load at which the matrix yield stress permits composite deformation to increase with no increase in load. The failure strength of a ductile matrix may be approximated by this limit load.

4.2.4 Strength under combined stress. It is possible to apply the micro-mechanical models for failure described above, to combined stresses in the principal directions. Little work of this type has been done however. Generally the strengths in principal directions have been used in a failure surface for a homogeneous, anisotropic material for estimation of strength under combined loads. The understanding of failure mechanisms resulting from the above micro-mechanical models can be used to define the general form of failure surface to be utilized. This approach is outlined in the following sections.

Knowledge of the different failure mechanisms and quantitative experimental data for a UDC under single stress components can be used to formulate practical failure criteria for combined stresses. Plane stress failure criteria are discussed below with references also given for more complicated stress

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systems. The stresses considered are averaged over a representative volume element. The fundamental assumption is that there exists a failure criterion of the form:

$$F(\sigma_{11}, \sigma_{22}, \sigma_{12}) = 1 \qquad 4.2.4(a)$$

which characterizes the failure of the UDC. The usual approach to construction of a failure criterion is to assume a quadratic form in terms of stress or strain since the quadratic form is the simplest form which can adequately describe the experimental data. The various failure criteria which have been proposed all use coefficients based on experimental information such as ultimate stresses under single load components (References 4.2.4(a) - (d)). For example, the general quadratic version of Equation 4.2.4(a) for plane stress would be:

$$C_{11}\sigma_{11}^2 + C_{22}\sigma_{22}^2 + C_{66}\sigma_{12}^2 + 2C_{12}\sigma_{11}\sigma_{22} + 2C_{16}\sigma_{11}\sigma_{12} + 2C_{26}\sigma_{22}\sigma_{12} + C_1\sigma_{11} + C_2\sigma_{22} + C_6\sigma_{12} = 1 \qquad 4.2.4(b)$$

The material has different strengths in uniaxial, longitudinal, and transverse tension and compression. Evidently the shear strength is not affected by the sign of the shear stress. It follows that all powers of shear stress in the failure criterion must be even. Consequently, the criterion simplifies to

$$C_{11}\sigma_{11}^2 + C_{22}\sigma_{22}^2 + C_{66}\sigma_{12}^2 + 2C_{12}\sigma_{11}\sigma_{22} + C_1\sigma_{11} + C_2\sigma_{22} = 1 \qquad 4.2.4(c)$$

The ultimate stresses under single component stress conditions for each of σ_{11}, σ_{22}, and σ_{12} determine the constants for the failure criterion.

$$C_{11} = \frac{1}{F_1^{tu} F_1^{cu}} \qquad C_{22} = \frac{1}{F_2^{tu} F_2^{cu}}$$

$$C_1 = \frac{1}{F_1^{tu}} - \frac{1}{F_1^{cu}} \qquad C_2 = \frac{1}{F_2^{tu}} - \frac{1}{F_2^{cu}} \qquad 4.2.4(d)$$

$$C_{66} = \frac{1}{(F_1^{su})^2}$$

However, C_{12} cannot be determined from the single component ultimate stresses. Biaxial stress tests must be performed to determine this coefficient. Frequently, the coefficient is established by relating Equation 4.2.4(c) to the Mises-Henky yield criterion for isotropic materials, yielding

$$C_{12} = -\frac{1}{2}(C_{11}C_{22})^{1/2} \qquad 4.2.4(e)$$

The above failure criterion is the two-dimensional version of the Tsai-Wu criterion (Reference 4.2.4(c)). Its implementation raises several problems; the most severe of these is that the failure criterion ignores the diversity of failure modes which are possible.

The identification of the different failure modes of a UDC can provide physically more realistic, and also simpler, failure criteria (Reference 4.2.4(e)). Testing a polymer matrix UDC reveals that tensile stress in the fiber direction produces a jagged, irregular failure surface. Tensile stress transverse to the surface produces a smooth, straight failure surface (See Figures 4.2.4(a) and (b)). Since the carrying capacity deterioration in the tensile fiber mode is due to transverse cracks and the transverse stress σ_{22} has no effect on such cracks, it is assumed that the plane tensile fiber mode is only dependent on the stresses σ_{11} and σ_{12}.

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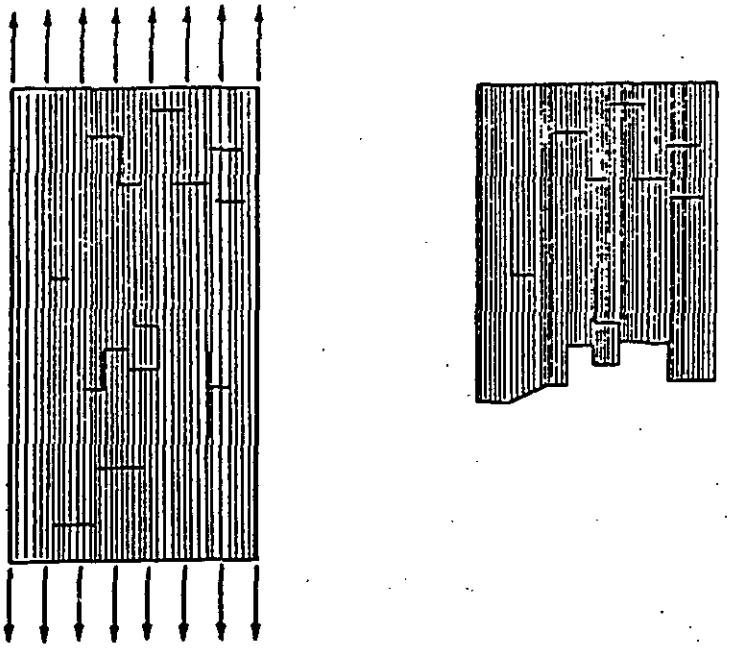


Figure 4.2.4(a) *Tensile fiber failure mode.*

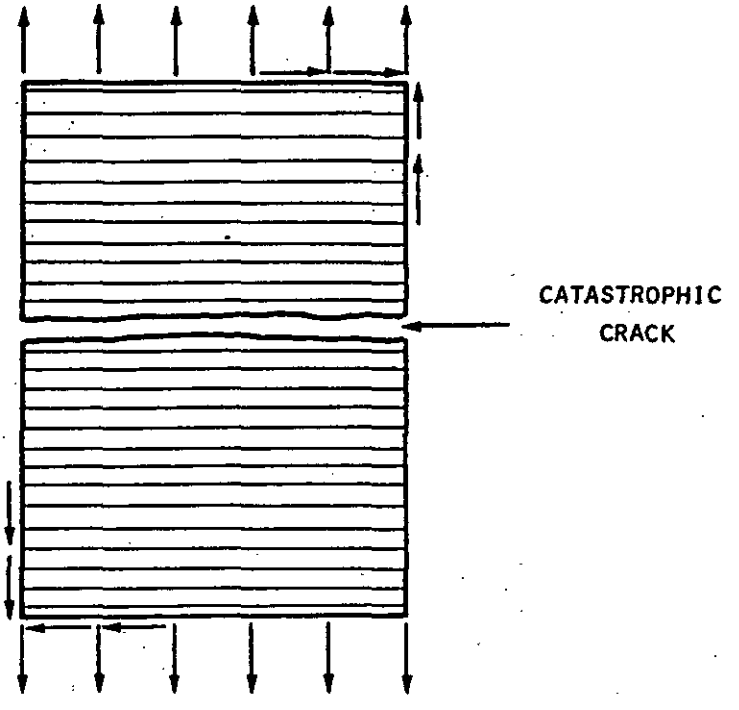


Figure 4.2.4(b) *Tensile matrix failure mode.*

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For compressive σ_{11}, failure is due to fiber buckling in the shear mode and the transverse stress σ_{22} has little effect on the compressive failure. In this *compressive fiber mode*, failure again depends primarily on σ_{11}. The dependence on σ_{12} is not known and arguments may be made for and against including it in the failure criterion.

For tension transverse to the fibers, the *tensile matrix mode*, failure occurs by a sudden crack in the fiber direction as shown in Figure 4.2.4(b). Since stress in the fiber direction has no effect on a crack in the fiber direction, this failure mode is dependent only on σ_{22} and σ_{12}.

For compressive stress transverse to the fibers, failure occurs on some plane parallel to the fibers, but not necessarily normal to σ_{22}. This *compressive matrix mode* is produced by normal stress and shear stress on the failure plane. Again, the stress σ_{11} does not effect this failure.

Each of the failure modes described can be modeled separately by a quadratic polynomial (Reference 4.2.4(e)). This approach provides four individual failure criteria. Note the choice of stress components included in each of these criteria, and the particular mathematical form used, are subjects which are not yet fully resolved. The following criteria appear to a reasonable set with which the different modes of failure can be handled separately.

Fiber modes

Tensile

$$\left(\frac{\sigma_{11}}{F_1^{tu}}\right)^2 + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(f)$$

Compressive

$$\left(\frac{\sigma_{11}}{F_1^{cu}}\right)^2 + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(g)$$

Matrix modes

Tensile

$$\left(\frac{\sigma_{22}}{F_2^{tu}}\right)^2 + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(h)$$

Compressive

$$\left(\frac{\sigma_{22}}{2F_{23}^{su}}\right)^2 + \left[\left(\frac{F_2^{cu}}{2F_{23}^{su}}\right)^2 - 1\right]\left(\frac{\sigma_{22}}{F_2^{cu}}\right) + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(i)$$

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Note that F_2^{cu} in Equation 4.2.4(i) should be taken as the absolute value. The ultimate transverse shear stress, $\sigma_{23} = F_{23}^{su}$, is very difficult to measure. A reasonable approximation for this quantity is the ultimate shear stress for the matrix. For any given state of stress, one each of Equations 4.2.4(f) and (g) and Equations 4.2.4(h) and (i) are chosen according to the signs of σ_{11} and σ_{22}. The stress components are introduced into the appropriate pair and whichever criterion is satisfied first is the operative criterion.

The advantages are Equations 4.2.4(f) - (i) are:

1. The failure criteria are expressed in terms of single component ultimate stresses. No biaxial test results are needed.

2. The failure mode is identified by the criterion which is satisfied first.

The last feature is of fundamental importance for analysis of fiber composite structural elements, since it permits identification of the nature of initial damage. Moreover, in conjunction with a finite element analysis, it is possible to identify the nature of failure in elements, modify their stiffnesses accordingly, and proceed with the analysis to predict new failures.

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4.3 ANALYSIS OF LAMINATES.

4.3.1 Lamina stress-strain relations. A laminate is composed of uni-directionally-reinforced laminae oriented in various directions with respect to the axes of the laminate. The stress-strain relations developed in the Section 4.2 must be transformed into the coordinate system of the laminate to perform the laminate stress-strain analysis. A new system of notation for the lamina elastic properties is based on x_1 in the fiber direction, x_2 transverse to the fibers in the plane of the lamina, and x_3 normal to the plane of the lamina.

$$\begin{aligned} E_1 &= E_1^* & \nu_{12} &= \nu_{12}^* \\ E_2 &= E_3 = E_2^* & \nu_{23} &= \nu_{23}^* \\ G_{12} &= G_1^* & G_{23} &= G_2^* \end{aligned} \qquad 4.3.1(a)$$

In addition, the laminae are now treated as effective homogeneous, transversely isotropic materials.

$$\begin{Bmatrix} e_{11} \\ e_{22} \\ e_{33} \\ 2e_{23} \\ 2e_{13} \\ 2e_{12} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E_1} & \frac{-\nu_{12}}{E_1} & \frac{-\nu_{12}}{E_1} & 0 & 0 & 0 \\ \frac{-\nu_{12}}{E_1} & \frac{1}{E_2} & \frac{-\nu_{23}}{E_2} & 0 & 0 & 0 \\ \frac{-\nu_{12}}{E_1} & \frac{-\nu_{23}}{E_2} & \frac{1}{E_2} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{23}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{12}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{Bmatrix} \qquad 4.3.1(b)$$

It has become common practice in the analysis of laminates to utilize engineering shear strains rather than tensor shear strains. Thus the factor of two has been introduced into the stress-strain relationship of Equation 4.3.1(b).

The most important state of stress in a lamina is plane stress, where

$$\sigma_{13} = \sigma_{23} = \sigma_{33} = 0 \qquad 4.3.1(c)$$

since it occurs from both in-plane loading and bending at sufficient distance from the laminate edges. The plane stress version of Equation 4.3.1(b) is

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$$\begin{Bmatrix} e_{11} \\ e_{22} \\ 2e_{12} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E_1} & \frac{-\nu_{12}}{E_1} & 0 \\ \frac{-\nu_{12}}{E_1} & \frac{1}{E_2} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{Bmatrix} \qquad 4.3.1(d)$$

which may be written as

$$\{e_\ell\} = [S]\{\sigma_\ell\} \qquad 4.3.1(e)$$

Here [S], the compliance matrix, relates the stress and strain components in the principal material directions. These are called laminae coordinates and are denoted by the subscript ℓ.

Equation 4.3.1(d) relates the in-plane strain components to the three in-plane stress components. For the plane stress state, the three additional strains can be found to be

$$e_{23} = e_{13} = 0$$

$$e_{33} = -\sigma_{11}\frac{\nu_{13}}{E_1} - \sigma_{22}\frac{\nu_{23}}{E_2} \qquad 4.3.1(f)$$

and the complete state of stress and strain is determined.

The relations 4.3.1(d) can be inverted to yield

$$\{\sigma_\ell\} = [S]^{-1}\{e_\ell\} \qquad 4.3.1(g)$$

or

$$\{\sigma_\ell\} = [Q]\{e_\ell\} \qquad 4.3.1(h)$$

The matrix [Q] is defined as the inverse of the compliance matrix and is known as the reduced lamina stiffness matrix. Its terms can be shown to be

$$[Q] = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix} \qquad 4.3.1(i)$$

$$Q_{11} = \frac{E_1}{1-\nu_{12}^2\frac{E_2}{E_1}} \qquad Q_{12} = \frac{\nu_{12}E_2}{1-\nu_{12}^2\frac{E_2}{E_1}}$$

$$Q_{22} = \frac{E_2}{1-\nu_{12}^2\frac{E_2}{E_1}} \qquad Q_{66} = G_{12}$$

In the notation for the [Q] matrix, each pair of subscripts of the stiffness components is replaced by a single subscript according to the following scheme.

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$$11 \rightarrow 1 \quad 22 \rightarrow 2 \quad 33 \rightarrow 3$$
$$23 \rightarrow 4 \quad 31 \rightarrow 5 \quad 12 \rightarrow 6$$

The reduced stiffness and compliance matrices 4.3.1(i) and (d) relate stresses and strains in the principal material directions of the material. To define the material response in directions other than these coordinates, transformation relations for the material stiffnesses are needed.

In Figure 4.3.1(a), two sets of coordinate systems are depicted. The 1-2 coordinate system corresponds to the principal material directions for a lamina, while the x-y coordinates are arbitrary and related to the 1-2 coordinates through a rotation about the axis out of the plane of the figure. The angle θ is defined as the rotation from the arbitrary x-y system to the 1-2 material system.

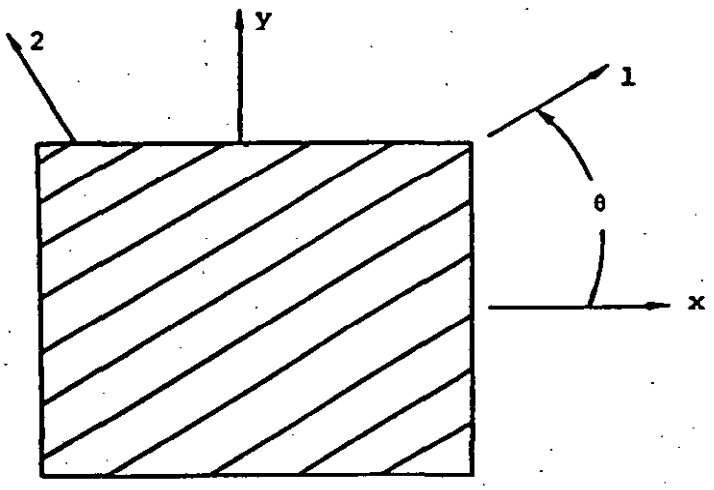
The transformation of stresses from the 1-2 system to the x-y system follows the rules for transformation of tensor components.

$$\begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} = \begin{bmatrix} m^2 & n^2 & -2mn \\ n^2 & m^2 & 2mn \\ mn & -mn & m^2-n^2 \end{bmatrix} \begin{Bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{12} \end{Bmatrix} \qquad 4.3.1(k)$$

or

$$\{\sigma_x\} = [\theta]\{\sigma_t\} \qquad 4.3.1(l)$$

where $m = \cos\theta$, and $n = \sin\theta$. In these relations, the subscript x is used as shorthand for the laminate coordinate system.



1,2 Principal Material Coordinates
x,y Laminate, or Arbitrary Coordinates

Figure 4.3.1(a) *Coordinate systems.*

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The same transformation matrix [θ] can also be used for the tensor strain components. However, since the engineering shear strains have been utilized, a different transformation matrix is required.

$$\begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ 2\epsilon_{xy} \end{Bmatrix} = \begin{bmatrix} m^2 & n^2 & -mn \\ n^2 & m^2 & mn \\ 2mn & -2mn & m^2-n^2 \end{bmatrix} \begin{Bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ 2\epsilon_{12} \end{Bmatrix} \qquad 4.3.1(m)$$

or

$$\{\epsilon_x\} = [\psi]\{\epsilon_t\} \qquad 4.3.1(n)$$

Given the transformations for stress and strain to the arbitrary coordinate system, the relations between stress and strain in the laminate system can be determined.

$$\{\sigma_x\} = [\bar{Q}]\{\epsilon_x\} \qquad 4.3.1(o)$$

The reduced stiffness matrix $[\bar{Q}]$ relates the stress and strain components in the laminate coordinate system.

$$[\bar{Q}] = [\theta][Q][\psi]^{-1} \qquad 4.3.1(p)$$

The terms within $[\bar{Q}]$ are defined to be

$$\begin{aligned}
\bar{Q}_{11} &= Q_{11}m^4 + Q_{22}n^4 + 2m^2n^2(Q_{12} + 2Q_{66}) \\
\bar{Q}_{12} &= m^2n^2(Q_{11} + Q_{22} - 4Q_{66}) + (m^4+n^4)Q_{12} \\
\bar{Q}_{16} &= [Q_{11}m^2 - Q_{22}n^2 - (Q_{12} + 2Q_{66})(m^2 - n^2)]mn \\
\bar{Q}_{22} &= Q_{11}n^4 + Q_{22}m^4 + 2m^2n^2(Q_{12} + 2Q_{66}) \\
\bar{Q}_{26} &= [Q_{11}n^2 - Q_{22}m^2 + (Q_{12} + 2Q_{66})(m^2 - n^2)]mn \\
\bar{Q}_{66} &= (Q_{11} + Q_{22} - 2Q_{12})m^2n^2 + Q_{66}(m^2 - n^2)^2
\end{aligned} \qquad 4.3.1(q)$$

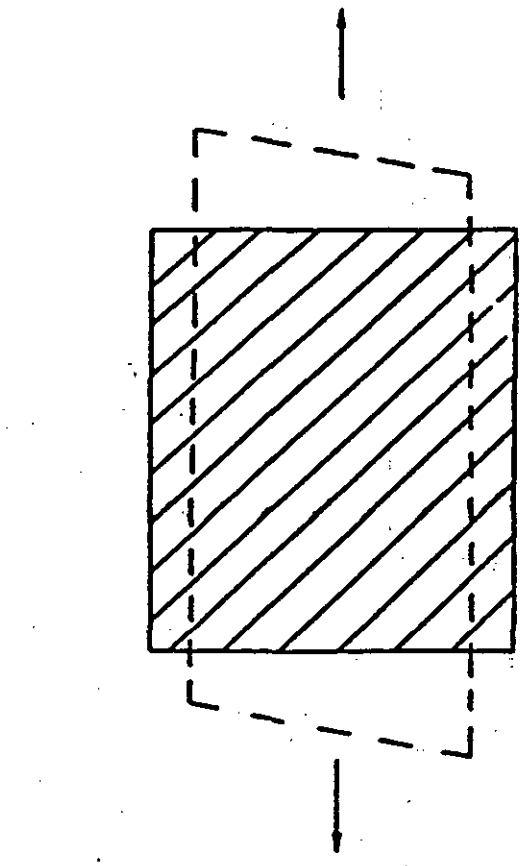
$$\bar{Q}_{21} = \bar{Q}_{12} \qquad \bar{Q}_{61} = \bar{Q}_{16} \qquad \bar{Q}_{62} = \bar{Q}_{26}$$

where the subscript 6 has been retained in keeping with the discussion following Equation 4.3.1(j).

$$[\bar{Q}] = \begin{bmatrix} \bar{Q}_{11} & \bar{Q}_{12} & \bar{Q}_{16} \\ \bar{Q}_{21} & \bar{Q}_{22} & \bar{Q}_{26} \\ \bar{Q}_{16} & \bar{Q}_{26} & \bar{Q}_{66} \end{bmatrix} \qquad 4.3.1(r)$$

A feature of $[\bar{Q}]$ matrix which is immediately noticeable is that $[\bar{Q}]$ is fully-populated. The additional terms which have appeared in $[\bar{Q}]$, $\bar{Q}_{16}$ and $\bar{Q}_{26}$, relate shear strains to extensional loading and vice versa. This effect of a shear strain resulting from an extensional strain is depicted in Figure 4.3.1(*b*). *From Equations* 4.3.1(q), these terms are zero for θ equal to 0° or 90°. Physically, this

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— — — — Deformed Shape under Extensional Loading

Figure 4.3.1(b) *Extensional-shear coupling.*

means that the fibers are parallel or perpendicular to the loading direction. For this case, extensional-shear coupling does not occur for an orthotropic material since the loadings are in the principal directions. The procedure used to develop the transformed stiffness matrix can also be used to find a transformed compliance matrix.

$$\{e_t\} = [S]\{\sigma_t\} \qquad 4.3.1(s)$$

$$\{e_x\} = [\psi][S][\theta]^{-1}\{\sigma_x\} \qquad 4.3.1(t)$$

$$\{e_x\} = [\bar{S}]\{\sigma_x\} \qquad 4.3.1(u)$$

Noting that the stress-strain relations are now defined in the laminate coordinate system, lamina stiffnesses can also be defined in this system. Thus, expanding the last of Equations 4.3.1(s) - (u):

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$$\begin{Bmatrix} e_{xx} \\ e_{yy} \\ 2e_{xy} \end{Bmatrix} = \begin{bmatrix} \bar{S}_{11} & \bar{S}_{12} & \bar{S}_{16} \\ \bar{S}_{21} & \bar{S}_{22} & \bar{S}_{26} \\ \bar{S}_{16} & \bar{S}_{26} & \bar{S}_{66} \end{bmatrix} \begin{Bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{xy} \end{Bmatrix} \qquad 4.3.1(v)$$

The engineering constants for the material can be defined by specifying the conditions for an experiment. For $\sigma_{yy} = \sigma_{xy} = 0$, the ratio σ_{xx}/e_{xx} is Young's modulus in the x direction. For this same stress state, $-e_{yy}/e_{xx}$ is Poisson's ratio. In this fashion, the lamina stiffnesses in the coordinate system of Equations 4.3.1(s) - (u) are found to be:

$$E_x = \frac{1}{\bar{S}_{11}} \qquad E_y = \frac{1}{\bar{S}_{22}}$$

$$G_{xy} = \frac{1}{\bar{S}_{66}} \qquad \nu_{xy} = \frac{-\bar{S}_{21}}{\bar{S}_{11}} = \frac{-\bar{S}_{12}}{\bar{S}_{11}} \qquad 4.3.1(w)$$

It is sometimes desirable to obtain elastic constants directly from the reduced stiffnesses, $[\bar{Q}]$, by using Equations 4.3.1(o). In the general case where the $\bar{Q}_{ij}$ matrix is fully populated, this can be accomplished by using Equations 4.3.1(w) and the solution for $\bar{S}_{ij}$ as functions of $\bar{Q}_{ij}$ obtained from the inverse relationship of the two matrices. An alternative approach is to evaluate extensional properties for the case of zero shear strain. For single stress states and zero shear strain, the elastic constants in terms of the transformed stiffness matrix terms are:

$$E_x = \bar{Q}_{11} - \frac{\bar{Q}_{12}^2}{\bar{Q}_{22}}$$

$$E_y = \bar{Q}_{22} - \frac{\bar{Q}_{12}^2}{\bar{Q}_{11}} \qquad 4.3.1(x)$$

$$\nu_{xy} = \frac{\bar{Q}_{12}}{\bar{Q}_{22}} = \frac{\bar{Q}_{21}}{\bar{Q}_{22}}$$

Also,

$$G_{xy} = \bar{Q}_{66}$$

From the terms in the $[\bar{Q}]$ matrix (Equation 4.3.1(q)) and the stiffness relations (Equation 4.3.1(x)), the elastic constants in an arbitrary coordinate system are functions of all the elastic constants in the principal material directions as well as the angle of rotation.

The variation of elastic modulus E_x with angle of rotation is depicted in Figure 4.3.1(c) for a typical graphite/epoxy material. For demonstration purposes, two different shear moduli have been used in generating the figure. The differences between the two curves demonstrate the effect of the principal material shear modulus on the transformed extensional stiffness. The two curves are identical at 0° and 90°, as expected since E_x is simply E_1 or E_2. Between these two endpoints, substantial differences are present. For the smaller shear modulus, the extensional stiffness is less than the E_2 value from approximately 50° to just less than 90°. For these angles, the material stiffness is more strongly governed by the principal material shear modulus than by the transverse extensional modulus.

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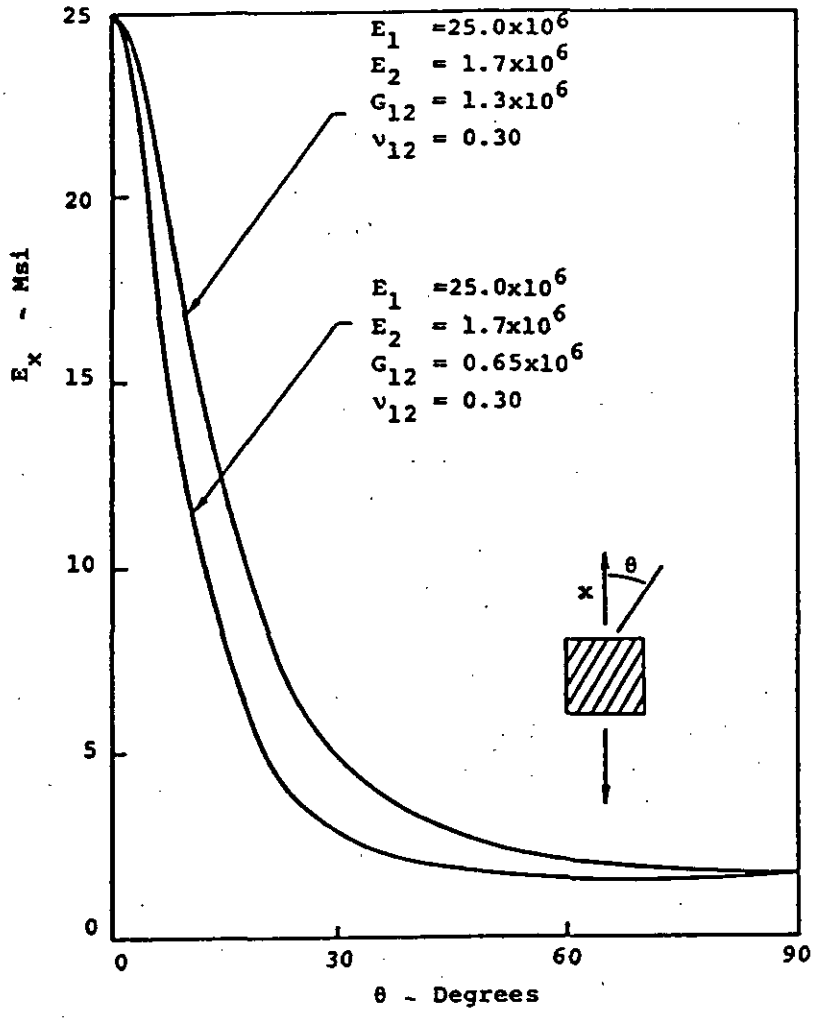


Figure 4.3.1(c) *Variation of E_x with angle and G_{12} for typical graphite/epoxy materials.*

The curves of Figure 4.3.1(c) can also be used to determine the modulus E_y by simply reversing the angle scale.

With the transformed stress-strain relations, it is now possible to develop an analysis for an *assemblage of plies, i.e., a laminate.*

4.3.2 Lamination theory. The development of procedures to evaluate stresses and deformations of laminates is crucially dependent on the fact that the thickness of laminates is much smaller than the in-plane dimensions. Typical thickness values for individual plies range between 0.005 and 0.010 inch (0.13 - 0.25 mm). Consequently, laminates using from 8 to 50 plies are still generally thin plates and, therefore, can be analyzed on the basis of the usual simplifications of thin plate theory.

In the analysis of isotropic thin plates it has become customary to analyze the cases of in-plane loading and bending separately. The former case is described by plane stress elastic theory and the

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latter by classical plate bending theory. This separation is possible since the two loadings are uncoupled for symmetric laminates; when both occur, the results are superposed.

The classical assumptions of thin plate theory are:

1) The thickness of the plate is much smaller than the in-plane dimensions;
2) The shapes of the deformed plate surface are small compared to unity;
3) Normals to the undeformed plate surface remain normal to the deformed plate surface;
4) Vertical deflection does not vary through the thickness; and
5) Stress normal to the plate surface is negligible.

On the basis of assumptions (2) - (4), the displacement field can be expressed as:

$$\begin{aligned} u_z &= \overset{\circ}{u_z}(x,y) \\ u_x &= \overset{\circ}{u_x}(x,y) - z\frac{\partial u_z}{\partial x} \\ u_y &= \overset{\circ}{u_y}(x,y) - z\frac{\partial u_z}{\partial y} \end{aligned} \qquad 4.3.2(a)$$

with the x-y-z coordinate system defined in Figure 4.3.2(a). These relations (Equation 4.3.2(a)) indicate that the in-plane displacements consist of a mid-plane displacement, designated by the *superscript (°), plus a linear variation* through the thickness. The two partial derivatives are bending rotations of the mid-surface. The use of assumption (4) prescribes that u_z does not vary through the thickness.

The linear strain displacement relations are

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x} \qquad \varepsilon_{yy} = \frac{\partial u_y}{\partial y}$$
$$\varepsilon_{xy} = \frac{1}{2}\left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right) \qquad 4.3.2(b)$$

and performing the required differentiations yields

$$\begin{aligned} \varepsilon_{xx} &= \overset{\circ}{\varepsilon_{xx}} + z\kappa_{xx} \\ \varepsilon_{yy} &= \overset{\circ}{\varepsilon_{yy}} + z\kappa_{yy} \\ 2\varepsilon_{xy} &= 2\overset{\circ}{\varepsilon_{xy}} + 2z\kappa_{xy} \end{aligned} \qquad 4.3.2(c)$$

or

$$\{\varepsilon_x\} = \{\varepsilon^\circ\} + z\{\kappa\} \qquad 4.3.2(d)$$

where

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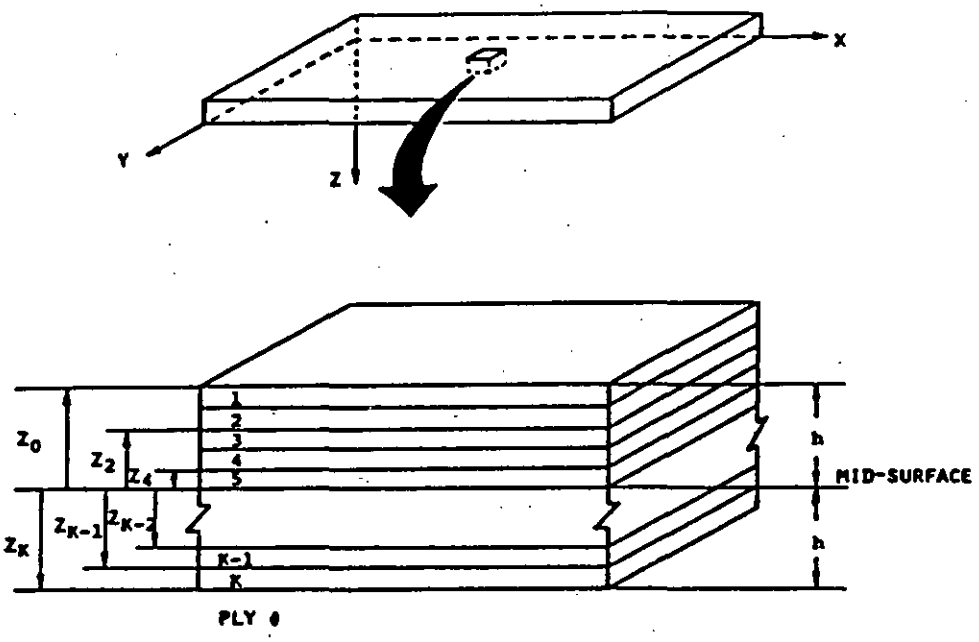


Figure 4.3.2(a) *Laminate construction.*

$$\{e^*\} = \begin{Bmatrix} \partial \dfrac{u_x^*}{\partial x} \\ \dfrac{\partial u_y^*}{\partial y} \\ \left[\dfrac{\partial u_x^*}{\partial y} + \dfrac{\partial u_y^*}{\partial x}\right] \end{Bmatrix} \qquad 4.3.2(e)$$

and

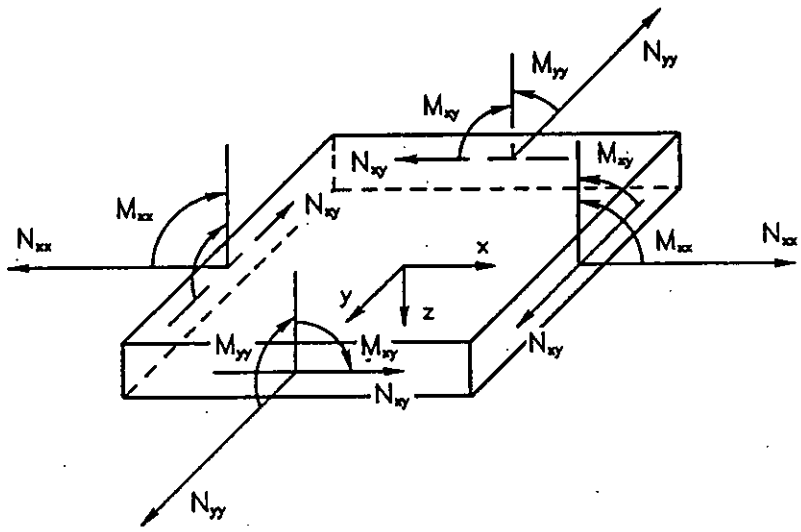
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$$\{\kappa\} = \begin{Bmatrix} \kappa_{xx} \\ \kappa_{yy} \\ 2\kappa_{xy} \end{Bmatrix} = \begin{Bmatrix} -\dfrac{\partial^2 u_z}{\partial x^2} \\ -\dfrac{\partial^2 u_z}{\partial y^2} \\ -2\dfrac{\partial^2 u_z}{\partial x \partial y} \end{Bmatrix} \qquad 4.3.2(f)$$

The strain at any point in the plate is defined as the sum of a mid-surface strain $\{\epsilon^o\}$, and a curvature $\{\kappa\}$ multiplied by the distance from the mid-surface.

For convenience, stress and moment resultants will be used in place of stresses for the remainder of the development of lamination theory (see Figure 4.3.2(b)). The stress resultants are defined as

$$\{N\} = \begin{Bmatrix} N_{xx} \\ N_{yy} \\ N_{xy} \end{Bmatrix} = \int_{-h}^{h} \{\sigma_x\}\, dz \qquad 4.3.2(g)$$



$$\{N\} = \int_{-h}^{h} \{\sigma_x\}\, dz$$

$$\{M\} = \int_{-h}^{h} \{\sigma_x\} z\, dz$$

Figure 4.3.2(b) *Stress and moment resultants.*

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and the moment resultants are defined as

$$\{M\} = \begin{Bmatrix} M_{xx} \\ M_{yy} \\ M_{xy} \end{Bmatrix} = \int_{-h}^{h} \{\sigma_x\} z dz \qquad 4.3.2(h)$$

where the integrations are carried out over the plate thickness.

Noting Equations 4.3.1(o) and 4.3.2(c), relations between the stress and moment resultants and the mid-plane strains and curvatures can be written as

$$\{N\} = \int_{-h}^{h} \{\sigma_x\} dz = \int_{-h}^{h} [\bar{Q}] (\{\epsilon^*\} + z\{\kappa\}) dz \qquad 4.3.2(i)$$

$$\{M\} = \int_{-h}^{h} \{\sigma_x\} z dz = \int_{-h}^{h} [\bar{Q}] (\{\epsilon^*\} + z\{\kappa\}) z dz \qquad 4.3.2(j)$$

Since the transformed lamina stiffness matrices are constant within each lamina and the mid-plane strains and curvatures are constant with respect to the z-coordinate, the integrals in Equations 4.3.2(i) and (j) can be replaced by summations.

Introducing three matrices equivalent to the necessary summations, the relations can be written as

$$\{N\} = [A]\{\epsilon^*\} + [B]\{\kappa\} \qquad 4.3.2(k)$$

$$\{M\} = [B]\{\epsilon^*\} + [D]\{\kappa\} \qquad 4.3.2(l)$$

where the stiffness matrix is composed of the following 3x3 matrices:

$$[A] = \sum_{i=1}^{N} [\bar{Q}]^i (z_i - z_{i-1})$$

$$[B] = \frac{1}{2} \sum_{i=1}^{N} [\bar{Q}]^i (z_i^2 - z_{i-1}^2) \qquad 4.3.2(m)$$

$$[D] = \frac{1}{3} \sum_{i=1}^{N} [\bar{Q}]^i (z_i^3 - z_{i-1}^3)$$

where N is the total number of plies, z_i is defined in Figure 4.3.2(a) and subscript i denotes a property of the ith ply. Note that $z_i - z_{i-1}$ equals the ply thickness. Here the reduced lamina stiffnesses for the ith ply are found from Equations 4.3.2(k) and (l) using the principal properties and orientation angle for each ply in turn. Thus, the constitutive relations for a laminate have been developed in terms of stress and moment resultants.

Classical lamination theory has been used to predict the internal stress state, stiffness and dimensional stability of laminated composites (e.g., References 4.3.2(a) - (e)). The constitutive law for CLT couples extensional, shear, bending and torsional loads with strains and curvatures. Residual strains or warpage due to differential shrinkage or swelling of plies in a laminate have also been incorporated in lamination theory using an environmental load analogy (See Sections 4.3.3.3 and 4.3.3.4.). The combined influence of various types of loads and moments on laminated plate response can be described using the ABD matrix from Equations 4.3.2(k) and (l). In combined form:

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$$\begin{bmatrix} N_x \\ N_y \\ N_{xy} \\ M_x \\ M_y \\ M_{xy} \end{bmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} & B_{11} & B_{12} & B_{16} \\ A_{12} & A_{22} & A_{26} & B_{12} & B_{22} & B_{26} \\ A_{16} & A_{26} & A_{66} & B_{16} & B_{26} & B_{66} \\ B_{11} & B_{12} & B_{16} & D_{11} & D_{12} & D_{16} \\ B_{12} & B_{22} & B_{26} & D_{12} & D_{22} & D_{26} \\ B_{16} & B_{26} & B_{66} & D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{bmatrix} \epsilon_x \\ \epsilon_y \\ \epsilon_{xy} \\ \kappa_x \\ \kappa_y \\ \kappa_{xy} \end{bmatrix} \qquad 4.3.2(n)$$

where N are loads, M are moments, ϵ are strains, κ are curvatures and

A_{ij} = extensional and shear stiffnesses
B_{ij} = extension-bending coupling stiffnesses
D_{ij} = *bending and torsional stiffnesses*

Several observations regarding lay-up and laminate stacking sequence (LSS) can be made with the help of Equation 4.3.2(n). These include:

(1) The stiffness matrix A_{ij} in Equation 4.3.2(n) is independent of LSS. Inversion of the stiffness matrix [ABD] yields the compliance matrix [A'B'D']. This inversion is necessary in order to calculate strains and curvatures in terms of loads and moments. The inversion results in a *relationship between LSS and extension/shear compliances.* However, this relationship is eliminated if the laminate is symmetric.

(2) Nonzero values of A_{16} and A_{26} indicates that there is extension/shear coupling (e.g., *longitudinal loads will result in both extensional and shear strains*). *If a laminate is balanced* A_{16} and A_{26} become zero, eliminating extension/shear coupling.

(3) Nonzero values of B_{ij} indicates that there is coupling between bending/twisting curvatures and extension/shear loads. *Traditionally, these couplings have been suppressed for most* applications by choosing an LSS that minimizes the values of B_{ij}. All values of B_{ij} become zero for symmetric laminates. Reasons for designing with symmetric laminates include structural dimensional stability requirements (e.g., buckling, environmental warping), compatibility of structural components at *joints and the inability to test for strength* allowables of specimens that have significant values of B_{ij}.

(4) In general, the values of D_{ij} are nonzero and strongly dependent on LSS. The average plate bending stiffnesses, torsional rigidity and flexural Poisson's ratio can be calculated per unit width using components of the compliance matrix [A'B'D'],i.e.,

$1/D'_{11}$ = bending stiffness about y-axis
$1/D'_{22}$ = bending stiffness about x-axis
$1/D'_{66}$ = torsional rigidity about x- or y-axis
$-D'_{12}/D'_{11}$ = flexural Poisson's ratio.

The D'_{16} and D'_{26} terms should also be included in calculations relating midplane curvatures to moments except when considering a special class of balanced, unsymmetric laminates.

(5) Nonzero values of D_{16} and D_{26} indicates that there is bending/twisting coupling. These terms will vanish only if a laminate is balanced and if, for each ply oriented at $+\theta$ above the laminate midplane, there is an identical ply (in material and thickness) oriented at $-\theta$ at an equal distance below the midplane. Such a laminate cannot be symmetric, unless it contains

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only 0° and 90° plies. Bending/twisting coupling can be minimized by alternating the location of $+\theta$ and $-\theta$ plies through the LSS (Section 4.6.5.2.2, Recommendation 5).

Additional information on laminate stacking sequence effects is found in Section 4.6.5.

4.3.3 Laminate properties. The relations between the mid-surface strains and curvatures and the membrane stress and moment resultants are used to calculate plate bending and extensional stiffnesses for structural analysis. The effects of orientation variables upon plate properties are also considered. In addition to the mechanical loading conditions treated thus far, the effects of temperature changes upon laminate behavior must be understood. Further, for polymeric matrix composites, high moisture content causes dimensional changes which can be described by effective swelling coefficients.

4.3.3.1 *Membrane stresses.* Recalling Equations 4.3.2(k) and (l) and noting that for symmetric laminates the [B] matrix is zero, the relations can be rewritten as

$$\begin{pmatrix} N_{xx} \\ N_{yy} \\ N_{xy} \end{pmatrix} = \begin{bmatrix} A_{11} & A_{12} & A_{16} \\ A_{12} & A_{22} & A_{26} \\ A_{16} & A_{26} & A_{66} \end{bmatrix} \begin{pmatrix} \dot{e}_{xx} \\ \dot{e}_{yy} \\ 2\dot{e}_{xy} \end{pmatrix} \qquad 4.3.3.1(a)$$

and

$$\begin{pmatrix} M_{xx} \\ M_{yy} \\ M_{xy} \end{pmatrix} = \begin{bmatrix} D_{11} & D_{12} & D_{16} \\ D_{12} & D_{22} & D_{26} \\ D_{16} & D_{26} & D_{66} \end{bmatrix} \begin{pmatrix} \kappa_{xx} \\ \kappa_{yy} \\ 2\kappa_{xy} \end{pmatrix} \qquad 4.3.3.1(b)$$

Since the extensional and bending behavior are uncoupled, effective laminate elastic constants can be readily determined. Inverting the stress resultant mid-plane strain relations yields

$$\{\epsilon^\circ\} = [A]^{-1}\{N\} = [a]\{N\} \qquad 4.3.3.1(c)$$

from which the elastic constants are seen to be

$$E_x = \frac{1}{2ha_{11}} \qquad G_{xy} = \frac{1}{2ha_{66}}$$

$$E_y = \frac{1}{2ha_{22}} \qquad \nu_{xy} = -\frac{a_{12}}{a_{11}} \qquad 4.3.3.1(d)$$

where the divisor 2h corresponds to the laminate thickness.

Note that the [A] matrix is comprised of [Q] matrices from each layer in the laminate. It is obvious that the laminate elastic properties are functions of the angular orientation of the plies. This angular influence is illustrated in Figure 4.3.3.1 for a typical high modulus graphite/epoxy system which has the lamina properties listed in Table 4.3.3.1(a). The laminae are oriented in $\pm\theta$ pairs in a symmetric, balanced construction, creating what is called an angle-ply laminate.

The variation of shear modulus and Poisson's ratio are noteworthy in Figure 4.3.3.1. The shear modulus is equal to the unidirectional value for 0° and 90° and rises sharply to a maximum at 45°. The peak at 45° can be explained by noting that shear is equivalent to a combined state of tensile and compressive loads oriented at 45°. Thus, the shear loading on a $[\pm 45]_s$ laminate is equivalent to

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tensile and compressive loading on a $[0/90]_s$ laminate. Effectively, the fibers are aligned with the loading and, hence, with the large shear stiffness.

An even more interesting effect is seen in the variation of Poisson's ratio. The peak value in this example is greater than 1.5. In an isotropic material, this would be impossible. In an orthotropic material, the isotropic restriction does not hold and a Poisson's ratio greater than one is valid and realistic. In fact, large Poisson's ratios are typical for laminates constructed from unidirectional materials with the plies oriented at approximately 30°.

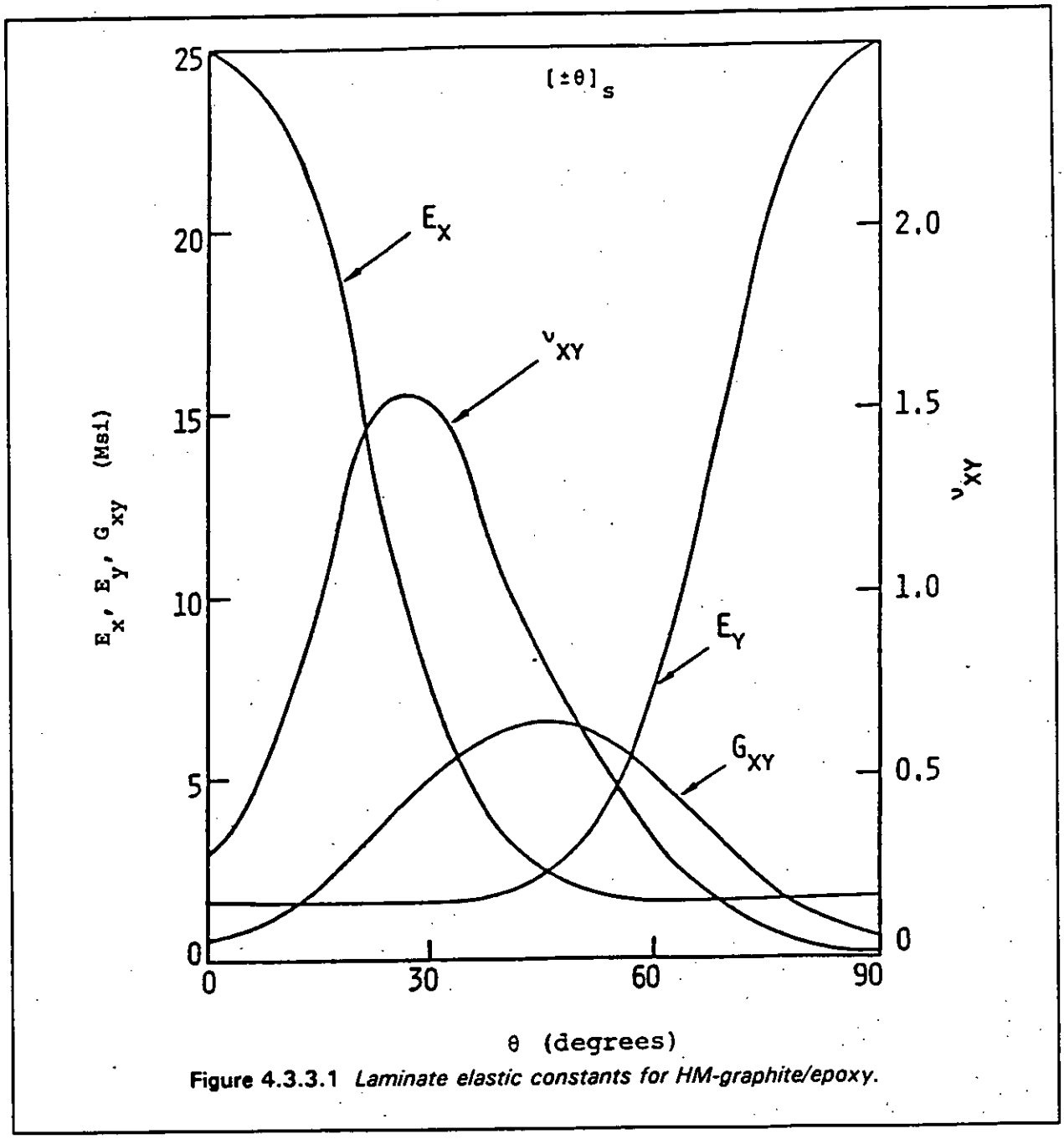


Figure 4.3.3.1 *Laminate elastic constants for HM-graphite/epoxy.*

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Table 4.3.3.1(a) *Properties of a high modulus graphite/epoxy lamina.*

$E_1 = 25.0 \times 10^6$ psi	$\alpha_1 = 0.30 \times 10^{-6}$ in/in/°F
$E_1 = 1.7 \times 10^6$ psi	$\alpha_2 = 19.5 \times 10^{-6}$ in/in/°F
$G_{12} = 0.65 \times 10^6$ psi	
$\nu_{12} = 0.30$	
$\rho = 0.056$ lb/in^3	
$F_1^{tu} = 110.0 \times 10^3$ psi	$F_1^{cu} = 110.0 \times 10^3$ psi
$F_2^{tu} = 4.0 \times 10^3$ psi	$F_2^{cu} = 20.0 \times 10^3$ psi
$F_{12}^{su} = 9.0 \times 10^3$ psi	
$v_f = 0.6$	$t_t = 0.0052$ in

Because of the infinite variability of the angular orientation of the individual laminae, one would assume that a laminate having a stiffness which behaves isotropically in the plane of the laminate could be constructed by using many plies having small, equal differences in their orientation. It can be shown that a symmetric, quasi-isotropic laminate can be constructed with as few as six plies as a $[0/\pm 60]_s$ laminate. A general rule for describing a quasi-isotropic laminate states that the angles between the plies are equal to π/N, where N is an integer greater than or equal to 3, and there is an identical number of plies at each orientation in a symmetric laminate. For plies of a given material, all such quasi-isotropic laminates will have the same elastic properties, regardless of the value of N.

A quasi-isotropic laminate has in-plane stiffnesses which follow isotropic relationships

$$E_x = E_y = E_\theta \qquad 4.3.3.1(e)$$

where the subscript θ indicates any arbitrary angle. Additionally,

$$G_{xy} = \frac{E_x}{2(1+\nu_{xy})} \qquad 4.3.3.1(f)$$

There are two items which must be remembered about quasi-isotropic laminates. First and foremost, only the elastic in-plane properties are isotropic; the strength properties, in general, will vary with directions. The second item is that two equal moduli $E_x = E_y$ do not necessarily indicate quasi-isotropy, as demonstrated in Table 4.3.3.1(b). The first two laminates in Table 4.3.3.1(b) are actually the same (a $[0/90]_s$ laminate rotated 45° is a $[\pm 45]_s$ laminate). Note that the extensional moduli of

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TABLE 4.3.3.1(b) *Elastic properties of laminates.*

	$E_x = E_y$ 10^6 psi	ν_{xy}	G_{xy} 10^6 psi
$[0°/90°]_s$	13.41	0.038	0.65
$[\pm 45°]_s$	2.38	0.829	6.46
$[0°/90°/\pm 45°/-45°/90°/0°]_s$	10.97	0.213	2.59

these laminates are not the same and that the shear modulus of each laminate is not related to the extensional modulus and Poisson's ratio. For these laminates, the π/N relation has not been satisfied and they are not quasi-isotropic. The third laminate has plies oriented at 45° to each other but there are not equal numbers of plies at each angle. This laminate is also not quasi-isotropic.

This discussion of symmetric laminates has centered on membrane behavior. Symmetric laminates can be constructed which are very well behaved in the membrane sense. The bending behavior of symmetric laminates is considerably more complex, primarily due to the arrangement of plies through the thickness of the laminate.

4.3.3.2 *Bending.* The equations for bending analysis of symmetric laminates has been developed with the extensional analysis. The first complication that arises in the treatment of laminate bending deals with relationships between the extensional (A) and bending (D) elastic properties. In composite laminates, there is no direct relationship between extensional and bending stiffnesses, unlike the case of a homogeneous material where

$$D = \frac{A(2h)^2}{12} \qquad 4.3.3.2$$

In determining the membrane stiffnesses (A), the position of the ply through the thickness of the laminate does not matter (Equation 4.3.2(m)). The relations for the bending stiffnesses are a function of the third power of the distance of the ply from the mid-surface. Therefore, the position of the plies with respect to the mid-surface is critical. The effects of ply position in a unit thickness laminate are shown in Table 4.3.3.2.

The three laminates shown in Table 4.3.3.2 are all quasi-isotropic. The membrane properties are isotropic and identical for each of the laminates. The bending stiffnesses can be seen to be a strong function of the thickness position of the plies. Additionally, bending stiffness calculations based on homogeneity (Equation 4.3.3.2) do not correspond to lamination theory calculations. Thus, the simple relations between extensional and bending stiffnesses are lost and lamination theory must be used for bending properties. Table 4.3.3.2 also demonstrates that quasi-isotropy holds only for extensional stiffnesses.

Another complication apparent in Table 4.3.3.2 involves the presence of the bending-twisting coupling terms, D_{16} and D_{26}. The corresponding extensional-shear coupling terms are zero because of the presence of pairs of layers at $\pm 60°$ orientations. Noting that the bending-twisting terms can

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TABLE 4.3.3.2 Extensional and bending stiffnesses.

	$[0/\pm 60]_s$	$[\pm 60/0]_s$	$[60/0/-60]_s$	Homogeneous Laminate
A_{11}	1.05×10^7	1.05×10^7	1.05×10^7	1.05x107
A_{12}	3.42×10^6	3.42×10^6	3.42×10^6	3.42×10^6
A_{22}	1.05×10^7	1.05×10^7	1.05×10^7	1.05×10^7
A_{66}	3.55×10^6	3.55×10^6	3.55×10^6	3.55×10^6
D_{11}	1.55×10^6	3.36×10^5	7.42×10^5	8.75×10^5
D_{12}	1.50×10^5	3.92×10^5	3.12×10^5	2.85×10^5
D_{16}	4.74×10^4	9.50×10^4	1.42×10^5	0.0
D_{22}	4.69×10^5	1.20×10^6	9.59×10^5	8.75×10^5
D_{26}	1.42×10^5	2.81×10^5	4.22×10^5	0.0
D_{66}	1.63×10^5	4.04×10^5	3.23×10^5	2.96×10^5

Lamina properties are from Table 4.3.3.1(a); unit thickness laminate.

[A] lb/in [D] in-lb

of the same order of magnitude as the principal bending terms, D_{11}, D_{22}, and D_{66}, the bending-twisting effect can be severe. This effect can be reduced by the proper selection of stacking sequence.

Another example that shows how the laminate stacking sequence (LSS) can significantly affect composite behavior in the bending stiffness of a laminated beam with rectangular cross-section (h ≡ laminate thickness). For the purpose of this example, define effective in-plane and bending moduli along the beam axis as

$$E_x = \frac{1}{A'_{11}h} \qquad 4.3.3.2(b)$$

$$E_x^b = \frac{12}{D'_{11}h^3} \qquad 4.3.3.2(c)$$

respectively. The relationship,

$$\Delta = \frac{E_x^b - E_x}{E_x} \times 100 \qquad 4.3.3.2(d)$$

provides a relative measure of the effect of LSS on beam bending stiffness. Bending moduli of laminated beams approach those of homogeneous beams as the number of plies increase provided that

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TABLE 4.3.3.2(b) *Stiffness predictions for seven different LSS for 16-ply, quasi-isotropic, carbon/epoxy, laminated beams*

Stacking Sequence	In-plane Modulus E_x (Msi)	Bending Modulus E_x^b (Msi)	Percent Difference Δ (%)
$[0_2/(\pm 45)_2/90_2]_s$	7.67	12.8	67
$[0/\pm 45/90]_{2s}$	7.67	10.1	32
$[\pm 45/0_2/\pm 45/90_2]_s$	7.67	7.80	1.7
$[\pm 45/0/90]_{2s}$	7.67	6.51	-15
$[(\pm 45)_2/0_2/90_2]_s$	7.67	4.45	-42
$[(\pm 45)_2/90_2/0_2]_s$	7.67	3.42	-55
$[90_2/(\pm 45)_2/0_2]_s$	7.67	3.25	-58

Properties for T300/934 (V_f = 0.63): E_{11} = 20.0 Msi, E_{22} = 1.4 Msi, G_{12} = 0.65 Msi, ν_{12} = 0.31, Ply Thickness = 0.0056 in.

there is no preferential stacking of ply orientations through the thickness.

Table 4.3.3.2(b) shows lamination theory predictions of in-plane and effective bending moduli for beams with seven different LSS variations of a 16-ply, carbon/epoxy, quasi-isotropic lay-up.[1] Note that the in-plane moduli are independent of LSS because all lay-ups are symmetric. Bending moduli are shown to vary significantly above or below the in-plane moduli depending on preferential stacking of 0° plies towards the surface or center of the laminate, respectively.

In general, the relationship between effective bending moduli and stacking sequence can be more complex than that shown in Table 4.3.3.2(b). Predictions in the table assumed that the basic lamina moduli were constant (i.e., linear elastic behavior). Depending on material type and the degree of accuracy desired, this assumption may lead to poor predictions. Lamina moduli for graphite/epoxy have been shown to depend on environment and strain level. Since flexure results in a distribution of tension and compression strains through the laminate thickness, *nonlinear elastic lamination theory* predictions may be more appropriate.

The example from Table 4.3.3.2(b) shows a significant effect of LSS on bending moduli of laminated beams. Similarly, calculations with Equation 4.3.2(n) can be used to indicate that LSS has a strong influence on the bending behavior of laminated plates. However, the bending response of common structures may depend more on the resulting moment of inertia, I, for a given geometry than

[1]The LSS used in Table 4.3.3.2(b) were chosen for illustrative purposes only and do not represent optimal LSS for a given application.

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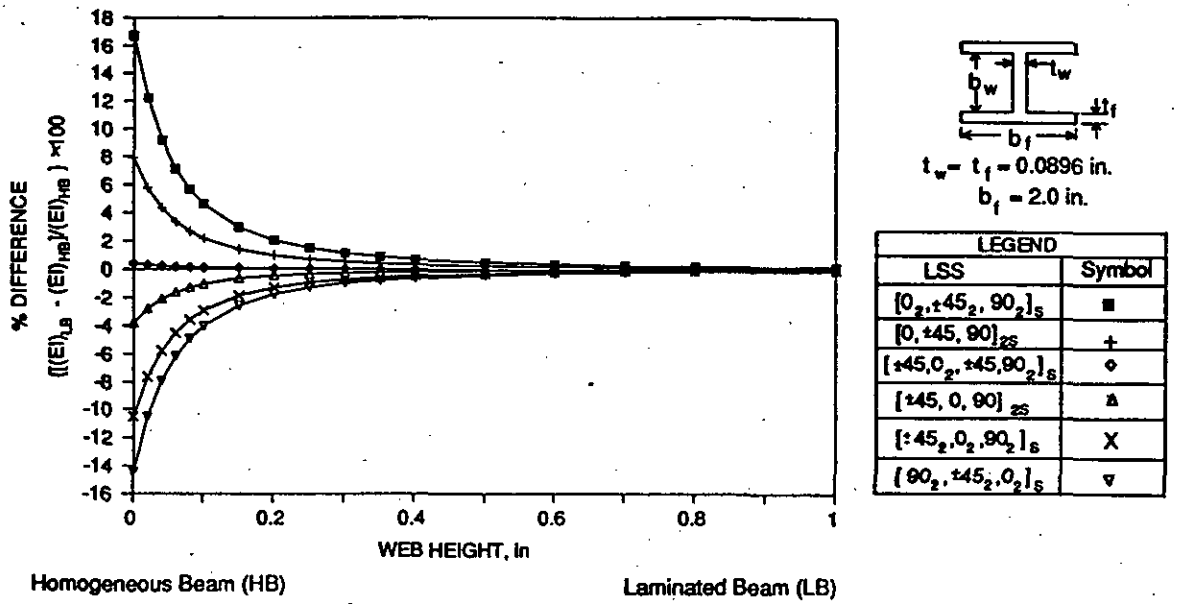
on LSS. This is particularly true for stringer geometries typically used to stiffen composite plates in aerospace structures.

Figure 4.3.3.2 illustrates how structural geometry of a beam section can overshadow the effects of LSS on bending. Web and flange members of each I-beam have LSS indicated in the legend of Figure 4.3.3.2[2] These LSS are the same as those used in Table 4.3.3.2(b). The ordinate axis of the figure indicates a percent difference between laminated and homogeneous beam calculations. As shown in Figure 4.3.3.2, the effect of LSS on the EI of an I-beam diminishes rapidly with increasing web height.

Additional information on laminate stacking sequence effects is found in Section 4.6.5.

4.3.3.3 *Thermal expansion.* As the use of composite materials becomes more commonplace, they are subjected to increasingly severe mechanical and environmental loading conditions. With the advent of high temperatures in systems, the range of temperatures over which composite systems can be used has increased. The response of laminates to temperature and moisture, as well as to applied loads, must be understood. Previously, laminate extensional and bending stiffnesses were determined; in this section laminate conductivities and expansion coefficients will be defined.

To determine the laminate thermal expansion coefficients and thermally-induced stresses quantitatively, begin at the ply level. The thermoelastic relations for strain in the principal material directions are



Homogeneous Beam (HB)

$$(EI)_{HB} = E_x\{2[b_f t_f^3/12 + b_f t_f (b_w+t_f)^2/4] + t_w b_w^3/12\}$$

Laminated Beam (LB)

$$(EI)_{LB} = 2[E_x^b b_f t_f^3/12 + E_x b_f t_f (b_w+t_f)^2/4] + E_x t_w b_w^3/12$$

E_x, E_x^b are listed in Figure 1.

FIGURE 4.3.3.2 *Laminated and homogeneous EI calculations for an I-beam stringer geometry with variable web height.*

[2]The LSS used in Figure 4.3.3.2 were chosen for illustrative purposes only and do not represent optimal LSS for a given application.

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$$\{e_t\} = \{e_t^M\} + \{\alpha_t\}\Delta T \qquad 4.3.3.3(a)$$

or

$$\{e_t\} = \{e_t^M\} + \{e_t^T\} \qquad 4.3.3.3(b)$$

where

$\{e_t^M\}$ = strain induced by stress

The change in temperature is represented by ΔT and the vector $\{\alpha_t\}$ represents the free thermal expansion coefficients of a ply. The individual components are

$$\{\alpha_t\} = \begin{Bmatrix} \alpha_1 \\ \alpha_2 \\ 0 \end{Bmatrix} \qquad 4.3.3.3(c)$$

The thermal strains, $\{\alpha_t\}\Delta T$, are the lamina free thermal expansions, which produce no stress in an unconstrained lamina. The thermal expansion coefficients α_1 and α_2 are the effective thermal expansion coefficients α_1^* and α_2^* of the unidirectional composite.

Substituting for the mechanical strain terms in Equation 4.3.3.3(a) and inverting yields

$$\{\sigma_t\} = [Q]\{e_t\} - \{\Gamma_t\}\Delta T \qquad 4.3.3.3(d)$$

where

$$\{\Gamma_t\} = [Q]\{\alpha_t\}$$

The components in the thermal stress coefficient vector $\{\Gamma_t\}$ are

$$\{\Gamma_t\} = \begin{Bmatrix} \dfrac{E_1\alpha_1 + \nu_{12}E_2\alpha_2}{\Delta} \\ \dfrac{E_2\alpha_2 + \nu_{12}E_1\alpha_1}{\Delta} \\ 0 \end{Bmatrix} \qquad 4.3.3.3(e)$$

where

$$\Delta = 1 - \frac{E_2}{E_1}\nu_{12}^2$$

The vector $\{\Gamma_t\}\Delta T$ physically represents a correction to the stress vector which results from the full constraint of the free thermal strains in a lamina. Both the thermal expansion vector, $\{\alpha_t\}\Delta T$, and

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the thermal stress vector, $\{\Gamma_t\}\Delta T$, can be transformed to arbitrary coordinates using the relations developed for stress and strain transformations, Equations 4.3.1(k) - (n).

With the transformed thermal expansion and stress vectors, the thermal elastic laminate relations can be developed. Following directly from the development of Equations 4.3.2(g) - (l), the membrane relations are:

$$\{N\} = [A]\{\epsilon^\circ\} + [B]\{\kappa\} + \{N^T\} \qquad 4.3.3.3(f)$$

where

$$\{N^T\} = -\int_{-h}^{h} \{\Gamma_x\}\Delta T\, dz \qquad 4.3.3.3(g)$$

Similarly, the bending relations are

$$\{M\} = [B]\{\epsilon^\circ\} + [D]\{\kappa\} + \{M^T\} \qquad 4.3.3.3(h)$$

where

$$\{M^T\} = -\int_{-h}^{h} \{\Gamma_x\}\Delta T\, z dz \qquad 4.3.3.3(i)$$

The integral relations for the thermal stress resultant vector $\{N^T\}$ and thermal moment resultant vector $\{M^T\}$ can be evaluated only when the change in temperature through the thickness is known. For the case of uniform temperature change through the thickness of a laminate, the term ΔT is constant and can be factored out of the integral, yielding:

$$\{N^T\} = -\Delta T \sum_{l=1}^{N} \{\Gamma_x\}^l (z_l - z_{l-1}) \qquad 4.3.3.3(j)$$

$$\{M^T\} = -\frac{1}{2}\Delta T \sum_{l=1}^{N} \{\Gamma_x\}^l \left(z_l^2 - z_{l-1}^2\right) \qquad 4.3.3.3(k)$$

With Equations 4.3.3.3(f) - (i), it is possible to determine effective laminate coefficients of thermal expansion and thermal curvature. These quantities are the extension and curvature changes resulting from a uniform temperature distribution.

Noting that for free thermal effects $\{N\} = \{M\} = 0$, and defining a free thermal expansion vector as

$$\{\alpha_x\} = \{\epsilon^\circ\}\frac{1}{\Delta T} \qquad 4.3.3.3(l)$$

and a free curvature vector as

$$\{\delta_x\} = \{\kappa\}\frac{1}{\Delta T} \qquad 4.3.3.3(m)$$

Equations 4.3.3.3(f) - (i) can be solved. After suitable matrix manipulations, the following expressions for thermal expansion and thermal curvature for symmetric laminates are found:

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$$\{\alpha_x\} = -\frac{1}{\Delta T}[A]^{-1}\{N^T\} \qquad 4.3.3.3(n)$$

$$\{\delta_x\} = -\frac{1}{\Delta T}[D]^{-1}\{N^T\} \qquad 4.3.3.3(o)$$

If the relation for $\{M^T\}$ in Equation 4.3.3.3(i) is examined, symmetry eliminates the $\{M^T\}$ vector. Therefore $\{\delta_x\} = 0$ and no curvatures occur due to uniform temperature changes in symmetric laminates.

The variation of the longitudinal thermal expansion coefficient for a symmetric angle-ply laminate is shown in Figure 4.3.3.3 to illustrate the effect of lamina orientation. At 0° the term α_x is simply the axial lamina coefficient of thermal expansion, and at 90°, α_x equals the lamina transverse thermal expansion coefficient. An interesting feature of the curve is the large negative value of α_x in the region of 30°. Referring to Figure 4.3.3.1, the value of Poisson's ratio also behaves peculiarly in the region of 30°. The odd variation of both the coefficient and Poisson's ratio stems from the magnitude and sign of the shear-extensional coupling present in the individual laminae.

Previously, classes of laminates were shown to have isotropic stiffnesses in the plane of the laminate. Similarly, laminates can be specified which are isotropic in thermal expansion within the plane of the laminate. The requirements for thermal expansion isotropy are considerably less restrictive than those for elastic constants. In fact, any laminate which has two identical, orthogonal thermal expansion coefficients and a zero shear thermal expansion coefficient is isotropic in thermal expansion. Therefore, $[0/90]_s$ and $[\pm 45]_s$ laminates are isotropic in thermal expansion even though they are not quasi-isotropic for elastic stiffnesses.

Laminates which are isotropic in thermal expansion have thermal expansions of the form:

$$\alpha_x = \begin{Bmatrix} \alpha_x \\ \alpha_y \\ \alpha_{xy} \end{Bmatrix} = \begin{Bmatrix} \alpha^* \\ \alpha^* \\ 0 \end{Bmatrix} \qquad 4.3.3.3(p)$$

where the term α^* can be shown to be a function of lamina properties only, as follows:

$$\alpha^* = \alpha_1 + \frac{(\alpha_2 - \alpha_1)(1 + \nu_{12})}{1 + 2\nu_{12} + \frac{E_1}{E_2}} \qquad 4.3.3.3(q)$$

Thus, all laminates of a given ply material, which are isotropic in thermal expansion, have identical thermal expansion coefficients.

4.3.3.4 *Moisture expansion.* The term hygroelastic refers to the phenomenon in resin matrix composites when the matrix absorbs and desorbs moisture from and to the environment. The primary effect of moisture is a volumetric change in the laminae. When a lamina absorbs moisture, it expands, and when moisture is lost, the lamina contracts. Thus, the effect is very similar to thermal expansion.

In a lamina, a free moisture expansion vector can be defined as

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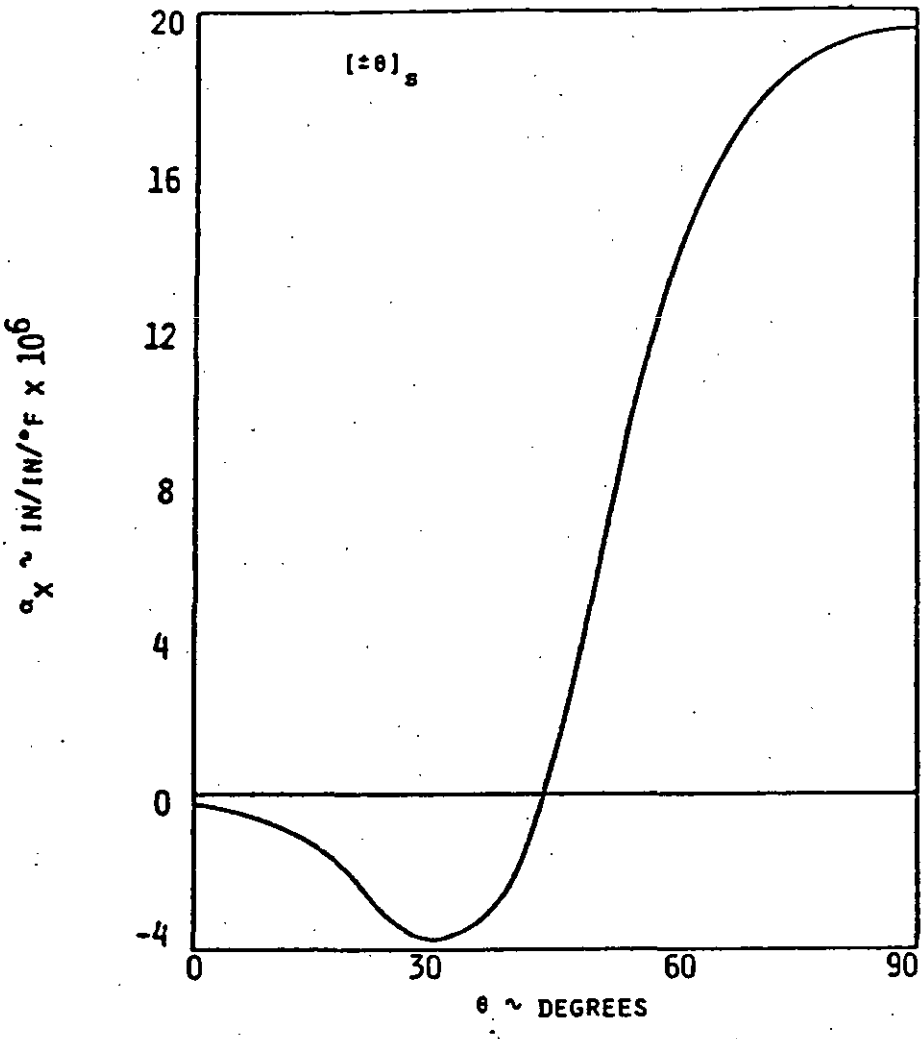


Figure 4.3.3.3 *Thermal expansion coefficients for HM-graphite/epoxy.*

$$\{e_t\} = \{\beta_t\}\Delta c \qquad \text{4.3.3.4(a)}$$

where

$$\{\beta_t\} = \begin{Bmatrix} \beta_1 \\ \beta_2 \\ 0 \end{Bmatrix} \qquad \text{4.3.3.4(b)}$$

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and Δc is the change in specific moisture. Noting that the relations 4.3.3.4(a) and (b) are identical to thermal expansion with $\{\beta_\ell\}$ substituted for $\{\alpha_\ell\}$ and Δc for ΔT, it can easily be seen that all the relations developed for thermal effects can be used for moisture effects.

4.3.3.5 *Conductivity.* The conductivity (thermal or moisture) of a laminate in the direction normal to the surface is equal to the transverse conductivity of a unidirectional fiber composite. This follows from the fact that normal conductivity for all plies is identical and unaffected by ply orientation.

In-plane conductivities will be required for certain problems involving spatial variations of temperature and moisture. For a given uniform state of moisture in a laminate, the effective thermal conductivities in the x and y directions can be obtained by methods entirely analogous to those used for stiffnesses in Section 4.3.2:

$$\mu_x = \frac{1}{2h} \sum_{i=1}^{N} (\mu_1 m^2 + \mu_2 n^2)\, t_t^i \qquad 4.3.3.5$$

where

μ_1 = conductivity in the fiber direction
μ_2 = conductivity transverse to the fibers
m = $\cos \theta^i$
n = $\sin \theta^i$
θ^i = orientation of ply i
t_t^i = thickness of ply i
N = the number of plies
$2h$ = laminate thickness

The results apply to both symmetric and unsymmetric laminates. The results for moisture conductivity are identical.

4.3.4 Thermal and hygroscopic analysis. The distribution of temperature and moisture through the thickness of a laminate influences the behavior of that laminate. The mathematical descriptions of these two phenomena are identical and the physical effects are similar. Some of these aspects have already been discussed in Sections 4.2.2.3 - 4.2.2.4 and 4.3.3.3 - 4.3.3.5.

A free lamina undergoes stress-free deformation due to temperature change or moisture swelling. In a laminate, stress-free deformation is constrained by adjacent layers producing internal stresses. In addition to these stresses, temperature and moisture content also affect the properties of the material. These effects are primarily related to matrix-dominated strength properties.

The principal strength-degrading effect is related to a change in the glass transition temperature of the matrix material. As moisture is absorbed, the temperature at which the matrix changes from a glassy state to a viscous state decreases. Thus, the elevated-temperature strength properties decrease with increasing moisture content. Limited data suggest that this process is reversible. When the moisture content of the composite is decreased, the glass transition temperature increases and the original strength properties return.

The same considerations also apply for a temperature rise. The matrix, and therefore the lamina, lose strength and stiffness when the temperature rises. Again, this effect is primarily important for the matrix-dominated properties such as E_2, G_{12}, F_2^{tu}, F_2^{cu}, and F_{12}^{su}.

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The differential equation governing time-dependent moisture sorption of an orthotropic homogeneous material is given by

$$D_1 \frac{\partial^2 c}{\partial x_1^2} + D_2 \frac{\partial^2 c}{\partial x_2^2} + D_3 \frac{\partial^2 c}{\partial x_3^2} = \frac{\partial c}{\partial t} \qquad 4.3.4(a)$$

where

t = time
x_1, x_2, x_3 = coordinates in principal material directions
c = specific moisture concentration
D_1, D_2, D_3 = moisture diffusivity coefficients

Equation 4.3.4(a) is based on Fick's law of moisture diffusion. The equation is analogous to the equation governing time dependent heat conduction with temperature ϕ replacing concentration c and thermal conductivities μ_1, μ_2, and μ_3 replacing the moisture diffusivities. For a transversely isotropic lamina with x_1 in the fiber direction, x_2 in the transverse direction, and $x_3 = z$ in the direction normal to the lamina,

$$D_2 = D_3 \qquad 4.3.4(b)$$

These quantities are analogous to the thermal conductivities of a unidirectional fiber composite and have been discussed in Section 4.2.2.4.

An important special case is one-dimensional diffusion or conduction through the thickness of a lamina. In this case, Equation 4.3.4(a) reduces to

$$D_3 \frac{\partial^2 c}{\partial z^2} = \frac{\partial c}{\partial t} \qquad 4.3.4(c)$$

This equation also applies to moisture diffusion or thermal conduction through a laminate, in the direction normal to its laminae planes, since all laminae are homogeneous in the z direction with equal diffusion coefficients, $D_3 = D_z$.

Equation 4.3.4(c) is applicable to the important problem of time-dependent moisture diffusion through a laminate where the two faces are in different moisture environments. After a sufficiently long time has elapsed, the concentration reaches a time-independent state. In this state, since c is no longer time-dependent, Equation 4.3.4(c) simplifies to

$$\frac{d^2 c}{dz^2} = 0 \qquad 4.3.4(d)$$

The specific moisture concentration is a linear function of z and, if the laminate faces are in environments with constant saturation concentrations, c_1 and c_2, then

$$c = \frac{1}{2}[(c_2 - c_1) z/h + c_2 + c_1] \qquad 4.3.4(e)$$

where the laminate thickness is 2h and z originates at the mid-surface. In the case where $c_1 = c_2$, Equation 4.3.4(e) reduces to $c = c_1 =$ constant as would be expected.

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The above discussion of moisture conduction also applies to heat conduction.

Solutions to the time-dependent problem are readily available and considerable work has been performed in the area of moisture sorption (Reference 4.3.4). The most interesting feature of the solutions relates to the magnitude of the coefficient D_z. This coefficient is a measure of how fast moisture diffusion can occur. In typical epoxy matrix systems, D_z is of the order of 10^{-8} (in^2/s, cm^2/s) to 10^{-10} (in^2/s, cm^2/s). The diffusion coefficient is sufficiently small that full saturation of a resin matrix composite may require months or years even when subjected to 100% relative humidity.

The approach typically taken for design purposes is to assume a worst case. If the material is assumed to be fully saturated, it is possible to compute reduced allowable strengths. This is a conservative approach, since typical service environments do not generate full saturation. This approach is used since it allows for inclusion of moisture effects in a relatively simple fashion. It is to be expected that as the design data base and analytical methodologies mature, more physically realistic methods will be developed.

For heat conduction, the time required to achieve the stationary, or time-independent, state is extremely small. Therefore, the transient time-dependent state is generally of little practical importance for laminates.

4.3.4.1 *Symmetric laminates.* The laminate stacking sequence (LSS) can be chosen to control the effects of environment on stiffness and dimensional stability. When considering the special case of constant temperature and moisture content distributions in symmetric laminates, the effect of environment on in-plane stiffness relates to the relative percentages of chosen ply orientations. For example, LSS dominated by 0° plies will have longitudinal moduli that are nearly independent of environment. Note that increasing the environmental resistance of one laminate in-plane modulus may decrease another.

Bending and torsional stiffnesses depend on both LSS and environment. Preferential stacking of outer ply groups having relatively high extensional or shear moduli will also promote high bending or torsional stiffness, respectively. As with in-plane moduli, the higher the bending or torsional stiffness the better the corresponding environmental resistance. When optimizing environmental resistance, compromises between longitudinal bending, transverse bending and torsion need to be made due to competing relationships with LSS.

Unsymmetric temperature and moisture content distributions will affect the components of the stiffness matrix [ABD] differently, depending on LSS. In general, coupling components which were zero for symmetric laminates having symmetric temperature and moisture content distributions become nonzero for an unsymmetric environmental state. This effect can be minor or significant depending on LSS, material type, panel thickness and the severity of temperature/moisture content gradients.

Environmentally-induced panel warpage will occur in symmetric laminates when conditions yield an unsymmetric residual stress distribution about the laminate midplane. This may occur during the cure process due to uneven heating or crystallization through the laminate thickness. Unsymmetric temperature and moisture content distributions can also lead to panel warpage in symmetric laminates. This is due to the unsymmetric shrinkage or swelling through the laminate thickness.

4.3.4.2 *Unsymmetric laminates.* The in-plane thermal and moisture expansion of unsymmetric laminated plates subjected to any environmental condition (i.e., constant, symmetric and unsymmetric temperature and moisture content distributions) is dependent on LSS (e.g., Reference 4.3.4.2(a)). In general, environmentally induced panel warpage occurs with unsymmetric laminates.

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Panel warpage in unsymmetric laminates depends on LSS and changes as a function of temperature and moisture content. Zero warpage will occur in unsymmetric laminates only when temperature and moisture content distributions result in either zero or symmetric residual stress distributions. Equilibrium environmental states that result in zero residual stresses are referred to as stress-free conditions (see Reference 4.3.2(e).

Since unsymmetric LSS warp as a function of temperature and moisture content, their use in engineering structures has generally been avoided. The warped shape of a given unsymmetric laminate has been found to depend on LSS and ratios of thickness to in-plane dimensions (e.g., References 4.3.4.2(b) and (c)). Relatively thin laminates tend to take a cylindrical shape rather than the saddle shape predicted by classical lamination theory. This effect has been accurately modelled using a geometrically nonlinear theory.

Additional information on laminate stacking sequence effects is found in Section 4.6.5.

4.3.5 Laminate stress analysis. The physical properties defined in Section 4.3.3 enable any laminate to be represented by an equivalent homogeneous anisotropic plate or shell element for structural analysis. The results of such analyses will be the definition of stress resultants, bending moments, temperature, and moisture content at any point on the surface which defines the plate. With this definition of the local values of state variables, a laminate analysis can be performed to determine the state of stress in each lamina to assess margins for each critical design condition.

4.3.5.1 *Stresses due to mechanical loads.* To determine stresses in the individual plies, the laminate mid-plane strain and curvature vectors are used. Writing the laminate constitutive relations

$$\begin{Bmatrix} N \\ \hline M \end{Bmatrix} = \begin{bmatrix} A & B \\ \hline B & D \end{bmatrix} \begin{Bmatrix} \epsilon^\circ \\ \hline \kappa \end{Bmatrix} \qquad 4.3.5.1(a)$$

a simple inversion will yield the required relations for $\{\epsilon^\circ\}$ and $\{\kappa\}$. Thus

$$\begin{Bmatrix} \epsilon^\circ \\ \hline \kappa \end{Bmatrix} = \begin{bmatrix} A & B \\ \hline B & D \end{bmatrix}^{-1} \begin{Bmatrix} N \\ \hline M \end{Bmatrix} \qquad 4.3.5.1(b)$$

Given the strain and curvature vectors, the total strain in the laminate can be written as

$$\{\epsilon_x\} = \{\epsilon^\circ\} + z\{\kappa\} \qquad 4.3.2(d)$$

The strains at any point through the laminate thickness are now given as the superposition of the mid-plane strains and the curvatures multiplied by the distance from the mid-plane. The strain field at the center of ply i in a laminate is

$$\{\epsilon_x\}^i = \{\epsilon^\circ\} + \frac{1}{2}\{\kappa\}(z^i + z^{i-1}) \qquad 4.3.5.1(c)$$

where the term

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$$\frac{1}{2}(z^{l} + z^{l-1})$$

corresponds to the distance from the mid-plane to the center of ply i. It is possible to define curvature induced strains at a point through the laminate thickness simply by specifying the distance from the mid-plane to the point in question.

The strains defined in Equation 4.3.5.1(c) correspond to the arbitrary laminate coordinate system. These strains can be transformed into the principal material coordinates for this ply using the *transformations developed previously* (Equation 4.3.1(m)). Thus

$$\{e_t\}^{l} = [\theta^{l}]^{-1}\{e_x\}^{l} \qquad 4.3.5.1(d)$$

where the superscript i indicates which layer and, therefore, which angle of orientation to use.

With the strains in the principal material coordinates defined, stresses in the same coordinates are written by using the lamina reduced stiffness matrix (Equation 4.3.1(h)).

$$\{\sigma_t\}^{l} = [Q^{l}]\{e_t\}^{l} \qquad 4.3.5.1(e)$$

Again, the stiffness matrix used must correspond to the correct ply, as each ply may be a different material.

The stresses in the principal material coordinates can be determined without the use of principal material strains. Using the strains defined in the laminate coordinates (Equation 4.3.5.1(c)) and the transformed lamina stiffness matrix (Equations 4.3.1(o,q,r)), stresses in the laminate coordinate system can be written as

$$\{\sigma_x\}^{l} = [\bar{Q}^{l}]\{e_x\}^{l} \qquad 4.3.5.1(f)$$

and these stresses are then transformed to the principal material coordinates using the relations 4.3.1(m). Thus

$$\{\sigma_t\}^{l} = [\theta^{l}]^{-1}\{\sigma_x\}^{l} \qquad 4.3.5.1(g)$$

By reviewing these relations, it can be seen that, for the case of symmetric laminates and membrane loading, the curvature vector is zero. This implies that the laminate coordinate strains are identical in each ply and equal to the mid-plane strains. The differing angular orientation of the various plies will promote different stress and strain fields in the principal material coordinates of each ply.

4.3.5.2 *Stresses due to temperature and moisture.* In Section 4.3.3.3, equations for the thermoelastic response of composite laminates were developed. It was indicated that thermal loading in laminates can cause stresses even when the laminate is allowed to expand freely. The stresses are induced because of a mismatch in thermal expansion coefficients between plies oriented in different directions. Either the mechanical stresses of the preceding section or the thermomechanical stresses can be used to evaluate laminate strength.

To determine the magnitude of thermally induced stresses, the thermoelastic constitutive relations (Equations 4.3.3.3(f) - (i)) are required. Noting that free thermal stress effects require that $\{N\} = \{M\} = 0$, these relations are written as

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$$\begin{Bmatrix} 0 \\ --- \\ 0 \end{Bmatrix} = \begin{bmatrix} A & | & B \\ --- & | & --- \\ B & | & D \end{bmatrix} \begin{Bmatrix} e^{\circ} \\ --- \\ \kappa \end{Bmatrix} + \begin{Bmatrix} N^T \\ --- \\ M^T \end{Bmatrix} \qquad 4.3.5.2(a)$$

Inverting these relations yields the free thermal strain and curvature vectors for the laminate. Proceeding as before, the strain field in any ply is written as

$$\{e_x\}^l = \{e^{\circ}\} + \frac{1}{2}(z + z^{l-1})\{\kappa\} \qquad 4.3.5.2(b)$$

Stresses in the laminate coordinates are

$$\{\sigma_x\}^l = [\bar{Q}^l]\{e_x\}^l - \{\Gamma_x\}^l \Delta T^l \qquad 4.3.5.2(c)$$

which can then be transformed to the principal material coordinates. Thus

$$\{\sigma_t\} = [\theta^l]^{-1}\{\sigma_x\}^l \qquad 4.3.5.2(d)$$

The stresses can also be found by transforming the strains directly to principal material coordinates and then finding the principal material coordinate stresses.

For uniform temperature fields in symmetric laminates, the coupling matrix, [B], and the thermal moment resultant vector, $\{M^T\}$, vanish and:

$$\{e^{\circ}\} = \{\alpha_x\}\Delta T \qquad 4.3.5.2(e)$$

and

$$\{\kappa\} = 0 \qquad 4.3.5.2(f)$$

In this case, the strains in the laminate coordinates are identical in each ply with the value

$$\{e_x\}^l = \{e^{\circ}\} = \{\alpha_x\}\Delta T \qquad 4.3.5.2(g)$$

and the stresses in the principal material coordinates are

$$\{\sigma_x\}^l = [\bar{Q}^l]\left(\{\alpha\} - \{\alpha_x\}^l\right)\Delta T \qquad 4.3.5.2(h)$$

These relations indicate that the stresses induced by the free thermal expansion of a laminate are related to the differences between the laminate and ply thermal expansion vectors. Therefore, the stresses are proportional to the difference between the amount the ply would freely expand and the amount the laminate will allow it to expand.

A further simplification can be found if the laminate under investigation is isotropic in thermal expansion. It can be shown that, for this class of laminates subjected to a uniform temperature change, the stresses in the principal material coordinates are identical in every ply. The stress vector is

$$\{\sigma_t\} = \frac{E_{11}(\alpha_{22} - \alpha_{11})\Delta T}{1 + 2\nu_{12} + \frac{E_{11}}{E_{22}}} \begin{Bmatrix} 1 \\ -1 \\ 0 \end{Bmatrix} \qquad 4.3.5.2(i)$$

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where it can be seen that the transverse direction stress is equal and opposite to the fiber direction stress.

Similar developments can be generated for moisture-induced stresses. All of the results of this section apply when moisture swelling coefficients, $\{\beta_i\}$, are substituted for thermal expansion coefficients, $\{\alpha_i\}$.

4.3.5.3 *Netting analysis.* Another approach to the calculation of ply stresses is sometimes used for membrane loading of laminates. This procedure is netting analysis and, as the name implies, treats the laminate as a net. All loads are carried in the fibers while the matrix material serves only to hold the geometric position of the fibers.

Since only fibers are assumed to load in this model, stress-strain relations in the principal material directions can be written as

$$\sigma_{11} = E_1 \epsilon_{11} \qquad 4.3.5.3(a)$$

or

$$\epsilon_{11} = \frac{1}{E_1} \sigma_{11} \qquad 4.3.5.3(b)$$

and

$$E_2 = G_{12} = \sigma_{22} = \sigma_{12} = 0 \qquad 4.3.5.3(c)$$

The laminate stiffnesses predicted with a netting analysis will be smaller than those predicted using lamination theory, due to the exclusion of the transverse and shear stiffnesses. This effect is demonstrated in Table 4.3.5.3 for a quasi-isotropic laminate comprised of high-modulus graphite/epoxy. The stiffness properties predicted using a netting analysis are approximately 10% smaller than lamination theory predictions. Experimental work has consistently shown that lamination theory predictions are more realistic than netting analysis predictions.

Although the stiffness predictions using netting analysis are of limited value, the analysis can be used as an approximation of the response of a composite with matrix damage. It may be considered as a worst case analysis and is frequently used to predict ultimate strengths of composite laminates.

4.3.5.4 *Interlaminar stresses.* The analytical procedures which have been developed can be used to predict stresses within each lamina of a laminate. The stresses predicted are planar due to the

TABLE 4.3.5.3 *Laminate elastic constants.*

Analysis	E_x psi	E_y psi	G_{xy} psi	ν_{xy}
Lamination Theory	9.42×10^6	9.42×10^6	3.55×10^6	0.325
Netting Analysis	8.33×10^6	8.33×10^6	3.13×10^6	0.333

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assumed state of plane stress. There are cases where the assumption of plane stress is not valid and a three-dimensional stress analysis is required.

An example of such a case exists at certain free edges in laminates where stress free boundary conditions must be imposed.

4.3.5.5 *Nonlinear stress analysis.* All the preceding material in this chapter has related to laminae which behave in a linear elastic fashion. Composites can behave in a non-linear manner due to internal damage or non-linear behavior of the matrix material. Matrix nonlinearity or micro-cracking can result in laminae which have non-linear stress-strain curves for transverse stress or axial shear stress. When this situation exists, the elastic laminate stress analysis of Section 4.3.5.1 must be replaced by a non-linear analysis. A convenient procedure for the non-linear analysis is presented in Reference 4.3.5.5.

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4.4 LAMINATE STRENGTH AND FAILURE. Methods of stress analysis of laminates subjected to mechanical loads, temperature changes, and moisture absorption were presented in Section 4.3.5. Such a stress analysis is required to determine or assess the strength of a laminate. Several modes of failure are possible in initial and final failure. For composite materials, the failure mode as well as the failure stress or strain must be predicted.

Ply level stresses are the commonly used approach to laminate strength. The average stresses in a given ply are used to calculate first ply failure and then subsequent ply failure leading to laminate failure. Two factors should be considered in assessing the validity of this method. The first is the question of which tests (or analyses) should be used to define the ply strength values. In particular, it must be recognized that a crack parallel to the fibers can result in failure of a transverse tensile test specimen, while the same crack may have an insignificant effect on a laminate test. The second factor is the assumption that local failures within a ply are contained within the ply. There is evidence that this assumption is not valid under fatigue loading, where a crack within one ply may well propagate into adjacent plies. In this case, the ply-by-ply model may not be the best analytical approach.

4.4.1 Initial failure. Consider stresses remote from the edges in a laminate which is loaded by in-plane forces and/or bending moments. If there is no external bending, if the membrane forces are constant along the edges, and if the laminate is balanced and symmetric, the stresses in the ith layer are constant and planar. With reference to the material axes of the laminae, fiber direction x_1 and transverse direction x_2, the stresses are written σ^l_{11}, σ^l_{22}, and σ^l_{12}. Failure will occur when one of the failure criteria in Equations 4.2.4(f) - (i) is satisfied.

Fiber modes

Tensile

$$\left(\frac{\sigma_{11}}{F_1^{tu}}\right)^2 + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(f)$$

Compressive

$$\left(\frac{\sigma_{11}}{F_1^{cu}}\right)^2 + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(g)$$

Matrix modes

Tensile

$$\left(\frac{\sigma_{22}}{F_1^{tu}}\right)^2 + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(h)$$

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Compressive

$$\left(\frac{\sigma_{22}}{2F_{23}^{su}}\right)^2 + \left[\left(\frac{F_2^{cu}}{2F_{23}^{su}}\right)^2 - 1\right]\left(\frac{\sigma_{22}}{F_2^{cu}}\right) + \left(\frac{\sigma_{12}}{F_{12}^{su}}\right)^2 = 1 \qquad 4.2.4(i)$$

The failure criteria must be used for each layer of the laminate. The layer for which the criteria are satisfied for the lowest external load will define the load which produces the initial laminate failure. The layer which fails and the nature of the failure (i.e. fiber failure or cracking along the fibers) are identified. This is called first-ply failure.

Bending occurs when there are external bending and/or twisting moments or when the laminate is not symmetric. In these cases the stresses σ_{11}^l, σ_{22}^l, and σ_{12}^l in a layer are symmetric in x_3. Consequently, the stresses assume their maximum and minimum values at the layer interfaces. The failure criteria must be examined at these locations for each layer.

The evaluation of failure as a result of the edge stresses is much more complicated as a result of the singularities in these stresses. Numerical methods cannot uncover the nature of the stress singularity but there are analytical treatments (e.g., Reference 4.4.1) which can. The implication of such edge stress singularities for failure of the laminate is difficult to assess. This situation is reminiscent of fracture mechanics in the sense that stresses at a crack tip are theoretically infinite. Fracture mechanics copes with this difficulty with a criterion for crack propagation based on the amount of energy required to open a crack, or equivalently, the value of the stress intensity factor. Similar considerations may apply for laminate edge singularities. This situation is more complicated since a crack initiating at the edge will propagate between anisotropic layers. It appears, therefore, that at the present time the problem of edge failure must be relegated to experimentation.

4.4.2 Subsequent failures. Often laminates have substantial strength remaining after the first ply has failed. A conservative approach for analyzing subsequent failure is to assume that the contribution of that first failed ply is reduced to zero. If failure occurs in the fiber mode, the stiffness in the fiber direction E_L is reduced to zero. If failure occurs in the matrix mode, the elastic properties E_T and G_L are reduced to zero. Generally, the progressive failures of interest are initial and subsequent failures in the matrix mode. In that case, the basic assumptions for netting analysis result where the ultimate load is defined by E_T and G_L vanishing in all laminae. The basic issues involved in modeling post-first-ply behavior are described in Reference 4.4.2.

4.4.3 Strength. In composites, there are two characteristic stress levels to be considered in the evaluation of strength. The first is stress at which first-ply failure occurs and the second is the maximum static stress which the laminate can carry. In the calculation of first-ply failure, consideration must be given to residual thermal stresses. The rationale for including residual thermal stresses in the analysis is obvious. The stresses exist after processing. Therefore, they can be expected to influence the occurrence of first-ply failure. However, matrix materials exhibit viscoelastic, or time-dependent, effects, and it may be assumed that a significant portion of the residual stresses will dissipate through a process of stress relaxation. Additionally, the processing stresses may be reduced through the introduction of transverse matrix microcracking. The question of whether to include residual stresses in the analysis is complicated by difficulties in measuring these stresses in a laminate and in observing first-ply failure during a laminate test.

It is common practice, in the aerospace industry, to neglect the residual thermal stresses in the calculation of ply failure. Data to support this approach do not appear to be available. However, at

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the present time, damage tolerance requirements limit allowable strain levels in laminates to 3000 to 4000 μin/in. This criterion becomes the dominant design restriction and obviates the need to resolve the effects of residual thermal stresses.

When laminae properties are known, it is possible to calculate a laminate stress-strain curve and determine first-ply failure, subsequent ply failures, and maximum stress at failure. It is also possible to measure a laminate stress-strain curve experimentally and determine proportional limit and maximum stress. Presently, the best procedure for design is to use analytical strength predictions in the preliminary design phase and experimental strength values for the final design.

When experimental strength values are used for laminates, the problem of obtaining values for combined loading conditions again arises. The difficulties of combined load testing preclude the experimental approach, and the uncertainties of failure mode cloud the choice of analytical interaction criteria. A reasonable approach is represented by the following sequence:

(1) analytical calculation of ultimate strength by netting analysis or any other straightforward method,

(2) use of this analytical results to define the shape of the interaction curve,

(3) use of the experimental data to define the amplitude of the interaction curve, and

(4) confirmation by limited combined stress testing for a condition of practical interest.

Design charts in the form of "carpet plots" are valuable for selection of the appropriate laminate. Figure 4.4.3 presents a representative carpet plot for the axial tensile strength of laminates having various proportions of plies oriented at 0°, ±45, and 90°. Appropriate strength data suitable for preliminary design can be found for various materials in References 4.4.3(a) and (b).

The development of laminate stacking sequence (LSS) optimization routines for strength-critical designs is a difficult task. Such a scheme must account for competing failure mechanisms that depend on material, load type (e.g., tension versus compression), environment (e.g., temperature and moisture content) and history (e.g., fatigue and creep). In addition, the load transfer must be adequately modeled to account for component geometry and edge effects. Even for a simple uniaxial load condition, the relationship between LSS and strength can be complex. Some qualitative rules currently exist for optimizing LSS for strength but they have been developed for a limited number of materials and load cases.

Relationships between LSS and laminate strength depend on several considerations. The initiation and growth of local matrix failures are known to depend on LSS. As these failures occur, internal stress distributions also depend on LSS through local stiffness and dimensional stability considerations. For example, delamination divides a base laminate into sublaminates having LSS that are generally unsymmetric. Reduced stiffness due to edge delaminations, causes load redistribution and can decrease the effective tensile strength of laminates. Likewise, local instability of sublaminates also causes load redistribution which can lower the effective compressive strength of laminates. As a result, both laminate and sublaminate LSS affect laminate strength.

Shear stress distributions play a significant role in determining the mechanical behavior and response of multi-directional laminates.As was the case for ply transverse tensile strength, ply shear strength depends on LSS. Laminates with homogeneous LSS have been found to yield higher in-situ ply shear strengths than those with ply orientations clumped in groups (Reference 4.4.3(b)). An

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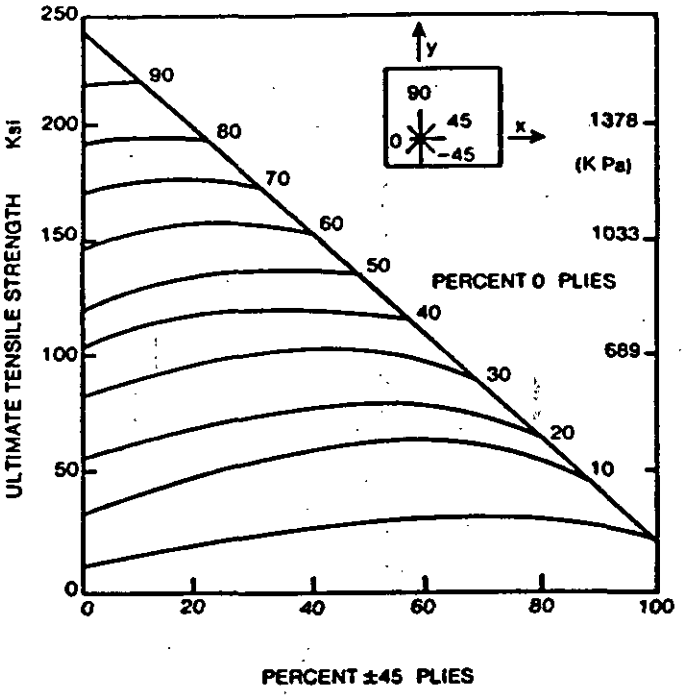


Figure 4.4.3 *Tensile strength of $[0_i/\pm 45_j/90_k]_s$ family of high strength graphite/epoxy laminates (Reference 4.4.3).*

inherent flaw density and interlaminar stresses appear to be major factors affecting the distribution of ply shear strengths in a LSS.

As was the case for bending stiffness, bending strength in composite laminates is strongly dependent on LSS. Failure mechanisms characteristic of tension, shear, and compression load conditions may all combine to affect bending strength. Table 4.3.3.2(b) showed that preferential stacking of plies in outer layers of the LSS increased bending stiffness. The bending strength performance of undamaged laminates may show similar trends; however, surface damage due to impact or other in-service phenomena would cause severe degradation to such laminates.

Additional information on laminate stacking sequence effects is found in Section 4.6.5.

4.4.4 Stress Concentrations. The presence of a hole or other discontinuity in a structure introduces local stress concentrations. These high local stresses can result in initial localized failure. The analysis of failure due to cracking, or fracture, which can result in this situation is complicated for composite materials because of material heterogeneity at the microscale and in a layer-to-layer basis. Effective in-plane laminate stiffnesses, E_x, E_y, and G_{xy}, may be calculated for any laminate by using the methods presented in Section 4.3.3. With these properties specified, a balanced symmetric laminate may be regarded as a homogeneous orthotropic plate, for structural analysis. Orthotropic elasticity theory may be used for the evaluation of stresses around a hole in such a plate (Reference 4.4.4(a)). Examples of the resulting stress concentrations are shown in Figure 4.4.4(a) for carbon/epoxy laminates. The laminae orientation combinations influence both the magnitude and the shape of the stress variation near the hole. The high stresses at the edge of the hole may initiate fracture.

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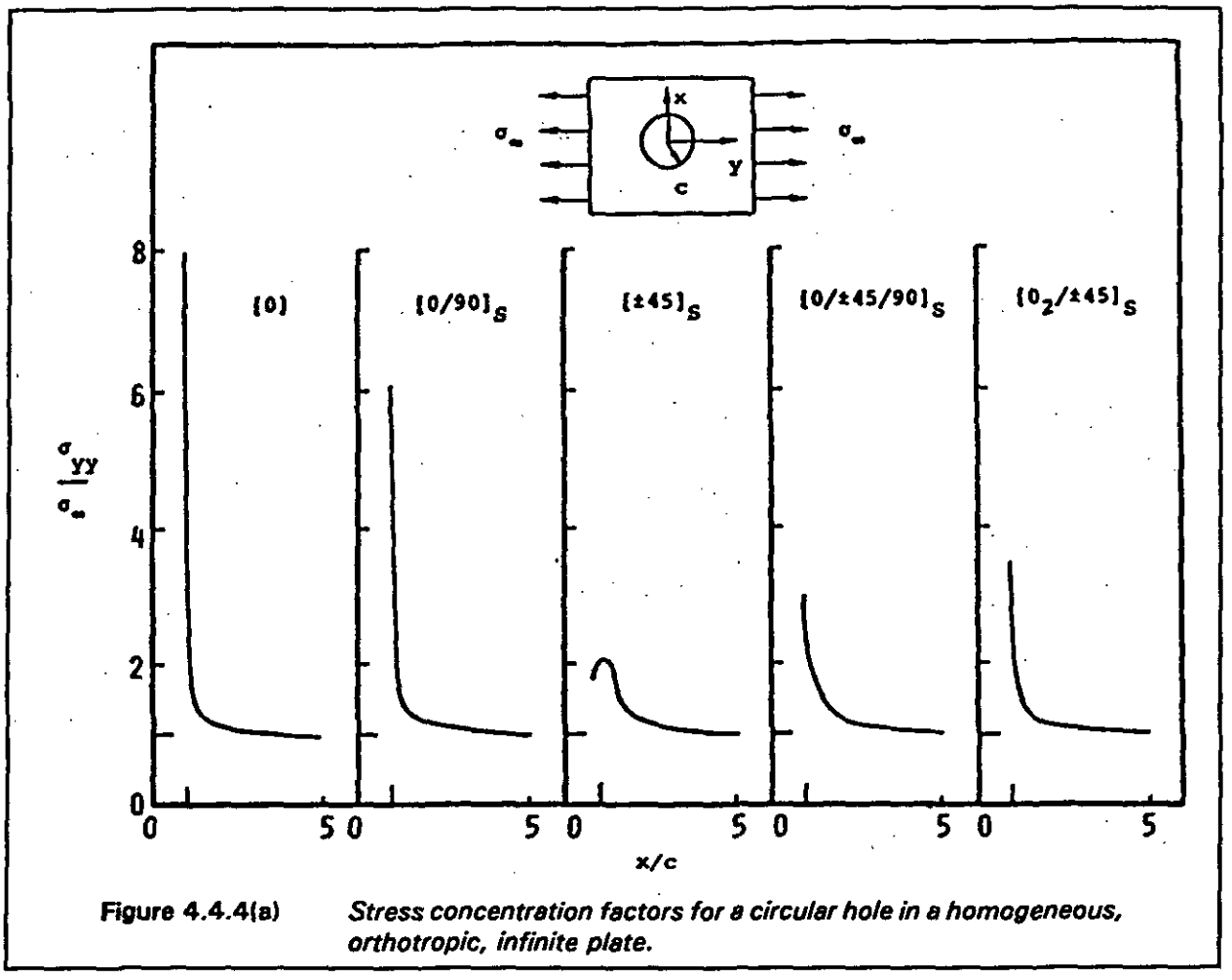


Figure 4.4.4(a) *Stress concentration factors for a circular hole in a homogeneous, orthotropic, infinite plate.*

If the laminate fails as a brittle material, fracture will be initiated when the maximum tensile stress at the edge of the hole equals the strength of the unnotched material. In a tensile coupon with a hole, as shown in Figure 4.4.4(a), failure will occur at the minimum cross-section. The failure will initiate at the edge of the hole, where the stress concentration is a maximum.

Consider the stress concentration factor in a finite width isotropic plate with a central circular hole. Stress distribution for this configuration are shown in Figure 4.4.4(b) for various ratios of hole diameter, a, to plate width, W. The basic stress concentration factor for this problem is the ratio of the axial stress at the edge of the hole ($x = a/2$, $y = 0$) to the applied axial stress, σ_∞. For small holes in an isotropic plate, this factor is three. The average stress at the minimum section, σ_n, is higher than the applied stress, σ_∞, and is given by the following relationship:

$$\sigma_n = \frac{\sigma_\infty}{(1 - \frac{a}{W})} \qquad 4.4.4(a)$$

The net section stress concentration factor, k_n, is the ratio of the maximum stress to this average stress.

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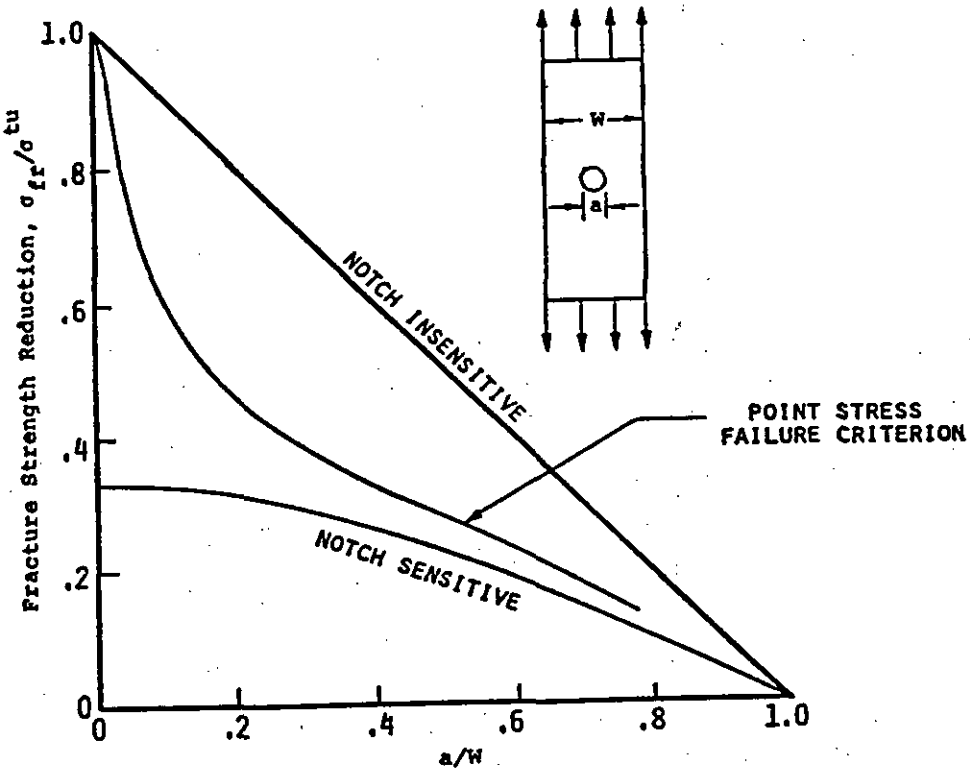


Figure 4.4.4(b) *Approximate failure theories.*

$$k_n = \frac{\sigma(\frac{a}{2},0)}{(1-\frac{a}{W})\sigma_\infty} \qquad 4.4.4(b)$$

Laminate fracture for the elastic-brittle case will occur at stress σ_{fr}:

$$\sigma_{fr} = F^{tu}/k_n \qquad 4.4.4(c)$$

A material which fails in this fashion is denoted a notch-sensitive material. In contrast, a ductile, or notch-insensitive, material will yield locally to alleviate the stress concentration effect.

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Various matrix damage effects are expected to occur at the maximum stress locations. This localized damage reduces the material stiffness and diminishes and spreads the stress concentration effects. Semi-empirical methods have been proposed to account for this reduction in the stress concentration.

The "point stress theory" (Reference 4.4.4(b)) proposes that the elastic stress distribution curve, e.g., Figure 4.4.4(a), be used, but that the stress concentration be evaluated at a distance, d_o, from the edge of the hole. The numerator of Equation 4.4.4(b) is evaluated at $x = a/2 + d_o$. The characteristic length, d_o, must be evaluated experimentally. The "average stress theory" (Reference 4.4.4(b)) takes a similar approach by proposing that the elastic stress distribution be averaged over a distance, a_o, to obtain the stress concentration.

$$k_n = \frac{\int_{a/2}^{(a/2)+a_o} \sigma_y dx}{\int_{a/2}^{W/2} \sigma_y dx} \qquad 4.4.4(d)$$

Again, the characteristic dimension, a, must be found experimentally. For both methods, the resulting stress concentration is used in Equation 4.4.4(c) to define the fracture stress. Representative results are plotted in Figure 4.4.4(b) to illustrate the differences associated with different types of material behavior.

The relationship between tensile strength and laminate stacking sequence (LSS) for laminates with holes, cutouts, and through-penetrations (i.e., a damage tolerance consideration) is complex (see Reference 4.4.4(c) - (g)). Certain combinations of ply splitting and delamination that occur at the tip of a notch can enhance residual strength by effectively reducing the stress concentration. Delaminations which uncouple plies, allowing individual plies to fail without fiber breaks, reduce the residual strength. Most existing analysis methods for predicting notched tensile strength are based on parameters determined by some notched laminate tests (e.g., characteristic dimension, fracture energy parameter). The effects of LSS on failure is included in the test parameter. Future analysis development that simulates progressive damage accumulation will provide a more efficient approach for studying the effects of LSS. Additional information on laminate stacking sequence effects is found in Section 4.6.5.

4.4.5 Delamination. The formation and growth of delaminations is generally related to LSS. Delaminations can have varying effects on tensile strength performance, depending on delamination location and the specific property of interest. Most studies performed to date have considered specimens with significant free edge surface area where interlaminar stresses are known to concentrate. Although all structures have some free edges, it is important to realize the limits of analysis and tests performed with specimen geometries. For example, the magnitude of interlaminar tensile stresses, which are crucial to edge delamination, approach zero for plate width to thickness ratios of 30 and greater (Ref. 4.4.5(a)).

As shown in Figure 4.4.5, laminated specimens prone to edge delamination have been shown to exhibit generally lower strength (ultimate stress level) when loaded in uniaxial tension (e.g., References 4.6.5.5.1.1(b), 4.4.5(b) - (f)). The reduction in strength has been directly tied to a drop in stiffness with increased edge delamination area for laminates exhibiting stable delamination growth (References 4.6.5.5.1(b), 4.4.5 (b) - (e)). The onset of edge delamination has been shown to relate to tensile strength for laminates exhibiting unstable delamination growth coupled with matrix cracks (Reference 4.4.5(f)).

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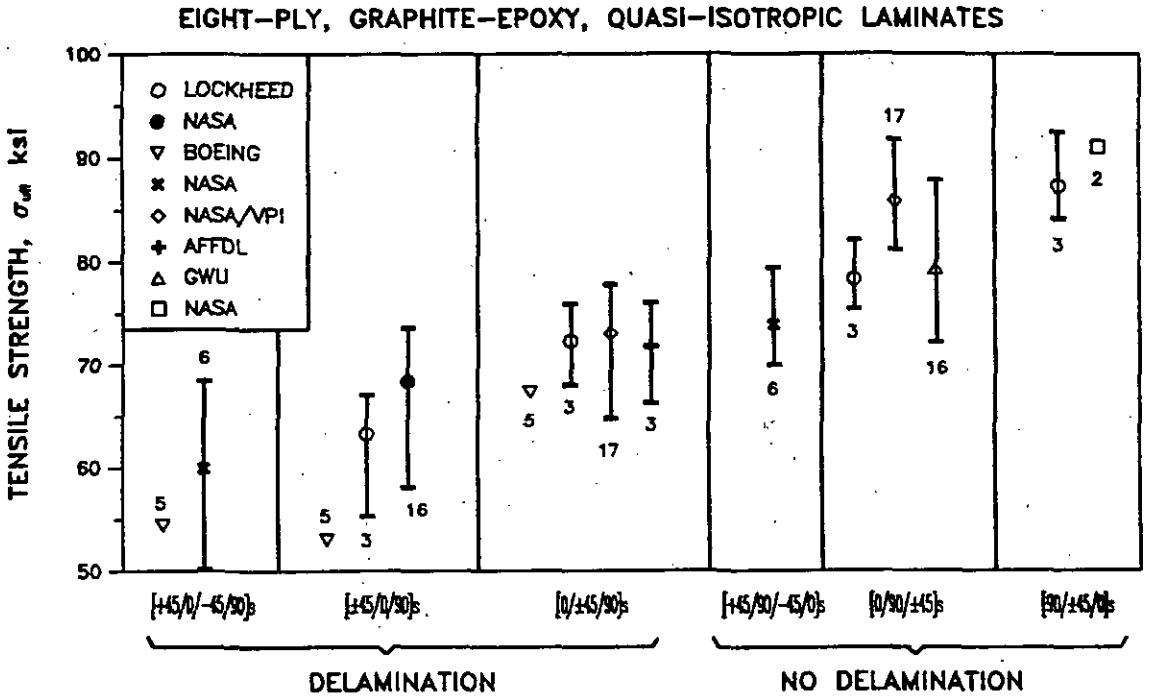


FIGURE 4.4.5 *Unnotched tensile strength variation with LSS (from Reference 4.4.5(d)).*

The reduced laminate stiffness due to edge delamination can affect the measured tensile strength in two distinct ways (e.g., Reference 4.4.5(e)). If all plies remain loaded after delamination, the ultimate laminate strain has been found to equal the critical strain of primary load bearing plies. In these cases laminate strength drops in proportion to the apparent axial modulus. However, if off-axis plies cease to carry loads because they have been isolated by an interconnected network of matrix cracks and delamination, a local strain concentration can form. When this occurs, the global laminate strain for failure can be less than the critical strain of primary load bearing plies.

Free edge delaminations split a laminate into sublaminates, each of which continue to carry tensile loads. The apparent modulus of this laminate depends on delamination length and the sublaminate moduli which may be calculated using lamination theory. These moduli will depend on LSS if unsymmetric sublaminates with strong extension/bending couplings are involved (References 4.4.5(g) and (h)). A simple rule-of-mixtures approach has been used to accurately calculate apparent moduli for edge delamination (References 4.4.5(e), (g) and (h)).

Local coupling between intralaminar matrix cracks and delaminations can cause complete or partial ply isolation. Note that complete ply isolation cannot occur unless associated damage extends the full laminate width. When this occurs, the apparent laminate stiffness and strain concentration can be calculated in a modified rule-of-mixtures approach which discounts isolated ply groups (References 4.6.5.5.1.1(c) and 4.4.5(e)). A local area of reduced stiffness also causes strain concentration (Reference 4.4.5(i)). The strain concentration depends on both the local reduced stiffness and global laminate stiffness. For example, hard laminates with strong anisotropy, such as lay-ups dominated by 0° plies and loaded uniaxially, will have large strain concentration factors. Consequently, hard laminates will be less tolerant of local damage than relatively soft laminates (e.g., quasi-isotropic).

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When high interlaminar shear stresses are present, coupled edge delamination and matrix crack growth are possible and may lead to catastrophic failure. Edge delamination behavior of laminates commonly used in design (e.g., quasi-isotropic laminates) become dominated by interlaminar shear stresses when subjected to off-axis loading. Note that for this problem the laminate lay-up is generally unbalanced relative to the loading axis. The measured tensile strength coincides with the onset of edge delamination for such laminates (Reference 4.4.5(f)). As a result, failure criteria that account for interlaminar stresses are needed to predict the tensile strength.

The use of a suitable analysis method is recommended to evaluate edge effects in composite materials (e.g., References 4.4.6.1.1(f), 4.4.5(d), (g), (h), (j) - (l)). Applied mechanical loads and environmental effects should be included in the free edge analysis. Two approaches have been successfully applied to quantify free edge stresses and predict edge delamination: (1) a fracture mechanics based method using strain energy release rates (References 4.6.5.5.1.1(f), 4.4.5(d), (g), (h), (j)), and (2) a strength of materials based approach using an average stress failure criterion (References 4.4.5(k) and (l)).

The combined use of resin interlayers between the plies in a laminate and specimen edge polishing have been found to be effective methods for suppressing edge delamination (References 4.4.5(f)). Materials with high interlaminar toughness have an inherent resistance to delamination. Other methods that have been used to suppress edge delamination include resin interlayer strips at critical interfaces along the edge of laminates (References 4.4.5(m)), termination of critical plies offset from the edge (Reference 4.4.5(n)), hybridization (References 4.4.5(o) and (p)), and serrated edges (Reference 4.4.5(p)).

Most of the above discussion on the effects of delamination suggest a decrease in tensile properties. This is generally true for unnotched specimen geometries prone to edge delamination. Isolated delaminations that occur away from the edge of a laminate (e.g., manufacturing defects) and are not coupled with matrix cracks have been shown to have little effects on tensile strength (Reference 4.4.5(r)). Theoretically, such delaminations do not result in local reduced laminate stiffness when loaded in tension due to compatibility considerations. Multiple delaminations located away from the edge of a laminate have been shown to cause a small reduction in tensile strength (Reference 4.4.5(r)). This was explained by coupling between delaminations and other matrix damage (eg., ply splits) that occurred during loading, resulting in partial ply isolation and local reduced stiffness. Most of the discussion in this section is related to free edge effects (Section 4.6.3) and laminate stacking sequence effects (Section 4.6.5).

4.4.5.1 *Compression.* Delaminations generally have a stronger affect on compressive strength than on tensile strength. As a result, the potential for delamination should always be considered when selecting a suitable LSS. The effect of delamination occurring due to manufacturing defects and/or in-service events such as impact needs to be included in this evaluation. For example, the best LSS for avoiding edge delamination in specimen geometries may not be best for suppressing the effects of delaminations occurring in structures due to impact.

Delamination breaks the laminate into sublaminates, each having associated stiffness, stability, and strength characteristics. Sublaminates are usually unsymmetric and, therefore, all of the sublaminate stiffnesses will depend on LSS. As shown in Figure 4.4.6.2, stability and local compressive performance of sublaminate ply groups ultimately determines catastrophic failure.

Compressive failure in composite laminates having delaminations is strongly tied to the stability of sublaminate plates. Since delaminations may occur at many different interfaces in a laminate, sublaminates LSS will generally not be balanced and symmetric. As discussed earlier, the

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bending/extension couplings characteristic of such LSS reduce buckling loads. The sublaminate boundary conditions and shape are also crucial to the relationship between LSS and stability.

Several methods exist for predicting sublaminate stability in composite laminates (e.g., References 4.4.5.1(a) - (e)). These models differ in assumed bending stiffness, boundary conditions, and sublaminate shape. Experimental data bases are needed to determine which assumption is appropriate for a given problem. The in-plane and out-of-plane stress redistribution due to a buckled sublaminate is crucial to compressive strength.

Environment can play a significant role in delamination growth and load redistribution if the environmental resistance of combined sublaminate stiffnesses are significantly different than those of the base laminate. The combined effects of environment and LSS on laminate dimensional stability were covered in earlier sections. The stability of unsymmetric sublaminates is expected to relate to warpage. The warp depends on both LSS and environmental conditions. Warp may be treated as an imperfection in stability analysis.

The initiation of free edge delamination in compressively loaded laminates can be predicted using methods similar to those used for tension (e.g., Reference 4.4.5.1(a)). Once initiated, delamination growth depends on sublaminate stability. An adequate sublaminate stability analysis model must, therefore, be coupled with the growth model (e.g., References 4.4.5.1(b) and (c)). Delamination growth can be stable or unstable, depending on sublaminate LSS, delamination geometry, structural geometry, and boundary conditions. Growth of multiple delaminations, characteristic of impact damage, is currently not well understood.

4.4.6 Damage and failure modes.

4.4.6.1 *Tension.* Tensile rupture of laminates with multidirectional plies normally involves a series of pre-catastrophic failure events, including both matrix damage and localized fiber breaks. Catastrophic failure is expected whenever the longitudinal tensile strength of any ply in a laminate is exceeded; however, laminates can separate without fiber failure by coupling various forms of matrix damage. Example laminates that can fail due to matrix damage include those with less than three distinct ply orientations (angle-ply laminates loaded in the 0° direction). Recommendation 2 in Section 4.6.5.2.1 is intended to avoid the low strengths associated with catastrophic failures occurring without fiber breaks.

Figure 4.4.6.1 shows the various failure mechanisms that can occur at micro and lamina dimensional scales for a multidirectional laminate loaded in tension. Depending on load conditions and material properties, matrix failure (e.g., transverse matrix cracks, delamination) or isolated fiber breaks occur at stress levels less than the static strength. Load redistributes around local failures until a critical level of damage is reached, upon which catastrophic fiber failure occurs. Resin is of secondary importance through its effect on resistance to matrix damage accumulation and local load transfer (i.e., near matrix damage and isolated fiber breaks). The LSS also plays a secondary role by affecting damage accumulation and load transfer.

Critical micro failure mechanisms shown in Figure 4.4.6.1 include localized fiber failure and fiber/matrix interfacial cracking. These mechanisms occur mostly in plies aligned with a major axis of tensile stress. The laminate stress levels at which these failures occur depend on load redistribution due to the characteristic damage state in adjacent plies. A limited number of fiber breaks are tolerated within a lamina before the entire ply fails, which can trigger catastrophic laminate failure.

Matrix failure mechanisms at the lamina scale for laminates with multidirectional plies are also shown in Figure 4.4.6.1. Intralaminar matrix cracks align parallel to the fiber direction and span the

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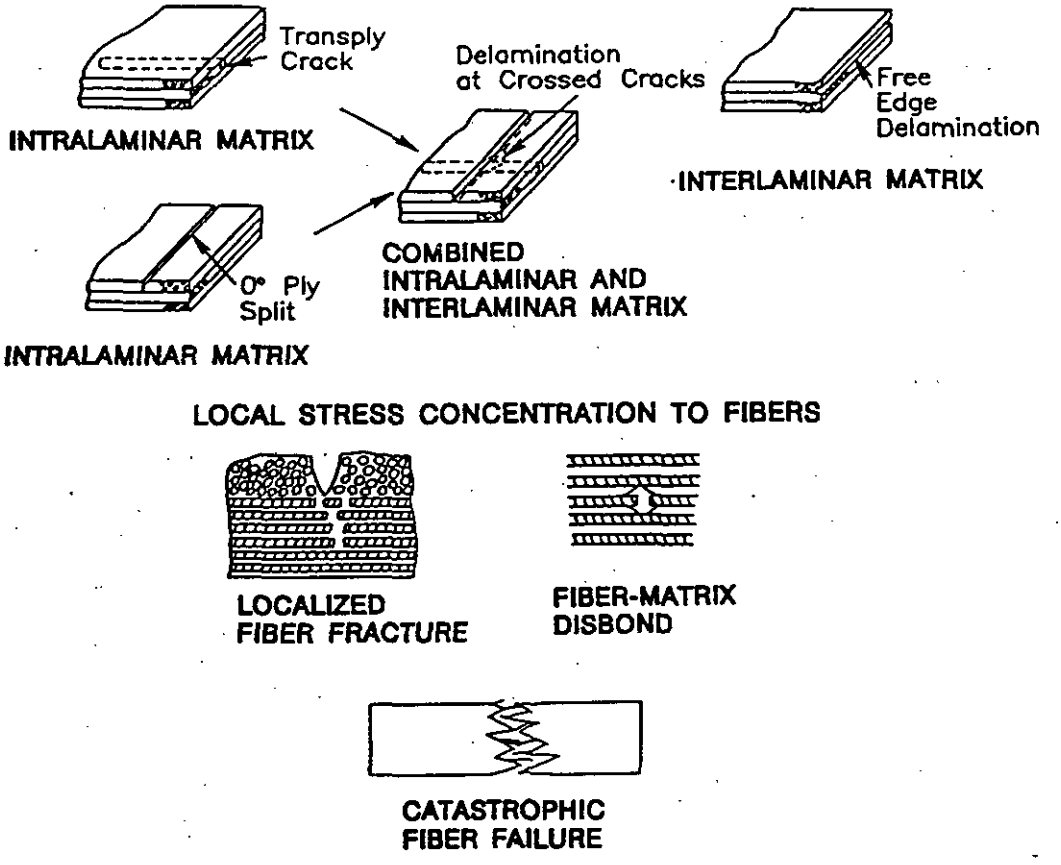


FIGURE 4.4.6.1 *Failure mechanisms for laminates loaded in tension.*

thickness of a ply or group of plies stacked with the same orientation. These have also been referred to as transply cracks or ply splits depending on whether a crack orients at an angle or parallel to the tensile load axis, respectively.

Interlaminar matrix failure, often referred to as delamination, can form near free edges or at intersections between intralaminar cracks. Delaminations form due to excessive interlaminar normal and shear stresses. The accumulation of intralaminar and interlaminar matrix failures depends strongly on LSS.

4.4.6.1.1 *Matrix Cracks.* Matrix cracks occur in plies of laminated composites due to combined mechanical and environmental stresses. These transverse cracks align with fibers and, when fully formed, span the thickness of individual plies or ply groups stacked together in the same orientation.

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Matrix cracks redistribute local stress in multidirectional laminates, allowing a crack density to develop in the ply of ply group as a function of load and environmental history. These cracks can also form prior to service exposure due to processing.

Studies with coupons loaded in uniaxial tension have shown that initial fiber failures found in 0° plies occur near intralaminar matrix cracks in neighboring off-axis plies (Ref. 4.4.6.1.1(a)). When matrix cracks span a single off-axis ply, the stress concentration in a neighboring 0° ply is generally small and localized over a small portion of the neighboring ply thickness. This has been found to influence the location of laminate failure, but has little effect on tensile strength (Refs. 4.4.6.1.1(b) and 4.4.6.1.1(c)).

Intralaminar matrix cracks normally span the full thickness of multiple off-axis plies that have been stacked together. The associated stress concentration in a neighboring ply increases with the thickness of a cracked group of stacked plies. The stress concentration in a 0° ply due to matrix cracks in a large group of stacked 90° plies was found to significantly decrease laminate tensile strength (Refs. 4.4.6.1.1(c) and 4.4.6.1.1(d)). This is one of the reasons for Recommendation 3, Section 4.6.5.2.1.

Even when strength is not altered by the presence of matrix cracks, it is important to understand the mechanics of matrix cracking for composite materials used in aerospace applications. For example, matrix cracks can play a fundamental role in the generation of delaminations. The increased surface area due to a network of matrix cracks can also alter physical properties such as composite thermal expansion, liquid permeability, and oxidative stability.

Residual stresses, that develop due to differences in thermal and moisture expansion properties of constituents, affect the formation of matrix cracks. In general, tensile residual stress develops in the transverse-fiber directions of lamina when multidirectional polymer matrix composites are exposed to temperatures below the residual stress free temperature. This occurs during a temperature drop because unconstrained shrinkage of tape lamina is much greater in transverse-fiber directions than in fiber directions. As moisture is absorbed into a laminate, matrix swelling counteracts thermal shrinkage, decreasing the transverse-fiber tensile stress.

The critical stress or strain causing the onset of matrix cracking in plies of a laminate has been referred to as in situ transverse lamina strength. This strength is not a material constant since it depends on LSS. Experiments and analysis have shown that in situ strength increases as the thickness of plies grouped together with the same orientation decreases (e.g., Refs. 4.4.6.1.1(e) - (i)). These studies have also shown that neighboring plies can impose differing constraints on matrix crack formation, depending on fiber orientation. Many materials currently used in the aerospace industry have resin-rich interlaminar layers (RIL). The magnitude of the in situ strengthening effect decreases if a RIL with significant thickness exists between plies (Ref. 4.4.6.1.1(j)). Relatively soft RIL eliminate some of the constraint imposed by neighboring plies.

4.4.6.2 *Compression.* Compressive strength is ultimately related to the local response of individual ply groups. Assuming no matrix damage exists due to impact or previous load history, the local stability and strength of plies aligned with the axis of loading will determine final failure. The location of load-carrying plies relative to the laminate surface can play a role in this instance. The short wavelength buckling load is reduced when critical plies are located in outer layers of the laminate stacking sequence. When matrix damage does exist, the combined local response of individual ply groups affects the compression strength. The stability and load redistribution within individual ply groups or sublaminates is crucial to the local response.

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Figure 4.4.6.2 shows three different types of local compressive failure mechanisms. These mechanisms were observed to occur as a function of θ for $(\pm\theta)_s$ type laminates (References 4.4.6.2(a) and (b)). When delamination occurs, all three of the local failure modes may combine to determine the compressive strength of a laminate stacking sequence. (Additional information on the effects of the laminate stacking sequence is found in Section 4.6.5.) In-plane matrix shearing and matrix compression failures were observed for $(\pm\theta)_s$ type laminates with $15° \leq \theta \leq 50°$ and $60° \leq \theta \leq 90°$, respectively. The shear mode of fiber microbuckling is most commonly observed for composites. This mode was shown to initiate compressive failure for $(\pm\theta)_s$ type laminates with $0° \leq \theta \leq 10°$. Depending on matrix and fiber combination, final local failures for such laminates involved some combination of fiber failure (shear, kinking, or bending) and matrix splitting or yielding (References 4.4.6.2(c) and (d)).

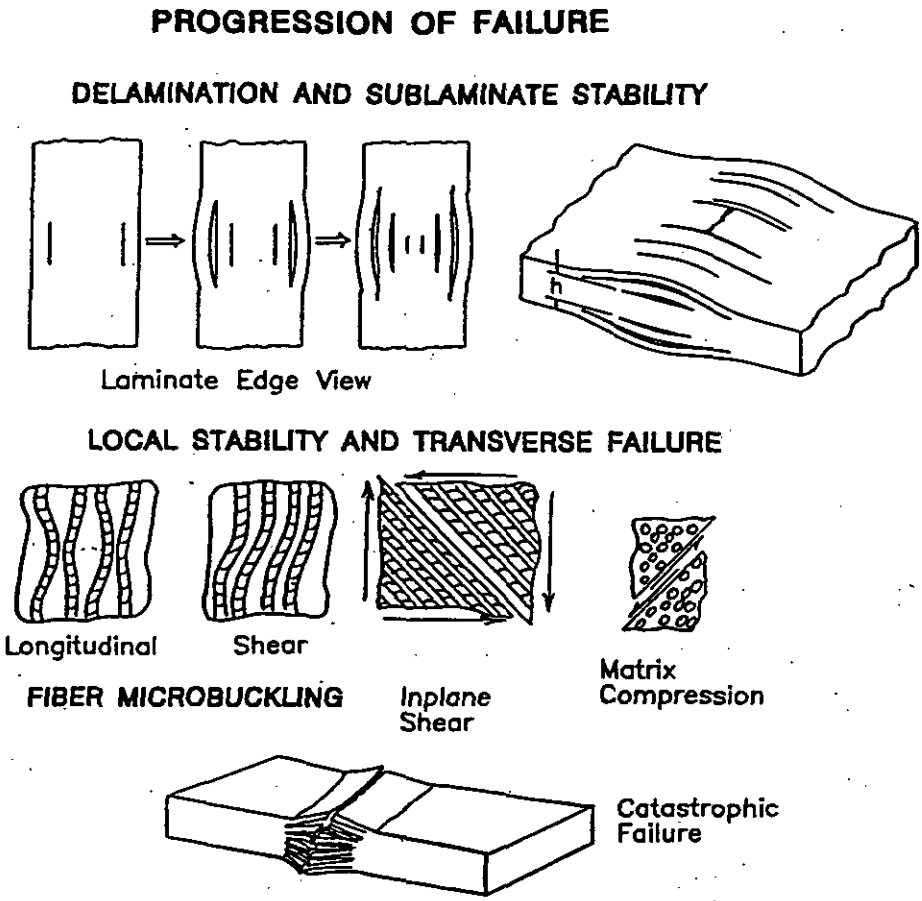


FIGURE 4.4.6.2 *Failure mechanisms for laminates loaded in compression.*

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4.5 COMPLEX LOADS.

4.5.1 Biaxial in-plane loads.

4.5.2 Out-of-plane loads.

4.6 LAMINA TO LAMINATE CONSIDERATIONS.

4.6.1 Residual stresses and strains. Residual curing stresses and strains have virtually no effect on fiber-dominated laminate properties. However, residual stresses in the resin can be greater than the mechanical stresses needed to cause failure. Neglecting these residual stresses therefore may be nonconservative. The residual stresses may be high enough that resin microcracking may occur before any mechanical load is applied. Consequently, the principle of superposition may not be applicable as the mechanical loading may result in nonlinear behavior. As an example, typical epoxy matrix residual strains at the microlevel, resulting from cool down after curing at 350°F, may be approximately 25 to 100% of the laminate failure strain.

4.6.2 Thickness effects. Much of the difference in properties found when comparing laminates with *different thicknesses* can be explained by the residual stresses developed during processing. Internal stresses developed during processing may produce voids, delaminations, and microcracks or cause residual stresses in the laminate that may affect material properties. Excessive porosity, generally caused by poor processing, or environmental effects due to temperature and moisture conditions may also degrade the material and affect its behavior.

Variations in material properties between thick laminate test data from different sources, for laminates having the same thickness, can generally be attributed to differences in processing. Such variations can be minimized by optimizing the cure cycle and by proper process control.

The residual stresses may be caused by non-isothermal conditions present during the solidification phase. Different layers of the laminate will undergo different degrees of volume contraction at any given time during the process cycle. This gives rise to a self-equilibrating force system producing tension stresses in the center and compression stresses in the surface layers of the laminate as reported by Manson and Seferis (Reference 4.6.2(a)). Thickness effects observed in composite laminates are primarily due to this phenomena.

In thermosetting materials, these through-the-thickness stress gradients can be virtually eliminated by modeling the total process, including cool-down, so isothermal conditions are present near the resin gelation point and are maintained for a sufficient period of time. In some high-temperature processing materials where rapid cooling is required, significant thermal stresses may build up in the laminate.

In their work, the authors in Reference 4.6.2(a) present a method to experimentally determine and analyze the internal stresses developed during processing of a composite laminate. This method consists of laying up a certain number of plies, separated by a release ply that can be removed after processing. The internal stresses in the laminate can then be analyzed by considering the deformations of the individual sublaminates.

In summary, variations in material properties in laminated composites are mostly the result of thermally induced residual stresses, although environmental effects and process parameters other than temperature may affect test data. True thickness effects are caused by temperature gradients across the thickness of the laminate. These effects may be minimized by mathematical modeling of the total process and can be virtually eliminated in thermosetting materials. Advance process models such as ROAST, described in Reference 4.6.2(b), may be used to optimize the process parameters.

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4.6.3 Edge effects. Consideration of edge effects in laminated composites is necessary due to behavior not observed in homogeneous solids. A complex stress state exists between the layers of different orientation at the free edge of a laminate, such as along a straight edge or around the perimeter of a hole. Where a fiber in a laminate has been subjected to thermal or mechanical strain, the end of the cut fibers must transfer the load to adjacent fibers. If these adjacent fibers have a different orientation, they will present a locally stiffer path and accept the load. The matrix is the only mechanism for this load transfer. The stresses due to this load, namely interlaminar stresses, can be sufficient to cause local microcracking and edge delamination. These interlaminar stresses, in general, include normal (peel stress σ_z) and shear components (τ_{yz}, τ_{xz}) and are only present in a small region near the free edge. A typical interlaminar stress distribution is shown in Figure 4.6.3. The high

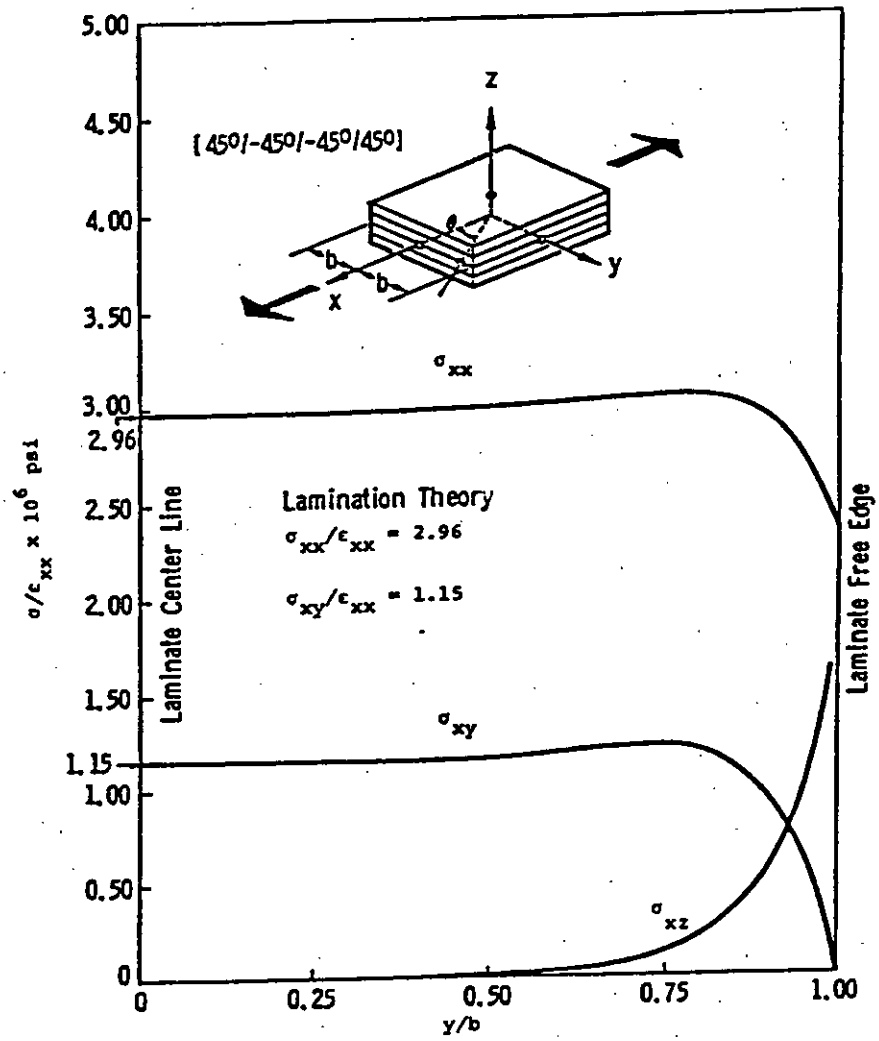


Figure 4.6.3 *Interlaminar stresses normalized with respect to the applied strain (reproduced by permission from Reference 4.6.3(e))*

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gradients of these stresses depend on differences in Poisson's ratio and in-plane shear stiffness that exist between the laminae groups in a laminate. The same kinds of stresses are induced by residual thermal stresses due to cool-down after cure at elevated temperatures.

Failure often occurs as a result of delamination at the locations of high interlaminar stresses because of low interlaminar strength. The effects of free edge stress are sufficient to reduce the strength of certain coupons in both static and fatigue tests significantly. This premature failure makes coupon data difficult to apply to large components because of the local effects of the free edge failure mode. Classical laminate theory which assumes a state of plane stress is incapable of predicting the edge stresses. However, determination of such stresses by higher order plate theory or finite element analysis is practical. Therefore, consideration of edge interlaminar stresses in a laminate design is feasible. The gradients of this stress can be reduced by such measures as 1) changing the laminate stacking order, 2) minimizing the mismatch of the Poisson's ratio, the coefficient of mutual influence, and coefficients of thermal and moisture expansion between adjacent laminae, and 3) by inserting an inner layer which has a lower shear modulus and a finite thickness between laminae, thus allowing greater local strain to occur (Reference 4.6.3(a)).

Edge effects may be analyzed by fracture mechanics, strength of materials, or other methods (References 4.6.3(a) - (d)). These methods can be used to provide a guideline for designers to select the laminate configuration and material system best suited for a particular application.

Very little work has been performed to date on free edge effects for load conditions other than uniaxial tension or compression. Some analysis results indicate that in-plane shear, out-of-plane shear/bending, in-plane bending, twisting moments, and combined loading yield a higher magnitude of interlaminar stress relative to those associated with axial load conditions (Reference 4.6.3(f)). For example, out-of-plane shear due to bending causes free edge interlaminar stresses that are an order of magnitude higher than that caused by axial tension. For more information on delaminations and free edge effects, see Section 4.4.5. Information on the laminate stacking sequence effects is found in Section 4.6.5.

4.6.4 Effects of transverse tensile properties in unidirectional tape. The transverse strength properties play only a minor role in establishing cross-plied laminate strengths. It is, however, well-known that the effective "in-situ" transverse strength of transverse plies is much greater than the strength measured on the lamina. This effect has been handled by post-first ply failure analysis methods.

In-plane shear tests on laminae exhibit relatively high strains to failure (4 -5%). The much lower transverse tensile strains to failure (1/2%) indicate a marked notch sensitivity that is suppressed in cross-plied laminates. The initial cracks that fail laminae are arrested by fibers in other directions; thus laminae with microcracks are still effective. Most laminae develop cracks due to residual thermal stresses and continue to function.

4.6.5 Laminate stacking sequence effects.

4.6.5.1 *Introduction.* Stacking sequence describes the distribution of ply orientations through the laminate thickness. As the number of plies with chosen orientations increase, more stacking sequences are possible. For example, a symmetric 8-ply laminate with four different ply orientations has 24 different stacking sequences. This presents a predicament when attempting to optimize composite performance as a function of stacking sequence.

Laminated composite structural properties such as stiffness, dimensional stability, and strength have all been found to depend on laminate stacking sequence (LSS). Generally, each property has a

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different relationship with LSS. Therefore, the choice of LSS for a particular design application may involve a compromise. Design optimization requires verified analysis methods and an existing materials data base. The development of verified analysis methods for predicting stiffness and stability of laminated composites is more mature than that for predicting strength.

Some simplified design guidelines for LSS are provided in Section 4.6.5.2. These guidelines are generally conservative; however, they limit design optimization, and may even be misleading for some special cases. As a result, a comment on the reason for each guideline is included in the discussion. Verified analysis methods should be used to help judge the effects of LSS whenever possible.

Additional discussion of stacking sequence effects on particular topics are provided in the sections noted in Table 4.6.5.1.

TABLE 4.6.5.1 *Additional discussions on stacking sequence effects.*

Topic	Section	Page
Bending	4.3.3.2,	4-43,
	4.4.3	4-60
Buckling	4.7.1.9	4-94
Compression after impact	4.11.1.4	4-102
Delamination	4.4.5	4-65
Free edge effects	4.6.3	4-74
Hygroscopic analysis	4.3.4	4-53
Lamination theory	4.3.2	4-38
Notched strength	4.4.4	4-65
Ply shear strength	4.4.3	4-61
Thermal analysis	4.3.4	4-53
Vibration	4.12.2	4-111

4.6.5.2 *Design guidelines.* Laminate design starts by selecting the number of plies and ply angles required for a given application. Once the number of plies and ply angles are selected, a LSS is chosen. A LSS is considered heterogeneous when there is preferential stacking of specific ply orientations in different locations through the thickness of the laminate. Thick laminates with heterogeneous LSS are created by clumping plies of similar orientation. A LSS is said to be homogeneous if ply angles are evenly distributed through the laminate thickness. The ability to generate homogeneous LSS depends on the number of plies and ply angles. For example, it is impossible to create a homogeneous LSS for a four-ply laminate consisting of four different ply angles.

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The following LSS guidelines are based on past experience from test and analysis. Guidelines are lumped under two categories; (1) strong recommendation, and (2) recommendation. Despite this classification, exceptions to the guidelines should be considered based on an engineering evaluation of the specific application.

4.6.5.2.1 *Strong recommendations.*

1. Homogeneous LSS are recommended for strength controlled designs (In other words, thoroughly intersperse ply orientations throughout the LSS).

   Comment: *Heterogeneous laminates should be avoided for strength-critical designs unless analysis and test data is available that indicates a clear advantage. In cases where heterogeneous laminates cannot be avoided (e.g., minimum gage laminates), it is generally best to stack primary load-carrying plies toward the laminate core. The best way to view possible strength problems with heterogeneous LSS is to consider the behavior of individual sublaminates (i.e., groups of plies separated by delaminations) that may be created during manufacturing or service exposure. This will be discussed later in greater detail.*

   Heterogeneous LSS can yield optimum stiffness or stability performance; however, the effects on all other aspects of the design (e.g., strength, damage tolerance, and durability) should be considered before ignoring Recommendation 1. For example, interlaminar stress distributions are affected by variations in the in-plane stress field around the periphery of holes and cutouts and the "effective" LSS (i.e., ply orientations relative to a tangent to the edge). Since it is difficult to optimize for a single lay-up in this case, the best solution is to make the LSS as homogeneous as possible.

2. A LSS should have at least four distinct ply angles (e.g., 0°, ± θ°, 90°) with a minimum of 10% of the plies oriented at each angle. Ply angles should be selected such that fibers are oriented with principal load axes.

   Comment: *This rule is intended to avoid the matrix-dominated behavior (e.g., nonlinear effects and creep) of laminates not having fibers aligned with principal load axes. Such behavior can lead to low strengths and dimensional stability problems.*

3. Minimize groupings of plies with the same orientation. For tape plies, stack no more than four plies of the same orientation together (i.e., limit stacked ply group thicknesses $\leq$0.03 in.). In addition, stacked ply group thicknesses with orientations perpendicular to a free edge should be limited to $\leq$0.015 in.

   Comment: *This guideline is used for laminate strength-critical designs. For example, it will help avoid the shear-out failure mode in bolted joints. It also considers relationships between stacked ply group thickness, matrix cracking (i.e., transverse tension and shear ply failures) and delamination.*

   In general, ply group thickness should be limited based on details of the design problem (e.g., loads, free edges, etc.) and material properties (e.g., interlaminar toughness). Note that the absolute level of ply group thickness identified in this guideline is based on past experience. It should be confirmed with tests for specific materials and design considerations.

4. If possible, LSS should be balanced and symmetric about the midplane. If this is not possible due to other requirements, locate the asymmetry or imbalance as near to the laminate

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midplane as possible. A LSS is considered symmetric if plies positioned at an equal distance above and below the midplane are identical (i.e., material, thickness, and orientation). Balanced is defined as having equal numbers of $+\theta$ and $-\theta$ plies, where θ is measured from the primary load direction.

Comment: *This guideline is used to avoid shear/extension couplings and dimensional stability problems (e.g., warpage which affects component manufacturing tolerances). The extension/bending coupling of unsymmetric laminates can reduce buckling loads. Note that some coupling may be desired for certain applications (e.g., shear/extension coupling has been used for aeroelastic tailoring).*

4.6.5.2.2 *Recommendations.*

5. Alternate $+\theta$ and $-\theta$ plies through the LSS except for the closest ply either side of the symmetry plane. A $+\theta/-\theta$ pair of plies should be located as closely as possible while still meeting the other guidelines.

   Comment: *This guideline minimizes the effect of bending/twisting coupling, which is strongest when angle plies are separated near the surface of a laminate. Modifications to this rule may promote more efficient stiffness and stability controlled designs.*

6. Shield primary load carrying plies from exposed surfaces.

   Comment: *The LSS for laminates primarily loaded in tension or compression in the 0° direction should start with angle and transverse plies. Tensile strength, microbuckling resistance, impact damage tolerance and crippling strength can all increase by shielding the main load bearing plies from the laminate surface. With primary load fibers buried, exterior scratches or surface ply delamination will not have a critical effect on strength. For laminates loaded primarily in shear, consideration should be given to locating +45° and -45° plies away from the surface. For cases in which an element is shielded by other structures (e.g., shear webs), it may not be necessary to stack primary load carrying plies away from the surface.*

7. Avoid LSS that create high interlaminar tension stresses (σ_z) at free edges. Analyses to predict free edge stresses and delamination strain levels are recommended to help select LSS.

   Comment: *Composite materials tend to have a relatively low resistance to mode I delamination growth. Edge delamination, followed by sublaminate buckling can cause premature failure under compressive loads. Edge delamination occurring under tensile loads can also effectively reduce stiffness and lower the load carrying capability. Since delaminations occurring at the core of the laminate can have the strongest effect on strength, avoid locating tape plies with fibers oriented perpendicular to a free edge at the laminate midplane.*

8. Minimize the Poisson's ratio mismatch between adjacent laminates that are cocured or bonded.

   Comment: *Excessive property mismatches between cobonded elements (e.g., skin and stringer flange) can result in delamination problems. In the absence of more sophisticated analysis tools, a general rule of thumb is*

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$$| \nu_{xy} \text{ (laminate 1)} - \nu_{xy} \text{ (laminate 2)} | < 0.1. \qquad 4.6.5.2.1$$

4.6.6 Lamina-to-laminate statistics.

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4.7 COMPRESSIVE BUCKLING AND CRIPPLING.

4.7.1 Plate buckling and crippling.

4.7.1.1 *Introduction.* Rectangular flat plates are readily found in numerous aerospace structures in the form of unstiffened panels and panels between stiffeners of a stiffened panel, and as elements of a stiffener. Closed form classical buckling solutions available in the literature are limited to orthotropic plates with certain assumed boundary conditions. These boundary conditions may be fixed, simply supported, or free. For expediency, the engineer may wish to assume the most appropriate boundary conditions and obtain a quick solution rather than resort to using a buckling computer program such as Reference 4.7.1.1(a). However, the closed form solutions of laminated orthotropic plates are appropriate only when the lay-ups are symmetrical and balanced. Symmetrical implies identical corresponding plies about the plate mid-surface. Balanced refers to having a minus θ ply for every plus θ ply on each side of the mid-surface. Symmetrical and balanced laminated plates have B_{ij} terms vanish and the D_{16} and D_{26} terms virtually vanish. However, the balanced plies ($\pm\theta$) should be adjacent; otherwise the D_{16} and D_{26} terms could become significant and invalidate the use of the orthotropic analysis. The buckling solutions could be significantly nonconservative for thin unbalanced or unsymmetric plates (see Reference 4.7.1.1(b)). Note that not all closed form solutions give direct answers; sometimes the equations must be minimized with respect to certain parameters as will be shown later.

The behavior of flat plates in compression involves initial buckling, postbuckling out-of-plane displacements, and crippling (ultimate postbuckling failure). Only at crippling does permanent damage occur, usually some form of delamination due to interlaminar tensile or shear stresses.

Nomenclature used to describe the buckling behavior of composite plates in Section 4.7.1 is given in Table 4.7.1.1.

4.7.1.2 *Initial buckling.* Initial buckling is defined to occur at a load that results in incipient out-of-plane displacements. The classical equations are elastic, and finite transverse shear stiffness effects are neglected. (Reference 4.7.1.2). The buckling of certain plate geometries, however, can be influenced by the finite shear stiffness effects as shown in Section 4.7.1.8.

4.7.1.3 *Uniaxial loading - long plate with all sides simply supported.* The case of a long plate (a/b > 4) with all sides simply supported (SS) and loaded uniaxially is shown in Figure 4.7.1.3(a) and described by Equation 4.7.1.3.

$$N^{cr}_{x,cl} = \frac{2\Pi^2}{b^2}\left[(D_{11}D_{22})^{1/2} + D_{12} + 2D_{66}\right] \qquad 4.7.1.3$$

Equation 4.7.1.3 is the most frequently used plate buckling equation. It can be shown by the use of the STAGS computer program (Reference 4.7.1.1(a)) that this equation is also valid for fixed boundary conditions (FF) on the loaded edges, which is important since all testing is performed with fixed boundary conditions on the loaded edges to prevent local brooming. Comprehensive testing has shown these equation to be valid except for very narrow plates. Figure 4.7.1.3(b) shows the comparisons between experiment and classical theory from References 4.7.1.3(a) and (b), where the test results are plotted as $N^{cr}_{x,i}/N^{cr}_{x,cl}$ versus the b/t ratios. Notice the discrepancy becomes worse at the low b/t ratios (narrow plates). Thus the equation should be used with caution at b/t ratios less than 35. In Figure 4.7.1.3(c) the same experimental data has been normalized by the buckling load prediction which includes the effects of transverse shear ($N^{cr}_{x,w}$ from References 4.7.1.3(c) and (d)). Note that most available computer buckling programs will not account for this transverse shear effect.

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TABLE 4.7.1.1 *Buckling and crippling symbols.*

SYMBOL	DEFINITION
a	length
b	width
B_{ij}	stiffness coupling terms of laminated plate
D_{ij}	flexural/twisting stiffness terms of laminated plate
$F^{cr}_{x,cl}$	classical orthotropic longitudinal compressive buckling stress
$F^{cr}_{x,i}$	initial longitudinal compressive buckling stress from test
F^{cc}_{x}	longitudinal crippling stress from test
F^{cu}_{x}	longitudinal ultimate compressive stress of laminate
$N^{cr}_{x,cl}$, $N^{cr}_{y,cl}$	classical orthotropic longitudinal and transverse compressive uniform buckling loads, respectively
$N^{cr}_{x,i}$	initial longitudinal uniform buckling load from test
$N^{cr}_{x,w}$	longitudinal compressive uniform buckling load based on anisotropic theory, including transverse shear effects
N_x, N_y	longitudinal and transverse applied uniform loads, respectively, on a plate
$P^{cr}_{x,i}$	total longitudinal initial buckling load form test
$P^{cc}_{x,i}$	total longitudinal crippling load from test
t	thickness

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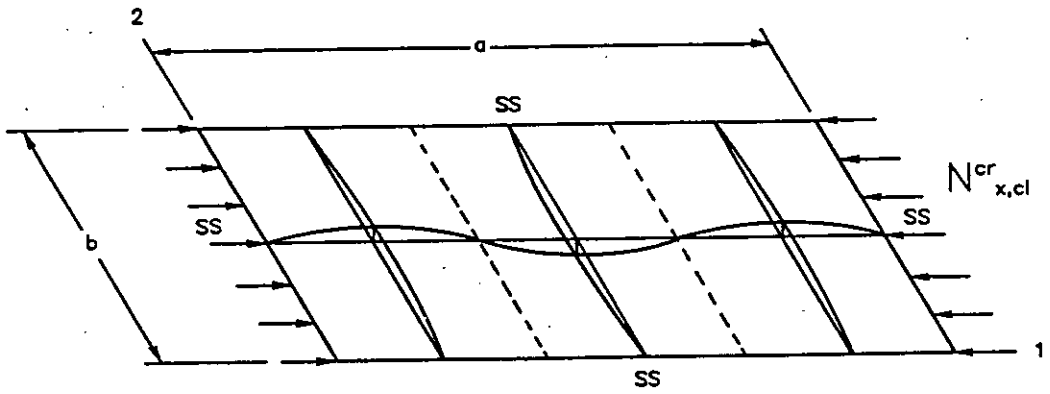


Figure 4.7.1.3(a) *Uniaxial loading - long plate (a/b > 4) with all sides simply supported (SS).*

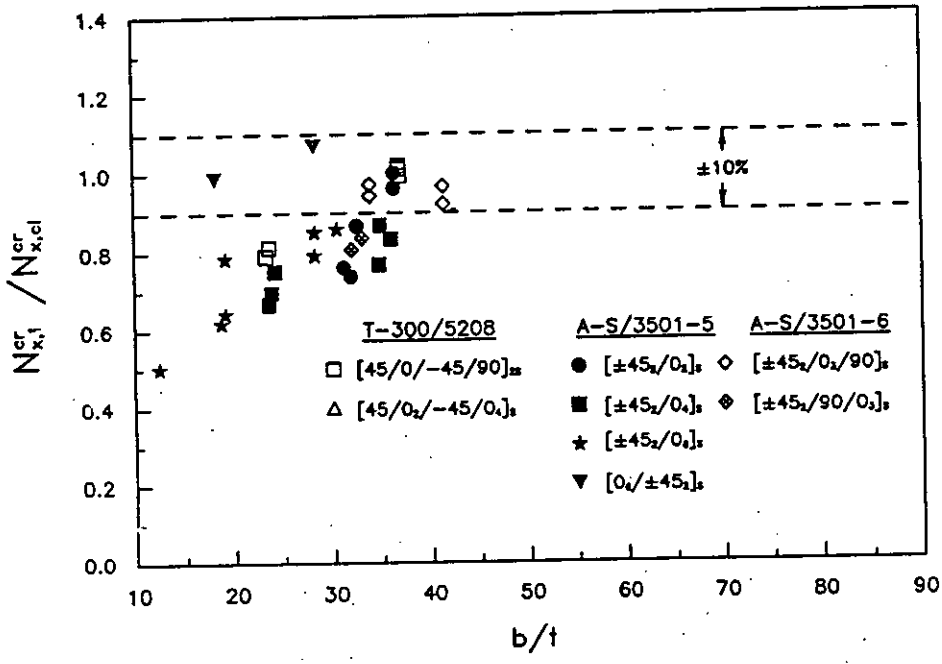


Figure 4.7.1.3(b) *Predicted classical buckling loads compared to experimental data (Reference 4.7.1.3(b)).*

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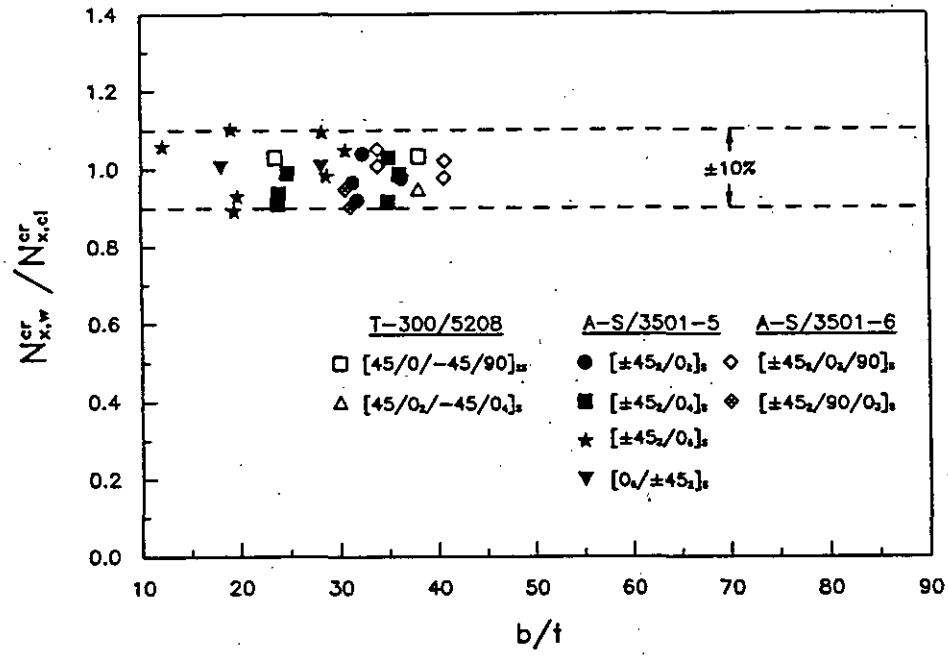


Figure 4.7.1.3(c) *Predicted buckling loads of the current theory compared to experimental data (Reference 4.7.1.3(b)).*

4.7.1.4 *Uniaxial loading - long plate with all sides fixed.* The case of a long plate (a/b > 4) with all sides fixed (FF) and loaded uniaxially is shown in Figure 4.7.1.4 and described by Equation 4.7.1.4.

$$N^{cr}_{x,cl} = \frac{\Pi^2}{b^2}\left[4.6\,(D_{11}\,D_{22})^{1/2} + 2.67\,D_{12} + 5.33\,D_{66}\right] \qquad 4.7.1.4$$

This equation has not had the comprehensive experimental study as has Equation 4.7.1.3. However, by conjecture the effect of transverse shear for narrow plates would be quite similar to that found for plates with all edges simply supported.

4.7.1.5 *Uniaxial loading - long plate with three sides simply supported and one unloaded edge free.* Figure 4.7.1.5 shows the case of a long plate (a/b > 4) with three sides simply supported and the remaining unloaded edge free. This plate is uniaxially loaded. This loading situation is described by Equation 4.7.1.5.

$$N^{cr}_{x,cl} = \frac{12\,D_{66}}{b^2} + \frac{\Pi^2\,D_{11}}{a^2} \qquad 4.7.1.5$$

where b/t must be greater than 20 because of transverse shear effects in narrow plates as discussed in Section 4.7.1.3.

4.7.1.6 *Uniaxial and biaxial loading - plate with all sides simply supported.* Biaxial and uniaxial loading of a simply supported plate is shown in Figure 4.7.1.6, where $1 < a/b < \infty$. The following classical orthotropic buckling equation must be minimized with respect to the longitudinal and transverse half-waves numbers, m and n :

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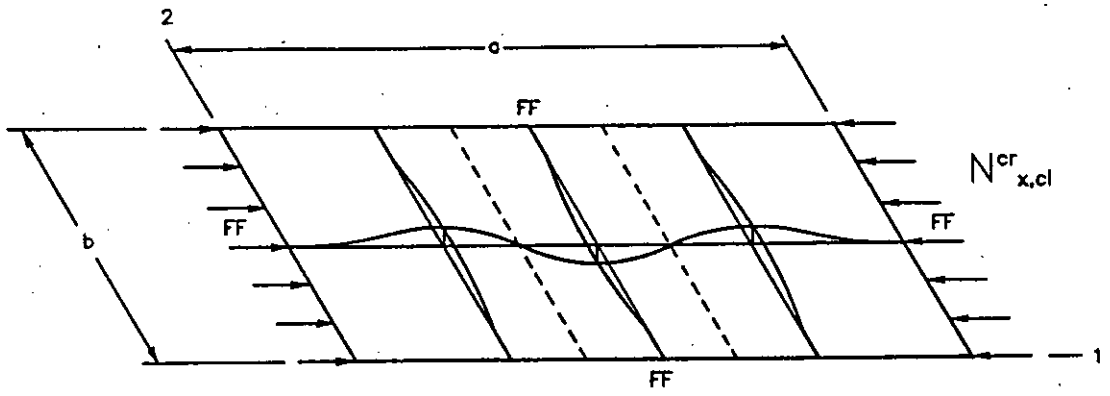


Figure 4.7.1.4 *Uniaxial loading - long plate (a/b > 4) with all sides fixed.*

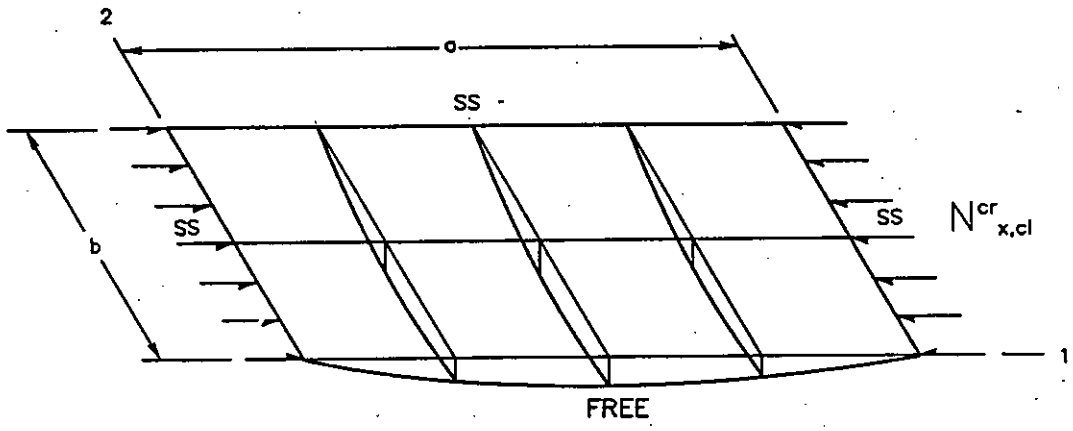


Figure 4.7.1.5 *Uniaxial loading - long plate with three sides simply supported and one unloaded edge free.*

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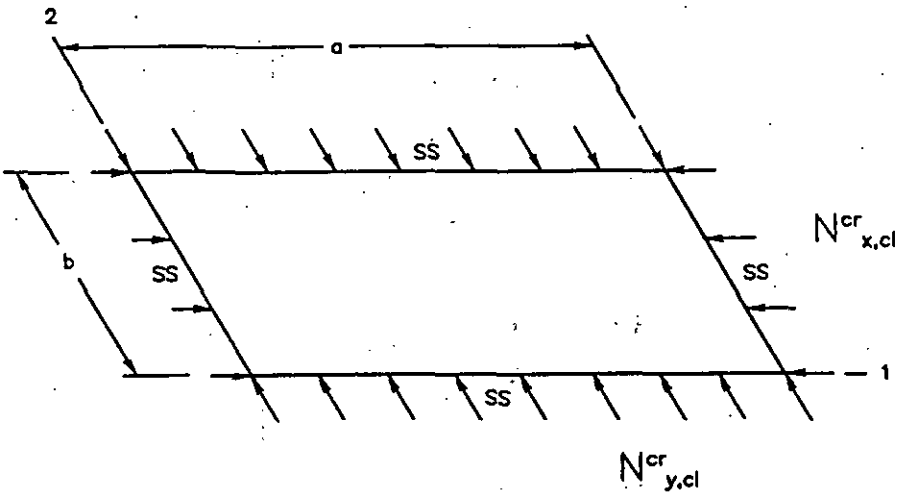


Figure 4.7.1.6 *Uniaxial and biaxial loading - plate with all sides simply supported.*

$$N^{cr}_{x,cl} = \frac{\Pi^2}{b^2} \frac{D_{11}m^4(b/a)^4 + 2(D_{12} + 2D_{66})\, m^2n^2(b/a)^4 + D_{22}n^4}{m^2(b/a)^2 + \phi n^2}, \text{ min} \qquad 4.7.1.6(a)$$

where

$$\phi = N_y/N_x \qquad 4.7.1.6(b)$$

which is the ratio of applied transverse to longitudinal loading. Accordingly, the corresponding transverse buckling load is

$$N^{cr}_{y,cl} = \phi N^{cr}_{x,cl} \qquad 4.7.1.6(c)$$

For uniaxial loading, let $\phi = 0$.

4.7.1.7 *Uniaxial loading - plate with loaded edges simply supported and unloaded edges fixed.* The case of a uniaxially loaded plate ($1 < a/b < \infty$) with the loaded sides simply supported (SS) and the unloaded sides fixed (FF) can also be considered. For this case, the following classical orthotropic buckling equation must be minimized with respect to the longitudinal half-wave number, m:

$$N^{cr}_{x,cl} = \frac{\Pi^2}{b^2}\left\{D_{11}m^2(b/a)^2 + 2.67\, D_{12} + 5.33\left[D_{22}(a/b)^2 + (1/m)^2 + D_{66}\right]\right\} \qquad 4.7.1.7$$

4.7.1.8 *Postbuckling/crippling.* Wide exploitation of advanced composites in stability critical structural designs will depend to a large degree on the ability of such heterogeneous anisotropic

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composites to support loads well beyond the initial buckling level. Unquestionably, the high stiffness-to-weight ratio of composites renders them potentially attractive up to initial buckling. However, since postbuckling design has been established over several decades for certain types of conventional metallic alloy construction, it should be anticipated that composites would demonstrate a similar tolerance to be competitive. Hence, this section addresses this vitally important issue as it pertains to the design of structural compression plates. At this point it is appropriate to state the following two definitions:

Postbuckling. The ability of a compression member or stiffened panel to carry loads well in excess of initial buckling load is widely accepted. The "postbuckling range" may be considered to exist between the initial buckling load and some higher load representing failure, e.g., delamination at the free edge of a compression member or the disbonding of a stiffener from the panel in a stiffened panel. For basic elements, comprising a simple structural constituent panel, the upper limit of the postbuckling range is sometimes termed "local crippling" or simply "crippling".

Failure. The load at which irreversible damage such as delamination or transverse cracking occurs, usually accompanied by an abrupt loss in load carrying capability. This event, a subset of which has been termed "crippling", may or may not be predicted.

The postbuckling behavior of composite plates presented here is limited to the empirical graphite tape data obtained from References 4.7.1.3(a) and (b). Relatively narrow plates, with simply supported unloaded edges or one edge free and fixed loading edges were tested and analyzed. At the time of this writing, there are insufficient postbuckling test data of plates with other boundary conditions. The simply supported unloaded edges were simulated by the use of steel V-blocks mounted on the compression test fixture. Specifically, the plates with both unloaded edges simply supported are defined as "no-edge-free". Plates with one unloaded edge simply supported and the other free are defined as "one-edge-free". A typical no-edge-free test in progress with the specimen in the postbuckling range is shown in Figure 4.7.1.8(a). In addition, a typical one-edge-free tests where crippling of the specimen has occurred is shown in Figure 4.7.1.8(b). Typical load-displacement curves of no-edge-free and one-edge-free tests are shown in Figures 4.7.1.8(c) and (d), respectively. The most convenient plot that exemplifies the postbuckling strength of the no-edge-free composite plates is shown in Figure 4.7.1.8(e). The value for F_{11}^{cu} is the ultimate compressive strength of the particular laminate, while a typical failed test specimen is shown in Figure 4.7.1.8(f). Finally an empirical design curve for determining the crippling strengths of no-edge-free plates is shown in Figure 4.7.1.8(g). With very limited test data, Figure 4.7.1.8(h) demonstrates the postbuckling strengths of one-edge-free plates. An empirical design curve for the crippling strengths of one-edge-free plates is shown in Figure 4.7.1.8(i). Noted that all the empirical data presented involved the testing of high strength graphite/epoxy tape. Other material systems or other forms of graphite/epoxy composites may yield considerably different results.

It was stated in Section 4.7.1.2 that initial buckling was more accurately determined by the use of References 4.7.1.3(c) and (d), which is due to accounting for transverse shear effects. Also, it is shown in these references that with the additional use of known material nonlinearity, the postbuckling and crippling loads can be determined. Some examples of test results vs. the theory of References 4.7.1.3(c) and (d) are shown in Figures 4.7.1.8(j) and (k). Unfortunately, the sophisticated computer programs available today do not have the features found in these references. Postbuckling fatigue may be permitted under certain circumstances without jeopardizing the structural integrity of the plate (References 4.7.1.3(b) and 4.7.1.8).

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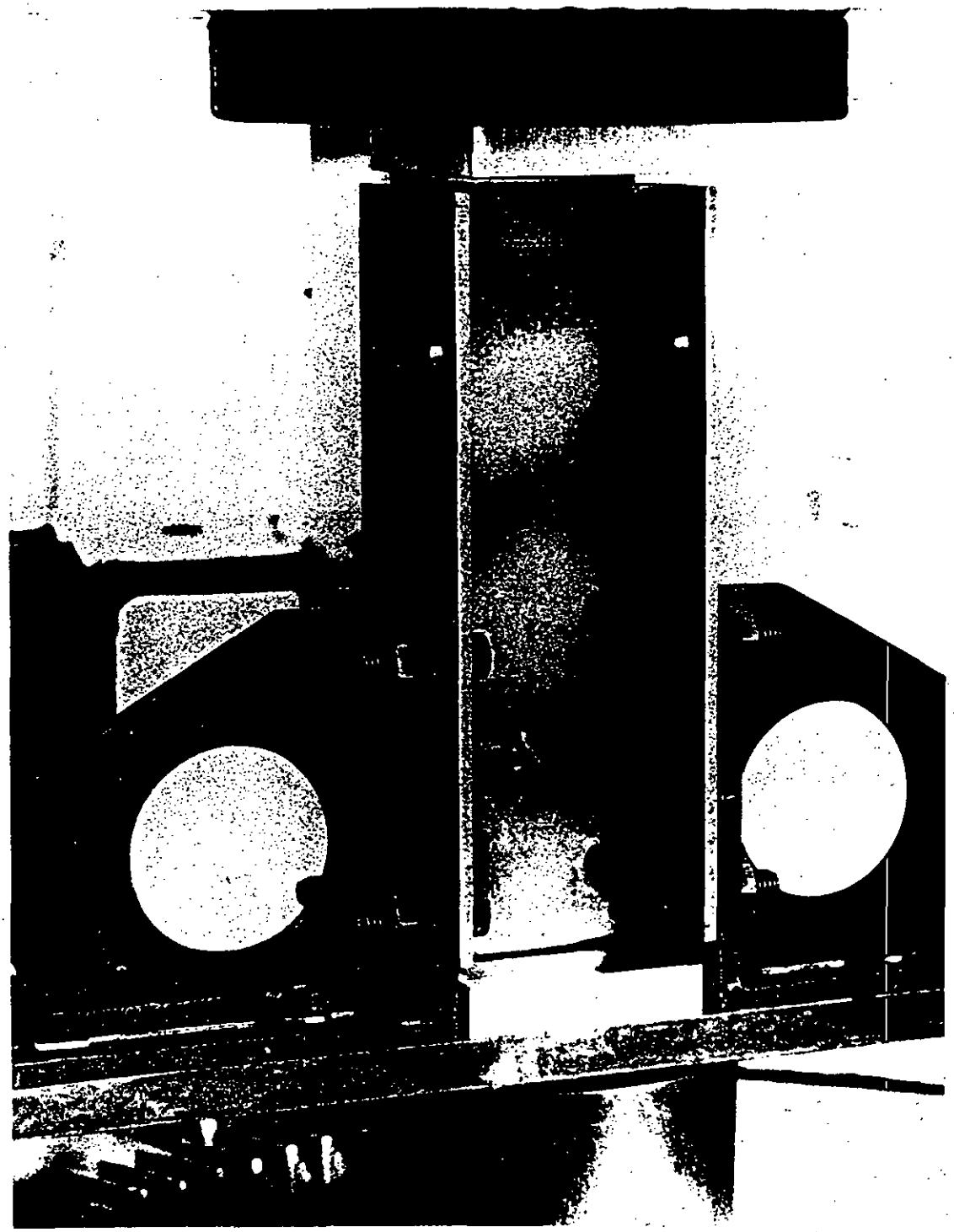


Figure 4.7.1.8(a) *No-edge-free graphite/epoxy test.*

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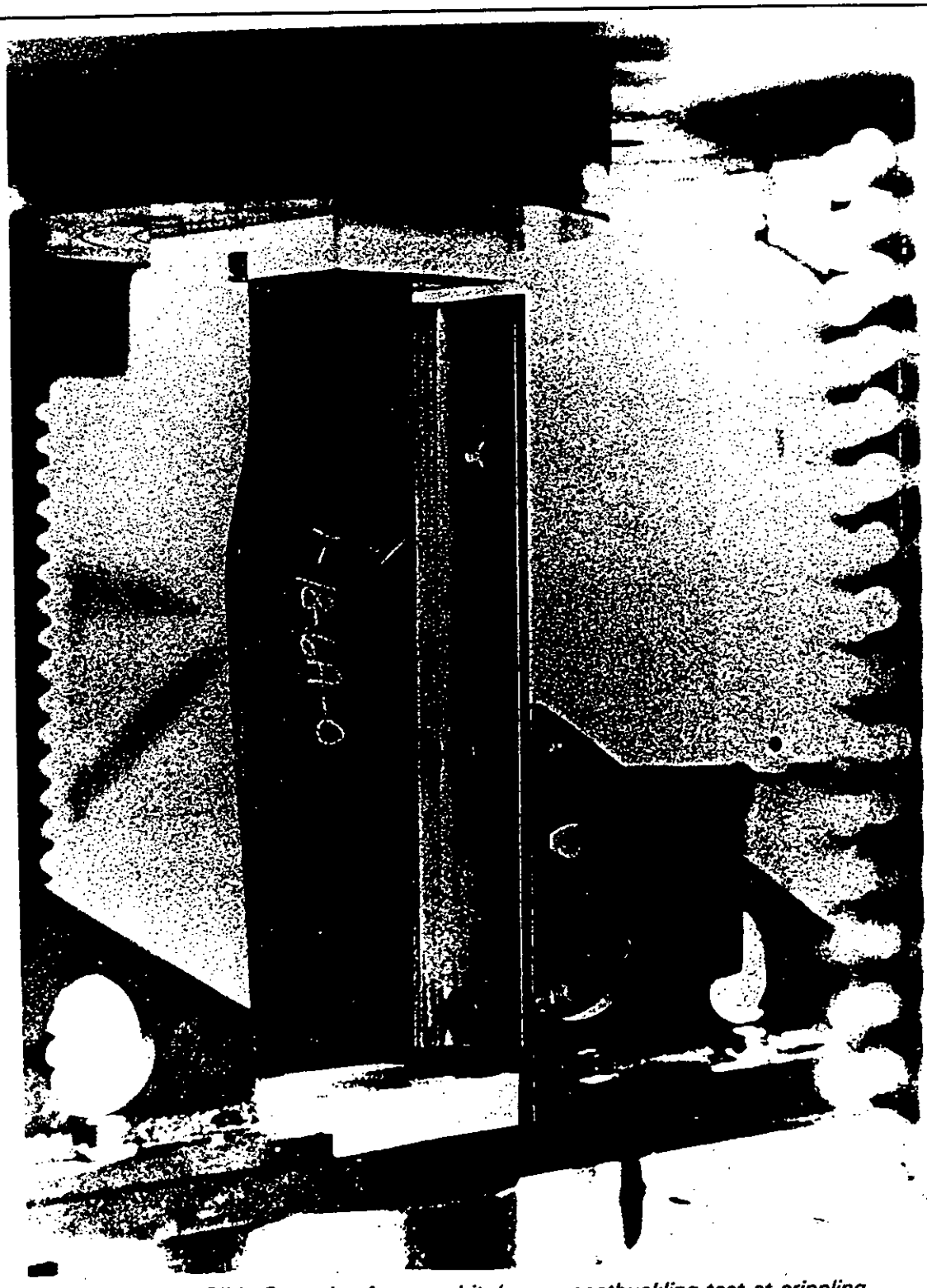


Figure 4.7.1.8(b) *One-edge-free graphite/epoxy postbuckling test at crippling.*

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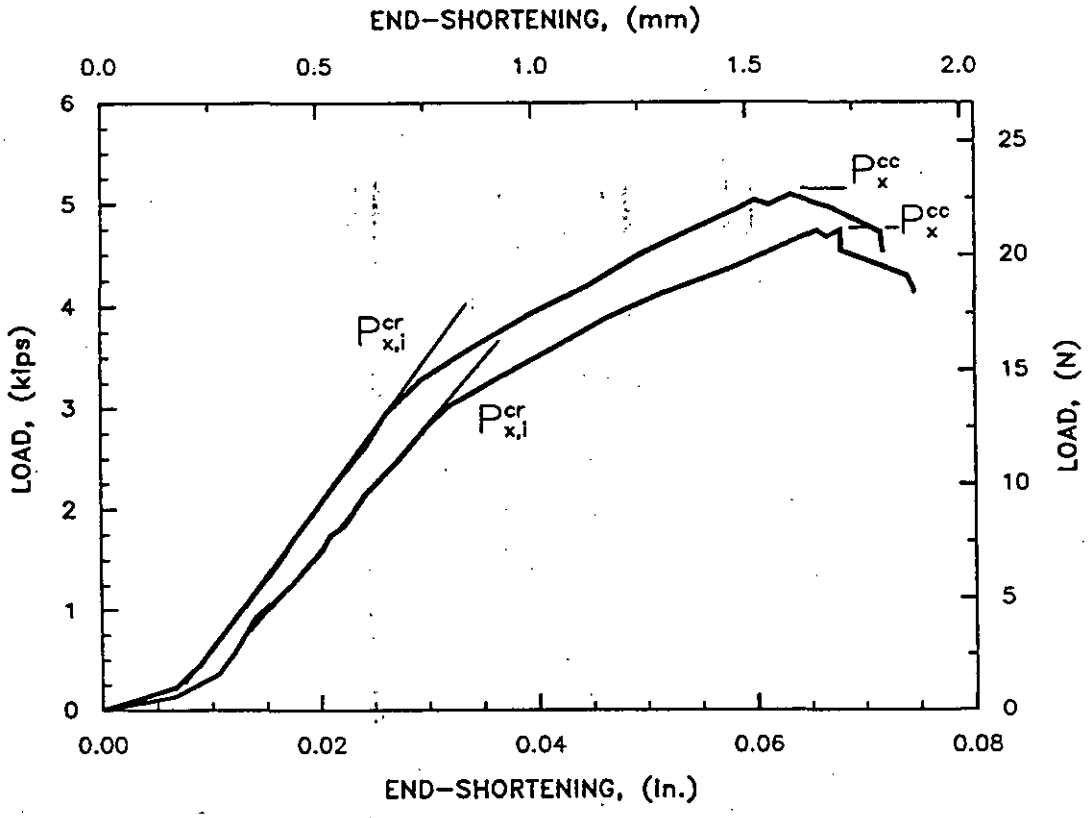


Figure 4.7.1.8(c) *No-edge-free plate. Crippling tests - AS/3501-6 $[\pm 45/90/0_3]_s$ - b/t ≈ 32.*

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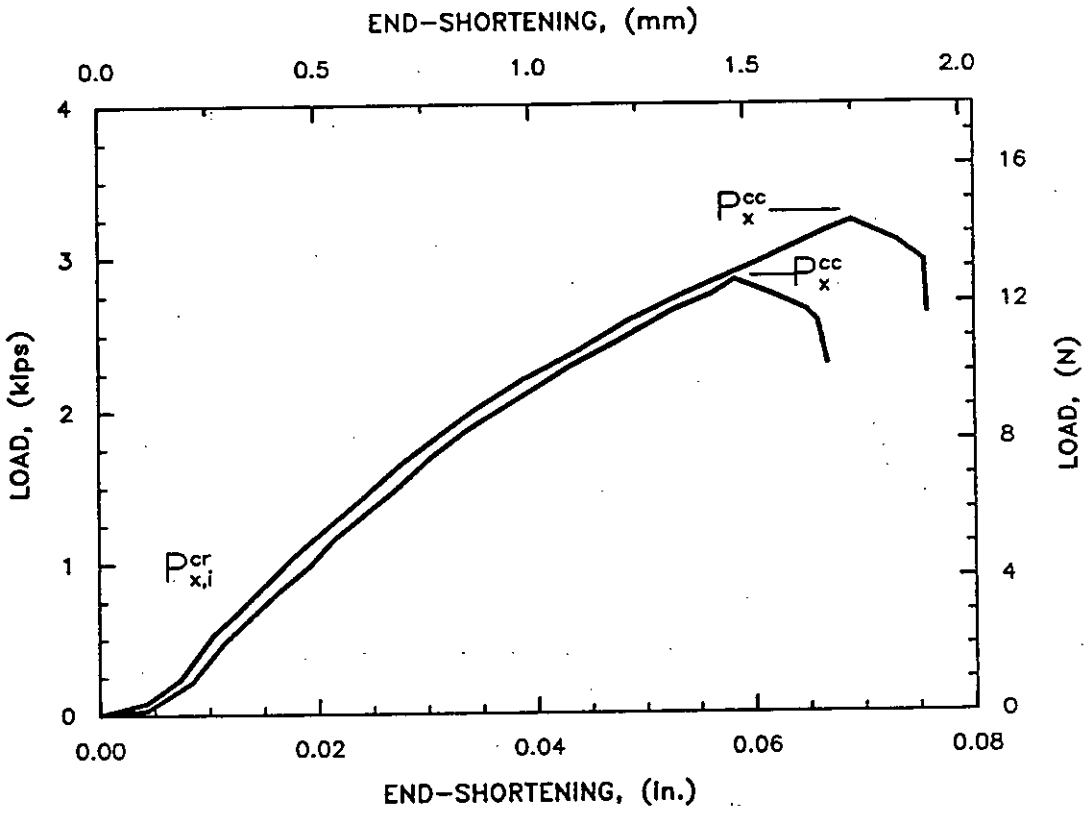


Figure 4.7.1.8(d) *One-edge-free plate. Crippling tests - AS/3501-6 $[\pm 45/90/0_3]_s$ - b/t ≈ 30.*

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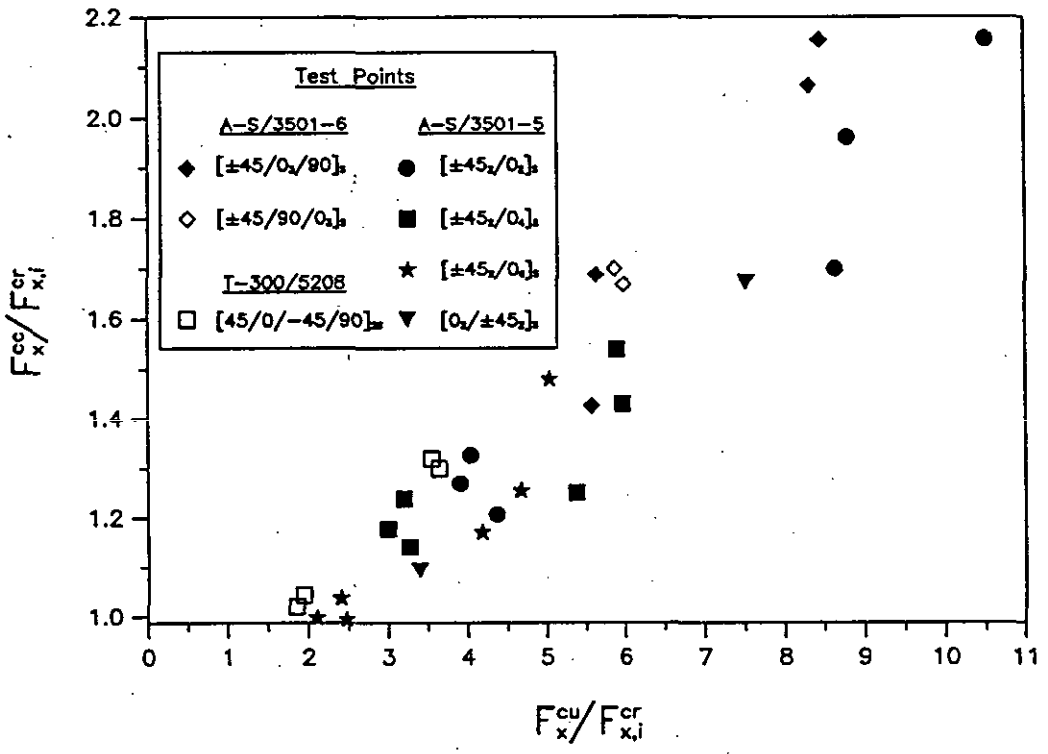


Figure 4.7.1.8(e) *Non-dimensional crippling plot of no-edge-free plates.*

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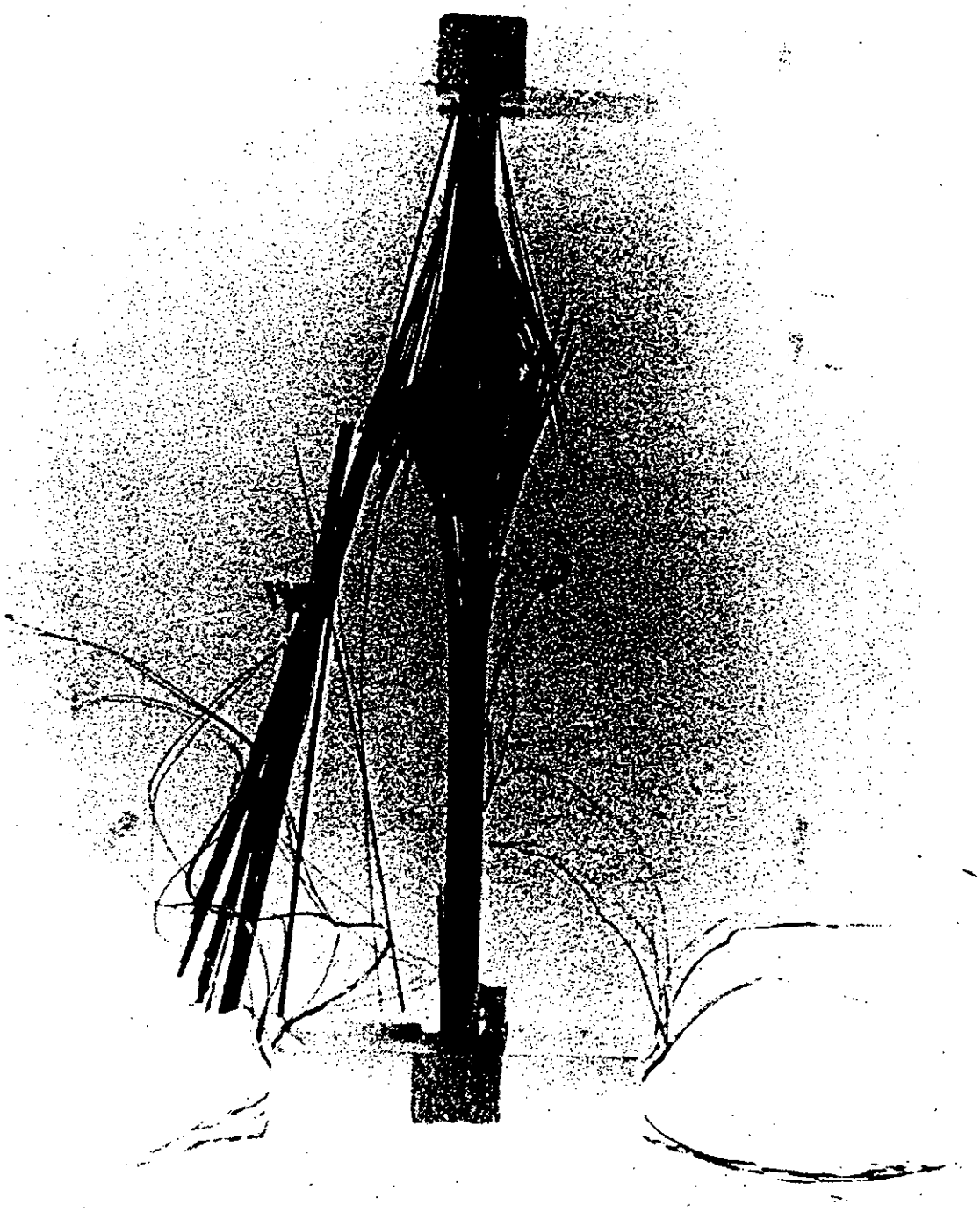


Figure 4.7.1.8(f) *Typical graphite/epoxy failed ultimate compression specimen.*

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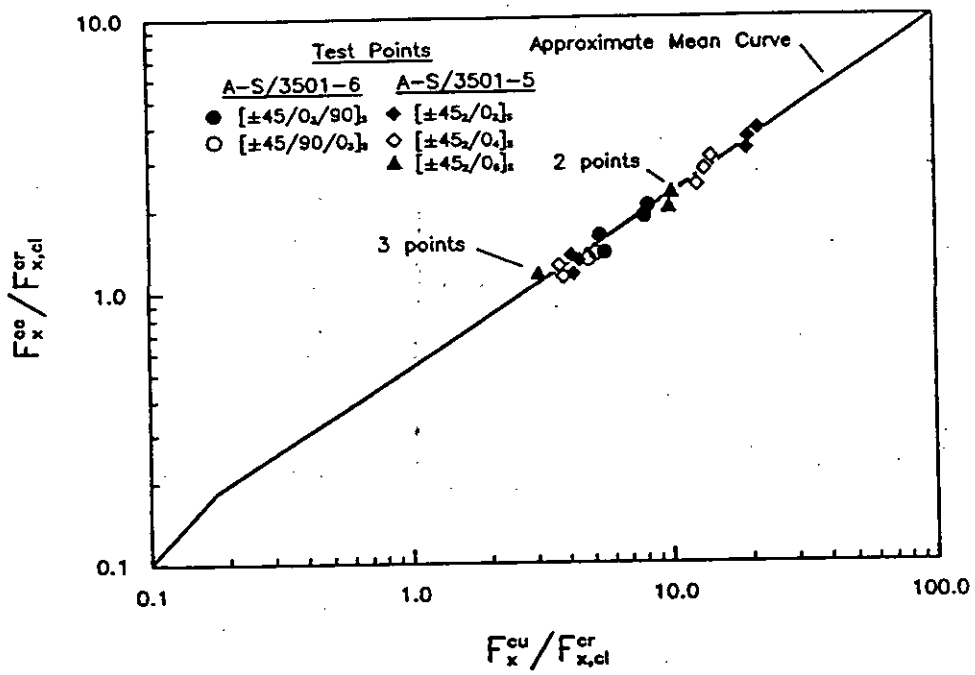


Figure 4.7.1.8(g) *Non-dimensional crippling plot of no-edge-free plates.*

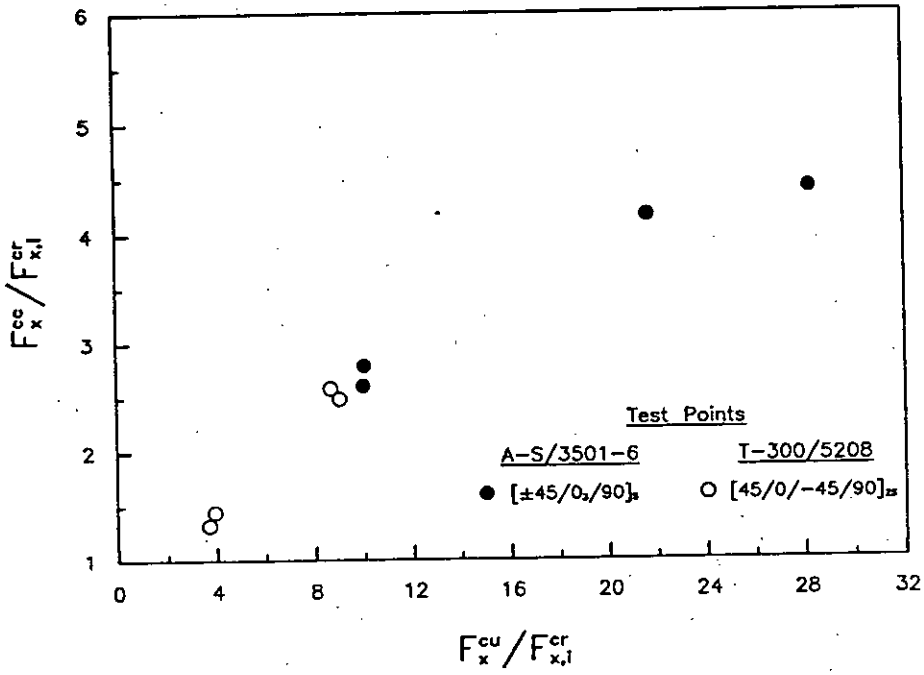


Figure 4.7.1.8(h) *Non-dimensional crippling plot of one-edge-free plates.*

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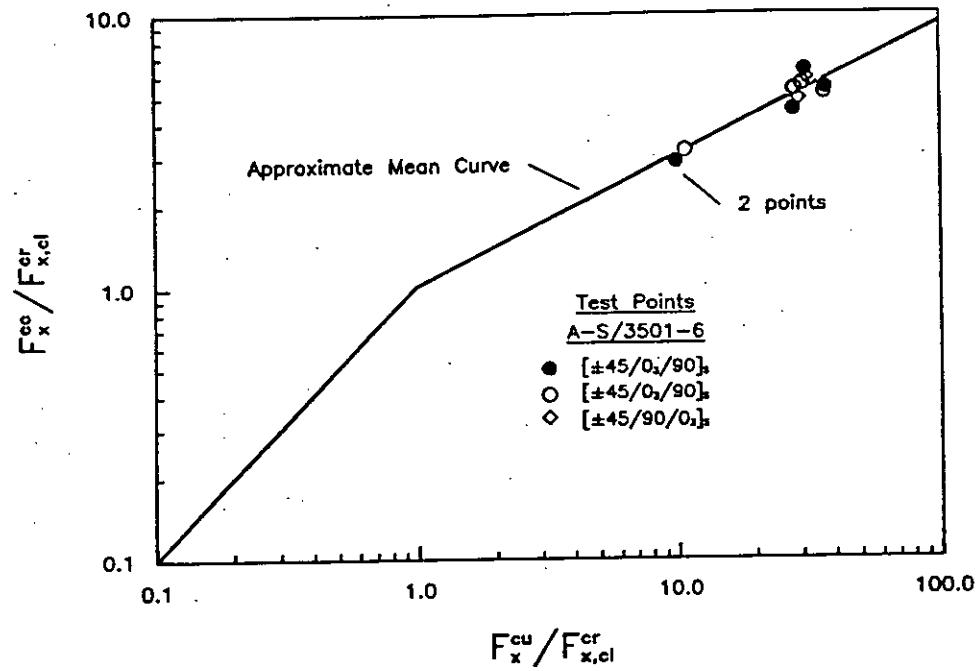


Figure 4.7.1.8(i) *Non-dimensional crippling plots of one-edge-free plates (Reference 4.7.1.3(b)).*

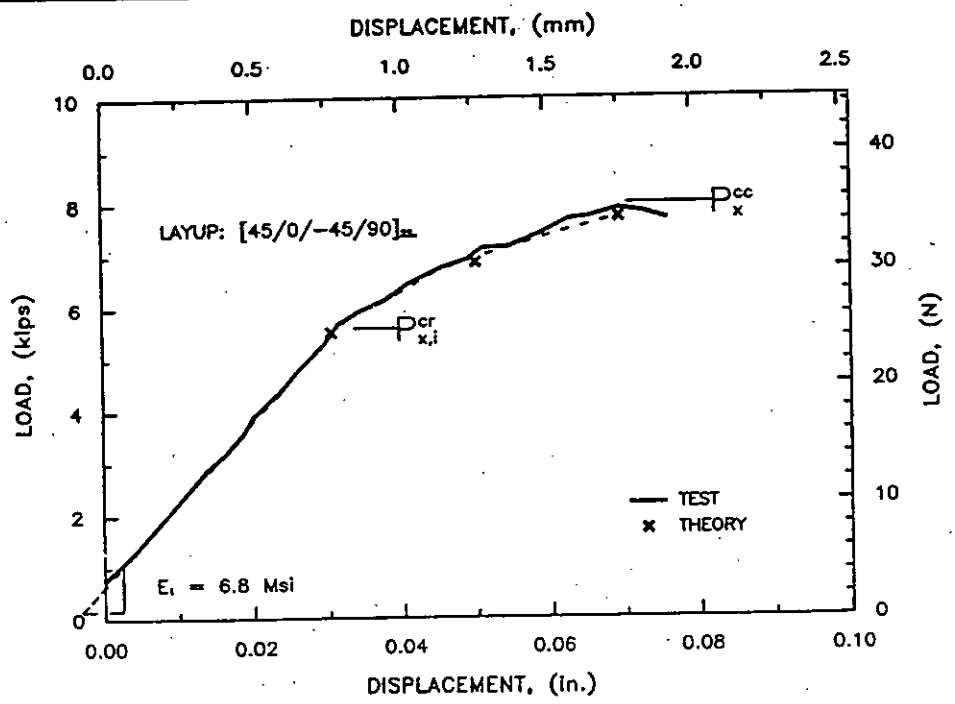


FIGURE 4.7.1.8(j) *Comparison of theory in References 4.7.1.3(b) and (c) with experiments for postbuckling curves and crippling strengths.*

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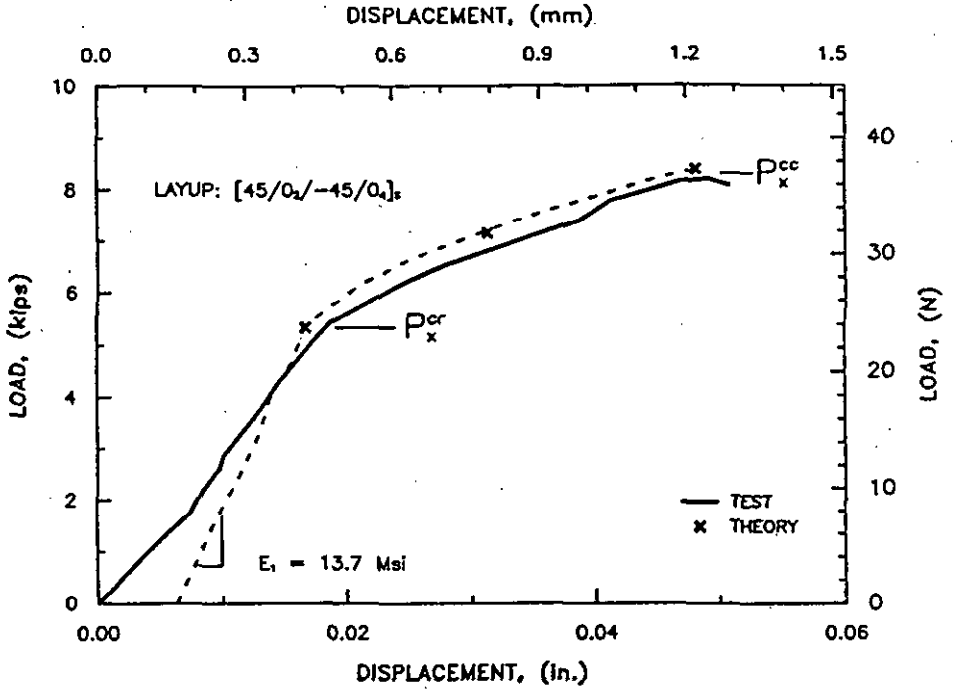


Figure 4.7.1.8(k) *Comparison of theory in References 4.7.1.3(b) and (c) with experiments for postbuckling curves and crippling strengths.*

4.7.1.9 *Stacking sequence effects in buckling.* Methods to accurately predict the stability of laminated plates have been documented (e.g., References 4.7.1.9(a)-(c)). Laminated plate stability can be strongly affected by LSS. However, factors such as plate geometry, boundary conditions and load type each contribute to the relationship between LSS and plate stability. As a result, general rules that define the best LSS for plate stability do not exist. Instead, such relationships must be established for specific structure and loading types. Three examples that illustrate this point will be shown in this section. Two different analysis methods were used in these examples. The first, utilized design equations from Reference 4.7.1.9(c) and bending stiffnesses as calculated using lamination theory. This method assumed the plate bending behavior to be "Specially orthotropic" (D_{16} and D_{26} terms were set equal to zero). The second method was a Boeing computer program called LEOTHA (an enhanced version of OTHA, Reference 4.7.1.9(a) which uses the Galerkin method to solve equations for buckling. This method allowed nonzero D_{16} and D_{26} terms.

Figures 4.7.1.9(a), (b), and (c) show plate buckling predictions for the seven LSS used in an earlier example (see Table 4.3.3.2(b)).[3] All plates were assumed to have simply-supported boundary conditions on the four edges. Figures 4.7.1.9(a) and (b) are rectangular plates loaded by uniaxial compression in long and short directions, respectively. Figure 4.7.1.9(c) shows shear buckling predictions for a square plate. Horizontal dashed lines on Figure 4.7.1.9(a) - (c) represent the results obtained when using the DOD/NASA design equations and assuming no LSS effect (i.e., a homogeneous orthotropic plate). The homogeneous plate assumption results in a buckling load that is roughly an average of the predictions for all LSS shown in the figures.

[3]*The LSS used in Figures 4.7.1.9(a), (b), and (c) were chosen for illustrative purposes only and do* not represent optimal LSS for a given application.

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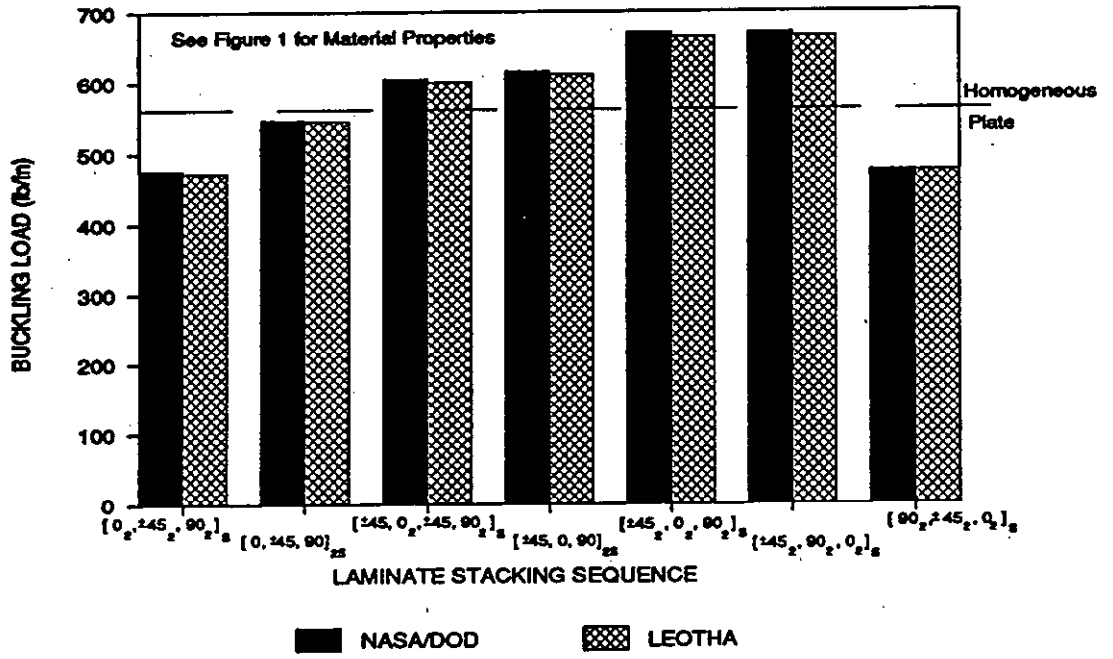


FIGURE 4.7.1.9(a) *Buckling analysis of 4 sides simply-supported, 24 in. by 6 in., laminated plates loaded in the long direction.*

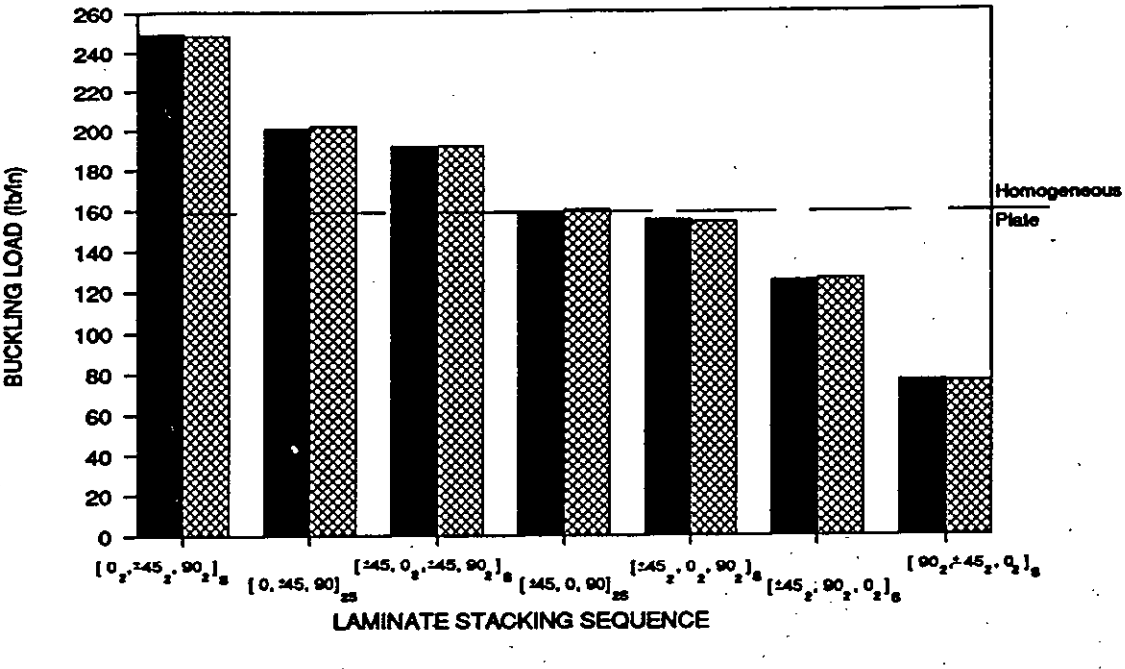


FIGURE 4.7.1.9(b) *Buckling analysis of 4 sides simply-supported, 6 in. by 24 in., laminated plates loaded in the short direction.*

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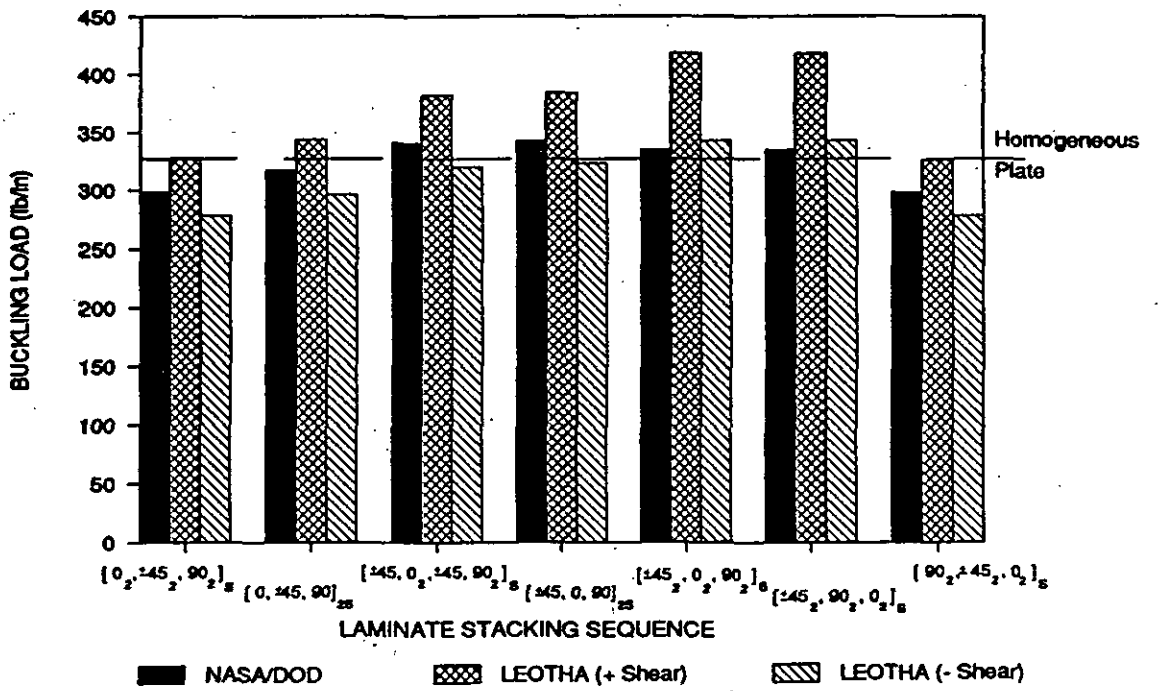


FIGURE 4.7.1.9(c) *Buckling analysis of 4 sides simply-supported, 12 in. by 12 in., laminated plates loaded in shear.*

The highest buckling loads for rectangular plates loaded in the long direction occur with preferential clumping of ±45° plies toward the surface layers (Figure 4.7.1.9(a)). Such is not the case for rectangular plates loaded in the short direction, where preferential stacking of 0° plies yield the highest buckling loads (4.7.1.9(b)). Note that predictions using the homogeneous plate assumption can be conservative or nonconservative depending on LSS. The DOD/NASA equations compare well with LEOTHA for conditions shown in Figures 4.7.1.9(a) and (b).

The highest buckling loads for square plates loaded in shear occur with preferential clumping of ±45° plies toward the surface layers (4.7.1.9(c)). Predictions using LEOTHA are different for positive and negative shear due to the relative positions of +45° and -45° plies. Predictions from DOD/NASA equations were generally lower than those of LEOTHA for positive shear loads. The opposite was true for negative shear loads. Differences may be attributed to the influence of D_{16} and D_{26} terms which were not included in the DOD/NASA design equations.

As with bending, structural geometry can overshadow the effects of LSS on stability (see the discussion pertaining to Figure 4.3.3.2). For example, the Euler buckling load of a laminated I-section used as a column is more strongly dependent on geometrical dimensions than on LSS of web and flanges. In fact, the effects of LSS on Euler buckling load diminishes sharply with increasing web height.

Design for local buckling and crippling of composite plates has typically relied on empirical data (e.g. Reference 4.7.1.9(c)). Local buckling and crippling have been found to relate to LSS. The lowest values for local buckling and crippling under uniaxial compression occurred with preferential stacking of 0° plies towards the outside surface of a laminate. Hence, when considering an I-section, Euler buckling loads may be independent of LSS while local buckling and crippling can relate to LSS.

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The effects of LSS on the stability of a stiffened panel is more complex. Assuming no local buckling and crippling, stiffener stability will not depend directly on LSS. However, post-buckling behavior of the skin and load redistribution to the stringer is strongly affected by the skin's LSS. As a result, overall stiffened panel stability can be influenced by the skin's LSS.

Basic information on laminate stacking sequence effects is found in Section 4.6.5.

4.8 CARPET PLOTS.

4.9 CREEP AND RELAXATION.

4.10 FATIGUE.

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4.11 OTHER STRUCTURAL PROPERTIES.

4.11.1 Damage tolerance.

4.11.1.1 *Background.* Damage tolerance is defined as a measure of the structure's ability to sustain a level of damage or presence of a defect and yet be able to perform its operating functions. Consequently, the concern with damage tolerance is ultimately with the damaged structure having adequate residual strength and stiffness to continue in service safely until the damage can be detected by scheduled maintenance inspection and repaired, or if undetected, for the remainder of the aircraft's life. Thus, safety is the primary goal of damage tolerance.

4.11.1.2 *Types of damage and concerns.* There are basically two types of damage which are categorized by their occurrence during the fabrication and use of the part, i.e., damage occurring during manufacturing or damage occurring in service. The former commonly includes consideration of defects such as porosity, microcracking, and delaminations resulting from processing discrepancies and also such items as inadvertent edge cuts, surface gouges and scratches, damaged fastener holes, and impact damage. The inadvertent (non-process) damage most commonly occurs in detail parts or components during assembly or transport.

It is expected that the occurrence of the majority of manufacturing associated damage, if beyond specification limits, will be detected by routine quality inspection. Nevertheless, some "rogue" defects or damage beyond specification limits may go undetected and consequently, their occurrence must be assumed in the design procedure. An example of such a damage tolerance design criteria would be the assumption of inadvertent delaminations existing in parts. Even though the capability to detect a 0.5 in diameter delamination by in-process ultrasonic inspection is considered typical, in designing to damage tolerance criteria, we may assume the presence of a larger delamination is missed by inspection. Quantifying the size of the "rogue" or missed flaw would be part of the contractor/customer criteria development process.

Another example of a manufacturing defect is when prepreg backing paper or separation film is inadvertently left in the laminate between plies during layup. Current inspection methods may not detect this type of discrepancy which may lead to a large delamination. Consequently, until more adequate inspection techniques are developed, in-process quality controls must be sufficiently rigid to preclude this occurrence. A damage tolerance criterion might be required to design for the occurrence of the flaw. Obviously a balance is needed between designing all possible "damage" into a part (with associated weight penalties) and what realistically can and should be eliminated from the fabrication/assembly process.

Service damage concerns are similar to those for manufacturing. Types of service damage include edge and surface gouges and cuts caused by improper tool use, or foreign object collision and blunt object impact damage caused by such incidences as dropped tools or contact with service equipment. Of these, cuts and gouges, are usually the more severe for tension loading while impact damage is generally the more severe damage type for compression, especially for thick laminates.

Delaminations can also be critical defects. However, unless they are very large, i.e., more than 2.0 inches in diameter, the problem is mostly with thin laminates. Effects of manufacturing defects such as porosity and flawed fastener holes are usually less severe. They are generally accounted for by the use of design allowable properties that have been obtained by testing specimens with notches, i.e., stress concentrations. Most commonly, these are specimens with a centered hole. Open holes are typically used for compression specimens while either open or filled holes (holes with an installed fastener) are used for tension testing. (Open holes are more critical then filled holes for compression. Filled holes may be more critical in tension, especially for laminates with ply orientations with a

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predominate number of plies oriented in the load direction). Consequently, the design allowables thus derived may be used to account for a nominal design stress concentration caused by an installed or missing fastener, at least to a 0.25 inch diameter, as well as many manufacturing defects.

One of the primary concerns with damage tolerance of composites is detection of the damage. Fortunately, most of the damage which is critical to tension loading such as cuts and gouges is, to some degree, visible. Tests have shown that for tension loading, the residual strength of a laminate with a cutout is primarily dependent on the width of the cutout and essentially independent of the cutout shape. Thus design values reduced to account for the presence of a 0.25 inch diameter hole also account for an equivalent length edge cut, however, and testing might be required to define appropriate design values. Cuts of this type that might be produced during manufacturing are a special problem since they may be filled with paint, and consequently, not detected. Sufficient testing should be done as part of design verification programs to ensure that cuts and gouges that are on the threshold of visibility will not degrade the structure enough to jeopardize its safety.

Low velocity impacts, e.g., impacts from dropped tools as opposed to ballistic impacts, present a special problem. Impacts on the laminate surface, especially those made by a blunt object, may cause considerable internal damage without producing visible indications on the surface. Damage to the resin may be particularly severe as evidenced by transverse shear cracks and delaminations. Consequently, the resin loses its ability to stabilize the fibers in compression and the local failure may initiate total structural collapse. Similarly, the impact may damage fibers and cause local stress concentrations which could result in significant loss of tensile strength. With conventional graphite/epoxy systems, which are quite brittle, losses in tension and compression strength for nondetectible impact may approach 50% and 60% respectively.

Low velocity impact damage, such as from dropped tools is a key problem in thin gage structures. The damage is characteristically different from damage in thick laminates. Laminate internal delamination is less of a problem because thin laminate damage is more typically in the form of fiber breakage and, possibly, penetration. For sandwich materials with thin faces, impact can result in visible core damage which has been shown to reduce the compressive and shear strengths. Impact damage which causes a break in the facesheet of the sandwich (as well as porosity, a manufacturing defect) also presents a long term durability problem in that it can allow water intrusion into the core.

It is considered possible that nonvisible damage might occur early in the aircraft's life and go undetected during subsequent service inspections. Thus, unless detection is ensured by more discriminating inspection procedures, the damage or defect must be assumed to be present for the entire life of the aircraft. This "duration of damage or defect" factor is important in determining the probability that the aircraft with a defect or damage will encounter a load which might cause failure. That is, structure having undetected damage, such as low-velocity impact damage, will have a probability of encountering a particularly high load that is directly proportional to the service time during which that load may occur. In contrast, structure that has damage that is discovered and repaired in a short time period will have less chance of encountering the same high load.

4.11.1.3 *Evolving military and FAA specifications.* The "duration of damage or defect" factor based on degree of detectability has been the basis for establishing minimum Air Force damage tolerance residual strengths for composite structures in requirements proposed for inclusion in AFGS-87221, "General Specification for Aircraft Structures". These strength requirements are identical to those for metal structure having critical defects or damage with a comparable degree of detectability. Requirements for cyclic loading prior to residual strength testing of test components are also identical. The nondetectable damage to be assumed includes a surface scratch, a delamination, and impact damage. The impact damage includes both a definition of dent depth, i.e., detectability, and a maximum energy cutoff. Specifically, the impact damage to be assumed is that "caused by the

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impact of a 1.0 inch diameter hemispherical impactor with a 100 ft-lb of kinetic energy, or that kinetic energy required to cause a dent 0.10 inch deep, whichever is least." For relatively thin structure, the detectability, i.e., the 0.1 inch depth, requirement prevails. For thicker structure, the maximum assumed impact energy becomes the critical requirement. The associated load to be assumed is the maximum load expected to occur in an extrapolated 20 lifetimes. This is a one time static load requirement. These requirements are coupled with assumptions that the damage occurs in the most critical location and that the assumed load is coincident with the worst probable environment.

In developing the requirements, the probability of encountering undetected or undetectable impact damage above the 100 ft-lb energy level was considered sufficiently remote that when coupled with other requirements a high level of safety was provided. For the detectability requirement, it is assumed that having greater than 0.10 inch in depth will be detected and repaired. Consequently, the load requirement is consistent with those for metal structure with damage of equivalent levels of detectability. Provisions for multiple impact damage, analogous to the continuing damage considerations for metal structure, and for the lesser susceptibility of interior structure to damage are also included.

In metal structure, a major concern about damage tolerance is not only initial damage but also growth of damage prior to the time of detection. Consequently, much development testing for metals has been focused on evaluating crack growth rates associated with defects and damage, and the time for the defect/damage size to reach residual strength criticality. Typically, the critical loading mode has been in tension. Crack growth, even at comparatively low stress amplitudes, may be significant. In general, damage growth rates for metals are consistent and, after test data has been obtained, can be predicted satisfactorily. Thus, knowing the expected stress history for the aircraft, inspection intervals have been defined that confidently ensure crack detection before failure.

By contrast, the fibers in composite laminates act to inhibit tensile crack growth. Through-thickness damage growth occurs only at relatively high stress levels. Consequently, through-the-thickness damage growth in composites has generally not been a problem. In-plane damage growth associated with delaminations or impact damage, however, must be considered. Detection again, is an important consideration. Unlike cracks in metal, growth of delaminations or impact damage in composites, will probably not be detected if it does occur. Additionally, interlaminar damage growth in composites, especially that associated with impact damage, cannot be predicted satisfactorily. This has been considered in development of the referenced Air Force specification. Hence, it is required that damage be monitored for growth during cyclic tests which are conducted during development and validation. Resulting design values must be established with sufficient margins to ensure that damage growth due to repeated loads will not occur.

The damage tolerance design procedures for civil/commercial aircraft are expressed more generally but with equal effectivity. Guidelines are addressed by the FAA in Federal Aviation Regulation (FAR) 25 and in Advisory Circular AC-107A. Relative to impact damage, the FAA guidelines states "It should be shown that impact damage that can be realistically expected from manufacturing and service, but not more than the established threshold of detectability for the selected inspection procedure, will not reduce the structural strength below ultimate load capability. This can be shown by analysis supported by test evidence, or by tests at the coupon, element, or subcomponent level." The guidance is to ensure that structure with undetected damage will still meet ultimate strength requirements. Again, it assumed that more obvious damage will be detected in a timely manner and will be repaired. The difference in the Air Force specification and the FAA guideline is primarily in the residual strength value. Also, while the Air Force specification assumes visual inspection, the FAA guideline leaves the inspection method to be selected. Consequently, since specifications and guidelines differ with the type of aircraft, the manufacturer must be aware of the differences and apply those guidelines and specifications appropriate to the situation.

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4.11.1.4 *Material effects on damage tolerance.* The ability of composite structures to resist or tolerate damage is strongly dependent on the constituent resin and fiber material properties and the material form. The properties of the resin matrix are most significant and include its ability to elongate and to deform plastically. The area under a resin's stress-strain curve indicates the material's energy absorption capability. Damage resistance or tolerance is also related to the material's interlaminar fracture toughness, G, as indicated by energy release rate properties. Depending on the application $G_{(I)}$, $G_{(II)}$, or $G_{(III)}$ may dominate the total G calculation. These parameters represent the ability of the resin to resist delamination, and hence damage, in the three modes of fracture. The beneficial influence of resin toughness on both damage resistance and tolerance, has been demonstrated by tests on new toughened thermoset laminates and with the tougher thermoplastic material systems.

Investigations have been conducted on the effect of fiber properties on impact resistance or tolerance. In general, laminates made with fabric reinforcement have better resistance to damage than laminates with unidirectional tape construction. Differences among the graphite fiber tape laminates, however, are small. Some studies have been made of composites with hybrid fiber construction, that is, composites in which two or more types of fibers are mixed in the layup. For example, a percentage of the graphite fibers are replaced with fibers with higher elongation capability, such as fiberglass or aramid. Studies in both cases have shown improvement in residual compression strength after impact. Basic undamaged properties, however, were usually reduced.

In thin gage structures such as a two- or three-ply fabric facesheet sandwich construction, materials can have a significant effect on damage tolerance. Investigations in general have shown that compression strength (both before and after impact) increases with the fiber strain-to-failure capability within a particular class of materials. Higher strain capability aramid or glass fiber structures tend to be more impact resistant than high-strength carbon fiber structure. However, the compressive strengths of the undamaged and damaged aramid and glass structures are lower than that of carbon. Structure incorporating high-modulus, intermediate-strength carbon fibers, with higher strain-to-failures offer a significant impact resistance while retaining higher strength.

In thermoset material systems, the nominal matrix toughness variations influence the impact resistance of thin gage structures but generally to a lesser extent than in thicker structures. For thermoplastic material systems, however, the generally much larger increase in the fracture toughness (G_{I_c}, G_{II_c}, etc.) of the resins do translate into significant impact resistance and residual strength improvements. For sandwich structures, where failure is not core-dependent, core density has been shown to have little effect on impact resistance. However, core density can have a significant effect on the residual strength of the sandwich if the sandwich mode is core-dependent (i.e., face wrinkling).

Methods such as through-thickness stitching have also been used to improve damage resistance. The effect has been to reduce the size of internal delaminations due to impact and similar to the hybrid high elongation fibers, arrest damage growth. Tests involving conventional graphite/epoxies have shown increases in the residual strength of up to 15% for comparative impact energy levels. The process is quite expensive, however, and probably should be considered for application in selected critical areas only. Additionally, the stitches tend to cause stress concentrations and the tensile strength, transverse to the stitching row, is usually reduced.

There are also other properties of composites that contribute to their damage resistance/tolerance. In contrast to metals, which have a Poisson's ratio typically near 0.3, the Poisson's ratio for composite laminates may vary widely, dependent on the particular combinations of ply layup orientations. Consequently, Poisson's ratios may be unbalanced between different sets of plies within a laminate. When the laminate is stressed, these imbalances may create high internal stresses. The consequence is that these stresses may add to damage-induced stresses thereby increasing the tendency for damage

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growth or causing a further reduction tensile or compression strength. Hence, ply stacking sequences that produce high internal stresses when the laminate is loaded, should be avoided.

As an example, consider the influence of laminate stacking sequence (LSS) on compression after impact properties. The LSS can affect compression after impact strength (CAI) in several ways. First, the bending stiffness of a laminate and failure mechanisms that occur during an impact event is strongly dependent on LSS. Load redistribution near the impact site is dependent on the distribution of damage through the laminate thickness (e.g., the LSS of sublaminates affects their stability). Finally, damage propagation leading to final failure also depends on LSS, as was the case for notched tensile strength.

When impact damage is dominated by fiber failure (e.g., Reference 4.11.1.4(a)), it is desirable to stack primary load carrying plies in locations shielded from fiber failure. Since fiber failure typically occurs first near outer surfaces, primary load carrying plies should be concentrated towards the center of the LSS.

Many of the impact damage states studied in the past have been dominated by matrix failures. The creation of matrix cracks and delaminations which combine to form sublaminates depends strongly on LSS (Reference 4.11.1.4(b)). Homogeneous sacking sequences have been found to lead to characteristic damage states which repeat through the laminate thickness. Alternatively, plies can be stacked in a sequence which concentrates damage in specific zones on the laminate.

Recent methods have been developed and verified to predict the effects of known impact damage states on CAI (References 4.11.1.4(c) - (e)). This analysis involves four steps. First, the damage state is characterized with the help of NDI and the damage is simulated as a series of sublaminates. Second, sublaminate stability is predicted with a model that includes the effects of unsymmetric LSS. Third, the in-plane load redistribution is calculated with a model that accounts for structural geometry (e.g., finite width effects). Finally, a maximum strain failure criterion is applied to calculate CAI. Figure 4.11.1.4(a) shows typical results from this analysis procedure.

Interlaminar toughness is crucial to the extent of damage created in a given impact event; however, the CAI of laminates with equivalent damage states (size and type) was found to be independent of material toughness (References 4.11.1.4(c) - (e)). The model from Reference 4.11.1.4(c), which accounts for the in-plane stress redistribution due to sublaminate buckling, has worked equally well for tough and brittle resin systems studied. Since delamination growth may be possible with some materials and LSS, a more general model would also account for out-of-plane stresses.

Experience to date suggests that a homogeneous LSS might be best for overall CAI performance dominated by matrix damage (Reference 4.11.1.4(b)). Figure 4.11.1.4(b) shows experimental data indicating that LSS has a strong effect on CAI. The combined influence of impact damage resistance and CAI performance is evident in the figure.

Basic information on laminate stacking sequence effects is found in Section 4.6.5.

4.11.2 Durability. Durability of a structure is its ability to maintain strength and stiffness throughout the service life of the structure. A structure must have adequate durability when subjected to the expected service loads and environment spectra to prevent excessive maintenance, repair or modification costs over the service life. Thus, durability is primarily an economical consideration.

The major factors limiting the life of metallic structure are corrosion and fatigue. In composites, it has been demonstrated that one of the most common damage growth mechanisms is intercracking and fiber breakage usually occurs. Also, in three-dimensional composite structures out-of-plane loadings

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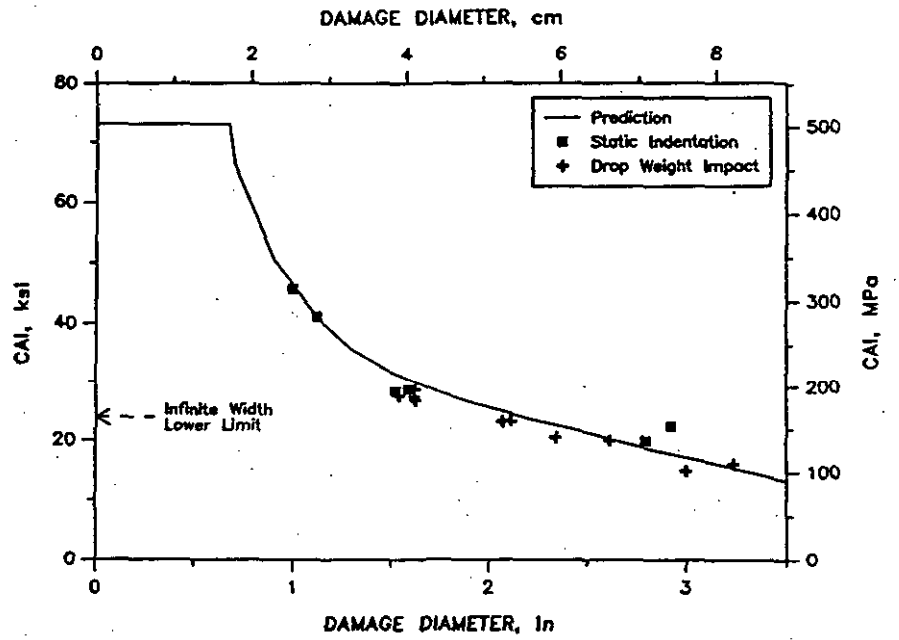


FIGURE 4.11.1.4(a) *Analysis and test results for 5 in. wide specimens, AS6/3501-6, [45/0/-45/90]$_{5S}$, and ply thickness = 0.0074 in. (from Reference 4.11.1.4(d).*

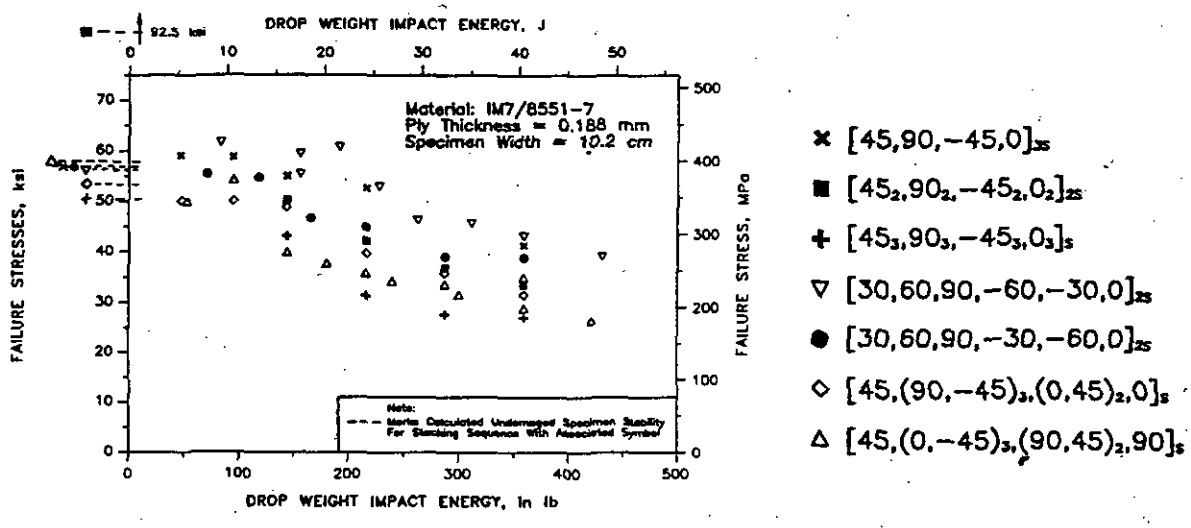


FIGURE 4.11.1.4(b) *Test data for CAI performance as a function of LSS (from Reference 4.11.1.4(e).*

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can either cause failure directly or induce failure indirectly. Because delamination growth is an important growth mechanism, composites are most sensitive to compression dominated fatigue loading.

A second common composite fatigue failure mode is fastener holes wear caused by high bearing stresses. In this failure mode, the hole gradually elongates may lead to a bearing failure or internal load redistribution.

Fatigue behavior of composites has been extensively investigated both experimentally and analytically. A partial list of these investigations is given as References 4.11.2(a) through 4.11.2(aa). In general, composites under in-plane loads have relatively flat fatigue stress-life (S-N) curves with high fatigue thresholds (endurance limits). This behavior is observed for various test specimen types from laminate to structural element. The fatigue threshold for a commonly used carbon/epoxy composite can be as high as 70% of the static strength under constant amplitude cyclic loading. The value of fatigue threshold is even higher when the composite is tested under applications spectra, such as that of a typical fighter aircraft spectrum (References 4.11.2(r) and (s)). Experimental data generated in Reference 19 showed that the typical fatigue failure mode was quasi-static failure when open hole specimens were tested under compression dominated fatigue spectrum loading. However, in high-cycle applications such as rotorcraft, fatigue structure endurance limits at 10^7 to 10^9 cycles must be established and these data are not readily available.

Accompanying the flat S-N curve is the significantly higher fatigue life scatter for composites compared to metals. Extensive statistical data analyses were conducted in References 4.11.2(ab) through (ae) to characterize the fatigue life data scatter of composites. The data compiled in these references included a variety of test variables such as geometry, environment and loading mode. The results of these analyses showed that the mean Weibull shape parameter for carbon/epoxy coupon data is approximately 2. This value is compared to 4.4 for commonly used aluminum alloys under constant amplitude fatigue loading and 7.7 under spectrum loading. This high variability in fatigue life is considered typical to most carbon fiber resin matrix composites and must be fully realized in order to assure adequate durability.

Traditionally, metal airframe structures are fatigue tested under spectrum loading to a minimum of two lifetimes to assure adequate durability. Based on the material variability observed in a commonly used aluminum alloys, a 2-4 lifetime fatigue test provides a very high structural reliability (approximately 0.999). However, because of the high variability of fatigue life test data observed in composites a 2-4 lifetime life factor provides a very low reliability (approximately 0.333). Therefore, the 2-4 lifetime fatigue test may not universally assure adequate durability for a composite structure. However, the use of significantly larger life factors in full-scale durability test programs may not be economically feasible. Alternate approaches may be used to reduce the test duration and at the same time demonstrate adequate reliability. Since these methods continue to evolve, early discussion with appropriate certification agency is recommended.

The load enhancement factor approach, evaluated in References 4.11.1 (ac) and (ad), is one of several alternative durability testing approaches. The objective of this approach is to increase the applied loads in the fatigue test so that the same high level of reliability can be achieved with a shorter test duration. In this approach, the fatigue life data variability, the residual strength scatter and the characteristics of composite fatigue S-N curves are integrated into a mathematical relationship. The required load enhancement factor and the test duration are then determined based on the material variability parameters and the number of tests. A simple method for determining the load enhancement factor for any composite material and failure mode is described in Reference 4.11.2.(ae).

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A second approach, the ultimate strength approach, takes advantage of the excellent fatigue response of composites and assuming no damage growth during the lifetime of the structure. The objective of the ultimate strength approach is to have operating stresses below the fatigue threshold or endurance limit, depending upon application. This is possible in practice because composites have flat S-N curves where fatigue threshold is a high proportion of ultimate strength. This approach and other durability testing approaches, such as residual strength approach, are described in detail in References 4.11.2(ac) and (ad).

4.11.2.1 *Design development/certification implications.* The high cost of full-scale structural test prohibits generation of sufficient data for reliability analysis. Thus, for meaningful interpretation of full-scale tests, a building-block approach is desirable for certification of composite structures. This approach fully utilized coupon, element, subcomponent and component level test data so that limited full-scale structural test data can be interpreted statistically. Use of a building-block approach is discussed in References 4.11.2.1(a) through (d). The number of tests decreases from the coupon level to the component level. A relatively large number of tests is required at the coupon level to establish the data scatter and design allowables for different loading modes, failure modes and environments. A smaller number of tests is required at the element and subcomponent level to determine failure mode interaction and to demonstrate the variability in structural response. This information is then used for design and interpretation of the full-scale structural test program. The three levels of fatigue tests--allowable, design development and full-scale--are discussed in the following paragraphs.

The purpose of design allowable tests is to evaluate the material scatter and to establish fatigue life parameters for structural design. Laminate lay-up, stacking sequence effects, ply drop-offs, out-of-plane loadings, fastener holes, cutouts, etc. should be addressed at this level. Because composites are environmentally sensitive, design allowables should be obtained for structure. In planning the fatigue allowable tests, the main consideration is the test environment. The test environment depends on the relationship between the load/temperature spectrum and the material operation limit. The proper approach is to use simple, conservative constant temperature tests with a constant moisture level. A minimum of three stress levels for each test condition is required to fully characterize the fatigue behavior within the fatigue sensitive stress range. Fatigue tests should be conducted until fatigue failure occurs, except at the lowest stress level. At this stress level, because the fatigue threshold is approached, long life is expected. To reduce the test time on extremely long fatigue tests (greater than 10^6 cycles of 24000 flight hours), test may be censored at a specified lifetime. The stress level used in the fatigue tests should be selected so that the fatigue threshold can be established. For common graphite/epoxy composites under typical spectra, the threshold stress level would be approximately 60% of the mean static strength. This may not be true for rotorcraft high cycle (10^7 - 10^9) fatigue structures which require special evaluation to establish thresholds. The test data should be interpreted for B-basis life/stress relationship (see for example Reference 4.11.2(q)).

Several factors determine the test complexity of composite design development tests. These are: structural geometry complexity, hygrothermal environment simulation, fatigue load spectrum simulation including load rates, and mixed composite/metal structure interaction. The levels of complexity in the design development testing should be functions of the design feature being validated and the failure mode. Special attention should be given to correct failure mode simulation since failure modes are frequently dependent on the test environment. In particular, the influence of complex loading on the local stress at a given design feature must be evaluated. In composites, out-of-plane stresses can be detrimental to structural integrity and, therefore, require careful evaluation.

The environmental complexity necessary for fatigue design development testing will depend on the structural hygrothermal history. Three factors must be considered. These are: structural temperature for each mission profile, the load/temperature relationships for the aircraft, and the moisture content as a function of the aircraft usage and structure thickness. In order to obtain these data, it is

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necessary to derive the real time load-temperature profiles for each mission in the aircraft's history, unless a simple end-of-lifetime moisture content and applicable temperature can be used. These relationships will have a significant influence on the environmental fatigue test requirements.

An example of this testing philosophy is given in Reference 4.11.1.1(e). These relationships strongly depend on the aircraft type, configuration and mission requirements and must be carefully developed on a case by case basis. The structural material should be selected to meet these mission requirements without exceeding the material operating limit.

No significant load sequence effect on fatigue life has been observed in composite materials. However, studies on fixed wing aircraft structural load spectrum variations have shown that carbon/epoxy composites are extremely sensitive to variation in the number of high loads in the fatigue spectrum (Reference 4.11.2(r)). In contrast, truncation of low load does not significantly affect fatigue life. Therefore, high loads in the fatigue spectrum must be carefully simulated in developing load spectrum for fatigue testing. Low loads may be truncated to save test time. It has been observed that the temperature spectrum has not significant effects on fatigue life, however, temperature spectrum truncation should be a consideration in planning fatigue tests. Rotorcraft exception from this generalization. Fatigue testing of mixed metal/composite structure may introduce conflicting requirements and should be evaluated on an individual basis.

The number of replicates for each design development test should be sufficient to verify the critical failure mode and provide a reasonable estimate of the required fatigue reliability. The test effort should be concentrated on the most critical design features of the structure. The number of replicates should be increased for these areas of concern.

Full-scale durability test may not always be necessary for structures with non-fatigue critical metal parts, provided the design development testing and full-scale static test are successful. For mixed composite/metal structures, with fatigue critical metal parts, a two lifetime ambient test should be required to demonstrate durability validation of the metal parts.

4.11.3 Damage resistance. In its normal operation, the aircraft can be expected to be subjected to potential damage from sources such as maintenance personnel and tools, runway debris, service equipment, exposure to hail and lightning, etc. Even during initial manufacturing and assembly, parts are subject to dropped tools, bumps during transportation to assembly locations, etc. The aircraft structure must be able to endure a reasonable level of such incidents without requiring costly rework or downtime. Providing this necessary damage resistance is an important design function. Unfortunately for the designer, providing adequate damage resistance may not always be the most popular task. Resistance to damage requires robustness. This requirement commonly necessitates the addition of extra material above that necessary to carry the structural loads. As a result, there are many pressures to compromise because of competing goals for minimum weight and cost.

In order to establish minimum levels of damage resistance, various requirements for aircraft structure have been identified. The Air Force requirements are defined in their General Specification for Aircraft Structures, AFGS-87221A. In general, the Specification defines the type and level of low energy impact which must be sustained without structural impairment, moisture ingestion or a requirement for repair. It provides provision for such incidents as dropped tools, hail, and impact from runway debris. The aircraft is zoned depending on whether the region has high or low susceptibility to damage and its required damage resistance defined by other government agencies, industry companies and the Aerospace Industries Association (AIA).

In defining the requirements for damage resistance, the type of structure is pertinent. For example, the level of impact energy which typically must be sustained by honeycomb sandwich control surfaces

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without requiring repair or allowing moisture ingestion is quite low, e.g., 4 to 6 in-lb. This is partially because these parts must be kept very light, consequently, damage resistance has admittedly been compromised. Repair is facilitated somewhat because these parts can usually be readily replaced with spares while repairs are being accomplished in the shop. Because of their light construction, however, they must be handled carefully to prevent further damage during processing or transport. By contrast, the damage resistance requirement for primary laminate structure, which is not normally readily removable from the aircraft, is much higher, e.g., 48 in-lb. In addition to such impact induced loads, there also needs to be requirements of resistance to damage from normal handling and step loads that might be encountered in manufacturing and airline environments. Suggested for these have been the following:

Handling loads:

Difficult access	50 lb over a 4 sq-in area
Overhead easy access	150 lb over a 4 sq-in area

"Difficult access" is interpreted as finger tips only.
"Overhead access" is interpreted as the ability to grip and hang by one hand.

Step loads:

Difficult access	300 lb over a 20 sq-in area
Easy access from above	600 lb over a 20 sq-in area

"Difficult access" is interpreted as allowing a foothold on a structure with difficulty.
"Easy access from above" is interpreted as allowing a 2g step or"hop" onto the structure. The area of 20 square inches represents a single foot pad.

There are certain damage susceptible regions of the airplane that require special attention. Examples of these are the lower fuselage and adjacent fairings, lower surfaces of the inboard flaps and areas around doors. These need to be reinforced with heavier structure and perhaps fiberglass reinforcement, i.e., in lieu of graphite. In addition to the above, structure in the wheel well area needs special attention because of damage susceptibility from tire disintegration. Similarly, structure in the vicinity of the thrust reversers is damage prone due to ice or other debris thrown up from the runway.

Minimum weight structure, such as that used for fairings, can cause excess maintenance problems if designed too light. For example, sandwich with honeycomb core that is too low density is unacceptable for commercial aircraft application. Also, face sheets must have a minimum thickness to prevent moisture entrance to the core. The design should not rely on the paint to provide the moisture barrier. Experience has shown that too often the paint erodes or is abraded and then moisture enters.

Honeycomb sandwich areas with thin skins adjacent to supporting fittings are particularly vulnerable to damage during component installation and removal. Consequently, use solid laminate construction within a reasonable working distance of fittings.

Trailing edges of control panels are most vulnerable to damage. The aft four inches are especially subject to ground collision and handling and also to lightning strike. Repairs in this region can be difficult because both the skins and the trailing edge reinforcement may be involved. A desirable approach for the design is to provide a load carrying member to react loads forward of the trailing edge, and material for the trailing edge, itself, that will be easily repairable and whose damage will not compromise the structural integrity of the component.

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In general, damage resistance is improved by thicker laminates and, in the case of sandwich, by the use of denser core. The use of a core with some minimum thickness is also desirable. The reason is that this provides a degree of protection of the inner face, which if it does become damaged, is difficult to repair. The use of reinforcement fibers with higher strain, such as fiberglass or aramid as opposed to graphite in laminates, also provides improved damage resistance. Similarly, improvement is gained by the use of higher modulus, higher strain graphite fibers.

A most effective method for obtaining improved damage resistance is by the use of tougher resin systems. Notable demonstrations have been made comparing tough thermoplastic resin laminates with less tough conventional thermosets. The energy required to initiate damage in the thermoplastics, as measured by instrumented impactors, was much higher. Additionally, when damage did occur, the damaged region was much smaller. Likewise, in service demonstrations, the thermoplastics have shown marked damage resistance improvement. An example is the tests on comparative landing gear doors conducted on F-5 and T-38 aircraft by Northrop. Similarly, substantial improvements are expected from the use of toughened epoxy or bismaleimide systems as opposed to the untoughened resins.

Other items that either improve damage resistance or aid in the repair include having a layer of fabric as the exterior ply of tape laminates. The outer fabric ply is more resistant to scratches and abrasion and allows drilling of the laminate without the otherwise occurrence of fiber breakout.

Laminate edges should not be positioned so they are directly exposed to the air stream since they are then subject to delamination. Options include:

a) Provide non-erosive edge protection such as a cocured metal edge member.
b) Provide an easily replaceable sacrificial material to wrap the edges.
c) Locate the forward edge below the level of the aft edge of the next panel forward.

High energy lightning strikes can cause substantial damage to composite surface structure. Resistance at fasteners and connections, in particular, generates heat that causes burning and delaminations. Minor attachments, also, can cause significant damage, particularly to the tips and trailing edges. The following are guidelines to reduce the repair requirement:

a) At critical locations, provide easily replaceable conductive material with adequate conductive area.
b) Provide protection at tips and along trailing edge spans.
c) Make all conductive path attachments easily accessible.

Another item, although not directly damage resistance, involves the judicious use of blind fasteners. Such fasteners that must be drilled out for removal result in a high percentage of damaged holes. A suggestion is to use nut plates with reusable screws for closeout panel attachment, where access to the far side of a panel is restricted or otherwise where use of a normal type fastener is prevented.

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4.12 CRITICAL FLUIDS SENSITIVITY AND EVALUATION.

4.12.1 Background. Historically, the concern over exposure of structural composite materials to commonly encountered service related fluids other than water or moisture has not been a major concern. This is because the majority of structural composites have had an epoxy resin matrix which has inherently been very fluid resistant. With the epoxies, in general, the allowance for property degradation caused by absorption of atmospheric moisture has been sufficient to cover degradation which might be caused by other pertinent fluids, e.g., fuel, hydraulic oil, etc. Although epoxy resin systems are subject to accelerated degradation in the presence of highly acidic media, the majority of service fluids tend to be basic in nature, e.g., cleaning solvents and hydraulic fluids. The poor resistance of epoxies to methylene chloride, a common ingredient in paint strippers, is an exception. Methylene chloride also aggressively attacks other structural polymers. Consequently, the use of chemical paint strippers on polymer matrix composites is generally not allowed.

With due consideration of the above, it is still important to evaluate the resistance of new polymer materials to fluids with which they might come in contact. Many new epoxies have components, added to improve properties such as toughness, which might affect their solvent resistance. Many other polymers, which have different solvent sensitivities, are also now being used or are being considered for use. An example of a problem encountered in the past was that associated with the developmental evaluation of polysulfone thermoplastic structural parts and their abandonment due to poor resistance to phosphate ester based hydraulic fluids (Reference 4.12.1(a)). Some other structural thermoplastics, although they have excellent resistance to moisture and hydraulic oils, have poorer resistance to fuels. Fuels with higher aromatic content, e.g. JP-4 as compared to JP-8, seem to cause the worst problems (Reference 4.12.1(b)). In the referenced case, the fuel exposure seemed mostly to effectively lower, the material's (PEEK) glass transition temperature (Reference 4.12.1(c)). The result was comparable lowering of the material's maximum use temperature.

Higher service temperature resin systems such as bismaleimides (BMI's) and polyimides are susceptible to degradation by fluids with high alkalinity. Both polymer formulations are susceptible to a cleavage of the functional imide rings in the presence of high concentrations of hydroxide ions. This is significant for two reasons. First, cleaning solvents and hydraulic fluids used by most airlines are alkaline by nature and second, hydroxide ions are produced locally at the resin boundary during galvanic coupling between carbon fibers and active metals and can cause degradation. The galvanic corrosion situation should be satisfactorily manageable with an attentive design. The incorporation of an isolation mechanism such as a resin/fiber ply between the carbon/resin and metal structure is one approach to mitigating the risk associated with the electrolytic driven degradation. Exposure can be lessened by providing drainage, etc. It is important that the laminate edges be well sealed if there is exposure in a sump area. In general, the exposure of these materials to alkaline solutions may be sufficiently incidental that this also may not be a problem.

The evaluation procedure given below is suggested to assess the suitability of polymer resin systems for application where they might be exposed to a harmful fluid environment.

4.12.2 Evaluation. The evaluation should account for different exposure levels of aircraft structure to fluids. Two fluid exposure classifications are suggested:

Group I — Fluids that have the potential for *pooling* or will contact the material for an extended period of time.

Group II — Fluids that are *wiped on and off* (or evaporate) or will not contact the material for an extended period of time.

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Group I

JP-4 Jet Fuel	MIL-T-8624
JP-5 Jet Fuel	MIL-T-8624
JP-8 Jet Fuel	MIL-T-83133
Hydraulic Fluid	MIL-83282
	MIL-H-8606
PAO (Poly Alphaolefin) Cooling Fluid	MIL-C-87252
Engine Lube Oil	MIL-L-23699
Lubricant	MIL-L-7808
Ethylene Glycol/Urea Deicer (Class I)	MIL-D-83411
Sump Water (JRF/3% Slat Water)	MIL-S-8802
Hydraulic Fluid	*
Methylene Chloride (Grade A)	MIL-D-6998**
SO_2/Salt Spray	**

* Monsanto low-density aviation hydraulic fluid, commercial
** Navy requirement

Group II

Alkaline Cleaner (Types 1 and 2)	MIL-C-87936
MEK Washing Liquid	TT-M-261
Dry Cleaning Solvent (Type 2)	PD 680
Hydrocarbon Washing Liquid	TT-S-735
Freon TF Cleaning Liquid (Type 2)	MIL-C-81302
1,1,1-Trichloroethane Degreasing Agent	MIL-T-81533
Polypropylene Glycol Deicer (Type 1)	MIL-A-8243
Isopropyl Alcohol Deicing Agent	TT-I-735
Decontaminating Agent	MIL-D-50030

Exposure by immersion prior to test or to evaluate weight loss is also recommended at two levels:

Group I — Immerse material in fluid until it reaches equilibrium weight gain (saturation).

Group II — Immerse material in fluid for 15 days to determine worst case effects. Then follow-up with tests that simulate a more realistic exposure including accidental extended exposure.

Both mechanical and physical testing should be done. Mechanical testing should include open hole compression tests on quasi-isotropic lay-up specimens and ±45° tension specimens. The open hole compression test has a meaningful relation to design values and is sensitive to matrix degradation. The ±45° tension test is an industry standard for comparison of matrix properties. It is a sensitive test which will identify "potentially" harsh fluids. It provides an indication of whether necessary shear stiffness has been retained to ensure acceptable resin to fiber property transfer. A shear modulus of 0.3 Msi is a recommended minimum. A minimum of five specimens should be tested after exposure at room temperature and at the maximum use temperature. The results should be compared with unexposed controls.

A more economical alternative to open hole compression and ±45° tension testing is interlaminar or short beam shear tests. These specimens are easily fabricated, machined, conditioned, and tested. Although not as generally related to design properties, short beam shear tests are sensitive to matrix

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degradation and can be valuable indicators for material evaluation. As with the ±45°tension tests, results after exposure should be compared to unexposed controls at room and elevated temperature to obtain fluid exposure effects.

Physical testing should include weighing to measure weight change, photomicrographs to examine for microcracks, and, where practical, scanning electron microscopy to examine for surface crazing. Relative to the former, it must be warned that because a saturation condition has apparently been reached, it *does not* automatically follow that further degradation of properties has ceased. Especially where new resin systems are involved, test with long term exposure to critical fluids should be conducted. An example of such testing is given in Reference 4.12.2. Due to the long exposure times involved, these tests should be started early in the evaluation process.

It has been the procedure in the past that if water or moisture has been proven to be the most property-degrading fluid, then fluid exposure tests involving other than moisture conditioning were not included in subsequent design testing. Such a procedure for Group I pooling fluids is presented in Figure 4.12.2(a). In effect, if the properties of the material after fluid exposure are better than after moisture exposure, then subsequent testing accounts for moisture only. If a fluid other than water is more critical, then subsequent testing must include evaluation with that fluid.

In the case of Group II wipe on/wipe off fluids, the procedure is somewhat different since water is not a good comparison. Consequently, comparison to a resin that has an acceptable service history is recommended. This is illustrated by the decision tree in Figure 4.12.2(b) where comparison with the performance of 3501-6 epoxy resin system is suggested.

4.13 VIBRATION.

4.13.1 Introduction.

4.13.2 Stacking sequence effects. Vibration characteristics of laminated plates are also sensitive to laminate stacking sequence (LSS). As was the case with bending and buckling of laminated plates, complex interactions between LSS, plate geometry and boundary conditions will not allow simple rules relating LSS to vibrations. Instead, such rules must be established for specific structure and boundary conditions. This indicates a need to use proven analysis methods as design tools for predicting dimensional stability of composite structure subjected to dynamic load conditions.

Figure 4.13.2 is one example of the complex interactions between LSS, plate geometry, and the natural frequency in the first vibrational mode.[4] A design equation from Reference 4.7.1.9(c) which was based on analysis from Reference 4.13.2 was used to make the predictions shown in the figure. Note that the relative difference in fundamental frequencies for various LSS changes with plate geometry. Higher frequencies occur for square plates with preferential stacking of ±45° plies in outer layers. The strongest effect of LSS occurs for rectangular plates in which preferential stacking of outer plies oriented perpendicular to the longest plate dimension have the highest fundamental frequencies. Basic information on laminate stacking sequence effects is found in Section 4.6.5.

4.14 COMPUTER PROGRAMS. Numerous programs for finite element analysis and prediction of composite material properties are available. Information on many of these programs can be found in Reference 4.14. In addition, there are programs available from NASA through COSMIC, Computer Software Management Information Center, 112 Barrow Hall, The University of Georgia, Athens,

[4]The LSS used in Figure 4.12.2 were chosen for illustrative purposes only and do not represent optimal LSS for a given application.

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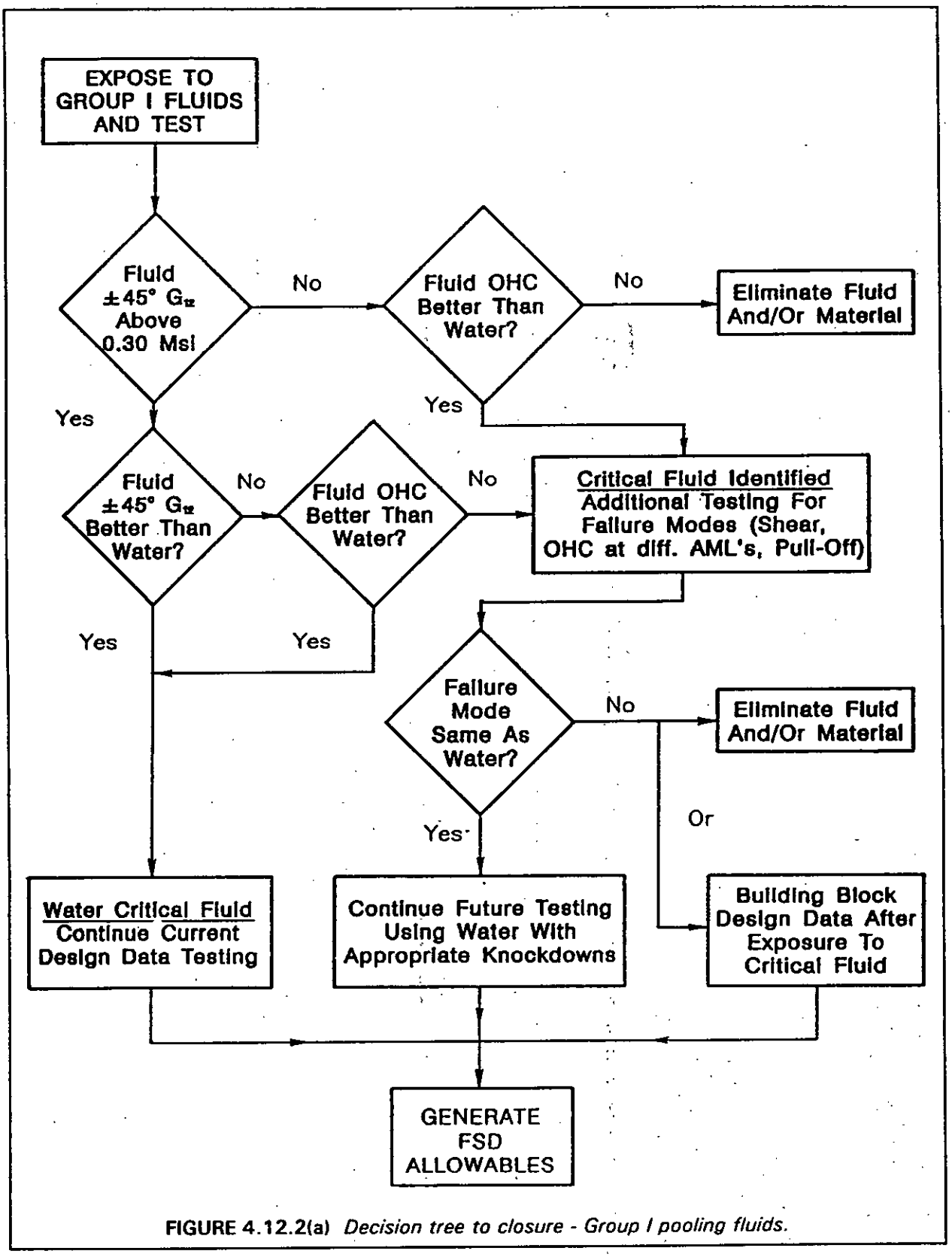


FIGURE 4.12.2(a) *Decision tree to closure - Group I pooling fluids.*

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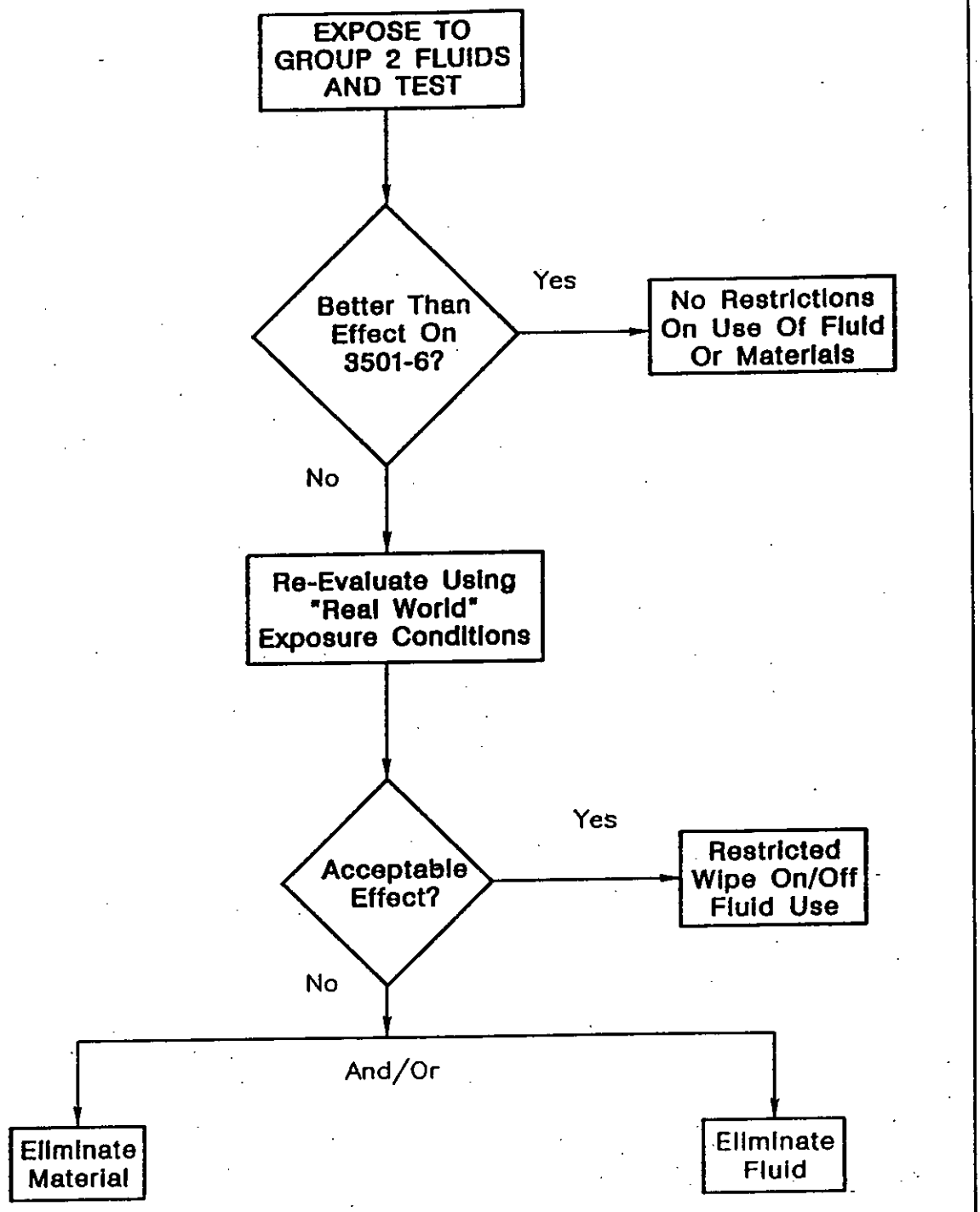


FIGURE 4.12.2(b) *Decision tree to closure - Group II fluids without long term exposure.*

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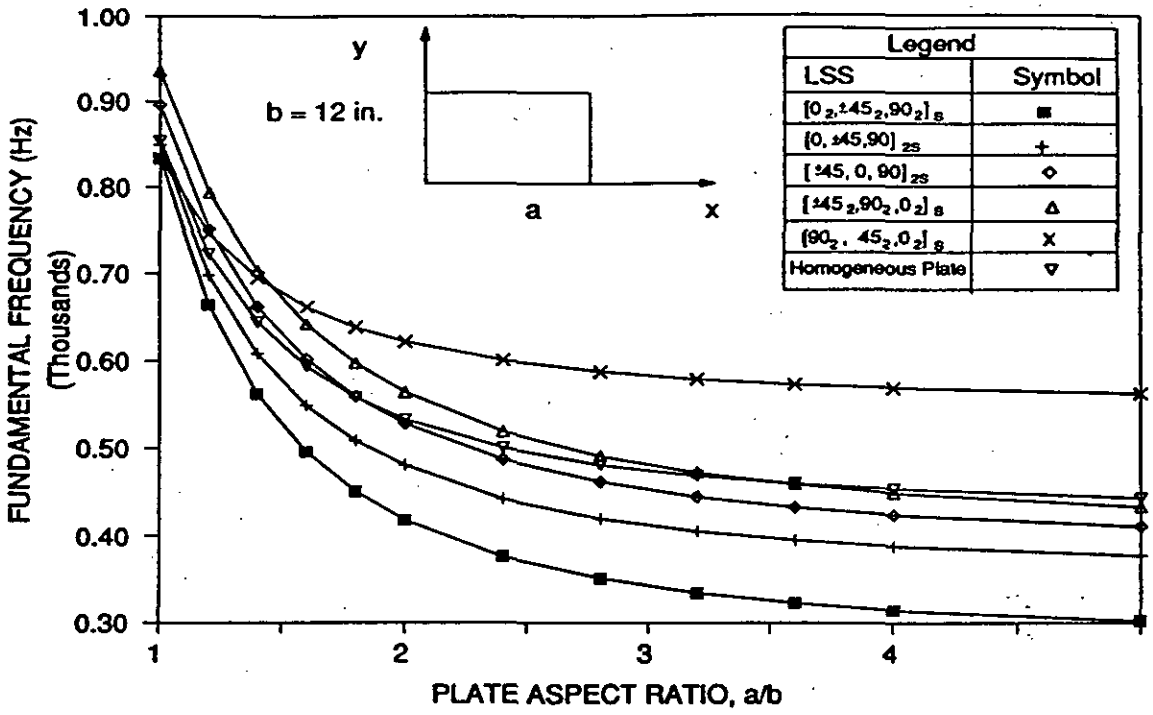


FIGURE 4.13.2 *Vibration analysis results for four sides simply-supported plates with variable aspect ratio.*

Georgia, 30602, (404) 542-3265. It should be noted that the use of and the results from these computer codes rely on the model developed, the material properties selected, and the experience of the user.

4.15 CERTIFICATION REQUIREMENTS.

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Chapter 6. STRUCTURAL RELIABILITY

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6.1 GENERAL INTRODUCTION Reliability is commonly defined (References 6.1(a) and (b)) as "the probability of a device performing its purpose adequately for the period of time intended under the operating conditions encountered". There are four elements to the definition that must be considered. First, *probability* refers to the *likelihood* that a device or structural component will work properly. These terms imply acceptance of some degree of uncertainty. The second element refers to adequate performance. In order to determine whether a component has performed adequately, a standard is needed to define what is meant by adequate performance. The third element is the intended period of time. This is the mission endurance or lifetime of the structure under consideration. The final element of the definition is the operating conditions. Environmental conditions play a large role in reliability of composite materials, particularly polymer matrix composites. Simply stated, structural reliability is a yardstick of the capability of a structure to operate without failure when put into service. In the broadest sense, structural reliability includes events that are safety and non-safety related.

Until recently, structural reliability was not routinely analyzed or quantified in the design process. Reliability was accounted for tacitly by the factor-of-safety approach to design. Also guidelines and lessons learned helped to improve reliability. The structural designer/analyst does not perform a formal risk analysis on newly designed structure. This task is performed by reliability specialists who employ methodologies that are empirically based. The reliability assessment is usually conducted after a drawing or concept is produced and bears little relationship to the structural margin-of-safety.

As implied in the definition, structural failure and, hence, reliability, is influenced by many factors. In its simplest form, the measure of reliability is made by comparing a component's stress to its strength. Failure occurs when the stress exceeds the strength. The larger this gap, the greater the reliability and the heavier the structure. Conversely, the smaller the gap, the lower the r, but the lighter the structure. The gap between stress and strength, enforced by the factor-of-safety, generally produces adequate although unmeasured reliability.

The complications that mask the ability to quantify reliability reside in the stochastic nature of design inputs. The calculations are relatively easy; statistical characterizations of the strength and stress distributions are compared mathematically and a probability of failure calculated. Definition of these distributions however, can be an imposing, if not impossible, task. Each is influenced by many considerations with relatively unknown effects.

The primary purpose for establishing a factor-of-safety for design is to ensure safety. Until recently, no objective analysis has gone into the choice for factor-of-safety. Consequently, no evaluations are performed on the factor-of-safety as new materials or technologies are developed. As suggested by methodologies developed in Reference 6.1(c), these evaluations can now be performed. This fact suggests that future design and design processes might benefit greatly by focusing on reliability targets rather than factors-of-safety. This may be particularly true for composite materials.

The following sections discuss some of the important factors that affect composite structure reliability.

6.2 FACTORS AFFECTING STRUCTURAL RELIABILITY.

6.2.1 Static strength. An aircraft structure's capability to sustain operational flight loads is commonly assessed by comparing material performance parameters to limit or ultimate loads. Limit loads are generally defined as the maximum load expected during the life of the aircraft. Ultimate loads are obtained by multiplying limit loads by the factor-of-safety. Limit loads are derived by considering the extremes of flight envelopes, gross weight, load factors, environments, and pilot inputs. In some cases, the likelihood of encountering limit load is very remote. The 1.5 factor-of-safety used to obtain ultimate design loads from applied loads has been widely accepted by generations of engineers, mostly

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without questioning the origin of the factor. Reference 6.2.1 provides an excellent historical review of the evolution of the 1.5 factor-of-safety in the United States.

From the beginning of flight, occupant safety has been a primary concern in designing manned vehicles and the "factor-of-safety" has been a prominent design criteria. Like many design requirements, the implementation of the 1.5 factor-of-safety evolved over a period of time and was influenced by many concerns.

Design criteria require structures to withstand ultimate loads without failure and limit loads with no permanent deformation. This has led to the impression that the 1.5 factor-of-safety was due to the performance of metals, 2024 in particular. At the time the 1.5 factor-of-safety was established, 2024 aluminum had a ratio of ultimate to yield stress of approximately 1.5. However, in the early 1930'2 when the 1.5 factor-of-safety was formally established by the Air Corps, material properties were not considered. Mr. A. Epstein, who worked for the United States Army Air Corps Material Center from 1929 to 1940, prepared the original Air Corps Structures Specification X-1803 in 1936. Mr. Epstein noted (Reference 6.2.1) that "the factor-of-safety of 1.5 has withstood many moves to alter it, but there was a period in 1939 when the Chief of the Structures Branch of Engineering Division at Wright Field thought seriously of reducing the value of the factor. Newer aluminum alloys were becoming available with higher ratios of yield to ultimate strength and he interpreted the factor as the ratio of ultimate to yield. However, no action was taken when the following explanation was offered: 'The factor-of-safety is not a ratio of ultimate to yield strength, but is tied in with the many uncertainties in airplane design, such as fatigue, inaccuracies in stress analysis, and variations of material gages from nominal values. It might also be considered to provide an additional margin of safety for an airplane subjected to shellfire.'" Thus, while the factor-of-safety does much to promote reliability, it was defined independently of any specific reliability goal.

Generally speaking, composite structures are sized by comparing ultimate internal stresses to statistically reduced material parameters (e.g., B-basis strengths). The internal stresses are a result of applied design ultimate loads (1.5 x DLL). In general the deterministic approach produces adequate reliability, but not necessarily the same as metallic structure. This is because composite materials exhibit different statistical distribution and variation from metals (see Figure 6.2.1). The result is that even though materials may have equivalent B-basis strengths, their reliabilities may be quite different. Reliability-based design procedures may be necessary to account for this difference (see Reference 6.1(c)).

6.2.2 Environmental effects. Composite material components are subjected to a wide range of environments. The operating conditions in which the aircraft must perform are not well characterized. Environmental factors of major importance include a combination of humidity and temperature. Many studies have been conducted to investigate moisture absorption as well as the reduction of mechanical properties due to temperature and moisture exposure. The current approach used to account for environmental factors is to define exposures that are extremes and selectively evaluate by test the effects on material properties. These extremes are then considered to be invariant during the lifetime of the structure. Strength values are reduced to coincide with the environmental extremes.

6.2.3 Fatigue. Composite materials exhibit higher fatigue threshold stresses than metals. Once this threshold is exceeded, composites show more scatter in fatigue than metals and might tend toward lower reliability performance if the composite structures were stressed that highly. Because of this high threshold stress, fatigue is not the limiting factor in the design of composite structures. Design criteria such as damage tolerance limit the stress levels in composite structures to such low values that fatigue does not generally represent a design constraint. However, this is not necessarily true for high-cycle fatigue (e.g., $n > 10^7$) dynamic system components in rotorcraft. (For more information on fatigue or durability of composite structures, see Volume 3, Sections 4.10 and 4.11.2).

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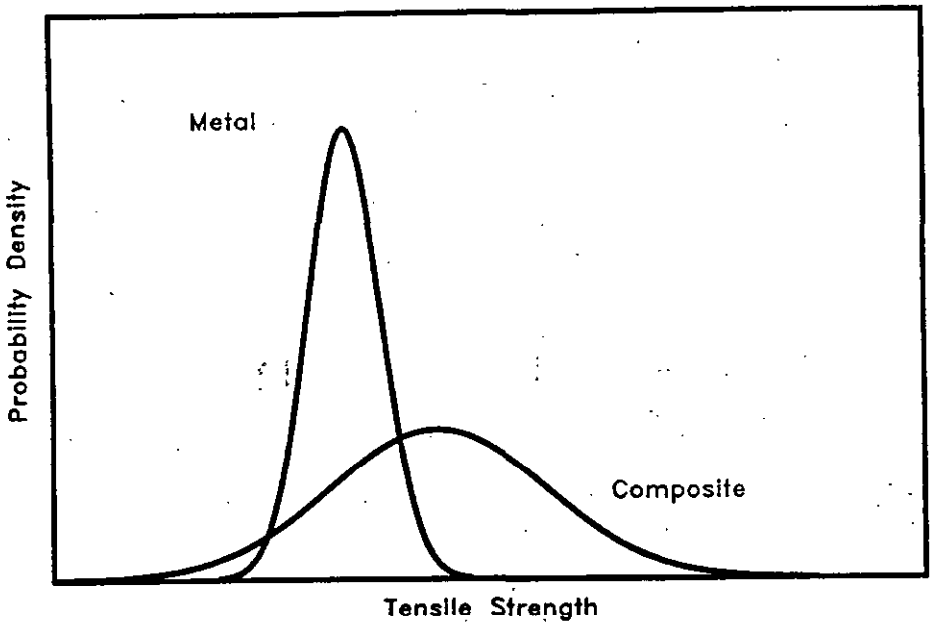


FIGURE 6.2.1 *Composites generally exhibit variation in material performance different from their metallic counterparts.*

6.2.4 Damage tolerance. As stated in Volume 3, Section 4.11.1, "damage tolerance is defined as a measure of a structure's ability to sustain a level of damage or presence if a defect and yet be able to safely perform its operating functions." Damage to composite structures can occur during manufacturing or operational usage. In order to design the structure to operate safely after sustaining such damage, a common practice is to limit the stress allowed in the composite structure. Typically, composite structures are designed to withstand the most severe of either of the following tow conditions: a 0.25-inch open hole in any location at ultimate load or damage sustained when objects of specified size strike the surface (representative of barely visible impact damage threats). Both criteria assume the defect exists for the life of the part. These criteria reduce the allowable strength.

6.3 *RELIABILITY ENGINEERING.* *Reliability is of such concern to the military that specific* requirements are defined in contractual documents. Industry satisfies these requirements by employing engineers who specialize in reliability to guide the design. Currently, structural reliability is being increasingly emphasized. The reasons are twofold. First, advances in materials technology have resulted in higher performance materials that often possess detrimental side effects (e.g., high strength steels that exhibit low fracture toughness). Second, the need for higher vehicle performance has pushed operating stresses to higher levels in order to reduce structural weight.

Structural designs are documented via engineering drawings. Drawings are not "released" until they undergo scrutiny by several technical disciplines. Reliability is one of the concerns that is dealt with by the technical disciplines. Reliability specialists ensure that these concerns are incorporated into the design.

Customer reliability criteria typically specify three goals. These are: Mean Time Between Failure (MTBF), mission reliability, and Failure Modes Effects Criticality Analysis (FMECA). MTBF is measured in unscheduled maintenance events per million flight hours. Mission reliability is an indication of the

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probability of having to abort a flight. FMECA determines the impact of specific failures on mission performance, safety, and utilization.

In addition to supplying input to design, reliability engineering output is supplied to maintainability groups for maintenance man-hour predictions. Their results are used by logistics persons to establish provisioning requirements for spare parts.

6.4 RELIABILITY DESIGN CONSIDERATIONS. The following is a list of general composite structures considerations which provide insight on improved reliability and causes of poor reliability:

- Eliminate/minimize potential galvanic corrosion and/or thermal expansion problems by selecting compatible materials.
- Allow for the difference in thermal expansion when mating composites to metals. The coefficient of thermal expansion for composites is low.
- Assess carefully the use of honeycomb sandwich panels which utilize thin facesheets in areas where Foreign Object Damage (FOD) and bird strikes are likely to occur. Thin facesheets are susceptible to impact damage.
- Protect the structure for possible lightning strikes. Good electrical contact between all metallic and carbon/epoxy structural components must provide for the dissipation of static and lightning-induced electrical currents.
- Fasteners: Use titanium alloy or other materials that are compatible with carbon/epoxy to prevent galvanic corrosion.
- The current ability to detect flaws in composite structures, especially honeycomb, is evolving. *Designs that enhance access for inspection* tend to promote reliability.
- Improved reliability can be obtained by avoiding anomalies such as wrinkling and porosity in integral stiffeners. The ability to detect such flaws is limited.
- Extreme care should be taken during the repair of composite structures. Avoid damaging additional plies during patch or repair operations as it may result in a decline in reliability.
- Variations in manufacturing processes such as curing and machining can be responsible for a range of part strengths thus influencing reliability.
- The supplier's prepreg material should be closely monitored (i.e., acceptance testing) to assure incoming material consistency and conformance to design values.

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Chapter 7. THICK SECTION COMPOSITES

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7.1 INTRODUCTION. Thick-section composites are ones where the effect of geometry (thickness-to-span ratio), material constituents (matrix and fiber stiffness/strength properties, lamination scheme, processing, and service loading exhibit three-dimensional states of stress. For instance, all loadings induce multiaxial stresses into individual plies of composite materials that are made of multi-directional ply laminates, even though the overall loadings may only be uniaxial. When transverse (through-thickness) stresses and strains occur to a significant degree, they must be accounted for in analysis, design, and testing. A significant degree is achieved when these effects contribute to failure (e.g., delamination), excessive deflection, or vibration. Frequently, these stresses and strains induce failures that cannot be accurately predicted by conventional two-dimensional analyses for thin laminates. These two-dimensional analyses are usually based solely on material response data obtained from traditional shear and uniaxial tensile/compressive testing techniques.

For example, thick-section composites made of high stiffness and strength fiber-reinforced plies often exhibit significant transverse shear and transverse normal deformations (the type of three-dimensional stress contributions that are negligibly small in thin laminates). The thickness effect can also be influenced by short wavelength loadings and, in dynamics, high frequency vibrations. These three-dimensional effects are considerably more pronounced in composites than in homogeneous isotropic materials due to their inherently high material compliance in the transverse directions relative to the axial fiber direction. Moreover, composite laminates exhibit much lower strength in the transverse directions and at the ply interfaces, making them particularly susceptible to matrix cracking and delaminations.

Thick-section composites can also be defined from the standpoint of fabrication effects associated with a large number of plies. These process induced stresses are significant and warrant special attention. Fabrication effects of special concern include residual stress, wrinkling, micro-cracking, exotherm, volatile removal, compaction, machining, and mechanical joining and/or adhesive bonding. To minimize these effects, special resins, processing, tooling, and cure cycles may be necessary.

The structural analyst needs to know the multiaxial strength and deformation characteristics for efficient thick composite material design. The full potential of thick composites cannot be realized until the material response under multiaxial service loadings can be established. The design, analysis, and associated material testing of thick composites is still much less developed than the generally accepted thin composite material characterization and applicability.

The step-by-step method for analysis of thick-section composites is illustrated by the flowchart in Figure 7.1.

7.2 MECHANICAL PROPERTIES REQUIRED FOR THICK SECTION COMPOSITE THREE-DIMENSIONAL ANALYSIS. The purpose of this section is to define the three-dimensional (3-D) orthotropic stiffness properties necessary to conduct a 3-D point stress analysis, and the failure strength and strain allowables required to calculate a margin of safety. This section will:

a) Define the stiffness properties currently used to conduct a conventional two-dimensional (2-D) analysis (Volume 1, Section 6.6).

b) Define the additional stiffness properties needed to conduct a three-dimensional (3-D) stress analysis.

c) Define the testing required to experimentally determine the 3-D stiffness properties and the failure strengths and strains for uniaxial loading (Section 7.2.3.1) and multiaxial loading (Section 7.2.3.2)

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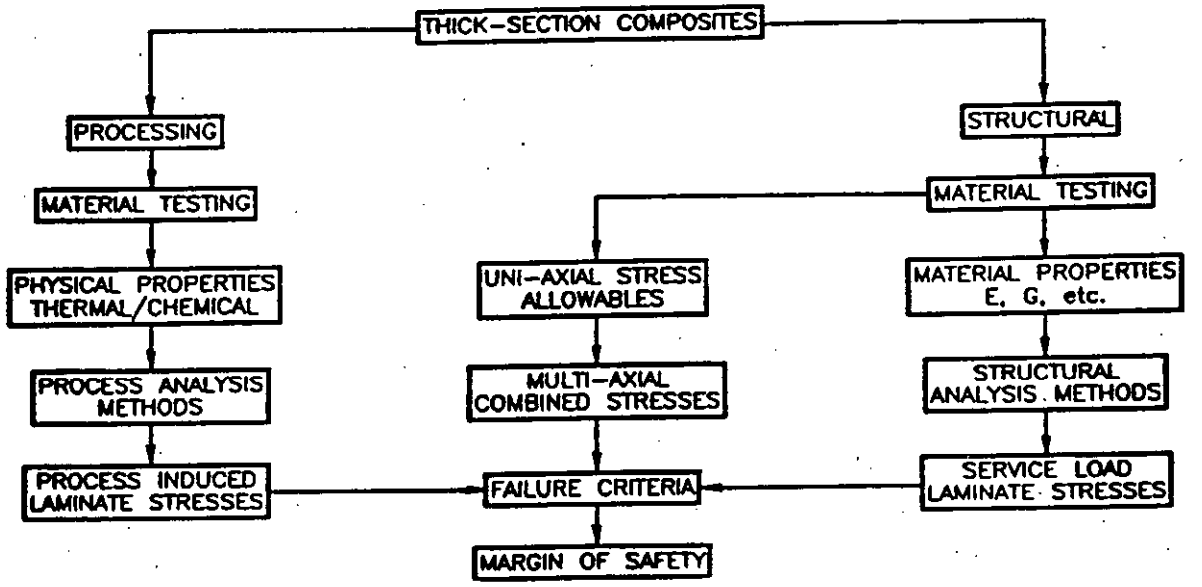


FIGURE 7.1 *Flowchart illustrating thick-section composites analysis method*

d) Discuss the methodology for predicting laminate stiffness properties through the thickness using the 3-D lamina properties (Section 7.2.4).

The symbols and nomenclature used in the handbook (Volume 3, Section 1.3.1) apply to 2-D and 3-D composites and utilize 1, 2, 3 for lamina axes and x, y, z for an oriented laminate axis directions.

7.2.1 2-D composite analysis. The two-dimensional composite analysis procedures (Volume 3, Section 4.3.1) apply when the through the thickness stresses are not significant. For unidirectional laminates that have low stresses in the thickness or 3-direction ($\sigma_3 = \tau_{23} = \tau_{13} = 0$, plane stress), the stress-strain relationship (Reference 7.2.1) is,

$$\{e_{ij}\} = [S_{ij}]\{\sigma_{ij}\} \qquad 7.2.1(a)$$

$$\begin{Bmatrix} e_1 \\ e_2 \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & 0 \\ S_{12} & S_{22} & 0 \\ 0 & 0 & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} \qquad 7.2.1(b)$$

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In terms of the engineering elastic constants obtained by simple tests

$$\begin{Bmatrix} e_1 \\ e_2 \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{12}}{E_1} & 0 \\ -\frac{\nu_{21}}{E_2} & \frac{1}{E_2} & 0 \\ 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{Bmatrix} \qquad 7.2.1(c)$$

The reciprocity relationships for stiffness is

$$\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2} \qquad 7.2.1(d)$$

For the plane stress two-dimensional analysis, the four independent elastic material properties are:

$$E_1,\ E_2,\ G_{12},\ \nu_{12}$$

In-plane failure stress and strain values can be obtained from the same test used for determining the stiffness as discussed in Section 7.2.3.1.

7.2.2 3-D composite analysis. When the stresses and strains in the thickness direction are significant, (applied values are approaching their allowables) the problem requires a three-dimensional orthotropic stress analysis. A 3-D analysis is frequently necessary as the section thickness of a composite increases or when thin sections have out-of-plane loading (bending moment, lateral pressures, etc.) which results in, for example, interlaminar tensile stresses in a corner radius or interlaminar shear stresses in a beam or plate.

7.2.2.1 *Unidirectional lamina 3-D properties.* For the orthotropic unidirectional lamina there are nine independent constants as shown by the following stress-strain relationship (Reference 7.2.1):

$$\begin{Bmatrix} e_1 \\ e_2 \\ e_3 \\ \gamma_{23} \\ \gamma_{31} \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} S_{11} & S_{12} & S_{13} & 0 & 0 & 0 \\ S_{12} & S_{22} & S_{23} & 0 & 0 & 0 \\ S_{13} & S_{23} & S_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & S_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & S_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & S_{66} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{31} \\ \tau_{12} \end{Bmatrix} \qquad 7.2.2.1(a)$$

or in terms of the engineering constants,

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$$\begin{Bmatrix} e_1 \\ e_2 \\ e_3 \\ \gamma_{23} \\ \gamma_{31} \\ \gamma_{12} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & -\frac{\nu_{31}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & -\frac{\nu_{32}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_1} & -\frac{\nu_{23}}{E_2} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{23}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{31}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{Bmatrix} \sigma_1 \\ \sigma_2 \\ \sigma_3 \\ \tau_{23} \\ \tau_{31} \\ \tau_{12} \end{Bmatrix} \qquad 7.2.2.1(b)$$

There are three reciprocal relationships that must be satisfied for an orthotropic material. They are

$$\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2}, \qquad \frac{\nu_{13}}{E_1} = \frac{\nu_{31}}{E_3}, \qquad \frac{\nu_{23}}{E_2} = \frac{\nu_{32}}{E_3} \qquad 7.2.2.1(c)$$

There are nine independent elastic material properties required for an orthotropic lamina

$$E_1,\ E_2,\ E_3,\ G_{12},\ G_{13},\ G_{23},\ \nu_{12},\ \nu_{13},\ \nu_{23}$$

When materials have a different stiffness in tension from in compression, it is common practice to use an average value when the difference is small. If the stiffness difference is significant, use the stiffness (tensile or compressive) that is representative of the application loading.

7.2.2.2 *Oriented orthotropic laminate 3-D properties.* The compliance matrix and associated nine elastic constants required to conduct a 3-D analysis are defined in this section and are for a oriented balanced and symmetric laminate loaded in the x, y, or z direction. Most practical composite laminate lay-ups generally are balanced and symmetric to prevent thermal warpage during processing. If the laminate is unbalanced and unsymmetric, or loaded "off-axis" to the principal orthogonal directions, then the matrix is fully populated with the Chentsov's coefficients ($\mu_{ij,kl}$) and coefficients of mutual influence ($\eta_{ij,}$ $\eta_{i,jl}$) (see References 7.2.1, 7.2.2.2).

The compliance matrix for the balanced and symmetric laminate loaded in the x, y, or z direction is

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$$\begin{Bmatrix} e_x \\ e_y \\ e_z \\ \gamma_{yz} \\ \gamma_{zx} \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} \bar{S}_{11} & \bar{S}_{12} & \bar{S}_{13} & 0 & 0 & 0 \\ \bar{S}_{12} & \bar{S}_{22} & \bar{S}_{23} & 0 & 0 & 0 \\ \bar{S}_{13} & \bar{S}_{23} & \bar{S}_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & \bar{S}_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & \bar{S}_{55} & 0 \\ 0 & 0 & 0 & 0 & 0 & \bar{S}_{66} \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{yz} \\ \tau_{zx} \\ \tau_{xy} \end{Bmatrix} \qquad 7.2.2.2(a)$$

In terms of the effective engineering elastic constants this relationship is,

$$\begin{Bmatrix} e_x \\ e_y \\ e_z \\ \gamma_{yz} \\ \gamma_{zx} \\ \gamma_{xy} \end{Bmatrix} = \begin{bmatrix} \frac{1}{E_x} & -\frac{\nu_{yx}}{E_y} & -\frac{\nu_{zx}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xy}}{E_x} & \frac{1}{E_y} & -\frac{\nu_{zy}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xz}}{E_x} & -\frac{\nu_{yz}}{E_y} & \frac{1}{E_z} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{yz}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{zx}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{xy}} \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{yz} \\ \tau_{zx} \\ \tau_{xy} \end{Bmatrix} \qquad 7.2.2.2(b)$$

There are three reciprocal relationships that must be satisfied by the effective laminate stiffnesses. They are,

$$\frac{\nu_{xy}}{E_x} = \frac{\nu_{yx}}{E_y}, \qquad \frac{\nu_{xz}}{E_x} = \frac{\nu_{zx}}{E_z}, \qquad \frac{\nu_{yz}}{E_y} = \frac{\nu_{zy}}{E_z} \qquad 7.2.2.2(c)$$

There are nine independent effective elastic material constants required for analysis of the oriented laminate,

$$E_x, E_y, E_z, G_{xy}, G_{xz}, G_{yz}, \nu_{xy}, \nu_{xz}, \nu_{yz}$$

7.2.3 Experimental property determination. The current approach for failure analysis of 2-D composites is to experimentally determine the strength and stiffness values for the unidirectional lamina from simple uniaxial tests and use a failure criterion to account for the various load direction interaction to calculate the margin of safety. These uniaxial tests are defined in Section 7.2.3.1 for 2-D and 3-D composites. Another approach is to conduct multi-axial tests that provide loading in proper proportion to simulate the actual application.

7.2.4 Theoretical property determination. In considering the use of theoretical procedures to determine the mechanical properties of composite materials the most fundamental level that can be addressed is that of the individual constituents, or the micromechanics level. A theoretical

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development of composite micromechanics is summarized in Volume 3 Section 4.2.2.1 of this Handbook and in Section 4 of Reference 7.2.4. 3-D laminate properties can be determined from constituent data using micromechanical analyses and these references should be consulted for additional information and references on this topic.

Since properties at the lamina or laminate level are typically used for the analysis of a composite structure, only these properties will be discussed in this section and all analyses considered are linear elastic.

7.2.4.1 *3D lamina property determination*. In Section 7.2, the nine independent elastic material properties required for a 3D lamina based analysis were listed as:

$$E_1,\ E_2,\ E_3\ ,G_{12},\ G_{13},\ G_{23},\ \nu_{12},\ \nu_{13},\ \nu_{23} \qquad 7.2.4.1\ (a)$$

Of these properties E_1, E_2, G_{12} and ν_{12} can be readily generated by conventional experimental methods. Methods for determining out-of-plane properties are discussed in Section 7.2.3. In the absence of experimental data for these properties, the assumption of transverse isotropy in the 2-3 plane is often reasonable. The validity of this assumption has been demonstrated by the experimental data available in References 7.2.4.1(a) - (c). The assumption of transverse isotropy implies

$$E_3 = E_2,\ G_{13} = G_{12},\ \nu_{13} = \nu_{12},\ \text{and}\ G_{23} = \frac{E_2}{2(1+\nu_{23})} \qquad 7.2.4.1\ (b)$$

Even with this simplifying assumption ν_{23} must be measured or estimated for full knowledge of the nine independent elastic material properties.

Values for ν_{23} that have been experimentally determined have been reported in References 7.2.4.1(a) - (c). A value for ν_{23} determined in compression for T300/5208 is reported in Ref. 7.2.4.1(a). A value for ν_{23} determined in tension and compression for T300/5208 is reported in Reference 7.2.4.1(b). Values for ν_{23} determined in compression for AS4/3501-6 and S2/3501-6 are reported in 7.2.4.1(c) and can be found in Table 7.2.4.1.

The need for all nine independent elastic constants does not imply that a 3-D analysis will be sensitive to the choice of the through-thickness material properties just discussed. For instance, a choice of ν_{23} of 0.50 versus 0.40 (a 20% difference) may only result in a 2% difference in the stress or strain results from a laminate or structural analysis. This sensitivity of a particular analysis to a particular material property should be evaluated through a parametric study if the value of the property is uncertain.

Likewise, the use of a linear analysis when certain material properties are extremely nonlinear (i.e., in-plane and through-thickness shear modulus) may not affect laminate or structural analysis and this too should be considered in 3-D analysis.

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Table 7.2.4.1 *3-D elastic constants for carbon and S2 glass reinforced epoxy (Reference 7.2.4.1(c)).*

	AS4/3501-6 59.5% FVF	S2/3501-6 56.5 FVF
E_1 (Msi)[1]	16.48 (3.7)[2]	7.15 (4.0)
E_2 (Msi)[1]	1.40 (3.6)	2.13 (2.2)
E_3 (Msi)	1.40[3]	2.13
ν_{12} [1]	0.334 (3.0)	0.296 (4.1)
ν_{13} [1]	0.328 (1.2)	0.306 (2.8)
ν_{23} [1]	0.540 (1.6)	0.499 (1.4)
G_{12} (Msi)	0.87[4]	0.98
G_{13} (Msi)	0.87[5]	0.98
G_{23} (Msi)	0.45[6]	0.71

[1] E_1, E_2, ν_{12}, ν_{13}, and ν_{23} determined from thick, flat, compression test specimens
[2] coefficient of variation (%)
[3] E_3 assumed equal to E_2
[4] G_{12} determined from $[\pm 45]_{2s}$ tension test
[5] G_{13} assumed equal to G_{12}
[6] G_{23} from assumption of transverse isotropy

7.2.4.2 *3D Laminate Property Determination.* As for the case of a 3D lamina properties, Section 7.2 lists the nine independent elastic material properties required for a 3D laminate based analysis as:

$$E_x, E_y, E_z, G_{xy}, G_{xz}, G_{yz}, \nu_{xy}, \nu_{xz}, \nu_{yz} \qquad 7.4.2.2\text{ (a)}$$

E_x, E_Y, G_{xY}, ν_{xy} can be readily determined by conventional experimental or theoretical methods. Theoretically they can be determined using classical lamination theory as presented in Volume 3 Section 4.3.2 of this Handbook. The determination of the remaining out-of-plane laminate properties present a much greater challenge than for the inplane properties. Little experimental data exists for out-of-plane laminate properties and the test methodologies used to generate them can be described as very specific to the programs they have been used for, such as those in References 7.2.4.1(a) - (c).

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• A number of methods have been developed to theoretically predict the out-of-plane properties based on in-plane lamina properties (References 7.2.4.2(a) - (h)). These methods basically replace a layered inhomogeneous media of orthotropic layers with a homogeneous anisotropic media. This replacement is termed "smearing" and the resulting effective material properties are referred to as "smeared properties". These smeared anisotropic properties are commonly used in the analysis of composite structures. If average, global stress states or average displacements are sufficient for the analysis being conducted then an analysis with smeared properties is all that would be needed. If local stress states are needed then other analyses techniques must be employed such as a "global-local" technique. In this approach smeared anisotropic properties are used to determine global stress states, then this information is used to interrogate stress states in specific regions of concern on a ply-by-ply basis, therefore avoiding the costly use of a ply-by-ply analysis for an entire structure made of a thick-section composite material. The use of this global-local analysis technique is commonly referred to as the most rational way to approach the problem of design and analysis for thick composite materials.

The solution methods available to generate smeared anisotropic 3-D properties range from approximate formulations (Reference 7.2.4.2(a)) to exact formulations not including bending-extensional coupling (Reference 7.2.4.2(c)). The exact solutions by Pagano (Reference 7.2.4.2(c)) and Sun (Reference 7.2.4.2(b)) lend themselves to simple programming on personal computers. In fact Trethewey et al. (Reference 7.2.4.2(d)) and Peros (Reference 7.2.4.2(e)) have encoded the Pagano solution while Sun has encoded his own solution for a personal computer.

Tables 7.2.4.2(a) and (b) contain 3D laminate elastic constants for six laminate configurations and two materials as determined by laminate plate theory (LPT), and by the Pagano, Sun, and Roy solutions (Reference 7.2.4.2(g), (h)). Table 7.2.4.1 lists the lamina input properties used in each of the analyses. The three exact solutions (LPT, Pagano, Sun) yield identical results for both inplane and through thickness properties for all of the cases presented. The results from the approximate solution by Roy differ from the others in the z-direction properties up to 12% in some cases.

Data verifying the results of these analyses are limited due to the difficulty in generating 3D experimental data. Data that does exist is documented in References 7.2.4.1(a) - (c), 7.2.4.2(h) and (i). Table 7.2.4.2(c) contains a comparison of theoretical predictions using the linear elastic theory by Pagano and experimental data from Reference 7.2.4.1(c) and Reference 7.2.4.2(i).

7.3 STRUCTURAL ANALYSIS METHODS FOR THICK-SECTION COMPOSITES.

7.3.1 Three-Dimensional Elasticity Methods.

7.3.2 Approximate Analytical Methods.

7.3.3 Finite Element Methods.

7.3.4 Case Studies.

7.4 PHYSICAL PROPERTY ANALYSIS REQUIRED FOR THICK-SECTION COMPOSITE THREE-DIMENSIONAL ANALYSIS.

7.4.1 Experimental Property Determination.

7.5 PROCESS ANALYSIS METHODS FOR THICK-SECTION COMPOSITES.

7.6 FAILURE CRITERIA.

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Table 7.2.4.2(a) *3D effective properties of various AS4/3501-6 laminates.*

Laminate Properties for AS4/3501-6, E and G in Msi												
	$[0_2/90]_s$				$[0/90]_{2s}$				$[0/90/\pm45]_s$			
	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy
E_x	11.5	11.5	11.5	11.5	9.01	9.00	9.00	9.00	6.68	6.68	6.68	6.67
E_y	6.48	6.47	6.47	6.47	9.01	9.00	9.00	9.00	6.68	6.68	6.68	6.68
E_z	--	1.80	1.80	1.65	--	1.82	1.82	1.60	--	1.82	1.82	1.61
ν_{xy}	.073	.073	.074	.072	.052	.052	.053	.052	.297	.297	.298	.296
ν_{xz}	--	.488	.489	.402	--	.506	.507	.438	--	.375	.376	.318
ν_{yz}	--	.519	.520	.465	--	.506	.508	.427	--	.375	.376	.317
G_{xy}	.870	.870	.870	.870	.870	.870	.870	.870	2.58	2.57	2.57	2.57
G_{xz}	--	.664	.664	.780	--	.593	.593	.612	--	.593	.593	.627
G_{yz}	--	.536	.536	.503	--	.593	.593	.573	--	.593	.593	.519

Laminate Properties for AS4/3501-6, E and G in Msi												
	$[\pm30]_{2s}$				$[\pm45]_{2s}$				$[\pm60]_{2s}$			
	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy
E_x	6.84	6.84	6.84	6.84	2.94	2.94	2.94	2.94	1.77	1.77	1.77	1.77
E_y	1.77	1.77	1.77	1.77	2.94	2.94	2.94	2.94	6.84	6.84	6.83	6.85
E_z	--	1.66	1.66	1.50	--	1.82	1.82	1.71	--	1.66	1.66	1.74
ν_{xy}	1.14	1.41	1.41	1.13	.691	.691	.691	.689	.295	.295	.295	.294
ν_{xz}	--	-.095	-.094	-.197	--	.165	.165	.211	--	.390	.390	.434
ν_{yz}	--	.390	.390	.434	--	.165	.165	.211	--	-.095	-.095	-.197
G_{xy}	3.43	3.43	3.42	3.42	4.28	4.28	4.27	4.27	3.43	3.43	3.42	3.42
G_{xz}	--	.705	.705	.708	--	.593	.593	.596	--	.512	.512	.515
G_{yz}	--	.512	.512	.515	--	.593	.593	.596	--	.705	.705	.708

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Table 7.2.4.2(b) *3D effective properties of various S2/3501-6 laminates.*

Laminate Properties for S2/3501-6, E and G in Msi												
	$[0_2/90]_s$				$[0/90]_{2s}$				$[0/90/\pm45]_s$			
	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy
E_x	5.52	5.52	5.52	5.52	4.68	4.68	4.68	4.68	3.89	3.89	3.89	3.89
E_y	3.83	3.83	3.83	3.83	4.68	4.68	4.68	4.68	3.89	3.89	3.89	3.89
E_z	--	2.38	2.38	2.30	--	2.40	2.40	2.29	--	2.40	2.40	2.32
ν_{xy}	.166	.166	.166	.165	.136	.136	.136	.135	.281	.281	.281	.280
ν_{xz}	--	.405	.405	.359	--	.435	.435	.393	--	.362	.362	.329
ν_{yz}	--	.459	.459	.427	--	.435	.435	.392	--	.362	.362	.329
G_{xy}	.980	.980	.980	.980	.980	.980	.980	.980	1.52	1.52	1.52	1.52
G_{xz}	--	.870	.870	.918	--	.823	.823	.830	--	.823	.823	.838
G_{yz}	--	.782	.782	.754	--	.823	.823	.811	--	.823	.823	.781

Laminate Properties for AS4/3501-6, E and G in Msi												
	$[\pm30]_{2s}$				$[\pm45]_{2s}$				$[\pm60]_{2s}$			
	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy	LPT	Pagano	Sun	Roy
E_x	4.45	4.45	4.45	4.54	2.88	2.88	2.88	2.88	2.26	2.26	2.26	2.26
E_y	2.26	2.26	2.26	2.26	2.88	2.88	2.88	2.88	4.45	4.45	4.45	4.45
E_z	--	2.30	2.30	2.16	--	2.40	2.40	2.33	--	2.30	2.30	2.44
ν_{xy}	.546	.546	.546	.545	.468	.468	.468	.467	.278	.278	.278	.277
ν_{xz}	--	.200	.200	.136	--	.267	.267	.284	--	.387	.387	.406
ν_{yz}	--	.387	.387	.406	--	.267	.267	.284	--	.200	.200	.136
G_{xy}	1.79	1.79	1.79	1.79	2.06	2.06	2.06	2.06	1.79	1.79	1.79	1.79
G_{xz}	--	.895	.895	.895	--	.823	.823	.823	--	.762	.762	.763
G_{yz}	--	.762	.762	.763	--	.823	.823	.823	--	.985	.985	.895

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Table 7.2.4.2(c) *Comparison of theoretical and experimental laminate results $[0_2/90]_{ns}$ from Reference 7.2.4.1(c), $[0_3/90]_{ns}$ from Reference 7.2.4.2(i).*

	AS4/3501-6 $[0_2/90]_{ns}$		S2 glass/3501-6 $[0_2/90]_{ns}$		AS4/3501-6 $[0_3/90]_{ns}$	
	Theor.	Exp.	Theor.	Exp.	Theor.	Exp.
E_x (Msi)	11.53	11.63[1] (4.0)[2](32)	5.52	5.82 (6.9)(32)	12.80	12.90[1]
E_y (Msi)	6.47		3.83		5.27	5.66[1]
E_z (Msi)	1.80		2.38		1.63	1.64[1]
ν_{xy}	0.073	0.069[1] (6.7)(7)[3]	0.166	0.166 (4.3)(7)	0.090	0.120[1]
ν_{xz}	0.488	0.469[1] (3.0)(14)	0.405	0.363 (2.7)(14)	0.440	---
ν_{yz}	0.519		0.459		0.452	---
G_{xy} (Msi)	0.87		0.98		0.87	0.70[4]
G_{xz} (Msi)	0.73		0.78		0.72	0.53[4]
G_{yz} (Msi)	0.63		0.64		0.54	0.66[4]

1 data from thick, flat, compression test specimens
2 coefficient of variation (%)
3 number of data points in average
4 data from Iosipescu shear test specimens

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Chapter 8. SUPPORTABILITY

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8.1 INTRODUCTION. The supportability of a structure in-service is to provide a function of maintaining the inherent strength and quality of a structure and to resist degradation due to *environmental hazards*.

Considerations in the design for and design of repairs in composite structures are based on maintainability and reliability factors. The maintainability of a composite structure is a measure of the ease with which it can be inspected, repaired, and if a separate part of the structure, replaced. Composite structures may possess characteristics that tend either to enhance or degrade maintainability. Considerations in the design of advanced structures basically involve fatigue life, damage resistance and damage tolerance. Desired in terms of reliability is a durable structure, capable of withstanding environmental and operational hazards, and substantially forgiving of routine damage. From a maintainability standpoint, inspectability and ease of repair are the main objectives.

The purpose of this section is to provide guidelines to the designer of new structures with supportability/maintainability in mind and to define guidance and information necessary to design simple, efficient and cost effective repair procedures.

8.2 DESIGN FOR REPAIR. Guidelines are provided to support the design of a new structure. Items for consideration are as outlined.

8.2.1 Inspectability. Designing of structures should include for the aspect of inspectability for potential damage. The methods chosen for inspection may be restricted due to the size and configuration of the component, limitations of the instrument for detecting certain types of damage, and operating characteristics of the instruments (environmental, health hazards, etc.).

8.2.1.1 *Methods of inspection.* A variety of nondestructive inspection techniques (NDI) are utilized as inspection tools for process-related and determining service-related defects in composite structures. However as in metallic structures, no single nondestructive inspection method can locate and isolate all defects. The designer should assess based on the structure and its configuration which methods of inspection are practical and usable to detect damage. See Volume I, Chapter 6.2 for description of inspection methods.

8.2.2 Material Selection. One of the major advantages in the utilization of advanced composite materials is the ability to tailor a structure by utilizing the specific material properties. Selective placement and orientation of fibers provides structural support and stiffness and in some cases, impact toughness. The matrix materials are for service and operational characteristics.

In the design of a given component, the designer needs to address when selecting materials that these materials have no limiting factor that would prevent it from being repairable. Such factors would include limited availability, poor handleability, deterioration due to subsequent processing and environmental degradation.

Specific tailoring of structures for fleet needs such as low observeability requires the selection of *materials that can perform the function of minimized signature for improved survivability. These* materials need to be repairable to restore structural integrity and signature of the component. The designer in the selection of the material must assess the electrical performance of the material if repaired and how easily it can be fielded.

For components that require materials that have a high-use temperature, the designer again must investigate the limiting factors of these material systems. Current practices do not afford repair concepts when utilizing these materials due to temperature processing degrading the surrounding

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parent and attaching structures. Materials used for the manufacture of components should be selected which will allow bonded repairs that do not have any limiting factors that would prevent a repair to be performed at temperatures at or below the service capability of the material.

Specific selection of materials for such properties as ballistic protection, extreme temperature exposure, hydrostatic pressure loading, EMI and RFI shielding, etc., requires the designer to analyze whether the material system chosen has the capabilities to be repaired, is functional after the repair or is it to be a replaceable article. Current practices for repair when utilizing these materials are remove and replace due to the lack of developed procedures for repair or the system when repaired no longer is fleet functional and should be avoided.

8.2.2.1 *Processability.* The selection of the processing of a composite component is based on the service functionality requirements of that component. The designer in selecting a fabrication process must address whether the process dictates that the component can be repaired if subsequently damaged.

Original processing may include heat and pressure, injection, thermal forming or wrapping which cannot in most practical applications be effected in a repair scheme. Such examples would include autoclave processing, filament winding, braiding, resin transfer or injection and hot-head laying.

Factors in defining a processing method based on repairability are developed repair schemes or remove and replace (module replacement) as the repair criteria but favors repairability. The fabrication process that is selected must be able to support cure conditions to attain the maximum required properties out of the system. The fabrication process must be able to afford repair processing without additional sophisticated equipment or loss of integrity to the component.

8.2.2.2 *Environmental requirements.* The environmental or service conditions can rapidly deteriorate the performance of a composite component if not properly accessed. The designer must investigate the operating or service characteristics of the structure individually and as part of the entire system to fully define the effect of the environmental parameters.

These environmental parameters include extreme temperature cycling, pressure cycling, exposure to moisture and other operating fluids, radiation or other bombardment e.g., erosion, EMI and RFI effects.

Protection schemes, methods of construction and material selection to protect the structure from environmental effects will require a secondary analysis for repairability. The environmental conditions may cause deterioration depending on the material system to the structure, permanently, by absorption and cannot be removed. Thus the structure must be designed to account for those losses and losses to the materials that are used in the repair.

8.2.2.3 *Resistance to damage.* In normal operating conditions, components can be expected to be subjected to potential damage from sources such as maintenance personnel, tools, runway debris, service equipment, hail, lightning, etc. During initial manufacturing and assembly, these components may be subject to the same or similar conditions. The given structure must be able to endure a reasonable level of such incidents without costly rework or downtime.

Minimum established levels of damage resistance are based on the type of structure and type of impact and the level of impact energy. In addition to strength and stiffness, the structure is zoned based on regions which have high or low susceptibility to damage and its required damage resistance. In defining the requirements, the type of structure; primary or secondary structure, construction

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method; sandwich or solid laminate, and whether its a removable or non-removable structure are pertinent.

As a general rule of thumb for design purposes, damage resistance is improved by utilizing thicker laminates and for sandwich applications, use of denser core materials and core materials with a minimum thickness - provides a degree of protection of the inner face. The selection of reinforcement fibers can also have a improved affect; use of high modulus, high strain fibers. Additionally, the selection of toughened matrix materials can greatly enhance damage resistance.

Other items include laminate layup; use of a layer of fabric as the exterior ply over tape to resist scratches, abrasion, softening of impact and reduction of fiber breakout. Laminate edge placement should not be positioned as exposed directly into the air stream - possibly subjected to delaminations. Utilize non-erosive edge protection, replaceable sacrificial materials or locate the forward edge below the level of the aft edge of the next panel forward. Resistance to damage requires structural robustness.

Areas prone to high energy lightning strike should utilize replaceable conductive materials, provide protection at tips and trailing edge surfaces and make all conductive path attachments easily accessible.

8.2.3 Damage tolerance. See Volume 3, Section 4.11.1 for damage tolerance criteria.

8.2.3.1 *In-service damage detectability*. See Volume 3, Section 4.11.1.2.

8.2.3.2 *Design strain allowables*. See Volume 3, Section 4 for criteria for design strain allowables.

8.2.4 Maintainability. The maintainability of a structure is achieved by developing schemes for methods of inspection and maintenance procedures during the design phase. The designer with the overall knowledge of the performance and operational characteristics of the structure should access, based on the construction method, and configuration, material selection, etc., whether the structure is maintainable. Such factors in assessment would include development of cradle-to-grave inspection methodology, techniques, protection schemes and defined inspection intervals for maintenance.

8.2.5 Removeability. An important aspect for the designer to utilize is the concept of removeability, the ability to remove a component or module from a structure for repair or replacement without damage to the existing and replacing structures.

The design would require the use of fastener systems that do not initiate damage upon removal and replacement. A bonded system may require the investigation of bonding agents that can be heated for removal and rebonding purposes.

8.2.6 Interchangeability. Interchangeability of high-unit cost, frequently-damaged or heavily loaded components that cannot be repaired should be a factor in the design of a structure. When damaged, a component is either repaired in place or removed, repaired and reinstalled. "Spares" or interchangeable components have not been traditionally utilized due to the lack of mandatory enforcement of MIL-I-8500, the use of interchangeable components.

Typically, components are trimmed and match drilled upon final assembly; essentially a custom, drilled installation. The designer in accessing this condition should tailor the structure so that it is easily removable. The process used for trimming and drilling of processed components should be standardized with the use of drill fixtures. The overall design should account for tolerancing factors

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which would prevent the use of interchangeable components. Interchangeability means no final trimming or match drilling of components.

8.2.7 Accessibility. Accessibility is an important factor when designing structures for repair. Sufficient access should always be provided to properly inspect, prepare the damage structure, fit and install the repair parts and use repair tools and bonding equipment. Limited access may dictate the repair approach, i.e., use of precured patches, use of mechanical fasteners in lieu of cocuring, etc. If feasible, two-sided access is preferred.

8.2.8 Repairability. *The repairability of a structure is influenced by several design factors:*

1. Type of Structure - The type of structure can be generally defined as primary, secondary or tertiary. Based on the structural loading condition, the type of repairs, and overall repair limitation can be assessed.

2. Design - The design of the overall structure, affects the repairability. Several of the state-of-the-art automated processing methods do not lend themselves to producing repairable structures but are primarily utilized to reduce production costs and maintain consistency in product properties. Repairability needs to be incorporated rather than simply removing and replacing. Methods of attachment are affected by design which could prevent repair of a structure.

3. Operating Parameters - Other parameters that can affect the repairability of a structure are its operating characteristics. In achieving such working parameters by the use of materials and construction methods, the repairability of a structure should be considered.

Structures that provide ballistic protection, armor, containment from radiation bombardment, chemical hazards or shielding requires special consideration to attempt to provide for repairability. In many cases, the protection is compromised or deteriorated beyond a repairable condition.

Other parameters would include for multiple repairs, flutter and mass balance. The design should allow to reestablish mass balance or allow for the increase of weight due to the repair.

8.2.5 Durability. See Volume 3, Section 4.11.2.

8.2.6 Margin of safety. The designer must take into account the margin of safety when designing a structure. Designs with a minimum or zero margin of safety should not be allowed. Margins of safety are based on such parameters as material properties, the effect of structural performance degradation due to exposure to moisture, light, thermal degradation and other operational characteristics. In turn, these margins of safety when coupled with a damaged structure must be sufficiently adequate to perform the mission requirements without failure. Margins need to be assessed for processing changes to affect a repair, change of materials, i.e., matrix materials and degradation due to undetected damage of a structure with environmental conditioning.

Current design practice is to use a fixed design allowable for compressively loaded laminated components and a fixed value for tension loaded laminated components. A location that is more susceptible to damage during service should be designed using lower strain levels. And locations where such threats are absent may be designed to larger strain levels, resulting in lighter, more efficient structures. The margin of safety should be based on the type of repair in a given location for a given type of structure.

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8.3 DESIGN OF REPAIRS. A repair has the objective of restoring a damaged structure to an acceptable capability in terms of strength, stiffness, functional performance, safety, cosmetic appearance or service life. Ideally, the repair will return the structure to original capability and appearance.

The design assessment of a repair for a given loading condition involves the selection of a repair concept, the choice of the appropriate repair materials and processes, then specifying the detailed configuration and size of the repair. Most repairs are basically designed as a joint to transfer load into and out of a patch. To ensure that the repair configuration will have adequate strength and stiffness, the repair joint must be analyzed to predict its strength.

The selection of the type of load-transfer joint to be used for a patch/strap is a tradeoff between simplicity, strength and stiffness. The easier configurations are generally not as strong as the more difficult ones. It is critical that the materials and process information is available prior to the system being put into place.

8.3.1 Damage assessment/classification.

8.3.2 Typical concepts.

8.3.3 Nondestructive inspection.

8.3.4 Effect of defects.

8.3.5 Repair/replace criteria.

8.3.5.1 *Levels of repair (zoning of structure).*

8.3.6 Design criteria.

8.3.7 Thermal zoning.

8.3.8 Thermal aging due to multiple/subsequent cure cycles - repair related.

8.3.9 Corrosion. The use of metallic hardware, e.g., fasteners, attaching hardware or materials, e.g., metal-filled adhesives, metallic core when contacted with graphite materials has resulted in galvanic corrosion. The repair materials shall be chosen so that they are galvanically compatible with themselves and the parent structure. Barrier plies and sealants are utilized to "break" the direct contact between these materials and minimizes corrosion. In addition, barrier ply materials should be included in the repair scheme.

8.3.10 Mean-time-to-repair.

8.3.11 Weight/mass balance.

8.3.12 Conductivity restoration. Composite structures are generally poor conductors of electricity. Electrically conductive materials may be incorporated in a structure to provide ground planes for antennas, discharge paths for lightning strikes and similar purposes. The materials used include tapes, fabrics, foils, mesh materials and coatings. When conductive treatments are damaged, they must be restored, either by replacing the original material or by using a substitute material to bridge the

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disrupted path. Complicated designs should not be considered; if utilized, have acceptable and workable procedures for repair.

8.3.13 Stress/strength criteria.

8.3.14 Compatibility with existing and surrounding structure.

8.3.15 Allowables.

8.3.16 Durability. Durability is another consideration in the design of repair joints. It is desirable that the repair joints withstand the static and fatigue stresses of the operating environment for the life of the structure without suffering environmental degradation. Some of the ways in which durability can be achieved are:

1. Avoidance of thermal mismatches between adherends which can produce residual bond stresses.
2. Avoidance of structural eccentricities which can cause stress risers and lead to repair joint deterioration and failure.
3. Attention to fastener fit and spacing to properly distribute fastener bearing loads.
4. Application of knockdown factors to account for environmental effects.
5. Adequate sealing to prevent moisture absorption and entrapment.
6. Repair should not degrade design fatigue life.
7. Bond durability is achieved by proper preparation of surfaces and cleanliness precautions are applied.

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